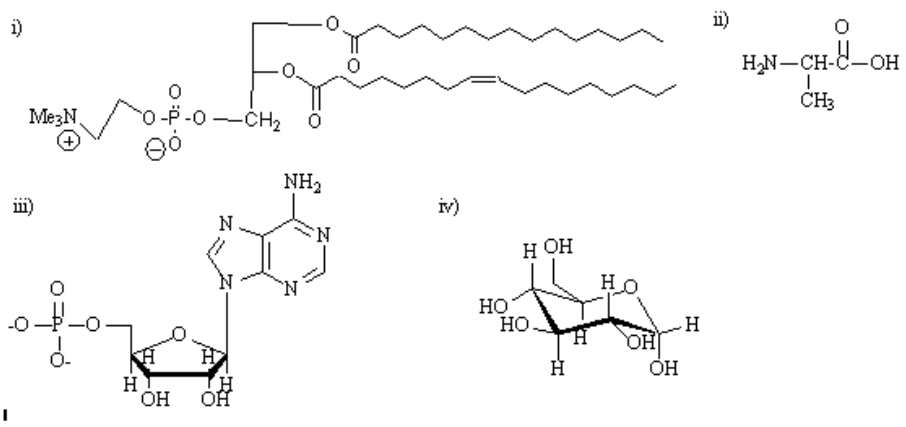
Medicinal Chemistry

Full-Test Bank

Prepared by: Amr Al-najjar

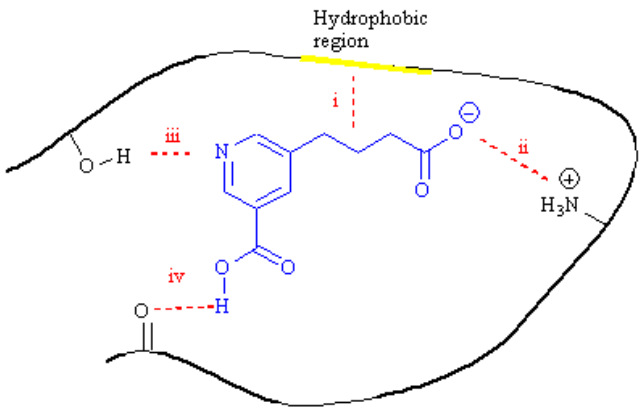
Type: multiple choice		
Title: Chapter 01 - Question 01		
01)	Which of the following molecules is a phospholipid?  	
Feedback: The 'phospho' of phospholipid indicates the presence of a phosphate group. The lipid part of the name refers to the two long hydrocarbon chains. The structure is also known as a phosphoglyceride. Structure ii) is an amino acid, structure iii) is an oligonucleotide, and structure iv is a carbohydrate.		
Correct	**a.**#	i
incorrect	**b.**#	ii
incorrect	**c.**#	iii
incorrect	**d.**#	iv
END OF QUESTION		

Type: multiple choice		
Title: Chapter 01 - Question 02		
02)	Which of the following statements is false regarding the phospholipid bilayer in cell membranes?	
Feedback: The cell membrane is made up of a bilayer of phospholipid molecules where the hydrophobic tails interact with each other in the centre of the membrane by van der Waals interactions. The polar head groups interact with water at the outer and inner surfaces of the membrane. The tails of the phospholipid molecules are hydrophobic (water hating) and repel water and ions. The phospholipid molecules are certainly fluid, but no pores are formed as a result of that.		
incorrect	**a.**#	It is made up of two layers of phospholipid molecules with the tails interacting with each other.
incorrect	**b.**#	Water and ions are unable to cross the bilayer due to the hydrophobic tails of the phospholipid molecules.
incorrect	**c.**#	There are charged groups at the inner and outer surfaces of the cell membrane.
Correct	**d.**#	The molecules in the bilayer are fluid and so the cell membrane is porous allowing the passage of ions and water across the cell membrane.
END OF QUESTION		

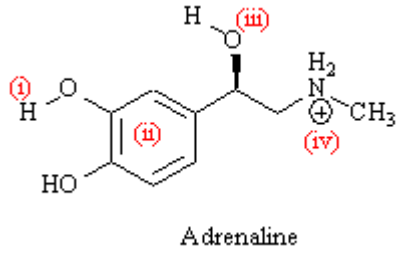
Type: multiple choice
Title: Chapter 01 - Question 03

03)	Which of the following statements is false?	
Feedback: Some drugs do form covalent bonds to their targets, but most drugs interact by intermolecular binding interactions. The other statements are true. Most drug targets are large molecules (macromolecules) that are far bigger than the drugs that interact with them. There are some inorganic compounds used in medicine, but the vast majority of drugs are organic molecules.		
incorrect	**a.**#	Drug targets are normally macromolecules.
incorrect	**b.**#	Drugs are generally smaller than drug targets.
incorrect	**c.**#	Drugs are generally organic molecules.
Correct	**d.**#	Drugs usually act by forming covalent bonds with drug targets.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 01 - Question 04		
04)	Which of the following statements is false with respect to a binding site?	
Feedback: The binding site is normally a hydrophobic hollow or cleft on the surface of a macromolecular target. In the binding site, there are functional groups and regions that form binding interactions with complementary groups on a drug.		
incorrect	**a.**#	It is part of a macromolecule that acts as a drug target.
incorrect	**b.**#	It contains binding regions.
incorrect	**c.**#	It is normally a hollow or cleft on the surface of the drug target.
Correct	**d.**#	It is usually hydrophilic in nature.
END OF QUESTION		

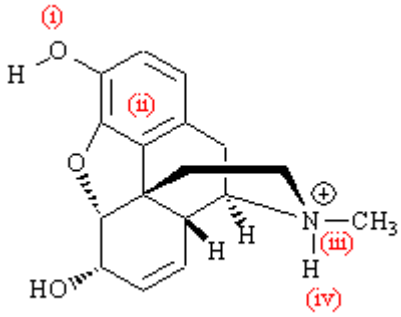
Type: matching			
Title: Chapter 01 - Question 05			
05)	Consider the molecule in blue bound to a binding site and identify the binding interactions (i-iv) that are formed by the various parts of the molecule to the binding site.		
			
Feedback: Van der Waals interactions take place between regions of molecules consisting solely of carbon and hydrogen atoms. Ionic bonds occur between charges of opposite sign. Hydrogen bonds occur between a molecule acting as a hydrogen bond donor (RXH; where X is normally oxygen or nitrogen), and a molecule acting as hydrogen bond acceptor where a heteroatom is present (normally oxygen or nitrogen).			
a.#	i	=	Van der Waals interactions

b.#	ii	=	Ionic bond
c.#	iii	=	Hydrogen bond acceptor
d.#	iv	=	Hydrogen bond donor
END OF QUESTION			

Type: matching	
Title: Chapter 01 - Question 06	
06)	Identify the binding interactions that might be possible at the specified positions of the following drug:  Adrenaline -

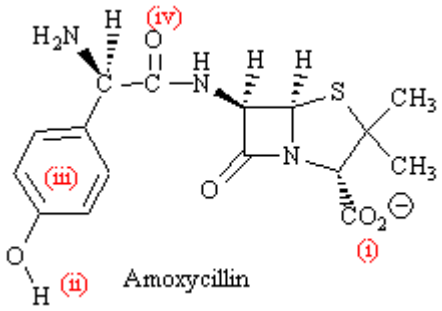
Feedback: Van der Waals interactions take place between molecular regions consisting solely of carbon and hydrogen atoms; for example alkyl groups or aromatic rings. Ionic bonds occur between charges of opposite sign. Hydrogen bonds occur between a molecule acting as hydrogen bond donor (RXH; where X is normally oxygen or nitrogen), and a molecule acting as hydrogen bond acceptor where a heteroatom is present (normally oxygen or nitrogen).

a.#	i	=	Hydrogen bond donor
b.#	ii	=	Van der Waals interactions
c.#	iii	=	Hydrogen bond acceptor
d.#	iv	=	Ionic interactions
END OF QUESTION			

Type: matching	
Title: Chapter 01 - Question 07	
07)	Identify the binding interactions that might be possible at the specified positions of the following drug.  Morphine

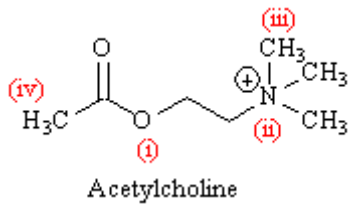
Feedback: Van der Waals interactions take place between molecular regions consisting solely of carbon and hydrogen atoms; for example alkyl groups or aromatic rings. Ionic bonds occur between charges of opposite sign. Hydrogen bonds occur between a molecule acting as hydrogen bond donor (RXH; where X is normally oxygen or nitrogen), and a molecule acting as hydrogen bond acceptor where a heteroatom is present

(normally oxygen or nitrogen).			
a.#	i	=	Hydrogen bond acceptor
b.#	ii	=	Van der Waals interactions
c.#	iii	=	Ionic interactions
d.#	iv	=	Hydrogen bond donor
END OF QUESTION			

Type: matching	
Title: Chapter 01 - Question 08	
08)	Identify the binding interactions that might be possible at the specified positions of the following drug.  Amoxycillin

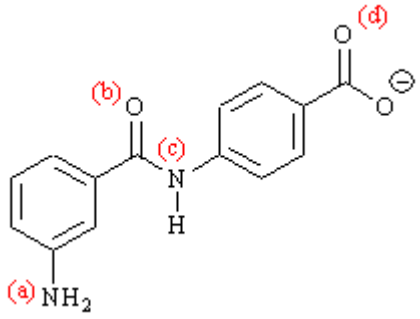
Feedback: Van der Waals interactions take place between molecular regions consisting solely of carbon and hydrogen atoms; for example alkyl groups or aromatic rings. Ionic bonds occur between charges of opposite sign. Hydrogen bonds occur between a molecule acting as hydrogen bond donor (RXH; where X is normally oxygen or nitrogen), and a molecule acting as hydrogen bond acceptor where a heteroatom is present (normally oxygen or nitrogen).

a.#	i	=	Ionic interactions
b.#	ii	=	Hydrogen bond donor
c.#	iii	=	Van der Waals interactions
d.#	iv	=	Hydrogen bond acceptor
END OF QUESTION			

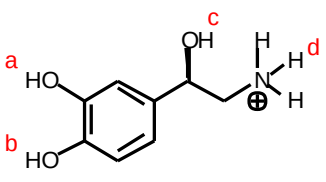
Type: matching	
Title: Chapter 01 - Question 09	
09)	Identify the binding interactions that might be possible at the specified positions of the following drug.  Acetylcholine

Feedback: Van der Waals interactions take place between molecular regions consisting solely of carbon and hydrogen atoms; for example alkyl groups or aromatic rings. Ionic bonds occur between charges of opposite sign. Hydrogen bonds occur between a

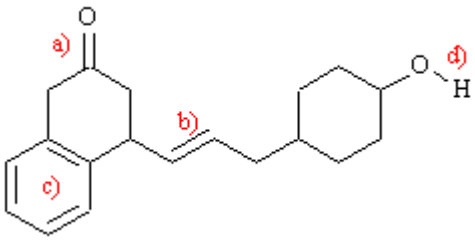
molecule acting as hydrogen bond donor (RXH; where X is normally oxygen or nitrogen), and a molecule acting as hydrogen bond acceptor where a heteroatom is present (normally oxygen or nitrogen).			
a.#	i	=	Hydrogen bond acceptor
b.#	ii	=	Ionic interactions
c.#	iii	=	Van der Waals interactions (1)
d.#	iv	=	Van der Waals interactions (2)
END OF QUESTION			

Type: multiple choice	
Title: Chapter 01 - Question 10	
10)	Which of the following atoms in the structure below is likely to be the strongest hydrogen bond acceptor? 

Feedback: Generally, the more electron rich the heteroatom, the better it will be as a hydrogen bond acceptor. Thus, a negatively charged oxygen will be the best hydrogen bond acceptor. The negative charge in the carboxylate ion is shared between the two oxygens by resonance, and so each of the carboxylate oxygens will be a better hydrogen bond acceptor than the amide oxygen. This in turn will be a better hydrogen bond acceptor than either of the nitrogens in the structure. The amide and aniline nitrogens will be poor hydrogen bond acceptors since the lone pair of electrons in each case interacts with a neighbouring group (carbonyl group and aromatic ring respectively). Therefore, it is not available to form hydrogen bonds.		
incorrect	**a.**#	The anilino nitrogen.
incorrect	**b.**#	The carbonyl oxygen.
incorrect	**c.**#	The amide nitrogen.
Correct	**d.**#	The carboxylate oxygen.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 01 - Question 11	
11)	Which of the following atoms is likely to be the strongest hydrogen bond donor? 

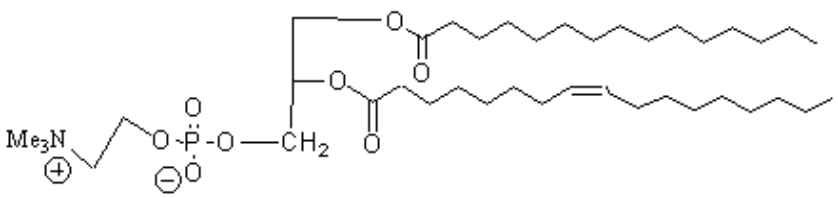
Feedback: The best hydrogen bond donor will be the one where the hydrogen is the most electron deficient. In the alkylammonium ion, the nitrogen has a positive charge and this in turn will make the attached hydrogens electron deficient. The phenol groups are likely to be the next best hydrogen bond donors since the oxygen's lone pair of electrons will interact with the aromatic ring giving it a slightly positive charge.		
incorrect	**a.**#	The phenol group *meta* to the side chain.
incorrect	**b.**#	The phenol group *para* to the side chain.
incorrect	**c.**#	The alcohol group.
Correct	**d.**#	The alkylammonium ion.
END OF QUESTION		

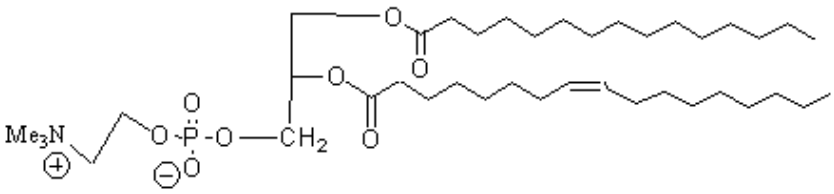
Type: multiple choice	
Title: Chapter 01 - Question 12	
12)	Which of the following functional groups is most likely to participate in a dipole-dipole interaction? 

Feedback: The ketone will have the strongest dipole moment due to the electronegative oxygen polarising the carbonyl bond. The alcohol will also have a significant dipole moment, but less than that of the ketone. The alkene and aromatic rings may have small or negligible dipole moments depending on their substituents.		
Correct	**a.**#	Ketone.
incorrect	**b.**#	Alkene.
incorrect	**c.**#	Aromatic ring.
incorrect	**d.**#	Alcohol.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 01 – Question 13		
13)	Which of the following statements is false?	
Feedback: Water molecules surrounding a hydrophobic region of a drug form an ordered layer of molecules with **low** entropy.		
incorrect	**a.**#	Desolvation is an energy expensive process which involves the removal of water from polar functional groups prior to a drug binding to its binding site.
Correct	**b.**#	Water molecules surrounding a hydrophobic region of a drug form an ordered layer of molecules with high entropy.
incorrect	**c.**#	Interaction between the non polar regions of a drug and the non polar regions of a target require the removal of an ordered water coat and represents a gain in binding energy due to increased entropy.
incorrect	**d.**#	An increase in entropy results in a greater negative value of ΔG and a greater chance of binding.

END OF QUESTION

Type: multiple choice		
Title: Chapter 01 - Question 14		
14)	Which of the following statements is true about the following structure? 	
Feedback: The head group includes a positively charged nitrogen and a negatively charged phosphate group. As a result, it is hydrophilic and polar. The hydrocarbon tails are ‘water hating' i.e. hydrophobic.		
incorrect	**a.**#	It has a polar head group and hydrophilic 'tails'.
Correct	**b.**#	It has a polar head group and hydrophobic 'tails'.
incorrect	**c.**#	It has a non polar head group and hydrophilic 'tails'
incorrect	**d.**#	It has a non polar head group and hydrophobic 'tails'
END OF QUESTION		

Type: multiple choice		
Title: Chapter 01 - Question 15		
15)	What is the role of the following structure in cell membranes? 	
Feedback: The cell membrane is made up of a bilayer of phospholipid molecules where the hydrophobic tails interact with each other in the centre of the membrane and the polar head groups interact with water at the outer and inner surfaces of the membrane. Protein receptors in the cell membrane are involved in communication. Carbohydrates can act as antigens. Adenosine triphosphate is an example of a molecule that acts as an energy source.		
incorrect	**a.**#	The structure plays an important role in communication between cells.
incorrect	**b.**#	The structure acts as an antigen or chemical 'fingerprint' for cells.
Correct	**c.**#	The structure is present as a structural component of cell membranes.
incorrect	**d.**#	The structure is an energy source.
END OF QUESTION		

Type: multiple choice
Title: Chapter 01 - Question 16

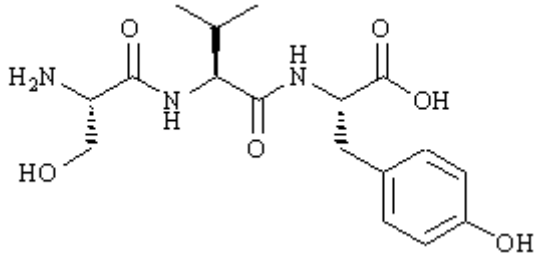
16)	What general term is used to describe large biologically important molecules?	
Feedback: Macro- means large and so the term macromolecule is used. Macromolecules can be considered to be polymeric in the sense that there may be repeating segments (e.g. repeating peptide bonds in proteins) but polymers are generally regarded as having repeating units (e.g. A-B-A-B-A-B etc). This is not true of many non-structural macromolecules. Proteins are examples of macromolecules, but not all macromolecules are proteins. Oligomers is a term used to describe a smaller version of a larger molecule (e.g. an oligonucleotide is a small portion of a larger nucleic acid.)		
incorrect	**a.**#	Polymers.
Correct	**b.**#	Macromolecules.
incorrect	**c.**#	Proteins.
incorrect	**d.**#	Oligomers.
END OF QUESTION		

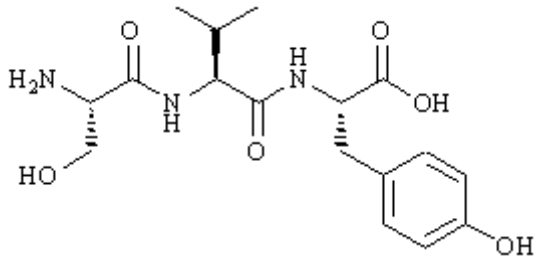
Type: matching			
Title: Chapter 01 - Question 17			
17)	Match the following terms to the descriptions.		
Feedback: The binding site is normally a hollow or cleft on the macromolecular target into which a drug can bind. Specific binding regions within that site 'recognise' specific binding groups on drugs (or ligands) to allow binding interactions to take place.			
a.#	A hollow or cleft present on the surface of a macromolecule to which a drug can bind	=	Binding site
b.#	Functional groups on a drug which are used to bind it to the drug target	=	Binding groups
c.#	A region of the drug target which can bind to a specific functional group present on a drug	=	Binding region
END OF QUESTION			

Type: multiple choice		
Title: Chapter 01 - Question 18		
18)	What are the relative strengths of different types of bonds?	
Feedback: Covalent bonds are strongest since they involve the sharing of electrons between two atoms. Ionic interactions are electrostatic in nature and involve interactions between opposite charges. Hydrogen bonding can be seen as a weak form of ionic interaction. Van der Waals interactions are weakest since they involve attractions between transient regions of charge.		
incorrect	**a.**#	Covalent > H-bond > Ionic > Van der Waals.
incorrect	**b.**#	Van der Waals > Ionic > H-bond > Covalent.
Correct	**c.**#	Covalent > Ionic > H-bond > Van der Waals.
incorrect	**d.**#	Ionic > Covalent > Van der Waals > H-bond.
END OF QUESTION		

Type: multiple choice
Title: Chapter 01 - Question 19

19)	Which of the following interactions has a directional property?	
Feedback: The hydrogen bond involves an interaction between the hydrogen of a hydrogen bond donor, and a lone pair of electrons on the heteroatom of a hydrogen bond acceptor. The orbital containing the lone pair of electrons will have a directional property as will the X-H bond of the hydrogen bond donor. This is not the case with the other interactions.		
incorrect	**a.**#	Van der Waals interactions.
incorrect	**b.**#	Hydrophobic interactions.
Correct	**c.**#	Hydrogen bond.
incorrect	**d.**#	Ionic bond.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 02 - Question 01		
01)	Identify the correct name for the following peptide. 	
Feedback: Peptides are named, starting from the amino end. L-tyrosyl-L-valyl-L-serine is wrong since it is named from the carboxyl end. L-seryl-L-valyl-L-phenylalanine is wrong since tyrosine has been wrongly identified as phenylalanine. L-phenylalanyl-L-valyl-L-serine is wrong since the labelling is from the wrong end of the molecule and tyrosine has been wrongly identified as phenylalanine.		
Correct	**a.#**	L-seryl-L-valyl-L-tyrosine.
incorrect	**b.#**	L-seryl-L-valyl-L-phenylalanine.
incorrect	**c.#**	L-phenylalanyl-L-valyl-L-serine.
incorrect	**d.#**	L-tyrosyl-L-valyl-L-serine.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 02 - Question 02		
02)	Which of the following single letter codes represents the structure below? 	
Feedback: The order of the amino acids from the *N*-terminal end of the peptide is serine (S), valine (V) and tyrosine (Y). YVS is wrong since the amino acids are labelled from the C-terminal end. SVF is wrong since it mislabels tyrosine as phenylalanine. FVS has the same mistakes as SVF and is also labelled from the wrong end.		
incorrect	**a.#**	YVS.
Correct	**b.#**	SVY.
incorrect	**c.#**	FVS.
incorrect	**d.#**	SVF.
END OF QUESTION		

Type: multiple choice
Title: Chapter 02 - Question 03

03)	What is unique about glycine with respect to other natural amino acids?	
Feedback: Glycine is the only amino acid not to have a residue attached to the alpha carbon. The other three characteristics are shared with the rest of the natural amino acids.		
incorrect	**a.#**	It can form peptide bonds with other amino acids.
Correct	**b.#**	It has no side chain.
incorrect	**c.#**	It contains an amine functional group and a carboxylic acid functional group attached to the same carbon.
incorrect	**d.#**	It exists as a zwitterion.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 02 - Question 04		
04)	Which of the following statements is true about a peptide bond (RCONHR')?	
Feedback: 'It is planar' is correct. The peptide bond has double bond character due to the interaction of the nitrogen lone pair with the carbonyl group. This prevents bond rotation and makes the bond planar. The other options are false. The NH of the amide can act as a hydrogen bond donor and the carbonyl group can act as a hydrogen bond acceptor. The **trans** isomer is favoured over the **cis** isomer. Single bond rotation is not allowed due to the double bond character of the group.		
Correct	**a.#**	It is planar.
incorrect	**b.#**	It is not capable of forming a hydrogen bond.
incorrect	**c.#**	The **cis** configuration is favoured over the **trans** configuration.
incorrect	**d.#**	Single bond rotation is permitted between nitrogen and the carbonyl group.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 02 - Question 05		
05)	Which of the following statements is untrue about protein secondary structure?	
Feedback: The steric influence of amino acid residues is important to secondary structure. In secondary structure, the residues are positioned to minimise interactions between each other and the peptide chain.		
incorrect	**a.#**	The alpha-helix, beta-pleated sheet, and beta-turns are examples of protein secondary structure.
incorrect	**b.#**	The ability of peptide bonds to form intramolecular hydrogen bonds is important to secondary structure.
Correct	**c.#**	The steric influence of amino acid residues is not important to secondary structure.
incorrect	**d.#**	The hydrophilic/hydrophobic character of amino acid residues is not important to secondary structure.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 02 - Question 06		
06)	Which of the following statements is untrue about protein tertiary structure?	
Feedback: In general, proteins fold up such that as many hydrophobic residues as		

possible are positioned in the centre of the protein such that they interact with each other rather than with water. Hydrophilic amino acids can interact favourably with water and are more likely to be on the outside. Several amino acid side chains interact with each other by means of ionic and hydrogen bonding.		
Correct	**a.#**	Proteins fold up into a tertiary structure such that most amino acids with hydrophobic residues are exposed to the aqueous surroundings.
incorrect	**b.#**	Proteins fold up into a tertiary structure such that most amino acids with hydrophilic residues are exposed to the aqueous surroundings.
incorrect	**c.#**	Proteins fold up into a tertiary structure such that most amino acids with hydrophobic residues are in the centre and hidden from the aqueous surroundings.
incorrect	**d.#**	Interactions between amino acid residues are important in protein tertiary structure.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 02 - Question 07	
07)	Which of the following descriptions best describes the primary structure of proteins?
Feedback: Primary structure refers to the order of the amino acids in a protein. Secondary structure indicates regions of ordered structure, and tertiary structure is the overall structure of the protein. Quaternary structure refers to the structure of a multi-protein complex.	
incorrect	**a.#** The arrangement of different protein subunits in a multiprotein complex.
Correct	**b.#** The order in which amino acids are linked together in a protein.
incorrect	**c.#** The overall three dimensional shape of a protein.
incorrect	**d.#** Regions of ordered structure within a protein.
END OF QUESTION	

Type: multiple choice	
Title: Chapter 02 - Question 08	
08)	Which of the following descriptions best describes the secondary structure of proteins?
Feedback: Primary structure refers to the order of the amino acids in a protein. Secondary structure indicates regions of ordered structure, and tertiary structure is the overall structure of the protein. Quaternary structure refers to the structure of a multi-protein complex.	
incorrect	**a.#** The arrangement of different protein subunits in a multiprotein complex.
incorrect	**b.#** The order in which amino acids are linked together in a protein.
incorrect	**c.#** The overall three dimensional shape of a protein.
Correct	**d.#** Regions of ordered structure within a protein.
END OF QUESTION	

Type: multiple choice	
Title: Chapter 02 - Question 09	
09)	Which of the following descriptions best describes the tertiary structure of proteins?

Feedback: Primary structure refers to the order of the amino acids in a protein. Secondary structure indicates regions of ordered structure, and tertiary structure is the overall structure of the protein. Quaternary structure refers to the structure of a multi-protein complex.		
incorrect	**a.#**	The arrangement of different protein subunits in a multiprotein complex.
incorrect	**b.#**	The order in which amino acids are linked together in a protein.
Correct	**c.#**	The overall three dimensional shape of a protein.
incorrect	**d.#**	Regions of ordered structure within a protein.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 02 - Question 10	
10)	Which of the following descriptions best describes the quaternary structure of proteins?

Feedback: Primary structure refers to the order of the amino acids in a protein. Secondary structure indicates regions of ordered structure, and tertiary structure is the overall structure of the protein. Quaternary structure refers to the structure of a multi-protein complex.		
Correct	**a.#**	The arrangement of different protein subunits in a multiprotein complex.
incorrect	**b.#**	The order in which amino acids are linked together in a protein.
incorrect	**c.#**	The overall three dimensional shape of a protein.
incorrect	**d.#**	Regions of ordered structure within a protein.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 02 - Question 11	
11)	Which of the following statements is not true regarding protein tertiary structure?

Feedback: Van der Waals interactions may be weak but they have an important role in protein tertiary structure since there are so many of them.		
Correct	**a.#**	Van der Waals interactions between hydrophobic residues are the least important factors in tertiary structure.
incorrect	**b.#**	Covalent bonds can have an influence on tertiary structure.
incorrect	**c.#**	Hydrogen bonds, ionic bonds and van der Waals interactions all have a role to play in tertiary structure.
incorrect	**d.#**	Planar peptide bonds have an indirect influence on protein tertiary structure .
END OF QUESTION		

Type: multiple choice	
Title: Chapter 02 - Question 12	
12)	Identify the strongest form of intermolecular bonding that could be formed involving the residue of the amino acid threonine.

Feedback: Threonine contains a hydroxyl functional group on its side chain and so the strongest possible interaction will be hydrogen bonding where the hydroxyl group could act as a hydrogen bond donor or hydrogen bond acceptor.		
incorrect	**a.#**	Ionic bond.
Correct	**b.#**	Hydrogen bond.
incorrect	**c.#**	Van der Waals interactions.

incorrect	**d.#**	None of the options are correct.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 02 - Question 13		
13)	Identify the strongest form of intermolecular bonding that could be formed involving the residue of the amino acid histidine	
Feedback: Histidine contains a heterocyclic ring with two nitrogen atoms. In the neutral form, there is an NH group where the proton can act as a hydrogen bond donor. However, the ring can be protonated resulting in a positively charged nitrogen which could form a stronger ionic interaction.		
Correct	**a.#**	Ionic bond.
incorrect	**b.#**	Hydrogen bond.
incorrect	**c.#**	Van der Waals interactions.
incorrect	**d.#**	Dipole-dipole interactions.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 02 - Question 14		
14)	Identify the strongest form of intermolecular bonding that could be formed involving the residue of the amino acid alanine	
Feedback: Alanine has no functional groups on its side chain. There is only a methyl group and so only van der Waals interactions are possible.		
incorrect	**a.#**	Ionic bond.
incorrect	**b.#**	Hydrogen bond.
Correct	**c.#**	Van der Waals interactions.
incorrect	**d.#**	Dipole-dipole interactions.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 02 - Question 15		
15)	Identify the strongest form of intermolecular bonding that could be formed involving the residue of the amino acid arginine	
Feedback: Arginine contains a basic functional group on its side chain which could take part in hydrogen bonding. However, it can also ionise allowing a stronger ionic interaction.		
Correct	**a.#**	Ionic bond.
incorrect	**b.#**	Hydrogen bond.
incorrect	**c.#**	Van der Waals interactions.
incorrect	**d.#**	Dipole-dipole interactions.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 02 - Question 16	
16)	Identify the strongest form of intermolecular bonding that could be formed involving the residue of the amino acid aspartic acid.
Feedback: Aspartic acid contains a carboxylic acid functional group which could act as a hydrogen bond donor or hydrogen bond acceptor. However, the group could lose its acidic proton to gain a negative charge allowing a stronger ionic interaction to take place.	

Correct	**a.#**	Ionic bond.
incorrect	**b.#**	Hydrogen bond.
incorrect	**c.#**	Van der Waals interactions.
incorrect	**d.#**	Dipole-dipole interactions.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 02 - Question 17		
17)	Identify the strongest form of intermolecular bonding that could be formed involving the residue of the amino acid asparagine.	
Feedback: Asparagine contains a primary amide functional group on its side chain and so the strongest possible interaction will be hydrogen bonding where the carbonyl group can act as a hydrogen bond donor and the NH_2 protons can act as hydrogen bond donors. The nitrogen cannot act as a hydrogen bond acceptor since the lone pair is interacting with the carbonyl group of the amide group to produce partial double bond character.		
incorrect	**a.#**	Ionic bond.
Correct	**b.#**	Hydrogen bond.
incorrect	**c.#**	Van der Waals interactions.
incorrect	**d.#**	Dipole-dipole interactions.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 02 - Question 18		
18)	Identify the strongest form of intermolecular bonding that could be formed involving the residue of the amino acid serine.	
Feedback: Serine contains a hydroxyl functional group on its side chain and so the strongest possible interaction will be hydrogen bonding where the hydroxyl group could act as a hydrogen bond donor or hydrogen bond acceptor.		
incorrect	**a.#**	Ionic bond.
Correct	**b.#**	Hydrogen bond.
incorrect	**c.#**	Van der Waals interactions.
incorrect	**d.#**	Dipole-dipole interactions.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 02 - Question 19		
19)	Which of the following statements is incorrect regarding transport proteins?	
Feedback: Transport proteins carry polar molecules across the cell membrane. Non-polar molecules can dissolve through the hydrophobic cell membrane.		
incorrect	**a.#**	They are present in cell membranes.
Correct	**b.#**	They serve to carry non polar molecules across the cell membrane.
incorrect	**c.#**	They are required to transport amino acids across cell membranes.
incorrect	**d.#**	They are not required to transport steroids across cell membranes.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 02 - Question 20		
20)	Which of the following statements best describes an amino acid residue?	
Feedback: The residue is the side chain that is attached to the alpha-carbon of the head group. The head group consists of the amino group, the carboxyl group, and the alpha-carbon atom. The portion of the amino acid remaining when a peptide bond is formed includes both the head group and the residue, since only water is lost from the condensation reaction. The structure formed by hydrolysis of a peptide or protein is the amino acid itself.		
incorrect	**a.#**	The portion of an amino acid that remains once a peptide bond has been formed.
incorrect	**b.#**	The structure that is formed following the hydrolysis of a peptide or protein.
Correct	**c.#**	The side chain of an amino acid.
incorrect	**d.#**	The head group of an amino acid.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 02 - Question 21		
21)	Which of the following statements is incorrect regarding transport proteins?	
Feedback: Steroids are hydrophobic molecules which can pass through the cell membrane without the need for a transport protein.		
incorrect	**a.#**	They are present in cell membranes.
incorrect	**b.#**	They serve to carry polar molecules across the hydrophobic cell membrane.
incorrect	**c.#**	They are required to transport amino acids across cell membranes.
Correct	**d.#**	They are required to transport steroids across cell membranes.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 03 - Question 01		
01)	Which of the following statements is not true regarding the active site of an enzyme?	
Feedback: An active site is normally a hollow or cleft on the surface of an enzyme, rather than the surface itself.		
Correct	**a.#**	An active site is normally on the surface of an enzyme.
incorrect	**b.#**	An active site is normally hydrophobic in nature.
incorrect	**c.#**	Substrates fit into active sites and bind to functional groups within the active site.
incorrect	**d.#**	An active site contains amino acids which are important to the binding process and the catalytic mechanism.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 03 - Question 02		
02)	Some enzymes have a binding site which is not recognised by the normal substrate, and affects the activity of the enzyme if it is occupied by a ligand. What term is used for such a binding site?	
Feedback: The allosteric binding site affects the activity of an enzyme, depending on whether it is occupied by a ligand or not. It is distinct from the active site where the enzyme catalysed reaction takes place. The terms secondary binding site and inhibitory binding site do not exist.		
incorrect	**a.#**	Active site.
Correct	**b.#**	Allosteric binding site.
incorrect	**c.#**	Secondary binding site.
incorrect	**d.#**	Inhibitory binding site.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 03 - Question 03		
03)	What term is used to describe the binding site of an enzyme where a reaction is catalysed?	
Feedback: The active site is where the reaction catalysed by the enzyme takes place. The allosteric binding site affects the activity of an enzyme, depending on whether it is occupied by a ligand or not. It is distinct from the active site. The terms secondary binding site and inhibitory binding site do not exist.		
Correct	**a.#**	Active site.
incorrect	**b.#**	Allosteric binding site.
incorrect	**c.#**	Secondary binding site.
incorrect	**d.#**	Inhibitory binding site.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 03 - Question 04		
04)	Which of the following terms refers to the process by which a substrate binds to an active site and alters the shape of that active site?	
Feedback: An induced fit is a consequence of the binding interactions taking place between a substrate and the active site. The term moulded fit does not exist. The lock and key theory is an older theory where it was proposed that the active site was already		

the correct shape for its substrate.		
incorrect	**a.#**	Moulded fit.
incorrect	**b.#**	Intermolecular bonding.
Correct	**c.#**	Induced fit.
incorrect	**d.#**	Lock and key theory.
END OF QUESTION		

Type: matching			
Title: Chapter 03 - Question 05			
05)	Identify which of the amino acids; glutamate, phenylalanine, threonine, and serine, would use the following interactions in an active site; hydrogen bonding, ionic bonding, van der Waals interactions, H-bonding, and van der Waals interactions.		
Feedback: Serine has a hydroxyl group which can take part in hydrogen bonding. Glutamate is the ionised form of glutamic acid and can take part in ionic interactions. Phenylalanine has an aromatic ring in its side chain that can interact by van der Waals interactions. Tyrosine has an aromatic ring that can interact by van der Waals interactions, and a phenol that can interact by hydrogen bonding.			
a.#	Hydrogen bonding	=	Serine
b.#	Ionic bonding	=	Glutamate
c.#	Van der Waals interactions	=	Phenylalanine
d.#	Hydrogen bonding and van der Waals bonding	=	Tyrosine
END OF QUESTION			

Type: multiple choice		
Title: Chapter 03 - Question 06		
06)	Which of the following descriptions best describes a cofactor?	
Feedback: A cofactor is a non-protein substance required by an enzyme to function properly. This term covers both organic molecules and metal ions. A co-enzyme is a non-protein organic molecule that is required by some enzymes in order to catalyse a reaction on a substrate. A prosthetic group is a non-protein organic molecule that is bound covalently to the active site of an enzyme, and which is required if the enzyme is to catalyse a reaction on a substrate. A substrate is a compound which is bound to the active site and undergoes a reaction.		
Correct	**a.#**	A non-protein substance that is required by an enzyme if it is to catalyse a reaction.
incorrect	**b.#**	A non-protein organic molecule that is required by some enzymes in order to catalyse a reaction on a substrate.
incorrect	**c.#**	A non-protein organic molecule that is bound covalently to the active site of an enzyme, and which is required if the enzyme is to catalyse a reaction on a substrate.
incorrect	**d.#**	A compound which is bound to the active site and undergoes a reaction.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 03 - Question 07	
07)	Which of the following descriptions best describes a coenzyme?

Feedback: A cofactor is a non-protein substance required by an enzyme to function properly. This term covers both organic molecules and metal ions. A co-enzyme is a non-protein organic molecule that is required by some enzymes in order to catalyse a reaction on a substrate. A prosthetic group is a non-protein organic molecule that is bound covalently to the active site of an enzyme, and which is required if the enzyme is to catalyse a reaction on a substrate. A substrate is a compound which is bound to the active site and undergoes a reaction.		
incorrect	**a.#**	A non-protein substance that is required by an enzyme if it is to catalyse a reaction.
Correct	**b.#**	A non-protein organic molecule that is required by some enzymes in order to catalyse a reaction on a substrate.
incorrect	**c.#**	A non-protein organic molecule that is bound covalently to the active site of an enzyme, and which is required if the enzyme is to catalyse a reaction on a substrate.
incorrect	**d.#**	A compound which is bound to the active site and undergoes a reaction.
END OF QUESTION		

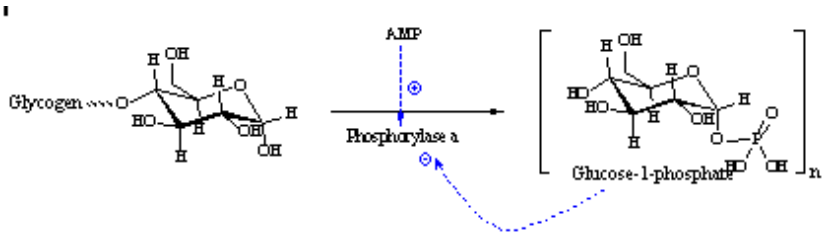
Type: multiple choice	
Title: Chapter 03 - Question 08	
08)	Which of the following descriptions best describes a prosthetic group?

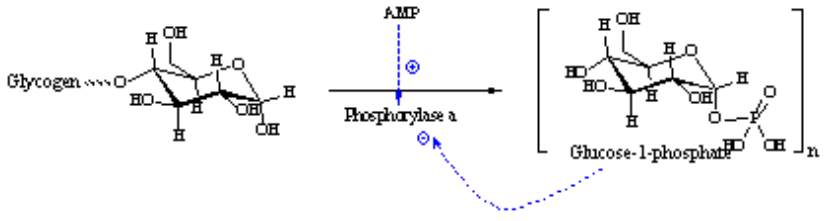
Feedback: A cofactor is a non-protein substance required by an enzyme to function properly. This term covers both organic molecules and metal ions. A co-enzyme is a non-protein organic molecule that is required by some enzymes in order to catalyse a reaction on a substrate. A prosthetic group is a non-protein organic molecule that is bound covalently to the active site of an enzyme, and which is required if the enzyme is to catalyse a reaction on a substrate. A substrate is a compound which is bound to the active site and undergoes a reaction.		
incorrect	**a.#**	A non-protein substance that is required by an enzyme if it is to catalyse a reaction.
incorrect	**b.#**	A non-protein organic molecule that is required by some enzymes in order to catalyse a reaction on a substrate.
Correct	**c.#**	A non-protein organic molecule that is bound covalently to the active site of an enzyme, and which is required if the enzyme is to catalyse a reaction on a substrate.
incorrect	**d.#**	A compound which is bound to the active site and undergoes a reaction.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 03 - Question 09	
09)	Which of the following descriptions best describes a substrate?

Feedback: A cofactor is a non-protein substance required by an enzyme to function properly. This term covers both organic molecules and metal ions. A co-enzyme is a non-protein organic molecule that is required by some enzymes in order to catalyse a reaction on a substrate. A prosthetic group is a non-protein organic molecule that is bound covalently to the active site of an enzyme, and which is required if the enzyme is to catalyse a reaction on a substrate. A substrate is a compound which is bound to the active site and undergoes a reaction.		
incorrect	**a.#**	A non-protein substance that is required by an enzyme if it is to catalyse a reaction.
incorrect	**b.#**	A non-protein organic molecule that is required by some

		enzymes in order to catalyse a reaction on a substrate.
incorrect	**c.#**	A non-protein organic molecule that is bound covalently to the active site of an enzyme, and which is required if the enzyme is to catalyse a reaction on a substrate.
Correct	**d.#**	A compound which is bound to the active site and undergoes a reaction.
END OF QUESTION		

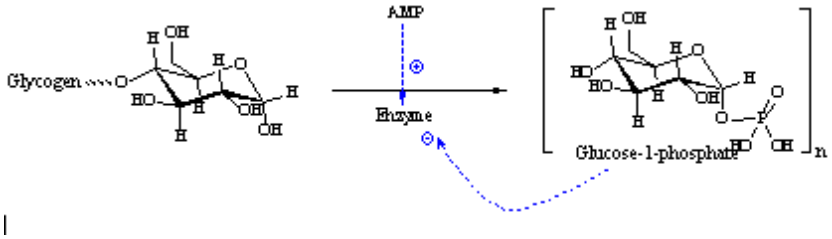
Type: multiple choice		
Title: Chapter 03 - Question 10		
10)	Consider the following enzyme-catalysed reaction carried out on glycogen. What is the product? 	
Feedback: Glycogen is the substrate for this reaction. Adrenaline is a hormone that can trigger a cascade of cellular events that will eventually activate phosphorylase *a*. AMP acts as an enzyme modulator to directly activate the enzyme. Glucose is the structure formed if the phosphate group is hydrolysed from glucose-1-phosphate.		
Correct	**a.#**	Glucose-1-phosphate.
incorrect	**b.#**	Adrenaline.
incorrect	**c.#**	Glucose.
incorrect	**d.#**	AMP.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 03 - Question 11		
11)	Consider the following enzyme catalysed reaction. What is the substrate? 	
Feedback: Glycogen is the substrate for this reaction. Phosphorylase *a* is the enzyme. AMP acts as an enzyme modulator to directly activate the enzyme. Glucose-1-phosphate is the product.		
incorrect	**a.#**	Glucose-1-phosphate.
incorrect	**b.#**	Phosphorylase *a*.
Correct	**c.#**	Glycogen.
incorrect	**d.#**	AMP.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 03 - Question 12		
12)	Consider the following enzyme-catalysed reaction. What is the enzyme?  	
Feedback: Glycogen is the substrate for this reaction. Phosphorylase *a* is the enzyme. AMP acts as an enzyme modulator to directly activate the enzyme. Glucose-1-phosphate is the product.		
incorrect	**a.#**	Glucose-1-phosphate.
Correct	**b.#**	Phosphorylase *a*.
incorrect	**c.#**	Glycogen.
incorrect	**d.#**	AMP.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 03 - Question 13		
13)	Consider the following enzyme-catalysed reaction. What is the enzyme modulator that activates the enzyme?  	
Feedback: Glycogen is the substrate for this reaction. Phosphorylase *a* is the enzyme. AMP acts as an enzyme modulator to directly activate the enzyme. Glucose-1-phosphate is the product.		
incorrect	**a.#**	Glucose-1-phosphate.
incorrect	**b.#**	Phosphorylase *a*.
incorrect	**c.#**	Glycogen.
Correct	**d.#**	AMP.
END OF QUESTION		

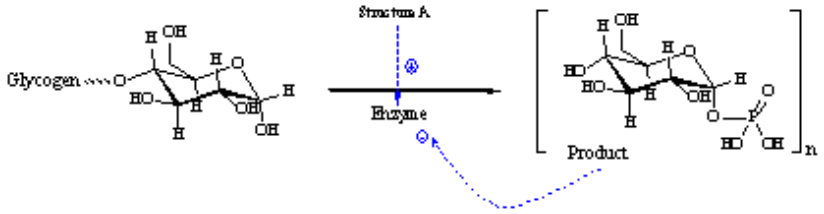
Type: multiple choice	
Title: Chapter 03 - Question 14	
14)	Consider the following enzyme-catalysed reaction. What sort of enzyme is involved?

Feedback: The reaction shown is a phosphorylation and so the enzyme involved is a phosphorylase. Esterases catalyse the hydrolysis of esters. Ligases catalyse the joining of two substrates. Transferases catalyse group transfer reactions.

incorrect	**a.#**	Esterase.
Correct	**b.#**	Phosphorylase .
incorrect	**c.#**	Ligase.
incorrect	**d.#**	Transferas.

END OF QUESTION

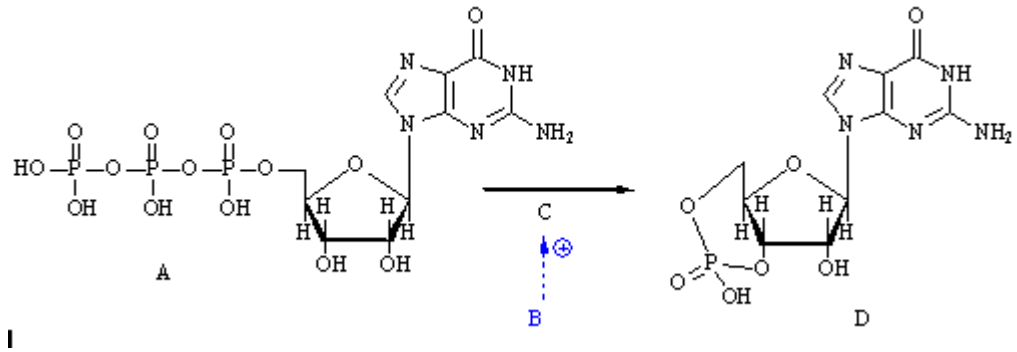
Type: multiple choice	
Title: Chapter 03 - Question 15	
15)	Consider the following enzyme-catalysed reaction. What chemical messenger triggers a cascade of cellular event that eventually activates the enzyme involved in this reaction? 

Feedback: Adrenaline is a hormone that can trigger the signalling cascade. The other agents mentioned are neurotransmitters which do not have an affect on this reaction.

incorrect	**a.#**	Acetylcholine.
incorrect	**b.#**	Noradrenaline.
incorrect	**c.#**	Serotonin.
Correct	**d.#**	Adrenaline.

END OF QUESTION

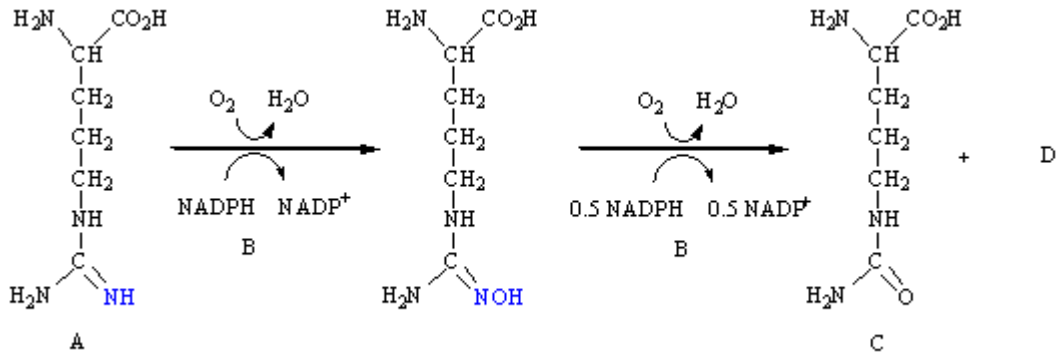
Type: matching	
Title: Chapter 03 - Question 16	
16)	Identify A-D in the following reaction.

	 I

Feedback: Nitrous oxide activates cyclase enzymes to generate the secondary messenger cyclic GMP from GTP.

a.#	A	=	Guanosine triphosphate
b.#	B	=	Nitric oxide
c.#	C	=	Cyclase
d.#	D	=	Cyclic GMP

END OF QUESTION

Type: matching	
Title: Chapter 03 - Question 17	
17)	Identify A-D in the following diagram. 

Feedback: Arginine contains a guanidine functional group that provides the nitrous oxide that is eventually formed in this reaction. The enzyme NO synthase converts arginine to citrulline in two steps with release of nitrous oxide.

a.#	A	=	Arginine
b.#	B	=	NO synthase
c.#	C	=	Citrulline
d.#	D	=	Nitric oxide

END OF QUESTION

Type: multiple choice
Title: Chapter 03 - Question 18

18)	Which of the following statements is not true regarding the active site of an enzyme?	
Feedback: An active site is normally hydrophobic in nature, not hydrophilic. Nevertheless, there are usually hydrophilic amino acids present which are important in binding the substrate in the active site.		
incorrect	**a.#**	An active site is normally a hollow or cleft on the surface of an enzyme.
Correct	**b.#**	An active site is normally hydrophilic in nature.
incorrect	**c.#**	Substrates fit into active sites and bind to functional groups within the active site.
incorrect	**d.#**	An active site contains amino acids which are important to the binding process and the catalytic mechanism.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 03 - Question 19		
19)	Which of the following amino acids commonly acts as a nucleophilic group in enzyme-catalysed reaction mechanisms?	
Feedback: Serine has a hydroxyl functional group that is frequently involved as a nucleophile in enzyme catalysed reactions. Cysteine is another amino acid that is commonly present as a nucleophile in enzyme active sites. Phenylalanine and valine are more likely to be involved in binding interactions. Histidine is commonly present as an acid-base catalyst.		
Correct	**a.#**	Serine.
incorrect	**b.#**	Phenylalanine.
incorrect	**c.#**	Histidine.
incorrect	**d.#**	Valine.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 03 - Question 20		
20)	Which of the following amino acids acts as a nucleophilic group in enzyme-catalysed reaction mechanisms?	
Feedback: Cysteine has a thiol functional group that is frequently involved as a nucleophile in enzyme catalysed reactions. Serine is another amino acid that is commonly present as a nucleophile in enzyme active sites. Threonine, tyrosine and glutamine are more likely to be involved in binding interactions.		
incorrect	**a.#**	Threonine.
incorrect	**b.#**	Tyrosine.
Correct	**c.#**	Cysteine.
incorrect	**d.#**	Glutamine.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 03 - Question 21		
21)	Which of the following amino acids acts as an acid-base catalyst in enzyme-catalysed reaction mechanisms?	
Feedback: Histidine is a weak base that can equilibrate between a protonated, ionised form and a non-protonated free base. It is, therefore, well placed to act as an acid-base catalyst. Phenylalanine, serine, and tryptophan can all be involved in substrate binding, while		

serine may act as a nucleophile.		
incorrect	**a.#**	Serine.
incorrect	**b.#**	Phenylalanine.
Correct	**c.#**	Histidine.
incorrect	**d.#**	Tryptophan.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 03 - Question 22		
22)	From which amino acid is nitrous oxide generated?	
Feedback: Arginine contains a guanidine functional group that provides the nitrous oxide. The enzyme NO synthase converts arginine to citrulline with release of nitrous oxide.		
Correct	**a.#**	Arginine.
incorrect	**b.#**	Aspartic acid.
incorrect	**c.#**	Asparagine.
incorrect	**d.#**	Lysine.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 03 - Question 23		
23)	What term is used for enzymes such as COX-1 and COX-2 which vary in structure and location but which catalyse the same reaction?	
Feedback: Isozymes is the correct term. Isosteres are chemical moieties that have similar size, shape, and valency. Isotopes are elements having the same number of protons, but a different number of neutrons. Isomers are molecules with the same molecular formula but differ in the way the atoms and bonds are linked together or are orientated with respect to each other.		
incorrect	**a.#**	Isosteres.
Correct	**b.#**	Isozymes.
incorrect	**c.#**	Isotopes.
incorrect	**d.#**	Isomers.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 03 - Question 24		
24)	Which of the following statements is true with respect to the Michaelis constant?	
Feedback: The Michaelis constant is determined by measuring how the enzyme-catalysed rate of reaction is affected by substrate concentration. Inhibitors are not involved and so options b) and c) are wrong. The Michaelis constant is the concentration of substrate at which the rate of reaction is half of its maximum value.		
Correct	**a.#**	It is equal to the concentration of substrate at which the reaction rate is half of its maximum value.
incorrect	**b.#**	It is equal to the concentration of inhibitor at which the reaction rate is half of its maximum value.
incorrect	**c.#**	It is equal to the concentration of substrate at which the reaction rate is at its maximum value.

incorrect	**d.#**	It is equal to the concentration of inhibitor at which the reaction rate is zero.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 04 - Question 01		
01)	Which of the following statements is **not** true about receptors?	
Feedback: Some protein receptors are certainly present inside cells, but the majority are present in the cell membrane. The binding site of receptors is analogous to the active site of enzymes. However, no reaction is catalysed. The ligand for the binding site acts as a chemical messenger. It binds and causes an induced fit that result in 'knock on effects' which lead to a message being received in the cell. Receptors do not catalyse reactions and so a messenger departs unchanged.		
Correct	**a.#**	Most receptors are proteins situated inside the cell.
incorrect	**b.#**	Receptors contain a hollow or cleft on their surface which is known as a binding site.
incorrect	**c.#**	Receptors bind chemical messengers such as neurotransmitters or hormones.
incorrect	**d.#**	Receptors do not catalyse reactions on chemical messengers.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 04 - Question 02		
02)	Which of the following statements is **not** true?	
Feedback: The receptors for neurotransmitters are located in the cell membranes of target cells and not within the cell itself.		
incorrect	**a.#**	Neurotransmitters are released by nerves.
incorrect	**b.#**	Neurotransmitters are required to carry a 'message' from a nerve to a target cell.
incorrect	**c.#**	Neurotransmitters only have small distances to cover to reach their target cells.
Correct	**d.#**	Neurotransmitters bind to receptors within target cells.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 04 - Question 03		
03)	Which of the following is **not** a neurotransmitter?	
Feedback: Cyclic GMP is a secondary messenger that is formed 'downstream' of a receptor-activated process.		
incorrect	**a.#**	Glycine.
Correct	**b.#**	Cyclic GMP.
incorrect	**c.#**	γ-Aminobutyric acid.
incorrect	**d.#**	Serotonin.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 04 - Question 04		
04)	Which of the following statements is **not** true regarding the binding site of a receptor?	
Feedback: The binding site is a hollow or cleft in the surface of the protein and is usually hydrophobic in character.		
incorrect	**a.#**	The binding site is normally a hollow or cleft in the surface of an receptor.
Correct	**b.#**	The binding site is normally hydrophilic in nature.

incorrect	**c.#**	Chemical messengers fit into binding sites and bind to functional groups within the binding site.
incorrect	**d.#**	The binding site contains amino acids which are important to the binding process.
END OF QUESTION		

Type: matching			
Title: Chapter 04 - Question 05			
05)	Which of the following amino acid residues is likely to form its strongest binding interaction by means of:		
Feedback: Valine has an alkyl side chain capable of forming van der Waals interactions. Glycine has no side chain or residue and so cannot form any interactions. Serine has an alcohol functional group capable of hydrogen bonding. Aspartate has an ionised carboxylate group which is capable of ionic bonding. It is also possible for this group to interact by hydrogen bonding as a hydrogen bond acceptor, but this will be weaker than an ionic bond.			
a.#	Hydrogen bonding	=	Serine
b.#	Ionic bonding	=	Aspartate
c.#	Van der Waals interactions	=	Valine
d.#	None of these options	=	Glycine
END OF QUESTION			

Type: multiple choice		
Title: Chapter 04 - Question 06		
06)	Which of the following statements best describes an induced fit?	
Feedback: An induced fit refers to the way the binding site changes shape as a result of the binding interactions formed between it and the chemical messenger. An induced fit cannot take place before the chemical messenger binds. The chemical messenger must be in the correct conformation to bind effectively, but this is a process which is not controlled by the binding site. The likelihood of a molecule being in the active conformation will depend on how stable the active conformation is relative to other inactive conformations.		
incorrect	**a.#**	The process by which a binding site alters shape such that it is ready to accept a chemical messenger.
incorrect	**b.#**	The process by which a chemical messenger adopts the correct binding conformation before entering a binding site.
Correct	**c.#**	The process by which binding of a chemical messenger to a binding site alters the shape of the binding site.
incorrect	**d.#**	The process by which a binding site alters the shape of the chemical messenger into the binding conformation before binding.
END OF QUESTION		

Type: matching			
Title: Chapter 04 - Question 07			
07)	Which of the following amino acid residues is likely to form its strongest binding interaction by means of:		
Feedback: Alanine has an alkyl side chain capable of forming van der Waals interactions. Glycine has no side chain or residue. Threonine has an alcohol functional group capable of hydrogen bonding. Lysine has an amino group which could be protonated to form an ionised centre capable of ionic bonding. It is also possible for this group to interact by hydrogen bonding, either in the ionised or non-ionised forms, but this will be weaker than an ionic bond.			
a.#	Hydrogen bonding	=	Threonine

b.#	Ionic bonding	=	Lysine
c.#	Van der Waals interactions	=	Alanine
d.#	None of the answers	=	Glycine
END OF QUESTION			

Type: multiple choice		
Title: Chapter 04 - Question 08		
08)	There is a fine balance required for the binding interactions of a neurotransmitter with its receptor. Which of the following statements best expands on this statement?	
Feedback: All the statements are accurate, but the following statement best describes the importance of having an equilibrium between the bound and unbound states of the chemical messenger: 'The binding interactions must be sufficiently strong that the neurotransmitter binds long enough to have an effect, but not too strong in case the neurotransmitter remains permanently bound.' If the interaction was too weak, the messenger would not be bound long enough to have an effect. If too strong, the receptor would be continually activated.		
incorrect	**a.#**	It is important that the binding interactions involve a mixture of van der Waals interactions, hydrogen bonds and ionic bonds since neurotransmitters have different functional groups.
incorrect	**b.#**	The binding interactions must be of the correct nature to match the functional groups of the neurotransmitter and the functional groups in the binding site.
Correct	**c.#**	The binding interactions must be sufficiently strong that the neurotransmitter binds long enough to have an effect, but not too strong in case the neurotransmitter remains permanently bound.
incorrect	**d.#**	There must be the correct balance of hydrophilic and hydrophobic interactions to ensure that the chemical messenger can enter a hydrophobic binding site.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 04 - Question 09		
09)	When a membrane-bound receptor binds its chemical messenger, an induced fit takes place which leads to secondary effects, allowing a chemical message to be received within the cell. Which of the following mechanisms is not involved in this process?	
Feedback: Transport proteins are involved in transporting molecules across cell membranes, and are not receptors. If a receptor is present in the cell membrane, there is no need for a chemical messenger to enter the cell. If the receptor is within the cell, the chemical messengers are generally hydrophobic and can pass through the cell membrane on their own account.		
Correct	**a.#**	The transport of the chemical messenger into the cell.
incorrect	**b.#**	The opening or closing of an ion channel.
incorrect	**c.#**	The activation of a signal protein.
incorrect	**d.#**	The activation of a membrane-bound enzyme.
END OF QUESTION		

Type: multiple choice

Title: Chapter 04 - Question 10		
10)	Which of the following statements is not true about a ligand-gated ion channel receptor?	
Feedback: Ligand-gated ion channels consist of five glycoproteins and not six. The other statements are correct. Receptors that are controlled by the membrane potential are known as voltage-gated ion channels.		
incorrect	**a.#**	Ligand-gated ion channel receptors are present in the cell membrane.
incorrect	**b.#**	Neurotransmitters can act as the chemical messengers for ligand-gated ion channels.
Correct	**c.#**	Ligand-gated ion channels consist of six glycoproteins.
incorrect	**d.#**	Some ion channel receptors are controlled by differences in membrane potential rather than by ligands.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 04 - Question 11		
11)	Which of the following statements is not true about ligand-gated ion channels?	
Feedback: The response time of ligand-gated ion channels to neurotransmitters is fast rather than slow. The other statements are correct.		
incorrect	**a.#**	There are different ligand-gated ion channels for specific ions.
Correct	**b.#**	The response time to a neurotransmitter is slow.
incorrect	**c.#**	The process by which an ion channel opens as a result of receptor-ligand binding is known as gating.
incorrect	**d.#**	The opening of cationic ion channels generally leads to depolarisation of the cell.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 04 - Question 12		
12)	Which of the following statements is true about the ion channel controlled by the nicotinic receptor?	
Feedback: Three of the options are wrong for the following reasons: Each protein subunit of the ion channel contains **four** hydrophobic regions which traverse the cell membrane **Two** of the protein subunits are identical (the α-subunits) The binding site for the nicotinic receptor is **shared** between the α-protein subunit and a neighbouring protein subunit. There are two binding sites per ion channel. Therefore the following option is correct: 'Each protein subunit of the ion channel has a lengthy *N*-terminal extracellular chain.'		
incorrect	**a.#**	Each protein subunit of the ion channel contains three hydrophobic regions which traverse the cell membrane.
incorrect	**b.#**	Three of the protein subunits are identical (the α-subunits).
incorrect	**c.#**	The binding site for the nicotinic receptor is located solely on the α-subunit which means that there are two binding sites per ion channel.
Correct	**d.#**	Each protein subunit of the ion channel has a lengthy *N*-terminal extracellular chain.
END OF QUESTION		

Type: multiple choice

Title: Chapter 04 - Question 13		
13)	The nicotinic receptor controls an ion channel. Which of the following statements is true about the α-subunit of this ion channel?	
Feedback: The *N*-terminal and *C*-terminal chains are both extracellular. There are two intracellular loops and only one extracellular loop.		
incorrect	**a.#**	The *N*-terminal and *C*-terminal chains are both intracellular.
incorrect	**b.#**	There is one intracellular loop.
Correct	**c.#**	The binding site for the neurotransmitter resides solely on the *N*-terminal chain.
incorrect	**d.#**	There are two extracellular loops
END OF QUESTION		

Type: multiple choice		
Title: Chapter 04 - Question 14		
14)	The mechanism of gating involves the rotation of five kinked α-helices which traverse the cell membrane. Which of the following statements is untrue?	
Feedback: It is the α-helix of the second transmembrane section that is involved and not the first. The other statements are correct.		
incorrect	**a.#**	Each protein subunit making up the ion channel contributes one of the kinked α-helices.
Correct	**b.#**	It is the α-helix of the first transmembrane section that is involved.
incorrect	**c.#**	Rotation of the helices opens up a central channel to allow the flow of ions.
incorrect	**d.#**	The neurotransmitter binds to the *N*-terminal chain to produce a rapid response.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 04 - Question 15		
15)	Which of the following statements is true about a G-protein coupled receptor?	
Feedback: There are **seven** transmembrane hydrophobic regions. The number of intracellular and extracellular loops is the **same**. The *N*-terminal chain is **extracellular** and the *C*-terminal chain is **intracellular**. The binding region for the G-protein involves one of the intracellular loops and the *C*-terminal chain, as stated.		
incorrect	**a.#**	It contains five transmembrane hydrophobic sections.
Incorrect	**b.#**	There are more extracellular loops than intracellular loops.
Correct	**c.#**	The binding region for the G-protein involves one of the intracellular loops and the *C*-terminal chain.
Incorrect	**d.#**	The *N*-terminal chain is intracellular and the *C*-terminal chain is extracellular.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 04 - Question 16		
16)	Which of the following is not a G-protein coupled receptor?	
Feedback: The nicotinic receptor is an ion channel.		
incorrect	**a.#**	The histamine receptor.
Correct	**b.#**	The nicotinic receptor.

incorrect	**c.#**	The dopaminergic receptor.
incorrect	**d.#**	The adrenergic receptor.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 04 - Question 17		
17)	Which of the following statements is not true about G-protein coupled receptors?	
Feedback: Growth factors are ligands for kinase-linked receptors.		
incorrect	**a.#**	They contain seven transmembrane sections.
incorrect	**b.#**	They mediate the action of some hormones.
incorrect	**c.#**	They activate signal proteins called G-proteins.
Correct	**d.#**	Growth factors can act as ligands for some G-protein coupled receptors.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 04 - Question 18		
18)	Which of the following pairs of receptors illustrates convergent evolution?	
Feedback: Both the H1 and H2 receptors bind histamine, but they have evolved from different branches of the evolutionary tree. The other pairings are examples of divergent evolution since they have evolved from the same branch.		
incorrect	**a.#**	The $D1_B$ and $D1_A$ dopaminergic receptor subtypes.
Correct	**b.#**	The H1 and H2 receptor subtypes.
incorrect	**c.#**	The α_{2a} and α_{2b} adrenoceptor subtypes.
incorrect	**d.#**	The M2 and M4 muscarinic receptor subtypes.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 04 - Question 19		
19)	Which of the following statements is true about a tyrosine kinase linked receptor?	
Feedback: The receptor is situated in the **cell membrane**. The *N*-terminal chain is **extracellular** and this is where the ligand binding site is. The receptor has one hydrophobic transmembrane region as stated.		
incorrect	**a.#**	It is situated in the cytoplasm.
incorrect	**b.#**	The *N*-terminal chain is intracellular.
incorrect	**c.#**	The ligand binding site is in the *C*-terminal chain.
Correct	**d.#**	It has one hydrophobic transmembrane region.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 04 - Question 20		
20)	Which of the following is a messenger for a tyrosine kinase linked receptor?	
Feedback: Adrenaline, dopamine, and histidine are neurotransmitters that activate G-		

protein coupled receptors. The epidermal growth factor activates a tyrosine kinase linked receptor.		
incorrect	**a.#**	Adrenaline.
incorrect	**b.#**	Dopamine.
Correct	**c.#**	Epidermal growth factor.
incorrect	**d.#**	Histidine.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 04 - Question 21	
21)	Which of the following statements is untrue regarding the epidermal growth factor receptor?
Feedback: Activation of the kinase results in the phosphorylation of tyrosine residues and not serine residues. It is a tyrosine kinase and not a serine-threonine kinase.	

incorrect	**a.#**	Binding of epidermal growth factor results in dimerisation of the receptor.
incorrect	**b.#**	Dimerisation activates the kinase active site.
Correct	**c.#**	Activation of the kinase results in phosphorylation of serine residues.
incorrect	**d.#**	One half of the receptor dimer catalyses reactions on residues present on the other half.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 04 - Question 22	
22)	Which of the following statements is true regarding intracellular receptors?
Feedback: Intracellular receptors are **single** proteins. They contain a ligand binding site near the ***C*-terminal** end. They contain a binding region for DNA near the middle of the protein and they are activated by hydrophobic molecules which are **extracellular** messengers and cross the cell membrane to reach their target.	

incorrect	**a.#**	They consist of three protein subunits.
incorrect	**b.#**	They contain a ligand binding site near the *N*-terminal end.
Correct	**c.#**	They contain a binding region for DNA near the middle of the protein.
incorrect	**d.#**	They are activated by hydrophobic molecules which are synthesised within the cell.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 04 - Question 23	
23)	Which of the following statements is true regarding the DNA binding region of intracellular receptors?
Feedback: The DNA binding region of intracellular receptors contains nine cysteine residues, **eight** of which are involved in binding two zinc ions. The receptor does not contain nucleotides. The DNA binding region is known as the **zinc finger** domains.	

Correct	**a.#**	It contains nine cysteine residues.
incorrect	**b.#**	Four cysteine residues are involved in binding two zinc ions.
incorrect	**c.#**	It contains particular nucleotide sequences that can base pair to DNA.

incorrect	**d.#**	The DNA binding region is known as having 'thiol fingers'.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 04 - Question 24		
24)		Which of the following statements is not true about a ligand-gated ion channel receptor?
Feedback: Ligand-gated ion channels are controlled by ligands and not by the membrane potential. Voltage-gated ion channels are controlled by membrane potential.		
incorrect	**a.#**	Ligand-gated ion channel receptors are present in the cell membrane.
incorrect	**b.#**	Neurotransmitters can act as the chemical messengers for ligand-gated ion channels.
incorrect	**c.#**	Ligand-gated ion channels consist of five glycoproteins.
Correct	**d.#**	Differences in membrane potential affect whether ligand-gated ion channel receptors open or close.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 04 - Question 25		
25)		Which of the following statements is not true about ligand-gated ion channels?
Feedback: The opening of cationic ion channels generally leads to **depolarisation** of the cell.		
incorrect	**a.#**	There are different ligand-gated ion channels for specific ions.
incorrect	**b.#**	The response time to a neurotransmitter is extremely fast.
incorrect	**c.#**	The process by which an ion channel opens as a result of receptor-ligand binding is known as gating.
Correct	**d.#**	The opening of cationic ion channels generally leads to polarisation of the cell.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 04 - Question 26		
26)		Which of the following statements is true about the ion channel controlled by the nicotinic receptor?
Feedback: There are **four** hydrophobic regions for each protein subunit. The binding site is **shared** between the α-protein subunit and a neighbouring protein subunit. However, there are two binding sites per ion channel. The *N*-terminal chain is **extracellular** and not intracellular.		
incorrect	**a.#**	Each protein subunit of the ion channel contains three hydrophobic regions which traverse the cell membrane.
Correct	**b.#**	Two of the protein subunits are identical (the α-subunit).
incorrect	**c.#**	The binding site for the nicotinic receptor is located solely on the α-subunit which means that there are two binding sites per ion channel.
incorrect	**d.#**	Each protein subunit of the ion channel has a lengthy *N*-terminal intracellular chain.
END OF QUESTION		

Type: multiple choice

Title: Chapter 04 - Question 27		
27)	The nicotinic receptor controls an ion channel. Which of the following statements is true about the α-subunit of this ion channel?	
Feedback: There are **two** intracellular loops. The binding site for the neurotransmitter resides solely on the ***N*-terminal** chain. There is only **one** extracellular loop.		
Correct	**a.#**	The *N*-terminal and *C*-terminal chains are both extracellular.
incorrect	**b.#**	There is one intracellular loop.
incorrect	**c.#**	The binding site for the neurotransmitter resides solely on the *C*-terminal chain.
incorrect	**d.#**	There are two extracellular loops.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 04 - Question 28		
28)	The mechanism of gating involves the rotation of five kinked α-helices which traverse the cell membrane. Which of the following statements is untrue?	
Feedback: The neurotransmitter produces a rapid response but it binds to the ***N*-terminal chain**.		
incorrect	**a.#**	Each protein subunit making up the ion channel contributes one of the kinked α-helices.
incorrect	**b.#**	It is the α-helix of the second transmembrane section that is involved.
incorrect	**c.#**	Rotation of the helices opens up a central channel to allow the flow of ions.
Correct	**d.#**	The neurotransmitter binds directly to the TM2 section to produce a rapid response.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 04 - Question 29		
29)	How many transmembrane sections are there in a G-protein coupled receptor?	
Feedback: G-Protein coupled receptors are also known as 7-TM receptors, where TM stands for transmembrane.		
incorrect	**a.#**	1
incorrect	**b.#**	3
incorrect	**c.#**	5
Correct	**d.#**	7
END OF QUESTION		

Type: multiple choice		
Title: Chapter 04 - Question 30		
30)	Which of the following statements is true about a G-protein coupled receptor?	
Feedback: The *N*-terminal chain is extracellular and the *C*-terminal chain is intracellular. However, there are **seven** transmembrane hydrophobic regions. There are the **same** number of intracellular and extracellular loops. The binding region of the G-protein involves **one of the intracellular loops and the *C*-terminal chain**.		
incorrect	**a.#**	It contains five transmembrane hydrophobic sections.

incorrect	**b.#**	There are more extracellular loops than intracellular loops.
incorrect	**c.#**	The binding region for the G-protein involves two extracellular loops.
Correct	**d.#**	The *N*-terminal chain is extracellular and the *C*-terminal chain is intracellular.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 04 - Question 31		
31)	Which of the following is not a G-protein coupled receptor?	
Feedback: The glycine receptor is an ion channel.		
incorrect	**a.#**	The muscarinic receptor.
Correct	**b.#**	The glycine receptor.
incorrect	**c.#**	The adrenergic receptor.
incorrect	**d.#**	The glutamate receptor.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 04 - Question 32		
32)	Which of the following statements is not true about G-protein coupled receptors?	
Feedback: Fast-acting neurotransmitters generally activate ion channels since the secondary effect is faster than that of G-protein coupled receptors.		
Correct	**a.#**	They generally mediate the action of fast acting neurotransmitters.
incorrect	**b.#**	They mediate the action of some hormones.
incorrect	**c.#**	They activate signal proteins called G-proteins.
incorrect	**d.#**	Calcium ions can act as a ligand for some G-protein coupled receptors.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 04 - Question 33		
33)	Which of the following pairs of receptors are likely to show the greatest structural similarity?	
Feedback: These receptors are closely related on the evolutionary tree of G-protein coupled receptors. The other pairings are more distantly related.		
incorrect	**a.#**	The dopamine receptor subtypes D_3 and D_5.
incorrect	**b.#**	The muscarinic receptor M_2 and the β-adrenergic receptor β_2.
Correct	**c.#**	The histamine H_2 receptor and the α_1 adrenoceptor.
incorrect	**d.#**	The histamine H_1 receptor and the β_2 adrenergic receptor.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 04 - Question 34	
34)	Which of the following statements is true about a tyrosine kinase

	linked receptor?	
Feedback: The ligand binding site is in the *N*-terminal chain is a true statement. The receptor is situated in the **cell membrane**. The *N*-terminal chain is **extracellular** and this is where the ligand binding site is. There is only **one** hydrophobic transmembrane region.		
incorrect	**a.#**	It is situated in the cytoplasm.
incorrect	**b.#**	The *N*-terminal chain is intracellular.
Correct	**c.#**	The ligand binding site is in the *N*-terminal chain.
incorrect	**d.#**	It has two hydrophobic transmembrane regions.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 04 - Question 35		
35)	Which of the following reactions is catalysed by a tyrosine kinase?	
Feedback: Esterifications are catalysed by esterases. A tyrosine kinase phosphorylates the phenol functional group present on the side chain of the amino acid tyrosine.		
incorrect	**a.#**	The phosphorylation of an alcohol functional group.
incorrect	**b.#**	The esterification of an alcohol functional group.
Correct	**c.#**	The phosphorylation of a phenol functional group.
incorrect	**d.#**	The esterification of a phenol functional group.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 04 - Question 36		
36)	Which of the following terms describes the epidermal growth factor?	
Feedback: The epidermal growth factor is a bivalent ligand which can bind to two binding sites on different proteins allowing dimerisation and activation of the protein receptor.		
incorrect	**a.#**	Divalent ligand.
incorrect	**b.#**	Heterovalent ligand.
incorrect	**c.#**	Covalent ligand.
Correct	**d.#**	Bivalent ligand.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 04 - Question 37		
37)	Which of the following statements is untrue regarding the epidermal growth factor receptor?	
Feedback: ATP and not GTP is required for phosphorylation. The other statements are accurate.		
incorrect	**a.#**	Binding of epidermal growth factor results in dimerisation of the receptor.
incorrect	**b.#**	Dimerisation activates the tyrosine kinase active site.
incorrect	**c.#**	Activation of tyrosine kinase results in phosphorylation of tyrosine residues on the other half of the dimer.
Correct	**d.#**	GTP is required for phosphorylation.
END OF QUESTION		

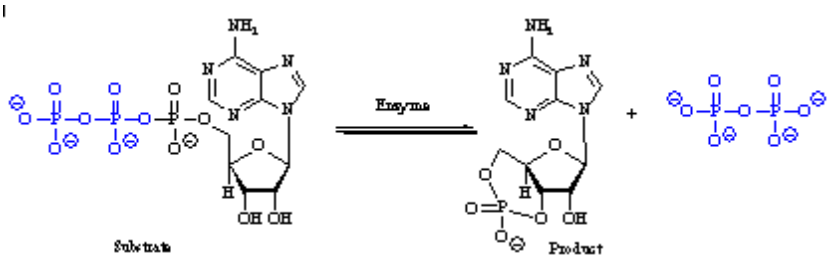
Type: multiple choice		
Title: Chapter 04 - Question 38		
38)	In what way does the insulin receptor differ from the EGF receptor?	
Feedback: The insulin receptor is preformed before the ligand binds whereas the EGF receptor requires the presence of its ligand to dimerise. All the other statements are true for both kinds of receptor.		
Correct	**a.#**	The receptor is preformed prior to ligand binding.
incorrect	**b.#**	A tyrosine kinase active site is activated.
incorrect	**c.#**	Phosphorylation of one half of the receptor complex is catalysed by enzyme active sites on the other half.
incorrect	**d.#**	ATP is required for phosphorylation.
END OF QUESTION		

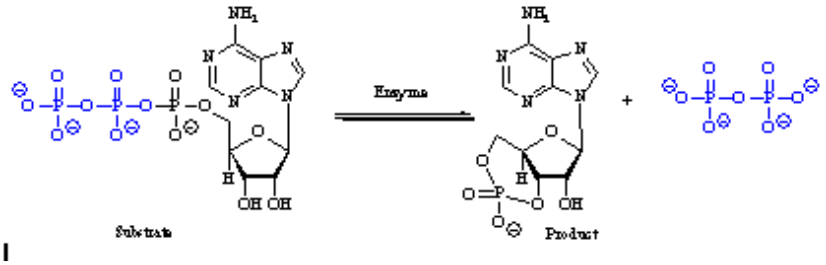
Type: multiple choice		
Title: Chapter 04 - Question 39		
39)	In what way does the growth hormone receptor differ from the EGF receptor?	
Feedback: The growth hormone receptor is formed as a result of ligand binding. Unlike the EGF-receptor, four protein subunits are assembled where the kinase enzymes are separate protein subunits from the protein subunits acting as receptors.		
incorrect	**a.#**	The receptor is formed as a result of ligand binding.
Correct	**b.#**	The kinase enzyme is a separate protein from the receptor protein.
incorrect	**c.#**	Phosphorylation of one half of the complex is catalysed by active sites on the other half.
incorrect	**d.#**	ATP is required for phosphorylation.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 04 - Question 40		
40)	What major role do intracellular receptors play?	
Feedback: The major role for intracellular receptors is to trigger transcription through the formation of a nuclear transcription factor.		
Correct	**a.#**	They trigger transcription.
incorrect	**b.#**	They trigger replication.
incorrect	**c.#**	They trigger translation.
incorrect	**d.#**	They trigger reproduction.
END OF QUESTION		

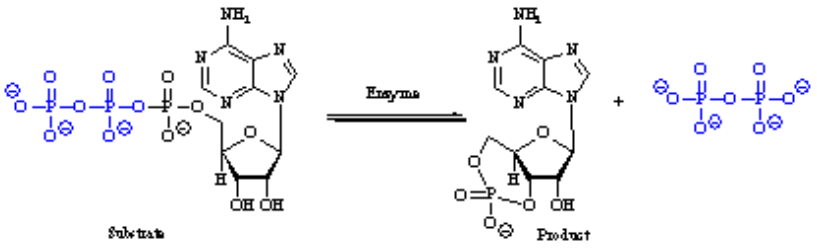
Type: multiple choice		
Title: Chapter 04 - Question 41		
41)	Which of the following statements is generally true regarding the chemical messengers for intracellular receptors?	
Feedback: Since the chemical messengers for intracellular receptors are extracellular, they have to be hydrophobic in order to cross the cell membrane.		
incorrect	**a.#**	They are hydrophilic in order to bind to zinc atoms.
incorrect	**b.#**	They are hydrophilic in order to dissolve in the cell's cytoplasm.

incorrect	**c.#**	They are hydrophobic in order to enter the nucleus of the cell.
Correct	**d.#**	They are hydrophobic in order to cross the cell membrane.
END OF QUESTION		

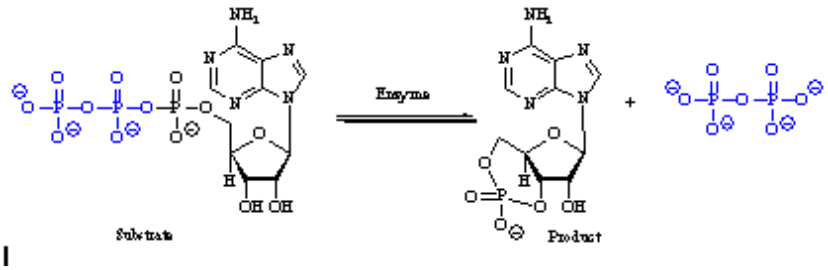
Type: multiple choice		
Title: Chapter 05 - Question 01		
01)	What is the name of the substrate in the following reaction?  	
Feedback: This reaction is the conversion of ATP to ADP, catalysed by the membrane-bound enzyme adenylate cyclase.		
Correct	**a.#**	ATP.
incorrect	**b.#**	CTP.
incorrect	**c.#**	GTP.
incorrect	**d.#**	UTP.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 05 - Question 02		
02)	What is the name of the product in the following reaction?  	
Feedback: This reaction is the conversion of ATP to ADP, catalysed by the membrane bound enzyme adenylate cyclase.		
Correct	**a.#**	Cyclic AMP.
incorrect	**b.#**	Cyclic CMP.
incorrect	**c.#**	Cyclic GMP.
incorrect	**d.#**	Cyclic UMP.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 05 - Question 03	
03)	What is the name of the enzyme in the following reaction?

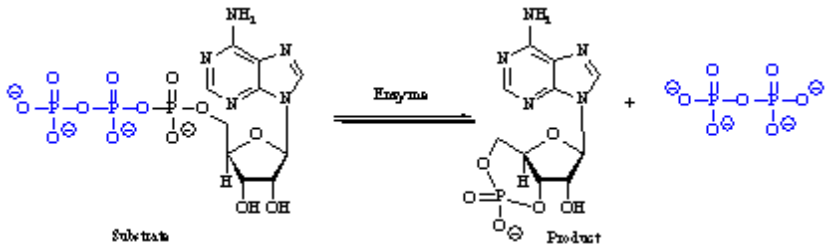
	

Feedback: This reaction is the conversion of ATP to ADP, catalysed by the membrane bound enzyme adenylate cyclase.		
incorrect	**a.#**	Phosphorylase *a*.
incorrect	**b.#**	Esterase.
Correct	**c.#**	Adenylate cyclase.
incorrect	**d.#**	Guanylate cyclase.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 05 - Question 04	
04)	What structure activates the enzyme that catalyses the following reaction?  **I**

Feedback: The enzyme catalysing this reaction is adenylate cyclase which is activated by the α_s-subunit of a G-protein.		
incorrect	**a.#**	The $\beta\gamma$-dimer of a G_s-protein.
Correct	**b.#**	The α_s subunit of a G_s-protein.
incorrect	**c.#**	A G_s-protein.
incorrect	**d.#**	The β-subunit of a G_s-protein.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 05 - Question 05	
05)	What structure deactivates the enzyme that catalyses the following reaction?

Feedback: The enzyme catalysing this reaction is adenylate cyclase which is activated by the α_s-subunit of a G-protein and deactivated by the α_i-subunit

incorrect	**a.#**	The $\beta\gamma$-dimer of a G_i-protein.
Correct	**b.#**	The α_i-subunit of a G_i-protein.
incorrect	**c.#**	A G_j-protein.
incorrect	**d.#**	The β-subunit of a G_i-protein.

END OF QUESTION

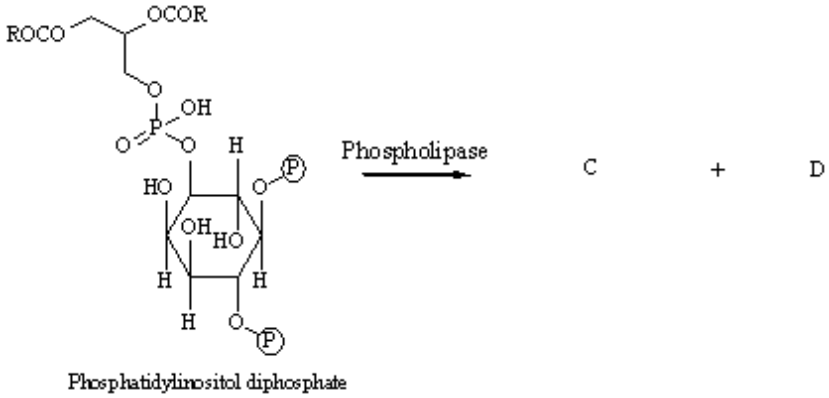
Type: multiple choice		
Title: Chapter 05 - Question 06		
06)	Which of the following enzymes catalyses a phosphorylation reaction?	
Feedback: Protein kinase enzymes are responsible for catalysing phosphorylations of alcohol or phenol groups on protein substrates. The hydrolysis of phosphates is catalysed by phosphorylases and not by kinases. Esterases catalyse the hydrolysis of esters. Ligases catalyse the linking together of two substrates.		
Correct	**a.#**	Protein kinases.
incorrect	**b.#**	Phosphorylases.
incorrect	**c.#**	Esterases.
incorrect	**d.#**	Ligases.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 05 - Question 07		
07)	Which of the following amino acids is phosphorylated by a protein kinase?	
Feedback: Protein kinases catalyse phenol or hydroxyl groups on the amino acid side chains of proteins. Tyrosine has a phenolic group which can be phosphorylated by such enzymes. The side chain of cysteine contains a thiol group which is not phosphorylated by protein kinases. Phenylalanine has an aromatic ring on its side chain and lysine has a primary amino group on its side chain. Neither of these groups is phosphorylated by protein kinases.		
Correct	**a.#**	Tyrosine.
incorrect	**b.#**	Cysteine.
incorrect	**c.#**	Phenylalanine.
incorrect	**d.#**	Lysine.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 05 - Question 08	
08)	Which of the following statements is true regarding protein kinase A?

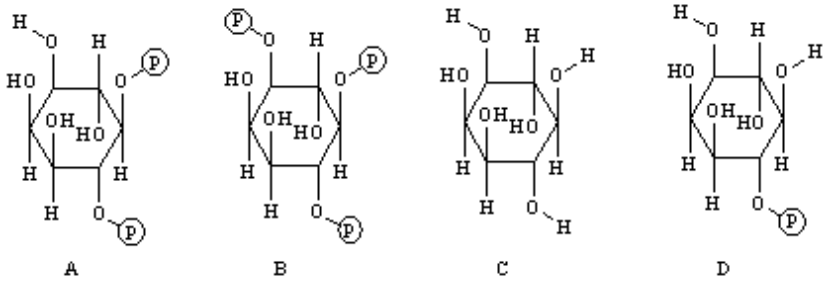
Feedback: Protein kinase A is **an intracellular** enzyme that contains **four** protein subunits in the resting state. It is a **serine-threonine kinase** and not a tyrosine kinase.		
incorrect	**a.#**	It is an extracellular enzyme.
incorrect	**b.#**	It is a tyrosine kinase.
incorrect	**c.#**	It contains two protein subunits in the resting state.
Correct	**d.#**	It is activated by cyclic AMP.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 05 - Question 09	
09)	Phospholipase catalyses the formation of two products from phosphatidylinositol diposphate.



Phosphatidylinositol diphosphate

Which of the following structures is one of these products?



Feedback: The reaction splits diacylglycerol from the substrate leaving all three phosphate groups unaffected. The product is inositol triphosphate.		
incorrect	**a.#**	Structure A.
Correct	**b.#**	Structure B.
incorrect	**c.#**	Structure C.
incorrect	**d.#**	Structure D.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 05 - Question 10	
10)	Which of the following statements is true for diacylglycerol?
Feedback: Diacylglycerol is a **hydrophobic** molecule that **does** act as a secondary messenger. It **remains** in the cell membrane and activates protein kinase C.	

incorrect	**a.#**	It is a hydrophilic molecule.
incorrect	**b.#**	It moves into the cytoplasm.
Correct	**c.#**	It activates an enzyme called protein kinase C.
incorrect	**d.#**	It is not a secondary messenger.
END OF QUESTION		

Type: matching			
Title: Chapter 05 - Question 11			
11)	Match the following secondary messengers to their actions.		
Feedback: Different secondary messengers trigger different signal transduction pathways			
a.#	Activation of protein kinase A	=	Cyclic AMP.
b.#	Activation of calcium ion release in the cell	=	Inositol triphosphate.
c.#	Activation of protein kinase C	=	Diacylglycerol.
d.#	Activation of protein kinase G	=	Cyclic GMP.
END OF QUESTION			

Type: multiple choice		
Title: Chapter 05 - Question 12		
12)	Which of the following statements is true regarding small G-proteins?	
Feedback: Small G proteins are **single** proteins and are **not** typical G-proteins (the latter are made up of three protein subunits). They are **not** directly activated by receptors, but are activated 'downstream' of activated receptors. They bind GDP in the resting state and GTP in the active state, just like the α-subunits of G-proteins.		
incorrect	**a.#**	Ras is an example of a typical G-protein.
incorrect	**b.#**	Small G-proteins have two subunits rather than three.
incorrect	**c.#**	Small G-proteins are directly activated by receptors.
Correct	**d.#**	Small G-proteins bind GDP in the resting state and GTP in the active state.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 05 - Question 13		
13)	How many subunits make up a G-protein?	
Feedback: The three subunits of a G-protein are labelled α, β and γ.		
incorrect	**a.#**	2
Correct	**b.#**	3
incorrect	**c.#**	4
incorrect	**d.#**	5
END OF QUESTION		

Type: multiple choice	
Title: Chapter 05 - Question 14	
14)	Which of the following molecules is bound to a G-protein in the

	inactivated state?	
Feedback: The G-protein binds GDP when it is in the resting, inactivated state. It binds GTP in the active state. GMP is not bound at all.		
incorrect	**a.#**	GMP
Correct	**b.#**	GDP
incorrect	**c.#**	GTP
incorrect	**d.#**	None of these.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 05 - Question 15	
15)	Which of the following molecules binds to the G-protein when the G-protein binds to the G-protein coupled receptor and is activated?
Feedback: The G-protein binds GDP when it is in the resting, inactivated state. It binds GTP in the active state. GMP is not bound at all.	

incorrect	**a.#**	GMP
incorrect	**b.#**	GDP
Correct	**c.#**	GTP
incorrect	**d.#**	None of these.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 05 - Question 16	
16)	Activation of the G-protein results in fragmentation of the G-protein. What does the G-protein fragment to?
Feedback: Feedback: The βγ-dimer and the α-subunit depart the G-protein binding site once fragmentation has taken place.	

Correct	**a.#**	A βγ-dimer and an α-subunit linked to GTP.
incorrect	**b.#**	A β-subunit, a γ-subunit, and an α-subunit linked to GTP.
incorrect	**c.#**	A β-subunit and a γ,α-dimer linked to GTP.
incorrect	**d.#**	A γ-subunit and a β,α-dimer linked to GTP.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 05 - Question 17	
17)	Which of the following explanations best describes the mechanism by which the G-protein fragments?
Feedback: Binding of the G-protein to the G-protein binding site does not in itself destabilise the G-protein. Binding alters the relative affinity of a binding site on the G-protein for GDP and GTP such that the latter is preferred over the former. Once GTP is bound, a further conformational change destabilises the G-protein leading to fragmentation. No phosphorylation reactions are involved in this process.	

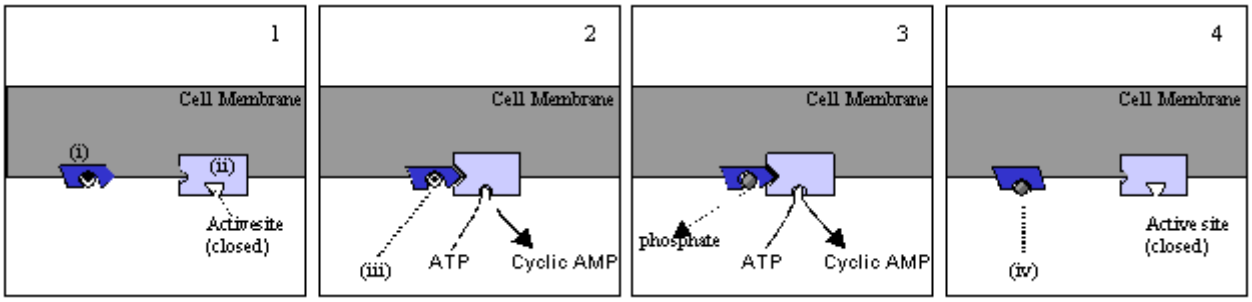
incorrect	**a.#**	Binding of the G-protein to the receptor results in a conformational change which destabilises the G-protein.
Correct	**b.#**	Binding of the G-protein to the receptor results in a conformational change which leads to the exchange of GDP for GTP, resulting in a further conformational change which destabilises the G-protein.
incorrect	**c.#**	Binding of the G-protein to the receptor results in a conformational change which leads to the exchange of GDP

		for GTP. GTP then phosphorylates tyrosine residues destabilising the G-protein.
incorrect	**d.#**	Binding of the G-protein to the receptor results in a conformational change which stabilises the $\beta\gamma$ dimer leading to dissociation of the α-subunit.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 05 - Question 18		
18)	Which of the following statements describes the signal amplification that results from activation of the G-protein coupled receptor?	
Feedback: All the statements are true, but only the correct answer explains how an amplification of the signal is achieved.		
incorrect	**a.#**	The binding of a ligand to the receptor increases the chances of the receptor binding a G-protein.
incorrect	**b.#**	The binding of a G-protein to the receptor increases the chances of it exchanging GDP for GTP.
Correct	**c.#**	The ligand is bound to the receptor sufficiently long such that the receptor can fragment several G-proteins.
incorrect	**d.#**	Fragmentation of the G-protein leads to an increase in entropy.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 05 - Question 19		
19)	Which of the following statements describes how signal amplification takes place with respect to the enzyme adenylate cyclase?	
Feedback: The α_s subunit binds to an allosteric binding site on adenylate cyclase and activates it. Several enzyme-catalysed reactions occur during the time that the subunit is bound, resulting in a signal amplification.		
incorrect	**a.#**	The α_s subunit of a G-protein binds to the enzyme and increases the chances of the enzyme catalysing a reaction.
Correct	**b.#**	Several enzyme-catalysed reactions are carried out during the period of time that the α_s subunit is bound to the enzyme.
incorrect	**c.#**	Each α_s subunit activates several different adenylate cyclase enzymes.
incorrect	**d.#**	Each $\beta\gamma$-dimer and α_s-subunit activates different adenylate cyclase enzymes.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 05 - Question 20		
20)	What term is used to describe the process by which an activated membrane-bound receptor passes a message into the cell to produce a change in cell chemistry?	
Feedback: Signal transduction is correct. The other terms are fictitious.		
Correct	**a.#**	Signal transduction.
incorrect	**b.#**	Signal translocation.
incorrect	**c.#**	Signal translation.
incorrect	**d.#**	Signal transcription.
END OF QUESTION		

Type: matching			
Title: Chapter 05 - Question 21			
21)	Identify (i) – (iv) in the following diagram. 		
Feedback: The diagram shows the activation of the enzyme adenylate cyclase by the α_s-subunit of a G_s protein. Once activated, the enzyme catalyses the synthesis of cyclic AMP from ATP. The α_s-subunit remains bound to the enzyme for as long as GTP is bound to the subunit. Eventually, a self-catalysed hydrolysis of GTP to GDP occurs which weakens the binding of the α_s-subunit to the enzyme. The α_s-subunit then departs and the enzyme is deactivated.			
a.#	(i).	=	α_s-subunit with bound GTP
b.#	(ii).	=	Adenylate cyclase
c.#	(iii).	=	GTP
d.#	(iv).	=	GDP
END OF QUESTION			

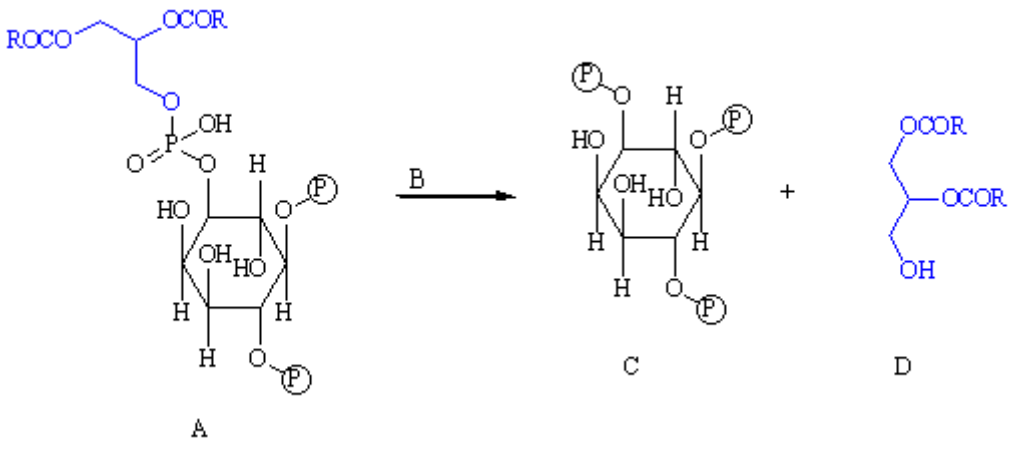
Type: multiple choice		
Title: Chapter 05 - Question 22		
22)	Which of the following events will eventually result in the termination of adenylate cyclase activation?	
Feedback: The α_s-subunit autocatalyses the hydrolysis of bound GTP to GDP with the release of a phosphate group. Once this occurs, the α_s-subunit has less binding affinity for the allosteric binding site on adenylate cyclase and departs. The enzyme then reverts to an inactive conformation.		
incorrect	**a.#**	Departure of GTP from the α_s-subunit.
incorrect	**b.#**	Departure of GTP from the α_s-subunit and replacement with GDP.
incorrect	**c.#**	Hydrolysis of bound GTP on the α_s-subunit to GMP.
Correct	**d.#**	Hydrolysis of bound GTP on the α_s-subunit to GDP.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 05 - Question 23		
23)	Which of the following inhibits adenylate cyclase?	
Feedback: The α_o-subunit inhibits calcium ion channels. The α_s-subunit activates adenylate cyclase. The α_i-subunit inhibits adenylate cyclase. The α_q-subunit affects the activity of phospholipase C.		
incorrect	**a.#**	An α_o-subunit.
incorrect	**b.#**	An α_s-subunit.
Correct	**c.#**	An α_i-subunit.

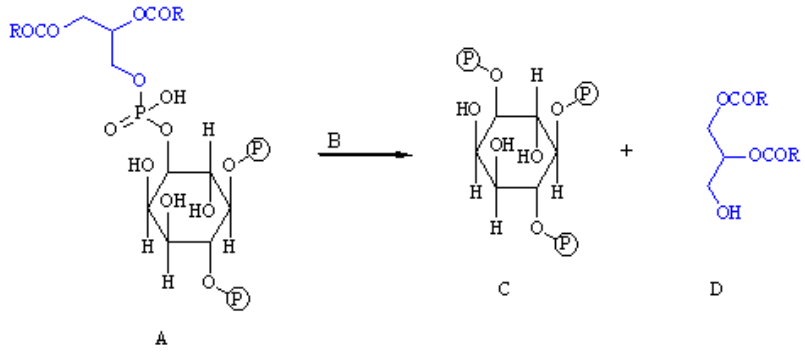
incorrect	**d.#**	An α_q-subunit.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 05 - Question 24		
24)	Which of the following statements is true regarding protein kinase A?	
Feedback: Protein kinase A is an **intracellular** enzyme that contains **four** protein subunits in the resting state and is activated by **cyclic AMP**. It is a serine-threonine kinase.		
incorrect	**a.#**	It is an extracellular enzyme.
Correct	**b.#**	It is a serine-threonine kinase.
incorrect	**c.#**	It contains two protein subunits in the resting state.
incorrect	**d.#**	It is activated by ATP.
END OF QUESTION		

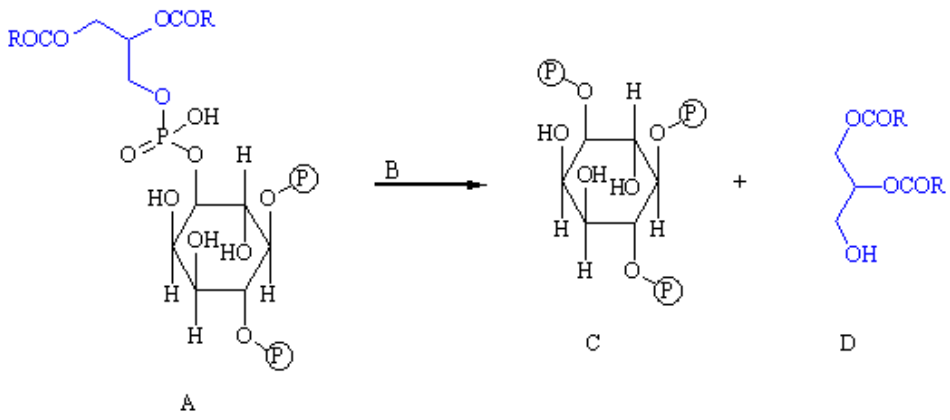
Type: multiple choice		
Title: Chapter 05 - Question 25		
25)	Which of the following is inactive when glycogen is broken down in a liver cell?	
Feedback: Glycogen synthase is the enzyme that catalyses the formation of glycogen and is phosphorylated to an inactive form when glycogen is being broken down. Activated α-adrenoceptor is the membrane bound receptor that triggers the process leading to glycogen breakdown. Phosphorylase kinase and phosphorylase *a* are enzymes involved in the signal transduction pathway leading to glycogen breakdown.		
incorrect	**a.#**	Phosphorylase kinase.
Correct	**b.#**	Glycogen synthase.
incorrect	**c.#**	The α-adrenoceptor.
incorrect	**d.#**	Phosphorylase *a.*
END OF QUESTION		

Type: multiple choice	
Title: Chapter 05 - Question 26	
26)	What enzyme (indicated by the letter B) catalyses the reaction in the following diagram? 
Feedback: Protein kinase A and protein kinase C catalyse the phosphorylation of serine and threonine	

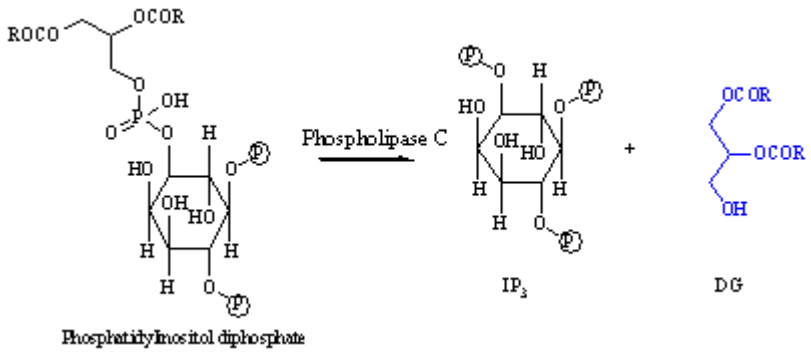
residues in protein substrates, while adenylate cyclase catalyses the formation of cyclic AMP from ATP.		
incorrect	**a.#**	Protein kinase A.
incorrect	**b.#**	Protein kinase C.
Correct	**c.#**	Phospholipase C.
incorrect	**d.#**	Adenylate cyclase.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 05 - Question 27	
27)	What is the substrate indicated by the letter A in the following diagram? 
Feedback: Inositol triphosphate is product C. Diacylglycerol is product D. Inositol diphosphate is not involved at all. Phosphatidylinositol diphosphate is the substrate A.	

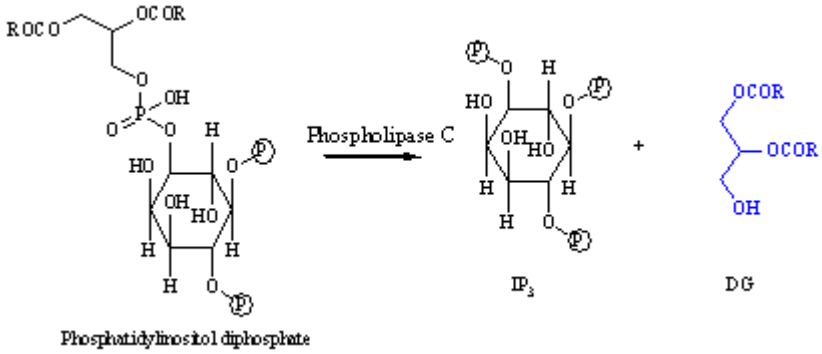
incorrect	**a.#**	Inositol triphosphate.
incorrect	**b.#**	Diacylglycerol.
incorrect	**c.#**	Inositol diphosphate.
Correct	**d.#**	Phosphatidylinositol diphosphate.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 05 - Question 28	
28)	What is the product indicated by the letter D in the following diagram? 
Feedback: Inositol triphosphate is product C. Diacylglycerol is product D. Inositol diphosphate is not involved at all. Phosphatidylinositol diphosphate is the substrate A.	

incorrect	**a.#**	Inositol triphosphate.
Correct	**b.#**	Diacylglycerol.
incorrect	**c.#**	Inositol diacylglycerate.
incorrect	**d.#**	Phosphatidylinositol diphosphate.
END OF QUESTION		

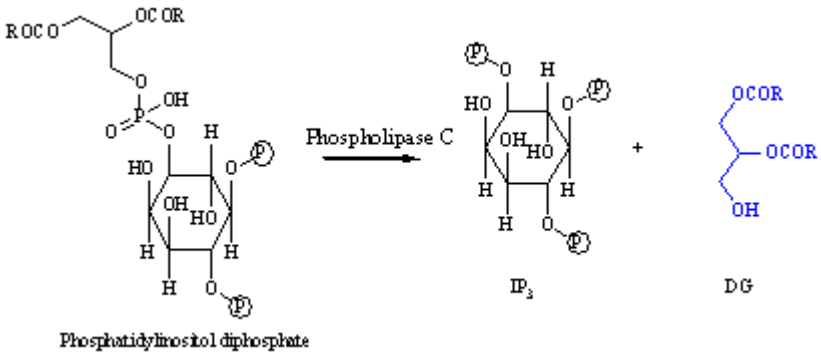
Type: multiple choice	
Title: Chapter 05 - Question 29	
29)	Which of the following affects the activity of phospholipase C in the following reaction?  Phosphatidylinositol diphosphate

Feedback: The α_o-subunit affects calcium ion channels. The α_q-subunit affects phospholipase C. The α_i-subunit and α_s-subunit affect adenylate cyclase.		
incorrect	**a.#**	An α_o-subunit.
Correct	**b.#**	An α_q-subunit.
incorrect	**c.#**	An α_i-subunit.
incorrect	**d.#**	An α_s-subunit.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 05 - Question 30	
30)	Which of the following statements is true for diacylglycerol (DG) formed in the following reaction?  Phosphatidylinositol diphosphate

Feedback: Diacylglycerol is a hydrophobic molecule that **does** act as a secondary messenger. It **remains** in the cell membrane and **activates** protein kinase C.		
Correct	**a.#**	It is a hydrophobic molecule.
incorrect	**b.#**	It moves into the cytoplasm.

incorrect	**c.#**	It deactivates an enzyme called protein kinase C.
incorrect	**d.#**	It is not a secondary messenger.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 05 - Question 31		
31)	Which of the following statements is true for inositol triphosphate (IP_3) formed in the below reaction?  Phosphatidylinositol diphosphate	
Feedback: Structure C is inositol triphosphate. It is a **hydrophilic** molecule that **does** act as a secondary molecule and moves into the cytoplasm to **activate** the release of calcium ions.		
incorrect	**a.#**	It is a hydrophobic molecule.
Correct	**b.#**	It moves into the cytoplasm.
incorrect	**c.#**	It deactivates the release of calcium ions.
incorrect	**d.#**	It is not a secondary messenger.
END OF QUESTION		

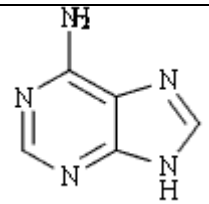
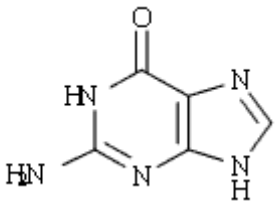
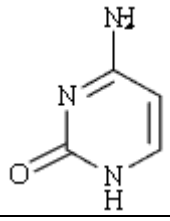
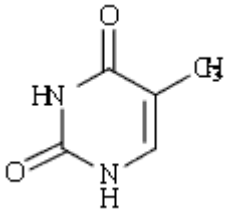
Type: multiple choice		
Title: Chapter 05 - Question 33		
33)	What is the molecular target for the antidepressant drug lithium?	
Feedback: Lithium inhibits phosphatase and prevents the resynthesis of phosphatidylinositol diphosphate.		
incorrect	**a.#**	Protein kinase A.
incorrect	**b.#**	Protein kinase C.
incorrect	**c.#**	Adenylate cyclase.
Correct	**d.#**	Phosphatase.
END OF QUESTION		

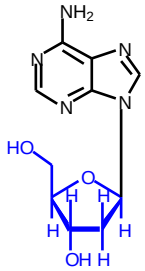
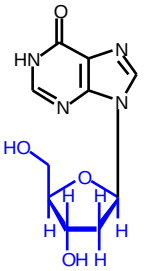
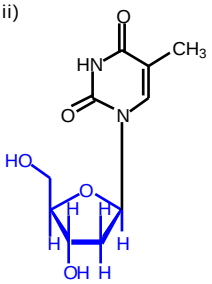
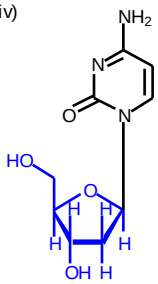
Type: multiple choice		
Title: Chapter 05 - Question 34		
34)	Which amino acids are phosphorylated by protein kinase C?	
Feedback: Protein kinase C is an example of a serine-threonine kinase which phosphorylates the alcohol groups present on the side chains of serine and threonine.		
incorrect	**a.#**	Tyrosine only.
incorrect	**b.#**	Serine only.

incorrect	**c.#**	Threonine only.
Correct	**d.#**	Serine and threonine.
END OF QUESTION		

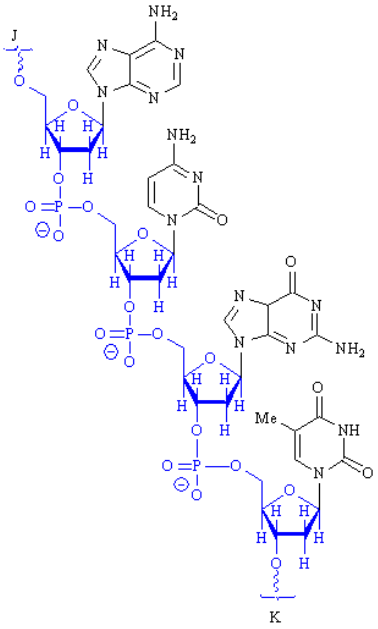
Type: multiple choice		
Title: Chapter 05 - Question 35		
35)	Identify the secondary messenger which activates protein kinase C (PKC).	
Feedback: Cyclic AMP activates protein kinase A. Inositol triphosphate activates the release of calcium ions in the cell. Diacylglycerol activates protein kinase C. Cyclic GMP activates protein kinase G.		
incorrect	**a.#**	Cyclic AMP.
incorrect	**b.#**	Inositol triphosphate.
Correct	**c.#**	Diacylglycerol.
incorrect	**d.#**	Cyclic GMP.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 05 - Question 36		
36)	Which of the following statements is true regarding small G-proteins?	
Feedback: Small G proteins are **single** proteins. They are **not** directly activated by receptors, but are activated 'downstream' of activated receptors. They bind **GDP** in the resting state and **GTP** in the active state, just like the α-subunits of G-proteins.		
Correct	**a.#**	Ras is an example of a small G-protein.
incorrect	**b.#**	Small G proteins have two subunits rather than three.
incorrect	**c.#**	Small G-proteins are directly activated by receptors.
incorrect	**d.#**	Small G-proteins bind GTP in the resting state and GDP in the active state.
END OF QUESTION		

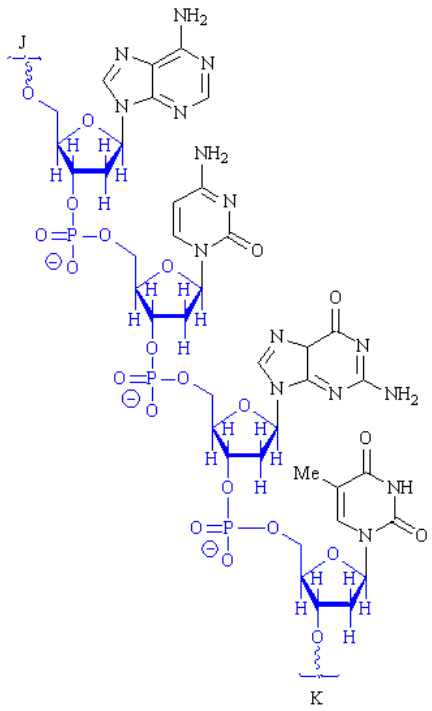
Type: matching			
Title: Chapter 06 - Question 01			
01)	Match the following structures to their names.		
Feedback: The structures shown are nucleic acid bases.			
a.#		=	Adenine
b.#		=	Guanine
c.#		=	Cytosine
d.#		=	Thymine
END OF QUESTION			

Type: matching			
Title: Chapter 06 - Question 02			
02)	Name the following structures (i) – (iv) i)  ii)  iii)  iv) 		
Feedback: The 'deoxy' portion of the name refers to the lack of a hydroxyl group at position 2 of the ribose ring.			
a.#	Deoxyadenosine	=	(i)
b.#	Deoxyguanosine	=	(ii)
c.#	Deoxythymidine	=	(iii)

d.#	Deoxycytidine	=	(iv)
END OF QUESTION			

Type: multiple choice		
Title: Chapter 06 - Question 03		
03)	Consider the following representation of DNA. What is the blue region called? 	
Feedback: The backbone consists of alternating deoxyribose sugars and phosphate groups. Therefore, it is the sugar phosphate backbone.		
incorrect	**a.#**	The carbohydrate backbone.
incorrect	**b.#**	The phosphate backbone.
Correct	**c.#**	The sugar-phosphate backbone.
incorrect	**d.#**	The nucleic acid base backbone.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 06 - Question 04	
04)	Consider the following representation of DNA. What are the groups shown in black?

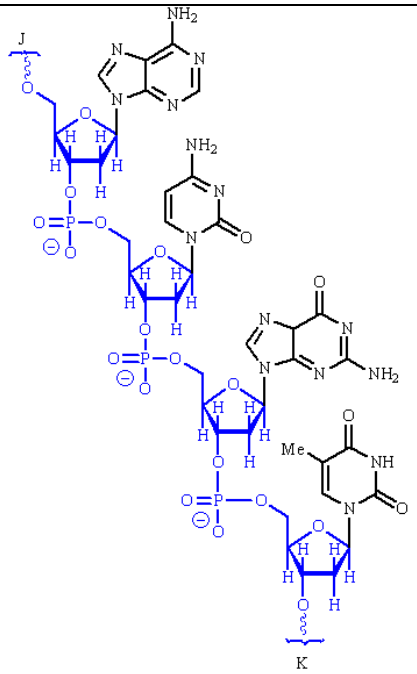
	

Feedback: The nucleic acid bases are linked to the sugar-phosphate backbone. Deoxyribose is the sugar in DNA. A nucleoside is a nucleic acid base linked to a sugar moiety. A nucleotide is a nucleoside linked to one or more phosphate groups.

Correct	**a.#**	Nucleic acid bases.
incorrect	**b.#**	Deoxyribose.
incorrect	**c.#**	Nucleosides.
incorrect	**d.#**	Nucleotides.

END OF QUESTION

Type: multiple choice	
Title: Chapter 06 - Question 05	
05)	Consider the following representation of DNA. What is the term used to describe the end of the chain marked J?

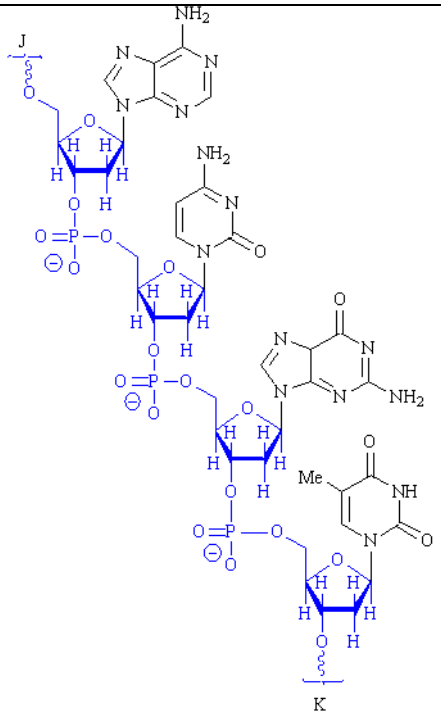
	

Feedback: The numbering relates to the numbering of the deoxyribose sugar moiety. The sugar-phosphate chain is extended from the 5’ and 3’ positions of each sugar moiety. The 5’ end of the chain is closest to the 5’ position of the final sugar moiety. The 3’ end is closest to the 3’ end of the final sugar moiety. The sugar phosphate backbone does not have a 2’ end or a 4’ end, since the chain is not extended from those positions of the sugar ring.

incorrect	**a.#**	The 2’ end.
incorrect	**b.#**	The 3’ end.
incorrect	**c.#**	The 4’ end.
Correct	**d.#**	The 5’ end.

END OF QUESTION

Type: multiple choice	
Title: Chapter 06 - Question 06	
06)	Consider the following representation of DNA. What type of DNA structure is shown here?

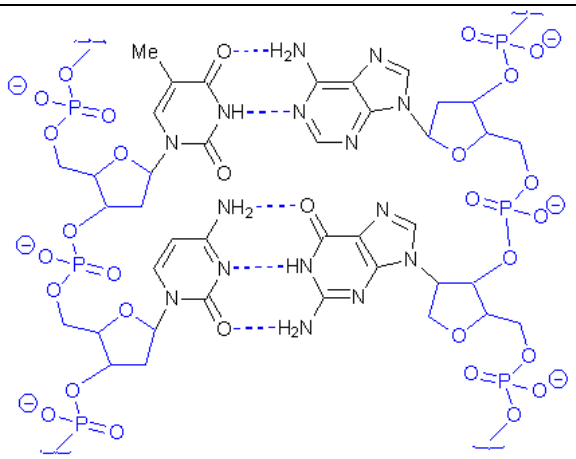
	

Feedback: The primary structure is the sequence in which the nucleotides are linked together in a DNA strand. The secondary structure refers to the double helix arrangement of two complementary DNA strands. The tertiary structure refers to the supercoiling of the DNA double helix into a more compact shape. There is no quaternary structure for DNA.		
Correct	**a.#**	Primary.
incorrect	**b.#**	Secondary.
incorrect	**c.#**	Tertiary.
incorrect	**d.#**	Quaternary.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 06 - Question 07	
07)	What type of structure is the double helix of DNA?

Feedback: The primary structure is the sequence in which the nucleotides are linked together in a DNA strand. The secondary structure refers to the double helix arrangement of two complementary DNA strands. The tertiary structure refers to the supercoiling of the DNA double helix into a more compact shape. There is no quaternary structure for DNA.		
incorrect	**a.#**	Primary.
Correct	**b.#**	Secondary.
incorrect	**c.#**	Tertiary.
incorrect	**d.#**	Quaternary.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 06 - Question 08	
08)	What does the following diagram demonstrate?

Feedback: The diagram shows base pairing between the nucleic acid bases adenine and thymine as well as between the nucleic acid bases guanine and cytosine. The interaction is through hydrogen bonding.		
incorrect	**a.#**	Dimerisation.
incorrect	**b.#**	Receptor binding.
Correct	**c.#**	Base pairing.
incorrect	**d.#**	Polymerisation.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 06 - Question 09	
09)	Which of the following statements is not true about DNA secondary structure?
Feedback: The phosphate groups are ionised and are positioned to the **outside** of the structure such that they interact favourably with surrounding water molecules. The base pairs are hydrophobic and are stacked on top of each other in the centre of the structure such that they interact with each other and not with water.	

incorrect	**a.#**	There is a minor groove and a major groove.
incorrect	**b.#**	A purine base pairs up with a pyrimidine base.
Correct	**c.#**	The phosphate groups are positioned to the inside of the structure.
incorrect	**d.#**	The base pairs are stacked.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 06 - Question 10	
10)	What non-covalent bonding interactions do the nucleic acid bases form in DNA secondary structure?
Feedback: Hydrogen bonds are involved in forming the base pairs. The base pairs are stacked and van der Waals interactions take place between the various stacks of base pairs. There are no ionic interactions involving nucleic acid bases. The phosphate groups on the outside of the double helix can form ionic interactions.	

incorrect	**a.#**	Ionic bonds only.
incorrect	**b.#**	Hydrogen bonds only.
incorrect	**c.#**	Van der Waals interactions only.
Correct	**d.#**	Hydrogen bonds and van der Waals interactions.

END OF QUESTION

Type: multiple choice		
Title: Chapter 06 - Question 11		
11)	What is the significance of base pairing to the function of DNA?	
Feedback: Base pairing results in two complementary chains. Each chain can be used as a template for a new chain, allowing genetic information to be passed on from generation to generation. There is an element of truth in the other statements but they serve no purpose in the function of DNA.		
incorrect	**a.#**	It neutralises basic groups and controls the cell pH.
Correct	**b.#**	It means that the chains of DNA are complementary allowing replication.
incorrect	**c.#**	It stabilises the molecule such that its lifetime in the cell is increased.
incorrect	**d.#**	It prevents water interacting with the nucleic acid bases.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 06 - Question 12		
12)	Which of the following is not true when RNA is compared with DNA?	
Feedback: RNA contains uracil, but it is in place of **thymine** and not adenine. Uracil can base pair with adenine in certain regions of RNA secondary structure.		
incorrect	**a.#**	RNA contains ribose and not deoxyribose.
Correct	**b.#**	RNA contains uracil and not adenine.
incorrect	**c.#**	RNA does not form a double helix.
incorrect	**d.#**	RNA is smaller than DNA.
END OF QUESTION		

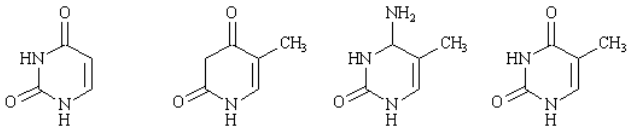
Type: multiple choice		
Title: Chapter 06 - Question 13		
13)	Which of the following mutations is most likely to be disruptive to protein synthesis or function?	
Feedback: UAU to UAC results in tyrosine being coded in each case. There is no disruption to protein synthesis or function. UAU to UUU results in a change from tyrosine to phenylalanine. This is a similar amino acid to tyrosine and so the chances of disruption are slight unless the tyrosine phenolic group is particularly important to function. UAU to UAA results in a stop code which prevents further synthesis of the protein. This will be the most drastic effect and will lead to a non-functioning protein. UAU to CAU results in a change from tyrosine to histidine. Only one amino acid is altered and the protein may still be functional if the tyrosine is not crucial to activity.		
incorrect	**a.#**	UAU to UAC.
incorrect	**b.#**	UAU to UUU.
Correct	**c.#**	UAU to UAA.
incorrect	**d.#**	UAU to CAU.
END OF QUESTION		

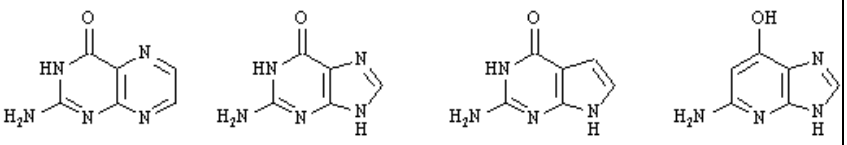
Type: matching			
Title: Chapter 06 - Question 14			
14)	Match the structure to the process described.		
Feedback: Different processes take place at different organelles in the body. Gene transcription takes place in the nucleus and the newly transcribed mRNA is modified by the spliceosome. The mRNA then departs the nucleus and moves to the endoplasmic reticulum where protein synthesis will take place. The actual synthetic process requires ribosomes, which are made up of protein and ribosomal RNA. The ribosomes bind the components required for protein synthesis and also catalyse the reaction.			
a.#	The organelle in the cell's cytoplasm where protein synthesis occurs.	=	Endoplasmic reticulum
b.#	The structure where transcription occurs.	=	Nucleus
c.#	The structure which catalyses protein synthesis and which is sometimes called the protein factory.	=	Ribosome
d.#	The structure which is responsible for modifying newly transcribed mRNA prior to translation.	=	Spliceosome
END OF QUESTION			

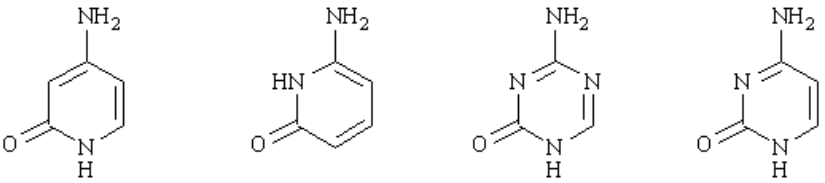
Type: multiple choice		
Title: Chapter 06 - Question 15		
15)	What is the term for the portion of messenger RNA that is excised when the molecule undergoes a splicing modification?	
Feedback: The intron is the excised region. The exons are the regions that are spliced together. Exclusion and inclusion are terms that are not used in descriptions of the splicing mechanism.		
Correct	**a.#**	Intron.
incorrect	**b.#**	Exon.
incorrect	**c.#**	Exclusion.
incorrect	**d.#**	Inclusion.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 06 - Question16		
16)	Which of the following molecules is not a nucleotide?	
Feedback: Adenosine is a nucleoside. All the others have phosphate groups linked to the nucleoside and are nucleotides.		
Correct	**a.#**	Adenosine.
incorrect	**b.#**	Adenosine monophosphate.
incorrect	**c.#**	Adenosine diphosphate.
incorrect	**d.#**	Adenosine triphosphate.
END OF QUESTION		

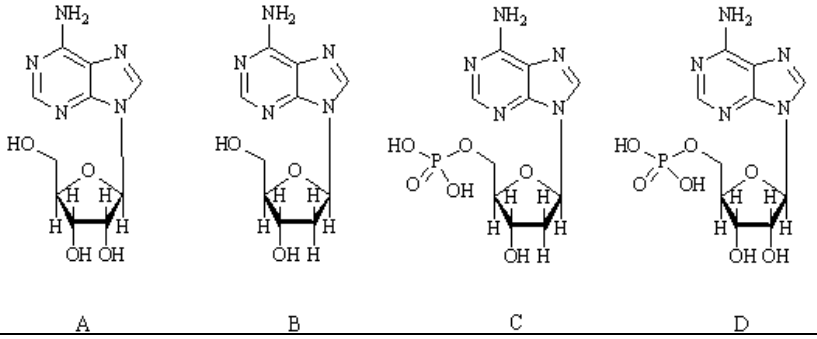
Type: multiple choice

Title: Chapter 06 - Question 17		
17)	Which of the following structures is uracil?  A B C D	
Feedback: Structure A is uracil. Structure D is thymine, but structures B and C are not present in DNA.		
Correct	**a.#**	Structure A.
incorrect	**b.#**	Structure B.
incorrect	**c.#**	Structure C.
incorrect	**d.#**	Structure D.
END OF QUESTION		

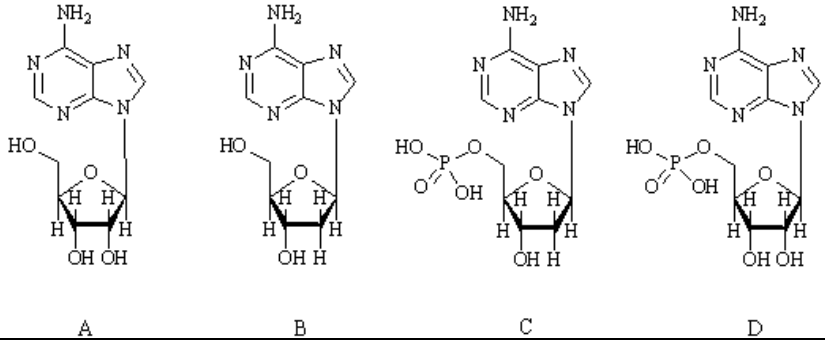
Type: multiple choice		
Title: Chapter 06 - Question 18		
18)	Which of the following structures is a purine structure?  A B C D	
Feedback: Structure B is guanine which is a purine structure. Structure A is a pteridine. Structure C is a pyrrolopyrimidine. Structure D is an imidazopyridine.		
incorrect	**a.#**	Structure A.
Correct	**b.#**	Structure B.
incorrect	**c.#**	Structure C.
incorrect	**d.#**	Structure D.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 06 - Question 19		
19)	Which of the following structures is a pyrimidine structure?  A B C D	
Feedback: Structure D is cytosine which is a pyrimidine structure. Structures A and B are pyridinones. Structure C is a triazinone.		

incorrect	**a.#**	Structure A.
incorrect	**b.#**	Structure B.
incorrect	**c.#**	Structure C.
Correct	**d.#**	Structure D.
END OF QUESTION		

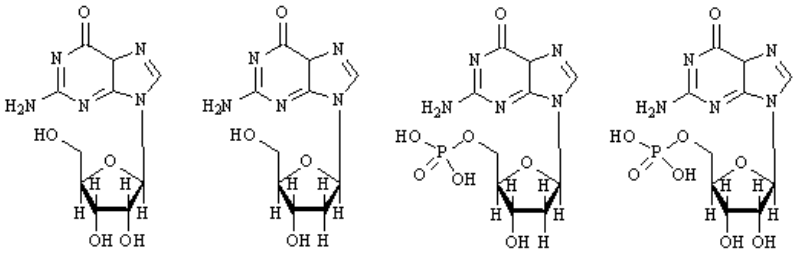
Type: multiple choice	
Title: Chapter 06 - Question 20	
20)	Which of the following structures is adenosine?  A B C D
Feedback: Structure A is adenosine. Structure B is deoxyadenosine. Structure C is deoxyadenosine monophosphate. Structure D is adenosine monophosphate (AMP).	

Correct	**a.#**	Structure A.
incorrect	**b.#**	Structure B.
incorrect	**c.#**	Structure C.
incorrect	**d.#**	Structure D.
END OF QUESTION		

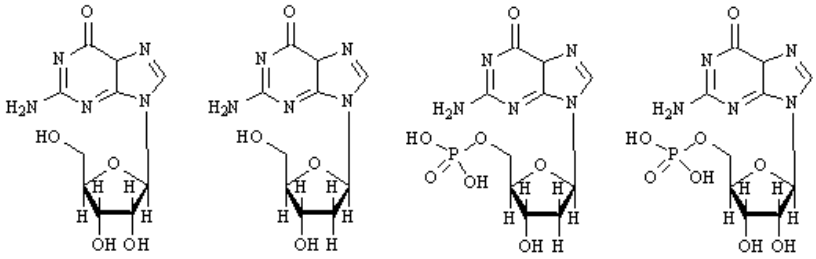
Type: multiple choice	
Title: Chapter 06 - Question 21	
21)	Which of the following structures is a nucleotide building block for DNA?  A B C D
Feedback: Structure C is deoxyadenosine monophosphate and is a nucleotide building block for DNA. Structure D (adenosine monophosphate) is also a nucleotide, but it is a building block for RNA. Structures A and B (adenosine and deoxyadenosine respectively) are nucleosides.	

incorrect	**a.#**	Structure A.
incorrect	**b.#**	Structure B.

Correct	**c.#**	Structure C.
incorrect	**d.#**	Structure D.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 06 - Question 22	
22)	Which of the following structures is guanosine?  A B C D
Feedback: Structure A is guanosine. Structure B is deoxyguanosine. Structure C is deoxyguanosine monophosphate. Structure D is guanosine monophosphate (GMP).	

Correct	**a.#**	Structure A.
incorrect	**b.#**	Structure B.
incorrect	**c.#**	Structure C.
incorrect	**d.#**	Structure D.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 06 - Question 23	
23)	Which of the following structures is a nucleotide building block for DNA?  A B C D
Feedback: Structure C is deoxyguanosine monophosphate and is a nucleotide building block for DNA. Structure D (guanosine monophosphate) is also a nucleotide but it is a building block for RNA. Structures A and B (guanosine and deoxyguanosine respectively) are nucleosides.	

incorrect	**a.#**	Structure A.
incorrect	**b.#**	Structure B.
Correct	**c.#**	Structure C.
incorrect	**d.#**	Structure D.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 06 - Question 24		
24)	To which of the following does adenine form hydrogen bonds in DNA?	
Feedback: Thymine base pairs with adenine. Guanine base pairs with cytosine.		
incorrect	**a.#**	Adenine.
Correct	**b.#**	Thymine.
incorrect	**c.#**	Cytosine.
incorrect	**d.#**	Guanine.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 06 - Question 25		
25)	To which of the following does cytosine form hydrogen bonds in DNA?	
Feedback: Thymine base pairs with adenine. Guanine base pairs with cytosine.		
incorrect	**a.#**	Adenine.
incorrect	**b.#**	Thymine.
incorrect	**c.#**	Cytosine.
Correct	**d.#**	Guanine.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 06 - Question 26		
26)	Which of the following statements is not true about DNA secondary structure?	
Feedback: A purine base always pairs up with a pyrimidine base.		
incorrect	**a.#**	There is a minor groove and a major groove.
Correct	**b.#**	Purine bases pair up with each other.
incorrect	**c.#**	The phosphate groups are positioned to the outside of the structure.
incorrect	**d.#**	The base pairs are stacked.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 06 - Question 27		
27)	Which of the following is not true when RNA is compared with DNA?	
Feedback: RNA molecules are smaller than DNA.		
incorrect	**a.#**	RNA contains ribose and not deoxyribose.
incorrect	**b.#**	RNA contains uracil and not thymine.
incorrect	**c.#**	RNA does not form a double helix.
Correct	**d.#**	RNA is larger than DNA.
END OF QUESTION		

Type: matching

Title: Chapter 06 – Question 28			
28)	Match the processes to their descriptions.		
Feedback: Transcription is the process by which the genetic code for a single protein is 'copied' in the form of messenger RNA. This molecule leaves the nucleus and moves to the endoplasmic reticulum and binds to ribosomes. Translation then occurs where the mRNA codes for the individual amino acids required for the protein. This takes place on ribosomes. Replication is the process by which one DNA molecule is copied to give two identical daughter DNA molecules, each of which contains one strand of the original DNA. Reproduction is the term that describes the overall physiological process by which new progeny are produced. Replication, Transcription and translation describe processes at the molecular level.			
a.#	The process by which DNA is copied to two daughter DNA molecules.	=	Replication
b.#	The process by which a segment of DNA is copied to produce a molecule of messenger RNA.	=	Transcription
c.#	The process by which proteins are synthesised from a genetic code.	=	Translation
d.#	The overall physiological process by which new progeny are produced in a particular species.	=	Reproduction
END OF QUESTION			

Type: multiple choice		
Title: Chapter 06 - Question 29		
29)	Which of the following descriptions applies to the process of translation?	
Feedback: Replication is the copying of DNA into two daughter DNA molecules. Translation is the process by which the genetic code for a specific protein (carried in messenger RNA) is used for the synthesis of a new protein. Transcription is the term used to describe how the genetic code relevant for the synthesis of one protein (a gene) is copied from DNA to form messenger RNA. Reproduction is the term used for the overall physiological process by which new progeny are produced in a particular species.		
incorrect	**a.#**	The process by which DNA is copied to produce two daughter DNA molecules.
incorrect	**b.#**	The process by which a segment of DNA is copied to produce a molecule of messenger RNA.
Correct	**c.#**	The process by which proteins are synthesised from a genetic code.
incorrect	**d.#**	The overall physiological process by which new progeny are produced in a particular species.
END OF QUESTION		

Type: matching	
Title: Chapter 06 - Question 30	
30)	Match the type of RNA to its function.
Feedback: The first stage in the generation of a new protein is the copying of the genetic information which codes for that protein. This is a process called transcription and results in a molecule of messenger RNA (mRNA). The mRNA may be modified in the nucleus of eukaryotic cells by structures called spliceosomes which include molecules of snRNA. The	

modified mRNA departs the nucleus to the endoplasmic reticulum where translation takes place on ribosomes. Ribosomes are made up of proteins and rRNA. The rRNA plays an important catalytic role in the mechanism of translation. mRNA acts as the code for the amino acids making up the protein and transfer RNA (tRNA) translates the code.			
a.#	The RNA which catalyses protein synthesis.	=	rRNA
b.#	The RNA which provides the genetic blueprint for a protein.	=	mRNA
c.#	The RNA which is responsible for translating the triplet code into a specific amino acid.	=	tRNA
d.#	The RNA which participates in modifying mRNA prior to protein synthesis.	=	snRNA
END OF QUESTION			

Type: multiple choice		
Title: Chapter 06 - Question 31		
31)	Which of the following types of RNA helps to catalyse the synthesis of proteins?	
Feedback: The process of protein synthesis is catalysed by ribosomal RNA (rRNA). Messenger RNA (mRNA) provides the genetic blueprint for the protein. Transfer RNA (tRNA) is responsible for translating the triplet code into a specific amino acid. Messenger RNA molecules are modified prior to protein synthesis by small nuclear RNA (snRNA).		
Correct	**a.#**	rRNA.
incorrect	**b.#**	mRNA.
incorrect	**c.#**	tRNA.
incorrect	**d.#**	snRNA.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 06 - Question 32		
32)	Which of the following types of RNA translates the genetic code for the synthesis of proteins?	
Feedback: The process of protein synthesis is catalysed by ribosomal RNA (rRNA). Messenger RNA (mRNA) provides the genetic blueprint for the protein. Transfer RNA (tRNA) is responsible for translating the triplet code into a specific amino acid. Messenger RNA molecules are modified prior to protein synthesis by small nuclear RNA (snRNA).		
incorrect	**a.#**	rRNA.
incorrect	**b.#**	mRNA.
Correct	**c.#**	tRNA.
incorrect	**d.#**	snRNA.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 06 - Question 33	
33)	What expression is given for the three nucleic acid bases present on transfer RNA that base pair with a triplet of bases on messenger RNA?

Feedback: The codon is a triplet of nucleic acid bases present on messenger RNA that will code for a specific amino acid. The anticodon is the complementary triplet of nucleic acid bases on transfer RNA that will link a specific transfer RNA to that codon. In doing so, a specific amino acid is carried to the site of protein synthesis by transfer RNA. There is no such thing as a translation factor.		
Correct	**a.#**	Anticodon.
incorrect	**b.#**	Triplet.
incorrect	**c.#**	Codon.
incorrect	**d.#**	Translation factor.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 06 - Question 34	
34)	How many nucleic acid bases on messenger RNA are responsible for the coding of one amino acid?
Feedback: Since three bases code for one amino acid, the genetic code is also called the triplet code.	

incorrect	**a.#**	1
incorrect	**b.#**	2
Correct	**c.#**	3
incorrect	**d.#**	4
END OF QUESTION		

Type: multiple choice	
Title: Chapter 06 - Question 35	
35)	What amino acid is coded by the triplet of bases UUU?
Feedback: The triplet code for phenylalanine can be UUU or UUC. For serine it can be UCU, UCC, AGU, or AGC. For tyrosine, it is UAU or UAC. For cysteine it can be UGU or UGC.	

Correct	**a.#**	Phenylalanine.
incorrect	**b.#**	Serine.
incorrect	**c.#**	Tyrosine.
incorrect	**d.#**	Cysteine.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 06 - Question 36	
36)	Which of the following mutations is most likely to be disruptive to protein synthesis or function?
Feedback: GAG to GAA results in glutamic acid being coded in each case. There is no disruption to protein synthesis or function. GAG to GAC or GAU results in a change from glutamic acid to aspartic acid. This is a similar amino acid to glutamic acid and so the chances of disruption are slight. GAG to UAG results in a stop code which prevents further synthesis of the protein. This will be the most drastic effect and will lead to a non-functioning protein.	

incorrect	**a.#**	GAG to GAA.
incorrect	**b.#**	GAG to GAC.
incorrect	**c.#**	GAG to GAU.

Correct	**d.#**	GAG to UAG.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 06 - Question 37		
37)	What sort of reaction is catalysed by a restriction enzyme in recombinant DNA technology?	
Feedback: An esterase enzyme cleaves ester bonds. A restriction enzyme cleaves sugar phosphate bonds in DNA. A reductase enzyme reduces ketones. A ligase enzyme forms sugar phosphate bonds.		
incorrect	**a.#**	The hydrolysis of an ester group.
Correct	**b.#**	The cleavage of a sugar phosphate bond.
incorrect	**c.#**	The reduction of a ketone.
incorrect	**d.#**	The formation of a sugar phosphate bond.
END OF QUESTION		

Type: matching			
Title: Chapter 06 - Question 38			
38)	Match the enzyme to the reaction catalysed.		
Feedback: Enzymes are often named according to the reaction they catalyse. Thus Esterases catalyse the hydrolysis of esters and reductases catalyse reductions. Restriction enzymes and ligases are used in genetic engineering. The former are used to cleave sugar phosphate bonds in a DNA chain, and the latter are used to rejoin them.			
a.#	The hydrolysis of an ester group.	=	Esterase
b.#	The cleavage of a sugar phosphate bond.	=	Restriction enzyme
c.#	The reduction of a ketone.	=	Reductase
d.#	The formation of a sugar phosphate bond.	=	Ligase
END OF QUESTION			

Type: multiple choice		
Title: Chapter 06 - Question 39		
39)	Which amino acid in the enzyme topoisomerase II is responsible for cleaving a phosphate-sugar link in a DNA strand?	
Feedback: Tyrosine contains a phenolic group which forms a bond to a phosphate group on the DNA backbone, resulting in cleavage of a phosphate-sugar bond in the DNA backbone.		
Correct	**a.#**	Tyrosine.
incorrect	**b.#**	Threonine.
incorrect	**c.#**	Serine.
incorrect	**d.#**	Cysteine.
END OF QUESTION		

Type: multiple choice

Title: Chapter 06 - Question 40		
40)	What role does the enzyme topoisomerase II have?	
Feedback: The enzymes responsible for the four actions described are telomerase, DNA polymerase, helicase, and topoisomerase II.		
incorrect	**a.#**	It catalyses the formation of protective units at the end of chromosomes.
incorrect	**b.#**	It catalyses the synthesis of DNA.
incorrect	**c.#**	It catalyses the unwinding of the DNA double helix.
Correct	**d.#**	It catalyses crossing over of DNA strands to relieve the tension created in DNA when the DNA double helix is unravelled.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 07 - Question 01		
01)	Which of the following descriptions best describes a competitive enzyme inhibitor?	
Feedback: A competitive or reversible inhibitor is in equilibrium between the enzyme-bound state and the unbound state. Therefore, increasing the substrate concentration will allow the substrate to compete for active site occupancy. This will result in a decrease in inhibition. The agent that binds to an active site and undergoes a reaction is called a **substrate**. An agent that binds to an active site and inhibits the enzyme, but which is not displaced by increasing the concentration of substrate is an **irreversible inhibitor**. An agent that binds to a different binding site from the active site and affects the activity of the enzyme is an **allosteric inhibitor**.		
incorrect	**a.#**	A drug that binds to an active site and undergoes a reaction.
Correct	**b.#**	A drug that binds to an active site and inhibits the enzyme, but which is displaced by increasing the concentration of substrate.
incorrect	**c.#**	A drug that binds to an active site and inhibits the enzyme, but which is not displaced by increasing the concentration of substrate.
incorrect	**d.#**	A drug that binds to a different binding site from the active site and affects the activity of the enzyme.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 07 - Question 02		
02)	Which of the following descriptions best describes an allosteric inhibitor?	
Feedback: An agent that binds to a different binding site from the active site and affects the activity of the enzyme is an allosteric inhibitor. An agent that binds to an active site and inhibits the enzyme, but which is displaced by increasing the concentration of substrate is called a **competitive or reversible inhibitor**. The agent that binds to an active site and undergoes a reaction is called a **substrate**. An agent that binds to an active site and inhibits the enzyme, but which is not displaced by increasing the concentration of substrate is an **irreversible inhibitor**.		
incorrect	**a.#**	A drug that binds to an active site and undergoes a reaction.
incorrect	**b.#**	A drug that binds to an active site and inhibits the enzyme, but which is displaced by increasing the concentration of substrate.
incorrect	**c.#**	A drug that binds to an active site and inhibits the enzyme, but which is not displaced by increasing the concentration of substrate.
Correct	**d.#**	A drug that binds to a different binding site from the active site and affects the activity of the enzyme.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 07 - Question 03	
03)	Which of the following is **not** true of suicide substrates?

Feedback: Suicide substrates form covalent bonds with the active site of the target enzyme and so act as irreversible inhibitors. When the enzyme binds the drug, the catalytic mechanism activates the suicide substrate and a strong covalent bond is formed between the drug and the active site. The drug is now permanently linked to the active site and causes irreversible inhibition. The enzyme ultimately commits suicide by binding to the suicide substrate.		
incorrect	**a.#**	They are activated by the enzyme-catalysed reaction mechanism.
Correct	**b.#**	They react reversibly with the enzyme.
incorrect	**c.#**	They form a covalent bond with the active site once activated.
incorrect	**d.#**	They are so called because they cause the target enzyme to commit suicide by reacting with them.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 07 - Question 04	
04)	Which statement best describes the relevance of an allosteric binding site to medicinal chemistry?

Feedback: Designing drugs to interact with allosteric binding site allows a method of regulating enzyme activity, other than using inhibitors which bind to the active site. Allosteric binding sites cannot be said to be more or less hydrophobic than active sites. Each binding site is different in that respect, regardless of whether it is an active site or an allosteric binding site. There is also no reason to suppose that allosteric binding sites should be capable of binding a wider variety of drug structures than an active site. The answer 'drugs can be designed based on the transition state of the enzyme-catalysed reaction' is wrong since this refers to drugs which bind to the active site and not the allosteric binding site.		
incorrect	**a.#**	It is more hydrophobic than normal binding sites and accepts hydrophobic drugs in preference to hydrophilic drugs.
incorrect	**b.#**	A larger variety of drug structures will bind to the allosteric site than to the active site.
Correct	**c.#**	Drugs can be designed based on the structure of the endogenous chemicals which bind to allosteric sites and control enzyme activity.
incorrect	**d.#**	Drugs can be designed based on the transition state of the enzyme-catalysed reaction.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 07 - Question 05	
05)	Identify the kind of bonding forces which are typically involved in binding a drug to an enzyme's active site.

Feedback: The active sites of different enzymes vary in the relative number of ionic interactions, hydrogen bonds, and van der Waals interactions that are used to bind a drug.		
incorrect	**a.#**	Predominantly van der Waals interactions.
incorrect	**b.#**	Predominantly ionic bonds.
incorrect	**c.#**	Predominantly hydrogen bonds.

Correct	**d.#**	A combination of different intermolecular binding interactions.
END OF QUESTION		

Type: matching			
Title: Chapter 07 - Question 06			
06)	Identify an amino acid in an active site whose residue could bind a drug, principally by:		
Feedback: Serine has a hydroxyl group which can take part in hydrogen bonding. Aspartic acid has a residue containing a carboxylic acid group that can take part in ionic interactions if it is ionised. Valine has an alkyl side chain that can interact by van der Waals interactions. Glycine has no residue available for bonding interactions.			
a.#	Hydrogen bonding	=	Serine
b.#	Ionic bonding	=	Aspartic acid
c.#	Van der Waals interactions	=	Valine
d.#	None of the above	=	Glycine
END OF QUESTION			

Type: multiple choice		
Title: Chapter 07 - Question 07		
07)	Which of the following descriptions is most accurate regarding binding sites and binding regions?	
Feedback: A binding region is part of the binding site and is capable of forming a specific binding interaction with a drug. The binding region may contain a particular functional group that can participate in a hydrogen bond or an ionic bond. Alternatively, it may be hydrophobic in nature and interact by van der Waals interactions.		
incorrect	**a.#**	A binding site is part of a binding region.
Correct	**b.#**	A binding region is part of a binding site.
incorrect	**c.#**	A binding region is the same as a binding site.
incorrect	**d.#**	A binding region is part of a drug whereas a binding site is part of a macromolecular target.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 07 - Question 08		
08)	What type of plots can be used to determine whether an enzyme inhibitor is competitive or non-competitive?	
Feedback: Michaelis-Menten plots are used to study how the enzyme-catalysed rate of reaction is affected by substrate concentration. Schild plots are used to study the effect of antagonists on receptors. Displacement curves are used to measure IC_{50} values for receptor antagonists. Lineweaver-Burk plots are used to determine whether an enzyme inhibitor is competitive or non competitive.		
incorrect	**a.#**	Michaelis-Menten plots.
incorrect	**b.#**	Schild plots.
incorrect	**c.#**	Displacement plots.
Correct	**d.#**	Lineweaver-Burk plots.

END OF QUESTION

Type: multiple choice		
Title: Chapter 08 - Question 01		
01)	Which of the following descriptions best fits an agonist?	
Feedback: An agonist binds to a receptor binding site and activates it in the same way as the endogenous chemical messenger. An antagonist can also bind to the receptor, but it results in a different induced fit which does not activate the receptor. A partial agonist binds but does not activate the receptor to the same extent as a full agonist. An inverse agonist binds to a receptor and stabilises the inactive form of the protein such that the endogenous messenger cannot bind. Any inherent activity associated with the receptor is also lost.		
Correct	**a.#**	A compound that has the same effect on a receptor as the endogenous chemical messenger.
incorrect	**b.#**	A compound that binds to a receptor, and activates it, but to a lesser extent than the endogenous chemical messenger.
incorrect	**c.#**	A compound that binds to a receptor fails to activate it and prevents the endogenous chemical messenger from binding.
incorrect	**d.#**	A compound that binds to a receptor fails to activate it and leads to a drop in inherent biological activity.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 08 - Question 02		
02)	Which of the following descriptions best fits an antagonist?	
Feedback: An agonist binds to a receptor binding site and activates it in the same way as the endogenous chemical messenger. An antagonist can also bind to the receptor, but it results in a different induced fit which does not activate the receptor. A partial agonist binds but does not activate the receptor to the same extent as a full agonist. An inverse agonist binds to a receptor and stabilises the inactive form of the protein such that the endogenous messenger cannot bind. Any inherent activity associated with the receptor is also lost.		
incorrect	**a.#**	A compound that has the same effect on a receptor as the endogenous chemical messenger
incorrect	**b.#**	A compound that binds to a receptor, and activates it, but to a lesser extent than the endogenous chemical messenger.
Correct	**c.#**	A compound that binds to a receptor fails to activate it and prevents the endogenous chemical messenger from binding.
incorrect	**d.#**	A compound that binds to a receptor fails to activate it and leads to a drop in inherent biological activity.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 08 - Question 03		
03)	Which of the following descriptions best fits an inverse agonist?	
Feedback: An agonist binds to a receptor binding site and activates it in the same way as the endogenous chemical messenger. An antagonist can also bind to the receptor, but it results in a different induced fit which does not activate the receptor. A partial agonist binds but does not activate the receptor to the same extent as a full agonist. An inverse agonist binds to a receptor and stabilises the inactive form of the protein such that the		

endogenous messenger cannot bind. Any inherent activity associated with the receptor is also lost.		
incorrect	**a.#**	A compound that has the same effect on a receptor as the endogenous chemical messenger.
incorrect	**b.#**	A compound that binds to a receptor, and activates it, but to a lesser extent than the endogenous chemical messenger.
incorrect	**c.#**	A compound that binds to a receptor fails to activate it and prevents the endogenous chemical messenger from binding.
Correct	**d.#**	A compound that binds to a receptor fails to activate it and leads to a drop in inherent biological activity.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 08 - Question 04	
04)	Which of the following terms applies to the study of how a drug interacts with its target binding site at the molecular level?
Feedback: Pharmacokinetics covers such topics as the absorption, distribution, metabolism, and excretion of a drug in a living system. Structure-based drug design is the study of how drugs can be designed using molecular modelling based on a drug's pharmacophore. Pharmacodynamics is the study of how a drug interacts with its target binding site at the molecular level. Structure-activity relationships is the study of which functional groups are important in binding a drug to its target binding site.	

incorrect	**a.#**	Pharmacokinetics.
incorrect	**b.#**	Structure based drug design.
Correct	**c.#**	Pharmacodynamics.
incorrect	**d.#**	Structure-activity relationships.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 08 - Question 05	
05)	Which of the following terms best describes the study of how drugs reach their target in the body and how the levels of a drug in the blood are affected by absorption, distribution, metabolism, and excretion?
Feedback: Pharmacokinetics covers such topics as the absorption, distribution, metabolism, and excretion of a drug in a living system. Structure-based drug design is the study of how drugs can be designed using molecular modelling based on a drug's pharmacophore. Pharmacodynamics is the study of how a drug interacts with its target binding site at the molecular level. Structure-activity relationships is the study of which functional groups are important in binding a drug to its target binding site.	

Correct	**a.#**	Pharmacokinetics.
incorrect	**b.#**	Structure based drug design.
incorrect	**c.#**	Pharmacodynamics.
incorrect	**d.#**	Structure-activity relationships.
END OF QUESTION		

Type: multiple choice
Title: Chapter 08 - Question 06

06)	Which of the following terms best describes the study of which functional groups are important in binding a drug to its target binding site, and the identification of a pharmacophore?	
Feedback: Pharmacokinetics covers such topics as the absorption, distribution, metabolism, and excretion of a drug in a living system. Structure-based drug design is the study of how drugs can be designed using molecular modelling based on a drug's pharmacophore. Pharmacodynamics is the study of how a drug interacts with its target binding site at the molecular level. Structure-activity relationships is the study of which functional groups are important in binding a drug to its target binding site.		
incorrect	**a.#**	Pharmacokinetics.
incorrect	**b.#**	Structure based drug design.
incorrect	**c.#**	Pharmacodynamics.
Correct	**d.#**	Structure-activity relationships.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 08 - Question 07		
07)	Which of the following statements best describes the efficacy of a drug?	
Feedback: Affinity is a measure of how strongly a drug binds to a receptor. Efficacy is a measure of the maximum biological effect resulting from a drug binding to its target. Potency is a measure of the amount of drug required to produce a defined biological effect. An indication of a drug's life time in the body is generally indicated by its half life - the time taken for blood levels to drop to 50% of the maximum level.		
Correct	**a.#**	The maximum biological effect resulting from a drug binding to its target.
incorrect	**b.#**	The measure of how strongly a drug binds to a receptor.
incorrect	**c.#**	The amount of drug required to produce a defined biological effect.
incorrect	**d.#**	The lifetime of the drug in the body.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 08 - Question 08		
08)	Which of the following statements best describes the affinity of a drug?	
Feedback: Affinity is a measure of how strongly a drug binds to a receptor. Efficacy is a measure of the maximum biological effect resulting from a drug binding to its target. Potency is a measure of the amount of drug required to produce a defined biological effect. An indication of a drug's life time in the body is generally indicated by its half life - the time taken for blood levels to drop to 50% of the maximum level.		
incorrect	**a.#**	The maximum biological effect resulting from a drug binding to its target.
Correct	**b.#**	The measure of how strongly a drug binds to a receptor.
incorrect	**c.#**	The amount of drug required to produce a defined biological effect.
incorrect	**d.#**	The lifetime of the drug in the body.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 08 - Question 09		
09)	Which of the following statements best describes the potency of a drug?	
Feedback: Affinity is a measure of how strongly a drug binds to a receptor. Efficacy is a measure of the maximum biological effect resulting from a drug binding to its target. Potency is a measure of the amount of drug required to produce a defined biological effect. The lifetime of the drug is related to its stability in the body and not to its biological effect.		
incorrect	**a.#**	The maximum biological effect resulting from a drug binding to its target.
incorrect	**b.#**	The measure of how strongly a drug binds to a receptor.
Correct	**c.#**	The amount of drug required to produce a defined biological effect.
incorrect	**d.#**	The lifetime of the drug in the body.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 08 - Question 10		
10)	Which of the following symbols represents the inhibitory or affinity constant?	
Feedback: The concentration of antagonist required to inhibit the binding of an agonist by 50% is known as the IC_{50} value. The inhibitory or affinity constant is denoted as K_i. The dissociation binding constant is denoted as K_d. The concentration of a drug required to produce 50% of a maximum possible effect is defined as the EC_{50} value.		
incorrect	**a.#**	IC_{50}
Correct	**b.#**	K_i
incorrect	**c.#**	K_d
incorrect	**d.#**	EC_{50}
END OF QUESTION		

Type: multiple choice		
Title: Chapter 08 - Question 11		
11)	Which of the following symbols represents the concentration of a drug required to produce 50% of a maximum possible effect?	
Feedback: The concentration of antagonist required to inhibit the binding of an agonist by 50% is known as the IC_{50} value. The inhibitory or affinity constant is denoted as K_i The dissociation binding constant is denoted as K_d. The concentration of a drug required to produce 50% of a maximum possible effect is defined as the EC_{50} value.		
incorrect	**a.#**	IC_{50}
incorrect	**b.#**	K_i
incorrect	**c.#**	K_d
Correct	**d.#**	EC_{50}
END OF QUESTION		

Type: multiple choice		
Title: Chapter 08 - Question 12		
12)	Which of the following symbols represents the concentration of antagonist required to inhibit the binding of an agonist by 50%?	
Feedback: The concentration of antagonist required to inhibit the binding of an agonist by 50% is known as the IC_{50} value. The inhibitory or affinity constant is denoted as K_i. The dissociation binding constant is denoted as K_d. The concentration of a drug required to produce 50% of a maximum possible effect is defined as by the EC_{50} value.		
Correct	**a.#**	IC_{50}
incorrect	**b.#**	K_i
incorrect	**c.#**	K_d
incorrect	**d.#**	EC_{50}
END OF QUESTION		

Type: multiple choice		
Title: Chapter 08 - Question 13		
13)	Which of the following symbols represents the dissociation binding constant?	
Feedback: The concentration of antagonist required to inhibit the binding of an agonist by 50% is known as the IC_{50} value. The inhibitory or affinity constant is denoted as K_i. The dissociation binding constant is denoted as K_d. The concentration of a drug required to produce 50% of a maximum possible effect is defined as the EC_{50} value.		
incorrect	**a.#**	IC_{50}
incorrect	**b.#**	K_i
Correct	**c.#**	K_d
incorrect	**d.#**	EC_{50}
END OF QUESTION		

Type: multiple choice		
Title: Chapter 08 - Question 14		
14)	Which of the following descriptions most accurately describes binding sites and binding regions?	
Feedback: The binding site is an area of the receptor into which a drug or a natural chemical messenger can bind. Specific binding regions within that site 'recognise' specific binding groups on drugs or ligands to allow binding interactions to take place.		
incorrect	**a.#**	A binding site is part of a binding region.
Correct	**b.#**	A binding region is part of a binding site.
incorrect	**c.#**	A binding region is the same as a binding site.
incorrect	**d.#**	A binding region is on a drug whereas a binding site is on a macromolecular target.
END OF QUESTION		

Type: multiple choice

Title: Chapter 08 - Question 15		
15)	Which of the following is not a necessary requirement when designing an agonist?	
Feedback: The correct binding groups and their relative orientation are essential to achieve binding. The size and shape of the molecule are crucial if the drug is to fit the binding site. It is not essential to have exactly the same functional groups as the endogenous chemical messenger. Different functional groups can be used if they interact in the same way as the original functional group. For example, it might be able to replace an amine functional group with an alcohol functional group since both can participate in hydrogen bonding.		
incorrect	**a.#**	The presence of the correct binding groups.
incorrect	**b.#**	The correct size and shape of the molecule.
incorrect	**c.#**	The correct relative orientation of the binding groups.
Correct	**d.#**	The identical functional groups present in the endogenous chemical messenger.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 08 - Question 16		
16)	What sort of agent binds to a receptor using a different binding site from that used by the endogenous chemical messenger, and distorts the binding site such that the endogenous messenger cannot bind?	
Feedback: An agonist binds in the same way as the natural messenger and activates the receptor. An allosteric antagonist binds to a different binding site, but the resulting induced fit distorts the binding site for the chemical messenger such that the latter is no longer recognised. The 'umbrella effect' occurs when an antagonist binds to a site next to the normal binding site and part of the molecule overlaps the normal binding site such that it acts as a steric shield. An inverse agonist binds to the receptor and stabilises an inactive conformation such that the natural messenger cannot bind. Furthermore, any inherent activity is lost. Such activity is associated with low levels of the active form of the receptor. These are present despite the absence of an endogenous messenger but equilibrate to inactive conformations in the presence of an inverse agonist.		
incorrect	**a.#**	An agonist.
Correct	**b.#**	An allosteric antagonist.
incorrect	**c.#**	An antagonist acting by the 'umbrella' effect.
incorrect	**d.#**	An inverse agonist.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 08 - Question 17	
17)	What sort of agent binds to a binding site that is next to the binding site for an endogenous chemical messenger, and sterically blocks the messenger from binding?
Feedback: An agonist binds in the same way as the natural messenger and activates the receptor. An allosteric antagonist binds to a different binding site, but the resulting induced fit distorts the binding site for the chemical messenger such that the latter is no longer recognised. The 'umbrella effect' occurs when an antagonist binds to a site next to	

the normal binding site and part of the molecule overlaps the normal binding site such that it acts as a kind of steric shield. An inverse agonist binds to the receptor and stabilises an inactive conformation such that the natural messenger cannot bind. Furthermore, any inherent activity is lost. Such activity is associated with low levels of the active form of the receptor. These are present despite the absence of an endogenous messenger but equilibrate to inactive conformations in the presence of an inverse agonist.		
incorrect	**a.#**	An agonist.
incorrect	**b.#**	An allosteric antagonist.
Correct	**c.#**	An antagonist acting by the 'umbrella' effect.
incorrect	**d.#**	An inverse agonist.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 08 - Question 18	
18)	What process takes place to produce desensitisation of a receptor to an agonist?
Feedback: Phosphorylation of an alcohol or a phenol places an ionised phosphate group at these positions. This can result in a change in receptor conformation to an inactive form since ionic interactions are now favoured over hydrogen bond interactions.	

incorrect	**a.#**	Esterification.
Correct	**b.#**	Phosphorylation.
incorrect	**c.#**	Peptide formation.
incorrect	**d.#**	Peptide hydrolysis.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 08 - Question 19	
19)	Which of the following terms best applies to the following; the situation where increasing doses of a drug are required to have the same biological effect?
Feedback: Sensitisation can occur when a receptor is exposed to an antagonist for long periods of time. The cell synthesises more receptors to counteract the effect of the antagonist. Tolerance refers to the growing dose levels of drug required to achieve the same effect, as the body adapts to the presence of the drug. Dependence relates to the situation where the body has adapted to the presence of the drug, and removal results in withdrawal symptoms. The patient does not feel 'normal' in the absence of the drug. Acceptance is not a term that is used in this context.	

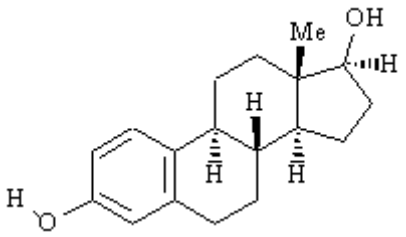
incorrect	**a.#**	Sensitisation.
Correct	**b.#**	Tolerance.
incorrect	**c.#**	Dependence.
incorrect	**d.#**	Acceptance.
END OF QUESTION		

Type: multiple choice

Title: Chapter 08 - Question 20		
20)	Which of the following terms best applies to the following; the situation where cessation of a drug results in withdrawal symptoms?	
Feedback: Sensitisation can occur when a receptor is exposed to an antagonist for long periods of time. The cell synthesises more receptors to counteract the effect of the antagonist. Tolerance refers to the growing dose levels of drug required to achieve the same effect, as the body adapts to the presence of the drug. Dependence relates to the situation where the body has adapted to the presence of the drug, and removal results in withdrawal symptoms. The patient does not feel 'normal' in the absence of the drug. Acceptance is not a term that is used in this context.		
incorrect	**a.#**	Sensitisation.
incorrect	**b.#**	Tolerance.
Correct	**c.#**	Dependence.
incorrect	**d.#**	Acceptance.
END OF QUESTION		

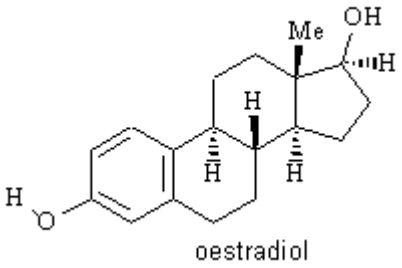
Type: multiple choice		
Title: Chapter 08 - Question 21		
21)	Which of the following situations is feasible as an explanation for tolerance and dependence?	
Feedback: As more receptors are produced, more antagonist is needed to block them, resulting in the need to increase dose levels. On stopping the antagonist, the cell has an excess of receptors and is supersensitive to the natural messenger. This results in withdrawal symptoms. If an increased production of receptors took place to counteract the presence of an agonist, the effect of agonists would be enhanced without changing the dose level. If a decreased production of receptors took place to counteract the presence of an antagonist, the antagonist would become more effective, not less. The same would be true if there was a decreased synthesis of chemical messenger in response to the presence of an antagonist. None of these effects are observed.		
Correct	**a.#**	An increased production of receptors to counteract the presence of an antagonist.
incorrect	**b.#**	An increased production of receptors to counteract the presence of an agonist.
incorrect	**c.#**	A decreased production of receptors to counteract the presence of an antagonist.
incorrect	**d.#**	A decreased synthesis of chemical messenger to counteract the presence of an antagonist.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 08 - Question 22	
22)	Which of the following statements is not true regarding the binding of estradiol to the estrogen receptor?

		
Feedback: The aromatic ring plays an important role in the binding process since it fits into a tight region of the binding site and thus orientates the rest of the molecule.		
incorrect	**a.#**	The alcohol and phenol groups of estradiol are important hydrogen bonding groups when estradiol binds to the binding site.
Correct	**b.#**	The aromatic ring is not important in the binding process.
incorrect	**c.#**	The hydrophobic skeleton of the molecule forms hydrophobic and van der Waals interactions with the binding site.
incorrect	**d.#**	The binding site into which estradiol fits is spacious except for the region accepting the aromatic ring.
END OF QUESTION		

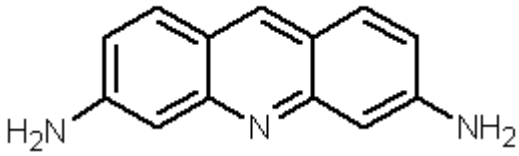
Type: multiple choice		
Title: Chapter 08 - Question 23		
23)	Which of the following statements is not true regarding the actions of the estrogen receptor?	
Feedback: The answer is accurate, except that a nuclear **transcription** factor is formed.		
incorrect	**a.#**	Binding of estradiol causes dimerisation of the receptor.
incorrect	**b.#**	Binding of estradiol causes a conformational change in the receptor.
incorrect	**c.#**	A helical section of the receptor protein shifts position to seal estradiol into the binding site and reveals a hydrophobic region called the deactivating function region.
Correct	**d.#**	A coactivator protein binds to complete a nuclear translation factor.
END OF QUESTION		

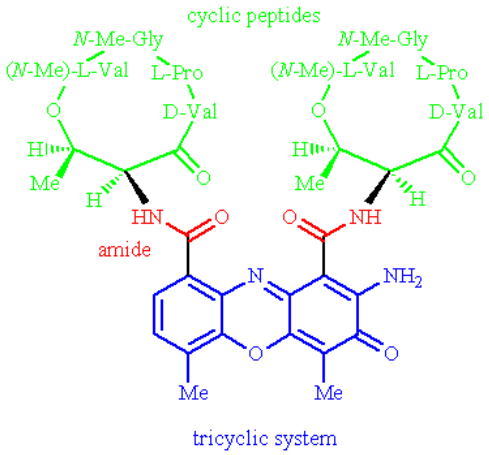
Type: multiple choice	
Title: Chapter 08 - Question 24	
24)	Which of the following statements is not true regarding the binding of estradiol to the estrogen receptor?

	 oestradiol

Feedback: The overall binding site is actually quite spacious, except for the region that accepts the aromatic ring of estradiol. All the other statements are true		
incorrect	**a.#**	The alcohol and phenol groups of estradiol are important hydrogen bonding groups when estradiol binds to the binding site.
incorrect	**b.#**	The aromatic ring is important in orientating the molecule in the binding site.
incorrect	**c.#**	The hydrophobic skeleton of the molecule forms hydrophobic and van der Waals interactions with the binding site.
Correct	**d.#**	The binding site is a narrow slot into which estradiol fits extremely snugly.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 08 - Question 25		
25)	Which of the following statements is not true regarding the actions of the estrogen receptor?	
Feedback: The answer is accurate, except that a **hydrophobic** binding site is revealed rather than a hydrophilic binding site.		
incorrect	**a.#**	Binding of estradiol causes dimerisation of the receptor.
incorrect	**b.#**	Binding of estradiol causes a conformational change in the receptor.
Correct	**c.#**	A helical section of the receptor protein shifts position to seal estradiol into the binding site and reveals a hydrophilic region called the deactivating function region.
incorrect	**d.#**	A coactivator protein binds to complete a nuclear transcription factor.
END OF QUESTION		

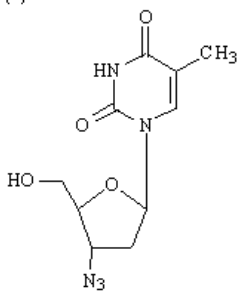
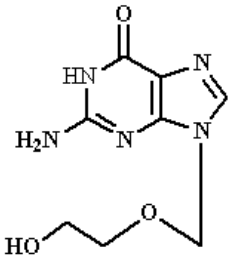
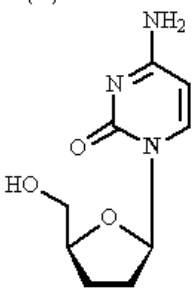
Type: multiple choice		
Title: Chapter 09 - Question 01		
01)	Proflavine is an intercalating agent which was used to treat wounded soldiers in the Far East during the second world war. Which of the following statements is false regarding proflavine?	
Feedback: Proflavine is too toxic to be used systemically since it does not distinguish between the DNA of host cells and those of an invading pathogen. It is used topically for the treatment of infected deep surface wounds. All the other statements are correct.  		
Correct	**a.#**	It a useful systemic antibacterial agent.
incorrect	**b.#**	Its planar shape is important to intercalation.
incorrect	**c.#**	The primary amine groups are protonated and interact with phosphate groups of the DNA backbone.
incorrect	**d.#**	When inserted into DNA, the tricyclic system can form van der Waals interactions with base pairs above and below it.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 09 - Question 02		
02)	The following antibiotic structure (Dactinomycin) is an intercalating agent. Which region of the molecule is chiefly responsible for its ability to intercalate into DNA?  	
Feedback: The tricyclic system is planar and is the correct size to intercalate into the DNA double helix. The macrocyclic peptide rings can interact with the sugar phosphate backbone through polar interactions.		
Correct	**a.#**	The tricyclic system.

incorrect	**b.#**	One of the macrocyclic peptide rings.
incorrect	**c.#**	Both of the macrocyclic peptide rings.
incorrect	**d.#**	The amide bonds.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 09 - Question 03	
03)	What term is used to describe the strategy where an oligonucleotide is designed such that it base pairs with a segment of messenger RNA and blocks protein translation?
Feedback: The other terms are not used. Anticodon therapy could have been a reasonable term to use for the therapy since the drugs used contain anticodons that can base pair with the codons present on messenger RNA. However, antisense therapy is the term that is used.	

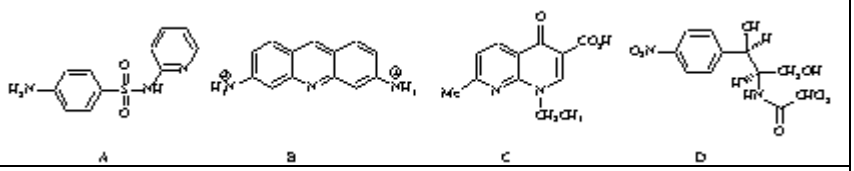
incorrect	**a.#**	Sense therapy.
incorrect	**b.#**	Anticodon therapy.
incorrect	**c.#**	Codon therapy.
Correct	**d.#**	Antisense therapy.
END OF QUESTION		

Type: matching	
Title: Chapter 09 - Question 04	
04)	The following compounds (i), (ii) and (iii) are antiviral drugs which mimic natural nucleosides. What nucleosides do they mimic? (i)  (ii)  (iii) 
Feedback: The antiviral drugs contain the nucleic acid bases adenine, guanine, and cytosine respectively. They all contain incomplete sugar moieties and if they are incorporated into a growing DNA chain, further chain growth is terminated.	

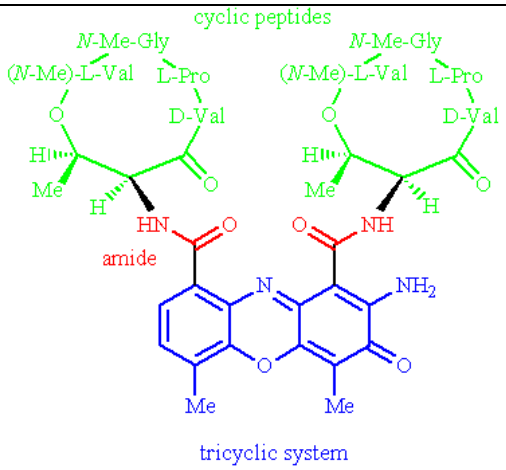
a.#	Deoxythymidine	=	(i)
b.#	Deoxyguanosine	=	(ii)
c.#	Deoxycytidine	=	(iii)
d.#	Deoxyadenosine	=	None of the answers

END OF QUESTION

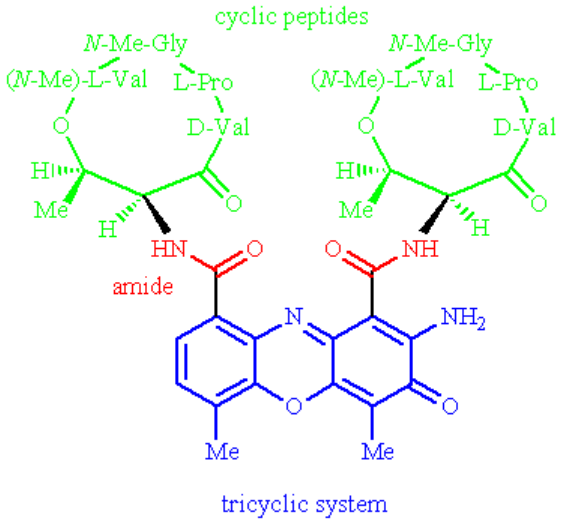
Type: multiple choice		
Title: Chapter 09 - Question 05		
05)	Which statement is false regarding the fluoroquinoline antibacterial agents?	
Feedback: The drugs target enzymes called topoisomerases. Helicases catalyse an unwinding process and the topoisomerases are required to relieve the strain resulting from this uncoiling.		
Correct	**a.#**	They target enzymes called helicases.
incorrect	**b.#**	They inhibit replication.
incorrect	**c.#**	They inhibit transcription.
incorrect	**d.#**	They stabilise an enzyme-DNA-drug complex where four fluoroquinoline molecules are stacked together.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 09 - Question 06		
06)	Proflavine is an antibacterial agent that was used in the second world war. Which one of the following structures is proflavine?  A B C D	
Feedback: The other structures are sulphapyridine (A), nalidixic acid (C), and chloramphenicol (D).		
incorrect	**a.#**	Structure A.
Correct	**b.#**	Structure B.
incorrect	**c.#**	Structure C.
incorrect	**d.#**	Structure D.
END OF QUESTION		

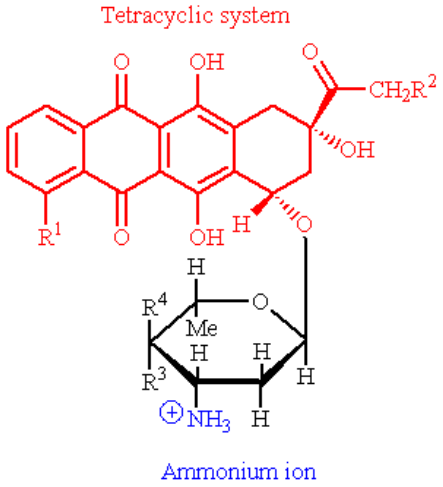
Type: multiple choice	
Title: Chapter 09 - Question 07	
07)	Dactinomycin is an intercalating anticancer agent.

	 Interactions with particular base pairs are favoured in the intercalation of this compound. Which base pairs are favoured?

Feedback: Note that adenine-uracil and thymine-guanine are not possible. First, uracil is present in RNA and not DNA. Second, thymine base pairs with adenine and not guanine. Similarly, guanine base pairs with cytosine and not thymine.		
Correct	**a.#**	Cytosine-guanine.
incorrect	**b.#**	Adenine-thymine.
incorrect	**c.#**	Adenine-uracil.
incorrect	**d.#**	Thymine-guanine.
END OF QUESTION		

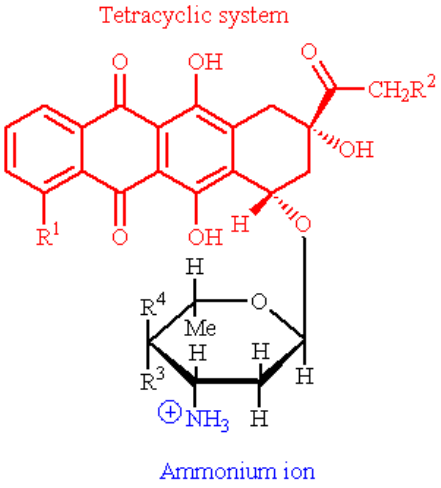
Type: multiple choice	
Title: Chapter 09 - Question 08	
08)	Dactinomycin is an intercalating anticancer agent.  Which region of the molecule interacts with the sugar phosphate backbone of DNA?
Feedback: The cyclic peptides have polar functional groups which can interact with	

the polar phosphate groups of the sugar phosphate chain.		
Correct	**a.#**	The cyclic peptides.
incorrect	**b.#**	The amide groups.
incorrect	**c.#**	The tricyclic system.
incorrect	**d.#**	All of the answers are correct.
END OF QUESTION		

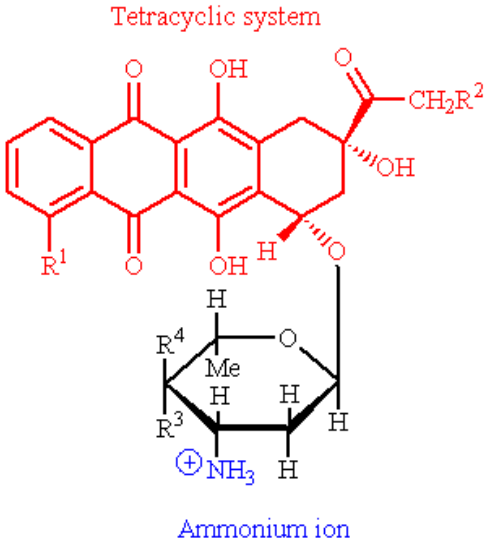
Type: multiple choice	
Title: Chapter 09 - Question 09	
09)	The following group of structures (tetracyclines) are intercalating anticancer agents.  To which family of compounds do these structures belong?

Feedback: The anthracyclines contain a tetracyclic ring system where all the rings are six membered. There is also a sugar substituent. The bleomycins are glycoproteins which include a bithiazole ring system. The podophyllotoxins have a tetracyclic ring system involving two five-membered rings and two six-membered rings. The camptothecins have a pentacyclic ring system involving four six-membered rings and a five-membered ring.		
incorrect	**a.#**	Podophyllotoxins.
incorrect	**b.#**	Camptothecins.
Correct	**c.#**	Anthracyclines.
incorrect	**d.#**	Bleomycins.
END OF QUESTION		

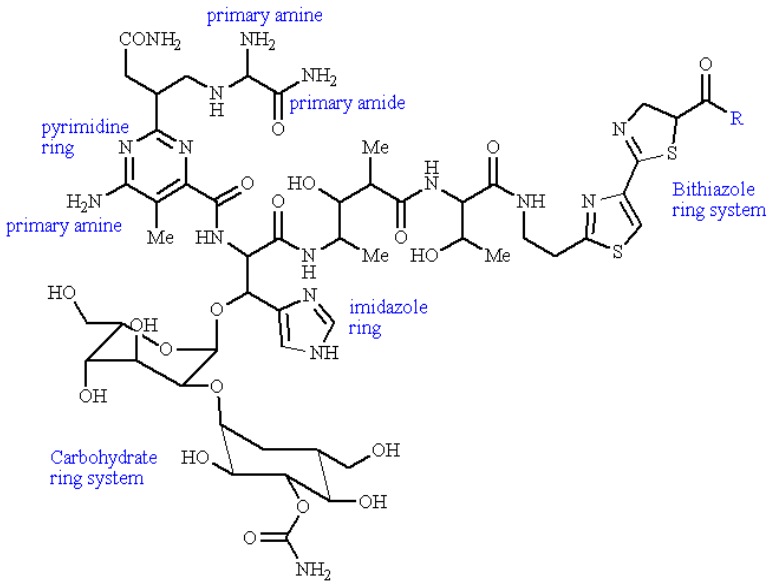
Type: multiple choice	
Title: Chapter 09 - Question 10	
10)	The following group of structures are intercalating anticancer agents.

	 How many rings of the tetracyclic system are involved in the intercalation?

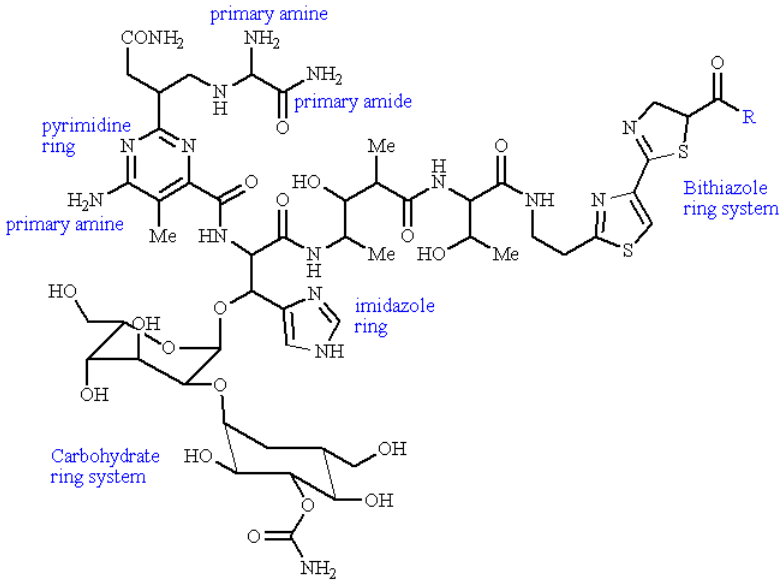
Feedback: The three 'left hand' rings are planar and are involved in the intercalation mechanism.		
incorrect	**a.#**	1
incorrect	**b.#**	2
Correct	**c.#**	3
incorrect	**d.#**	4
END OF QUESTION		

Type: multiple choice	
Title: Chapter 09 - Question 11	
11)	The following group of structures are intercalating anticancer agents.  What role does the ammonium ion play in the molecule's mode of action?

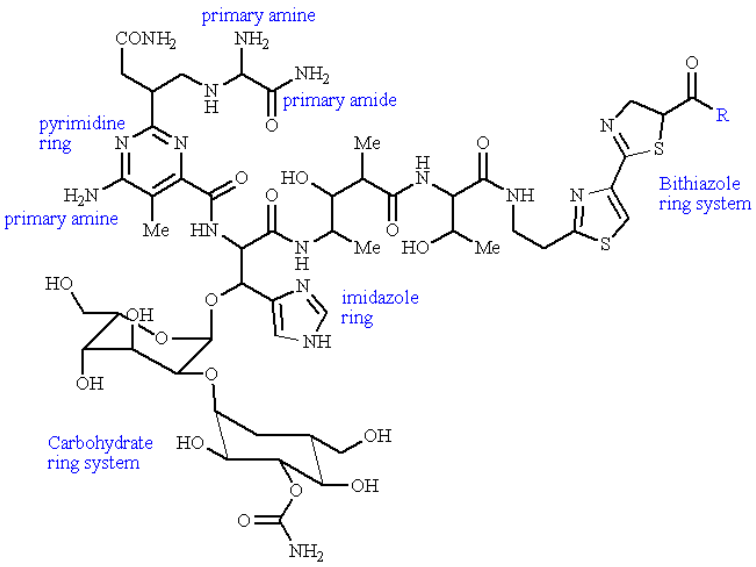
Feedback: The ammonium ion is positively charged and the phosphate ion on the DNA backbone is negatively charged and so a strong ionic bond can be formed between the two groups which helps to lock the molecule in position.		
incorrect	**a.#**	It makes the molecule water soluble.
incorrect	**b.#**	It prevents the molecule crossing cell membranes.
incorrect	**c.#**	It forms a hydrogen bond to a deoxyribose sugar moiety on the DNA backbone.
Correct	**d.#**	It forms an ionic bond to a phosphate group on the DNA backbone.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 09 - Question 12		
12)	The following is representative of a group of natural products which are intercalating anticancer agents.  What are these group of compounds called?	
Feedback: The anthracyclines contain a tetracyclic ring system where all the rings are six membered. There is also a sugar substituent. The bleomycins are glycoproteins which include a bithiazole ring system. The podophyllotoxins have a tetracyclic ring system involving two five-membered rings and two six-membered rings. The camptothecins have a pentacyclic ring system involving four six-membered rings and one five-membered ring.		
incorrect	**a.#**	Podophyllotoxins.
incorrect	**b.#**	Camptothecins.
incorrect	**c.#**	Anthracyclines.
Correct	**d.#**	Bleomycins.
END OF QUESTION		

Type: multiple choice

Title: Chapter 09 - Question 13		
13)	Bleomycins are natural products which are intercalating anticancer agents.  Which region of the molecule is involved in intercalation?	
Feedback: The bithiazole ring system is involved in intercalation. The carbohydrate ring system is too bulky to act as an intercalating unit. The pyrimidine and imidazole rings are important in chelating a ferrous ion and the mechanism of action.		
incorrect	**a.#**	The carbohydrate ring system.
incorrect	**b.#**	The pyrimidine ring.
incorrect	**c.#**	The imidazole ring.
Correct	**d.#**	The bithiazole ring system.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 09 - Question 14	
14)	Bleomycins are natural products which are intercalating anticancer agents.



They chelate a ferrous ion as part of their mechanism of action. Which region of the molecule is important in chelating ions?

Feedback: The nitrogen atoms present in all three groups are involved in chelating the ferrous ion.

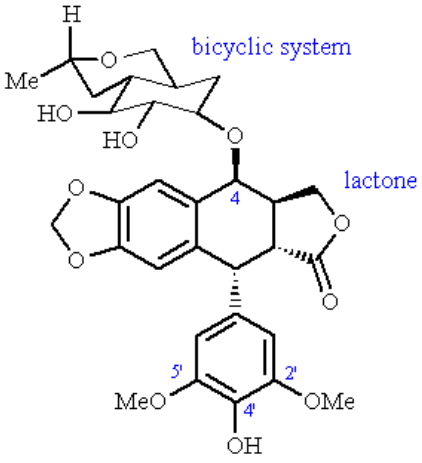
incorrect	**a.#**	The pyrimidine ring.
incorrect	**b.#**	The imidazole ring.
incorrect	**c.#**	The primary amine groups.
Correct	**d.#**	All of the answers are correct.

END OF QUESTION

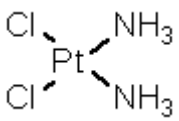
Type: matching

Title: Chapter 09 - Question 15

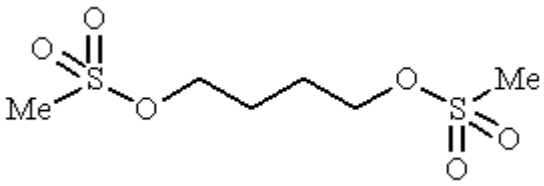
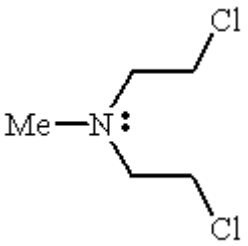
15) Identify the following structures:



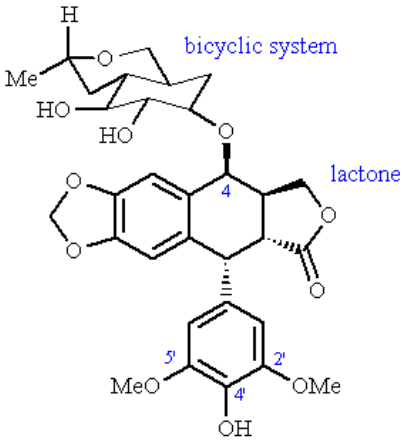
A

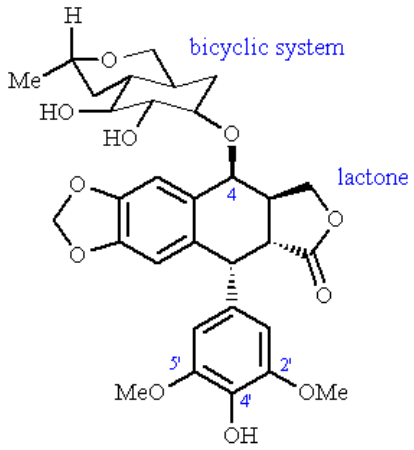


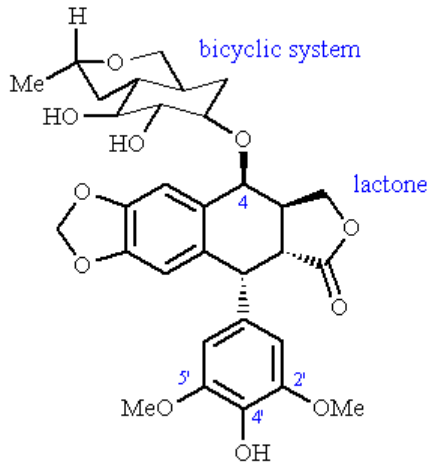
B

	 C  D

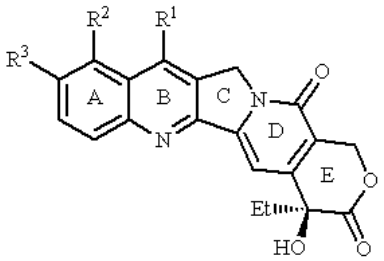
Feedback: Busulfan and chlormethine are alkylating agents. Cisplatin is a metallating agent. Etoposide is a topoisomerase poison.			
a.#	Structure A	=	Etoposide
b.#	Structure D	=	Chlormethine
c.#	Structure C	=	Busulfan
d.#	Structure B	=	Cisplatin
END OF QUESTION			

Type: multiple choice		
Title: Chapter 09 - Question 16		
16)	The following structure (etoposide) is used as an anticancer agent  Etoposide is a semi-synthetic derivative of which compound?	
Feedback: Etoposide is a semi-synthetic derivative of epipodophyllotoxin. Teniposide is a closely related analogue. Podophyllotoxin is a stereoisomer of epipodophyllotoxin.		
Correct	**a.#**	Epipodophyllotoxin.
incorrect	**b.#**	Teniposide.
incorrect	**c.#**	Etoposide.
incorrect	**d.#**	Podophyllotoxin.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 09 - Question 17		
17)	The following structure (etoposide) is used as an anticancer agent  Which enzyme is targeted by etoposide?	
Feedback: Topoisomerase II catalyses the temporary cleavage of DNA strands, the crossing of the strands and resealing of the strands. Etoposide stabilises the complex formed between DNA and the enzyme, thus preventing the completion of this process.		
Correct	**a.#**	Topoisomerase II.
incorrect	**b.#**	DNA polymerase.
incorrect	**c.#**	DNA ligase.
incorrect	**d.#**	Topoisomerase I.
END OF QUESTION		

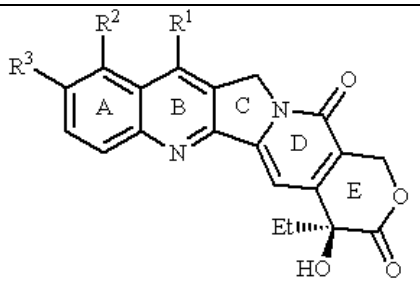
Type: multiple choice	
Title: Chapter 09 - Question 18	
18)	Etoposide is used as an anticancer agent  Which of the following groups is crucial to the mechanism of etoposide in producing radical species?

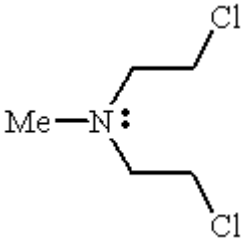
Feedback: Oxidation of the 4'-phenolic group can produce a semiquinone free radical.		
incorrect	**a.#**	The bicyclic substituent at position 4.
incorrect	**b.#**	The lactone.
incorrect	**c.#**	The methoxy substituents at positions 2' and 5'.
Correct	**d.#**	The phenolic substituent at position 4'.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 09 - Question 19	
19)	The following group of compounds are cytotoxic alkaloids used as anticancer agents.  What are the group of compounds illustrated above?

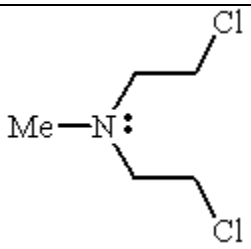
Feedback: The anthracyclines contain a tetracyclic ring system where all the rings are six membered. There is also a sugar substituent. The bleomycins are glycoproteins which include a bithiazole ring system. The podophyllotoxins have a tetracyclic ring system involving two five-membered rings and two six-membered rings. The camptothecins have a pentacyclic ring system involving four six-membered rings and one five-membered ring.		
incorrect	**a.#**	Bleomycins.
Correct	**b.#**	Camptothecins.
incorrect	**c.#**	Podophyllotoxins.
incorrect	**d.#**	Anthracyclines.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 09 - Question 20	
20)	Camptothecins are cytotoxic alkaloids used as anticancer agents.

	 What is the target for these compounds?	
Feedback: Topoisomerase I is an enzyme that relieves strain in DNA by cleaving one strand of the DNA double helix in order to pass one strand through the other.		
incorrect	**a.#**	Topoisomerase II.
incorrect	**b.#**	DNA polymerases.
incorrect	**c.#**	DNA ligase.
Correct	**d.#**	Topoisomerase I.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 09 - Question 21		
21)	The following agent is used for the treatment of Hodgkin's lymphoma as part of a multi-drug regime.  What sort of agent is it?	
Feedback: The agent alkylates guanine moieties of DNA.		
incorrect	**a.#**	A topoisomerase inhibitor.
Correct	**b.#**	An alkylating agent.
incorrect	**c.#**	An antimetabolite.
incorrect	**d.#**	A receptor antagonist.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 09 - Question 22	
22)	The following agent is used for the treatment of Hodgkin's lymphoma as part of a multi-drug regime.

	 What is the name of the activated species which is formed from the above structure and which acts as the actual alkylating agent?

Feedback: Neighbouring group participation by the lone pair of the nitrogen forms the aziridinium ion with loss of chloride ion. The aziridinium ion acts as the alkylating agent.

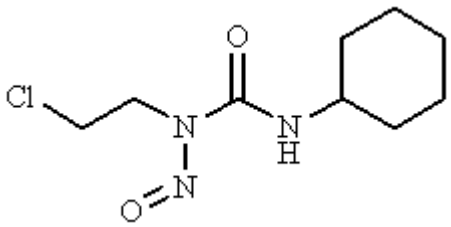
incorrect	**a.#**	A quaternary ammonium ion.
incorrect	**b.#**	An iminium ion.
incorrect	**c.#**	A cyclopropylamine ion.
Correct	**d.#**	An aziridinium ion.

END OF QUESTION

Type: multiple choice

Title: Chapter 09 - Question 23

23) The following structure is called lomustine, and is used in the treatment of brain tumours.



To which group of compounds does it belong?

Feedback: The structure shown contains a nitrosourea functional group and so it is a nitrosourea. The nitrogen mustards contain alkyl chloride groups attached to an amine. The camptothecins are pentacyclic structures. The podophyllotoxins are tetracyclic structures with an aromatic substituent.

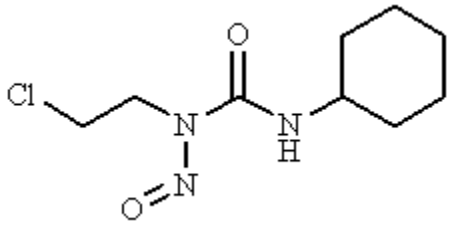
Correct	**a.#**	Nitrosoureas.
incorrect	**b.#**	Nitrogen mustards.
incorrect	**c.#**	Camptothecins.
incorrect	**d.#**	Podophyllotoxins.

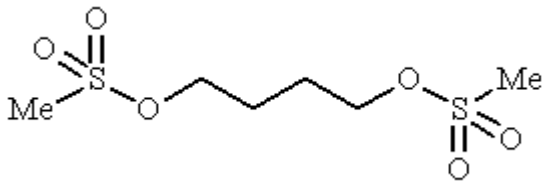
END OF QUESTION

Type: multiple choice

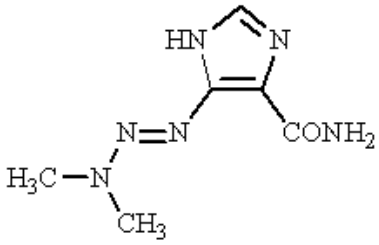
Title: Chapter 09 - Question 24

24) The following structure is called lomustine and is used in the treatment of brain tumours.

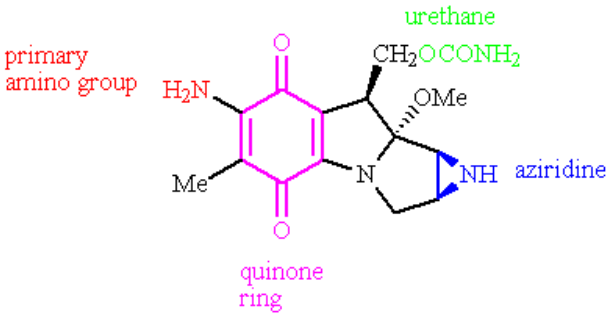
	 Which of the following statements is true regarding the above structure?	
Feedback: Lomustine is a lipid soluble agent that **can** cross the blood brain barrier, and <u>can</u> be given orally. It **decomposes spontaneously** in the body without the need of an enzyme. In doing so, it generates two species, one of which acts as an alkylating agent, and the other which acts as a carbamoylating agent.		
incorrect	**a.#**	It cannot cross the blood brain barrier.
incorrect	**b.#**	It is activated by an enzymatic metabolic reaction.
incorrect	**c.#**	It cannot be given orally.
Correct	**d.#**	It is a prodrug for both an alkylating agent and a carbamoylating agent.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 09 - Question 25		
25)	The following agent (busulfan) is used in the treatment of chronic granulocytic leukaemia:  Which of the following statements is not true regarding the above agent?	
Feedback: The mechanism is an S_N2 nucleophilic substitution.		
incorrect	**a.#**	The sulfonate groups are good leaving groups.
incorrect	**b.#**	It readily alkylates guanine residues.
incorrect	**c.#**	It produces interstrand cross linking of DNA.
Correct	**d.#**	The mechanism involves an S_N1 nucleophilic substitution.
END OF QUESTION		

Type: multiple choice
Title: Chapter 09 - Question 26

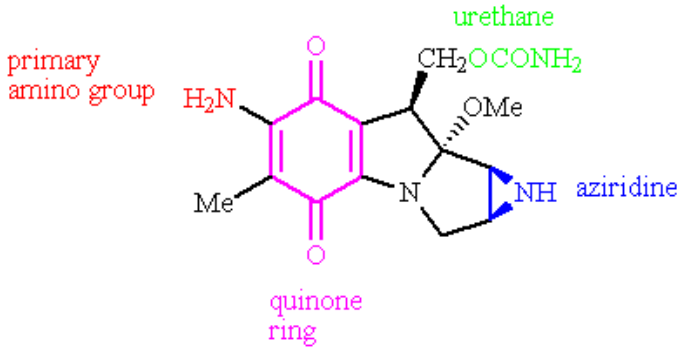
26)	The following agent (dacarbazine) is used in the treatment of a variety of cancers including melanoma.  Which of the following statements is true regarding dacarbazine?

Feedback: Dacarbazine is a prodrug which is activated by cytochrome P450 enzymes. Degradation results in the loss of **formaldehyde**. The active agent alkylates DNA.		
incorrect	**a.#**	It is a prodrug which is activated by peptidases.
Correct	**b.#**	The active agent is a methyldiazonium ion.
incorrect	**c.#**	Acetaldehyde is lost during degradation of the compound.
incorrect	**d.#**	The active agent acts as an enzyme inhibitor.
END OF QUESTION		

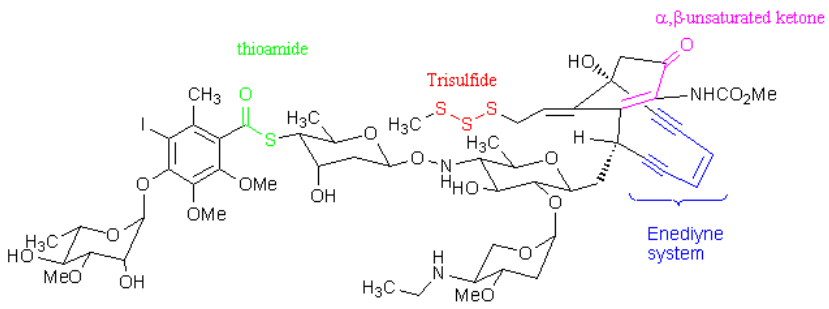
Type: multiple choice	
Title: Chapter 09 - Question 27	
27)	The following agent is called mitomycin C and is used as an anticancer drug against a variety of cancers.  The structure is a prodrug which is converted in the body to the active compound. What is the initial step in this activation process?

Feedback: Reduction of the quinone ring system generates a diphenol which then results in ring opening of the aziridine ring. Hydrolysis of the urethane group takes place subsequent to that. The primary amino group is not alkylated at all during the process. The drug is acting as an alkylating agent and so it is an amino group on DNA that gets alkylated.		
incorrect	**a.#**	Ring opening of the aziridine ring.
incorrect	**b.#**	Hydrolysis of the urethane group.

incorrect	**c.#**	Alkylation of the primary amino group.
Correct	**d.#**	Reduction of the quinone ring system.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 09 - Question 28		
28)	The following agent is called mitomycin C and is used as an anticancer drug against a variety of cancers:  Which of the following statements is true regarding the above drug?	
Feedback: Mitomycin is one of the most **toxic** anticancer drugs in clinical use. **Interstrand** cross linking takes place and not intrastrand cross linking. **Guanine** units are alkylated by the agent and not adenine units.		
incorrect	**a.#**	It is one of the safest anticancer agents used in clinical practice.
incorrect	**b.#**	It promotes intrastrand cross linking in DNA.
incorrect	**c.#**	Adenine units are alkylated by the agent.
Correct	**d.#**	It is possible that the drug may be more effective against tumours in oxygen starved environments.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 09 - Question 29	
29)	The following structure is calicheamicin γ^1 and is an anti tumour agent derived from a bacterium.

	 Which of the following functional groups is not involved in the mechanism of action?

Feedback: The trisulfide, α,β-unsaturated ketone, and the enediyne system are involved in a process which leads to cycloaromatisation and the generation of a diradical species. The thioamide group plays no role in the mechanism of action.

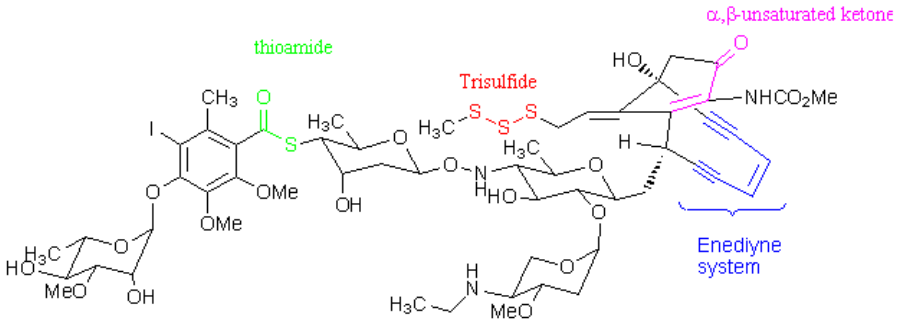
Correct	**a.#**	Thioamide.
incorrect	**b.#**	Trisulfide.
incorrect	**c.#**	α,β-unsaturated ketone.
incorrect	**d.#**	Enediyne system.

END OF QUESTION

Type: multiple choice

Title: Chapter 09 - Question 30

30) The following structure is calicheamicin γ^1 and it is an anti tumour agent derived from a bacterium:



What effect does the agent have?

Feedback: A diradical species is formed which abstracts two hydrogens from DNA and generates a diradical DNA species. Oxidative cleavage of the DNA chain then occurs.

incorrect	**a.#**	It alkylates DNA.
incorrect	**b.#**	It is an antimetabolite.
incorrect	**c.#**	It inhibits topoisomerases.
Correct	**d.#**	It cleaves the DNA chain by the production of radicals.

END OF QUESTION

Type: matching

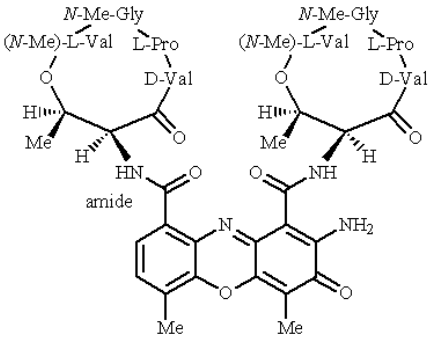
Title: Chapter 09 - Question 31

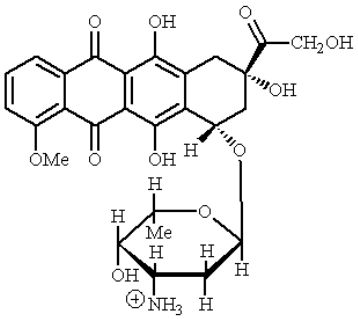
31)	Match the following agents (aciclovir, calicheamicin γ^1 chlormethine, doxorubicin) to their mechanism of action (alkylating agent, chain cutting agent, chain terminator, intercalating ag		
Feedback: Aciclovir has an incomplete sugar ring and can act as a chain terminator if it is added to the growing strand of a new DNA molecule. Doxorubicin has a tetracyclic ring sytem, of which three rings are planar and can intercalate into DNA. Chlormethine has two alkyl chlorides which will react with nucleophiles on DNA as an alkylating agent. Calicheamicin γ^1 generates free radical species which result in DNA chain cutting.			
a.#	Doxorubicin	=	Intercalating agent
b.#	Chlormethine	=	Alkylating agent
c.#	Calicheamicin γ^1	=	Chain cutting agent
d.#	Aciclovir	=	Chain terminator
END OF QUESTION			

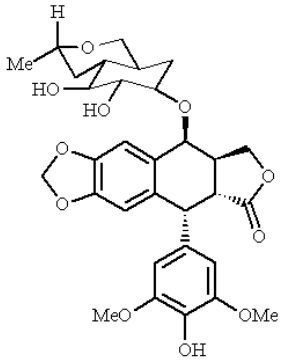
Type: multiple choice

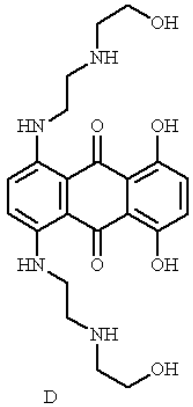
Title: Chapter 09 - Question 32

32) Which of the following structures is not an intercalating agent?





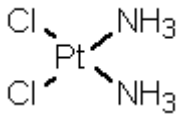


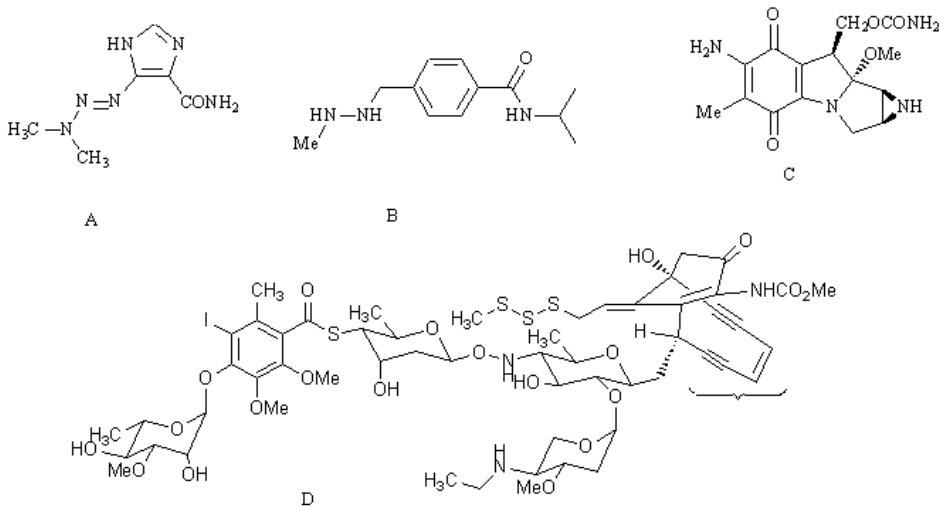


Feedback: The structures shown are dactinomycin, doxorubicin, etoposide, and mitoxantrone respecti

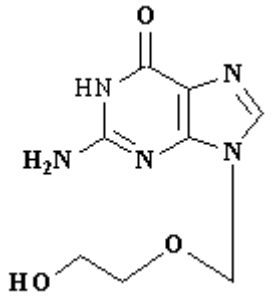
incorrect	**a.#**	Structure A.
incorrect	**b.#**	Structure B.
Correct	**c.#**	Structure C.

incorrect	**d.#**	Structure D.
END OF QUESTION		

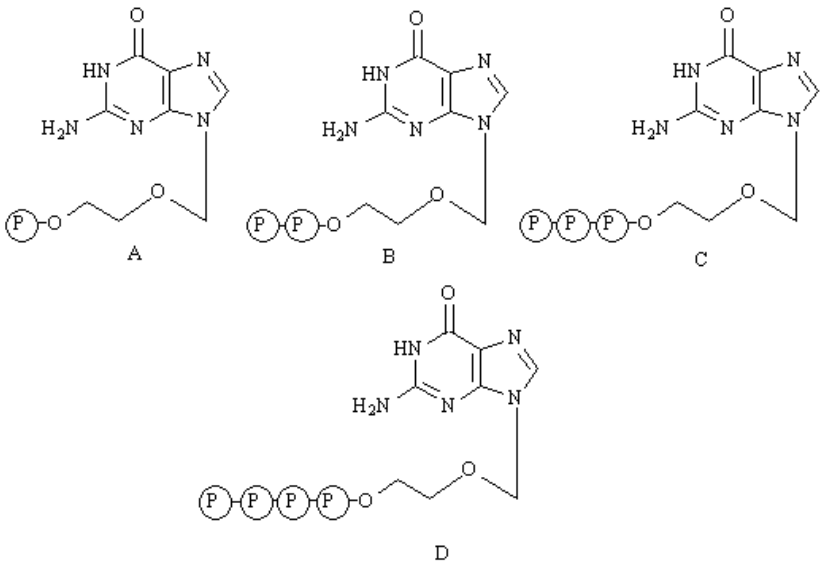
Type: multiple choice			
Title: Chapter 09 - Question 33			
33)	Cisplatin is used for the treatment of testicular and ovarian cancers.  Which of the following statements is untrue?		
Feedback: It is the chloro groups that are displaced.			
Correct	**a.#**	The amino groups are displaced before the drug becomes active.	
incorrect	**b.#**	Intrastrand cross linking of DNA results from the action of the agent.	
incorrect	**c.#**	Base pairing between cytosine and guanine is disrupted by the agent.	
incorrect	**d.#**	Ondansetron is often administered with the agent to reduce nausea.	
END OF QUESTION			

Type: matching			
Title: Chapter 09 - Question 34			
34)	Identify the following structures:  		
Feedback: Dacarbazine and procarbazine are related structures. Dacarbazine contains an imidazole ring, whereas procarbazine contains an aromatic ring and no imidazole ring. Calicheamicin is a complex structure which includes an unusual enediyne system. Mitomycin C is a tetracyclic structure which includes a three membered aziridine ring			
a.#	Procarbazine	=	Structure B

b.#	Calicheamicin γ^1	=	Structure D
c.#	Dacarbazine	=	Structure A
d.#	Mitomycin C	=	Structure C
END OF QUESTION			

Type: multiple choice		
Title: Chapter 09 - Question 35		
35)	What is the active form of aciclovir? 	
Feedback: The monophosphate and diphosphate of aciclovir are synthesised en route to the triphosphate, but it is the triphosphate which is the active form.		
incorrect	**a.#**	The compound itself.
incorrect	**b.#**	The monophosphate.
incorrect	**c.#**	The diphosphate.
Correct	**d.#**	The triphosphate.
END OF QUESTION		

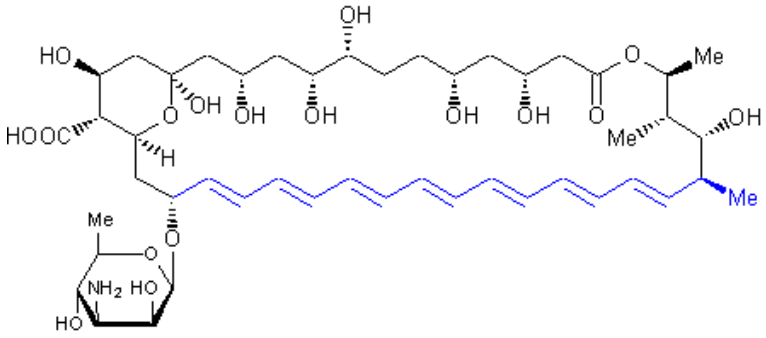
Type: multiple choice	
Title: Chapter 09 - Question 36	
36)	Aciclovir is a prodrug which is phosphorylated in virally infected cells to the active antiviral agent. Which of the following structures is the active agent?

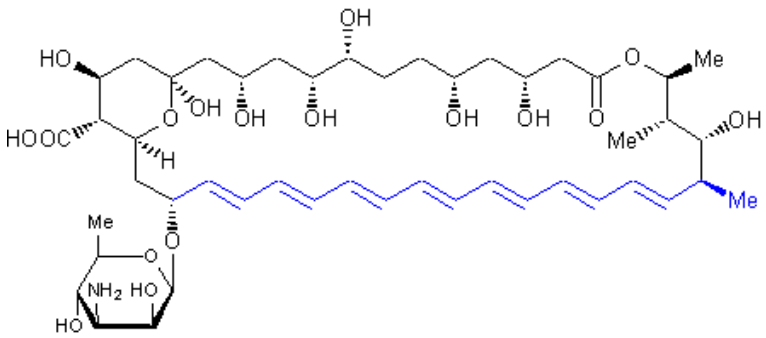
Feedback: It is the triphosphate which is the active compound. The monophosphate and diphosphate are biosynthetic intermediates to the triphosphate. The tetraphosphate is not formed.

incorrect	**a.#**	Structure A.
incorrect	**b.#**	Structure B.
Correct	**c.#**	Structure C.
incorrect	**d.#**	Structure D.

END OF QUESTION

Type: multiple choice		
Title: Chapter 10 – Question 01		
01)	Which of the following statements is true regarding amphotericin?  	
Feedback: The blue region is hydrophobic and interacts with the cell membrane as stated. Several molecules of amphotericin self assemble to form a **hydrophilic** tunnel through the cell membrane. The tunnel interior is lined with **polar** functional groups The amphotericin molecule is **not** the ideal length to span the cell membrane. It is **only** long enough to span half the width of the cell membrane and so the tunnel involves two clusters of amphotericin molecules arranged end to end. Amphotericin acts as an antifungal drug by **allowing** the uncontrolled passage of ions and water across the cell membrane		
Correct	**a.#**	The blue region is a hydrophobic region which interacts with the phospholipid bilayer of cell membranes by van der Waals interactions.
incorrect	**b.#**	Several molecules of amphotericin self assemble to form a hydrophobic tunnel through the cell membrane. The tunnel interior is lined with non polar functional groups.
incorrect	**c.#**	The amphotericin molecule is the ideal length to span the cell membrane.
incorrect	**d.#**	Amphotericin acts as an antifungal drug by preventing the passage of ions and water across the cell membrane.
END OF QUESTION		

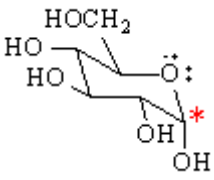
Type: multiple choice	
Title: Chapter 10 - Question 02	
02)	Identify the following antifungal agent:

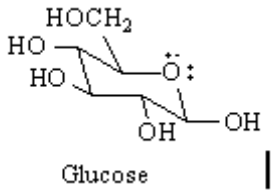
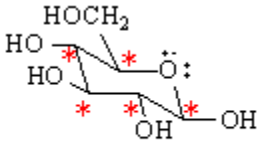
Feedback: Amphotericin is the correct answer. Ampicillin and amoxicillin are penicillins. Amprotropine is a cholinergic antagonist.		
incorrect	**a.#**	ampicillin.
Correct	**b.#**	amphotericin.
incorrect	**c.#**	amoxicillin.
incorrect	**d.#**	amprotropine.
END OF QUESTION		

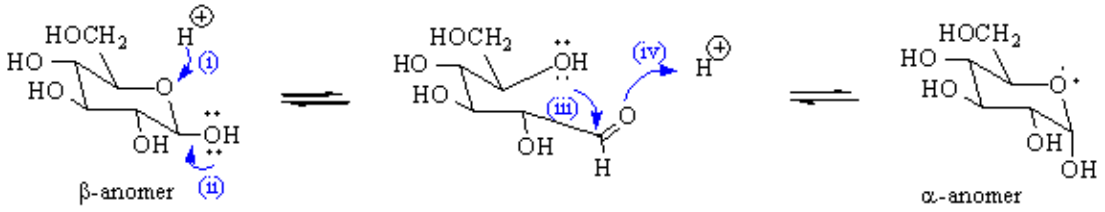
Type: multiple choice	
Title: Chapter 10 - Question 03	
03)	Which of the following statements is untrue about glycosphingolipids?
Feedback: Glycosphingolipids act as antigens and not antibodies. The latter are produced by the body's immune system in response to antigens.	

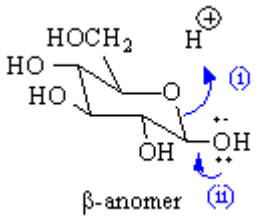
incorrect	**a.#**	Glycosphingolipids are responsible for labelling red blood cells as A, B, AB, or O.
Correct	**b.#**	Glycosphingolipids can act as antibodies.
incorrect	**c.#**	Glycosphingolipids are made up of a carbohydrate, a fatty acid, and sphingosine.
incorrect	**d.#**	Glycosphingolipids contain a hydrophilic carbohydrate group which acts as a molecular tag for red blood cells.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 10 - Question 04	
04)	Which of the following terms is untrue regarding the following molecule? 
Feedback: The molecule is chiral and contains five asymmetric centres	

incorrect	**a.#**	The carbon with the star is an anomeric centre.
incorrect	**b.#**	The molecule is capable of ring opening.
incorrect	**c.#**	The carbon with the star is an aldehyde in the ring opened structure.
Correct	**d.#**	The molecule is achiral.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 10 - Question 05	
05)	How many asymmetric centres are present in glucose?  Glucose
Feedback: The five asymmetric centres are shown in the diagram below. Each of these atoms has four different substituents.  Glucose	
incorrect	**a.#** 2.
incorrect	**b.#** 3.
incorrect	**c.#** 4.
Correct	**d.#** 5.
END OF QUESTION	

Type: multiple choice	
Title: Chapter 10 - Question 06	
06)	Which of the following curly arrows is wrong in the following mechanism? 

Feedback: Curly arrows indicate the movement of electrons and not atoms. It is not possible to draw a curly arrow from a proton since there are no electrons on the proton. Arrow (i) should point to the proton from the source of two electrons - in this case the bond that is being broken.  		
Correct	**a.#**	Curly arrow (i).
incorrect	**b.#**	Curly arrow (ii).
incorrect	**c.#**	Curly arrow (iii).
incorrect	**d.#**	Curly arrow (iv).
END OF QUESTION		

Type: multiple choice		
Title: Chapter 10 - Question 07		
07)	Which of the following is not an important cellular role for carbohydrates?	
Feedback: The most common biological catalysts are proteins (enzymes). Carbohydrates such as cellulose have a structural role. Glycogen is a carbohydrate that acts as an energy store, and carbohydrates acts as antigens in cell recognition.		
incorrect	**a.#**	Energy storage.
incorrect	**b.#**	Structure.
Correct	**c.#**	Catalysis.
incorrect	**d.#**	Cell recognition.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 10 - Question 08		
08)	Which of the following is the most accurate description of a glycoconjugate?	
Feedback: The term glycoconjugate is generally used to describe the situation where carbohydrates are linked to another type of macromolecule such as a protein or a lipid. It has nothing to do with conjugated functional groups or molecular interactions.		
incorrect	**a.#**	A polymer of glucose molecules.
incorrect	**b.#**	A carbohydrate with conjugated functional groups.
Correct	**c.#**	A carbohydrate linked to a macromolecule such as a protein or lipid.
incorrect	**d.#**	The interaction of a sugar molecule with a drug.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 10 - Question 09		
09)	Which of the following statements is not true?	
Feedback: Tricyclic antidepressants act by blocking the transport proteins that normally transport noradrenaline back into nerve cells.		
incorrect	**a.#**	Some polar drugs can be designed such that they are carried across cell membranes by transport proteins
Correct	**b.#**	Tricyclic antidepressants are transported into cells by transport proteins such that they can interact with intracellular targets.
incorrect	**c.#**	Designing drugs to block transport proteins can be useful in medicine
incorrect	**d.#**	Cocaine blocks the transport proteins responsible for transporting important neurotransmitters back into nerve cells.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 10 - Question 10		
10)	Which of the following statements is true regarding tubulin?	
Feedback: It is correct to say that tubulin is a structural protein. Tubulin is a component of **microtubules** and not ribosomes. All anticancer drugs that act on tubulin do so by inhibiting the **depolymerisation** of microtubules. Tubulin has an important role to play in **cell division**, not cell metabolism.		
Correct	**a.#**	It is a structural protein.
incorrect	**b.#**	It is a component of ribosomes.
incorrect	**c.#**	All anticancer drugs that act on tubulin do so by preventing it polymerising into microtubules.
incorrect	**d.#**	It has an important role to play in cell metabolism.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 10 - Question 11		
11)	Which of the following statements is not true?	
Feedback: The actions of cocaine are a result of cocaine blocking the transport proteins that carry the neurotransmitters, noradrenaline, and dopamine back into nerve cells.		
incorrect	**a.#**	Some polar drugs can be designed such that they are carried across cell membranes by transport proteins.
incorrect	**b.#**	Tricyclic antidepressants act by blocking the transport proteins that normally transport noradrenaline back into nerve cells.
incorrect	**c.#**	Designing drugs to block transport proteins can be useful in medicine.
Correct	**d.#**	The actions of cocaine are a result of it being carried into cells by transport proteins.
END OF QUESTION		

Type: multiple choice

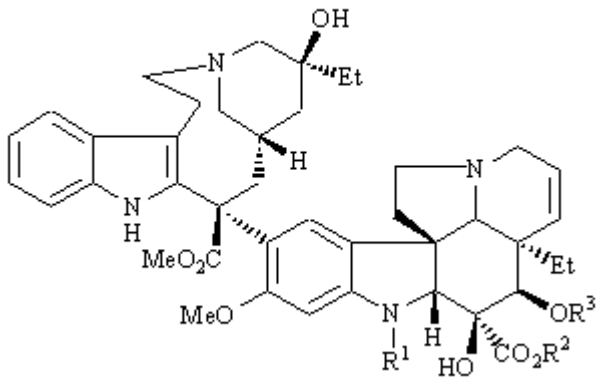
Title: Chapter 10 - Question 12		
12)	Which of the following statements is true regarding tubulin?	
Feedback: Tubulin is a **structural protein** and not an enzyme. It **stabilises** microtubules and thus **inhibits** the depolymerisation process. It has an important role to play in **cell division**, not cell metabolism. It is a component of microtubules.		
incorrect	**a.#**	It is an enzyme.
Correct	**b.#**	It is a component of structures known as microtubules.
incorrect	**c.#**	All anticancer drugs that act on tubulin do so by preventing it polymerising into microtubules.
incorrect	**d.#**	It has an important role to play in cell metabolism.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 10 - Question 13		
13)	Which of the following describes an antibody?	
Feedback: Foreign chemicals that trigger the immune response are called antigens. The molecules that 'recognise' antigens and bind to them are antibodies. These are produced by the body and are endogenous compounds. Antibodies are proteins but they are not structural proteins.		
incorrect	**a.#**	A foreign molecule which triggers an immune response.
Correct	**b.#**	A glycopeptide which identifies antigens and binds to them.
incorrect	**c.#**	A foreign chemical which is toxic to the body.
incorrect	**d.#**	A structural protein.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 10 - Question 14		
14)	What cell structure is targeted by valinomycin?	
Feedback: Valinomycin dissolves in the cell membranes of bacterial cells and can act as an ion carrier, leading to the uncontrolled movement of ions across the cell membrane.		
Correct	**a.#**	The cell membrane.
incorrect	**b.#**	The cell wall.
incorrect	**c.#**	Nucleic acids.
incorrect	**d.#**	Ribosomes.
END OF QUESTION		

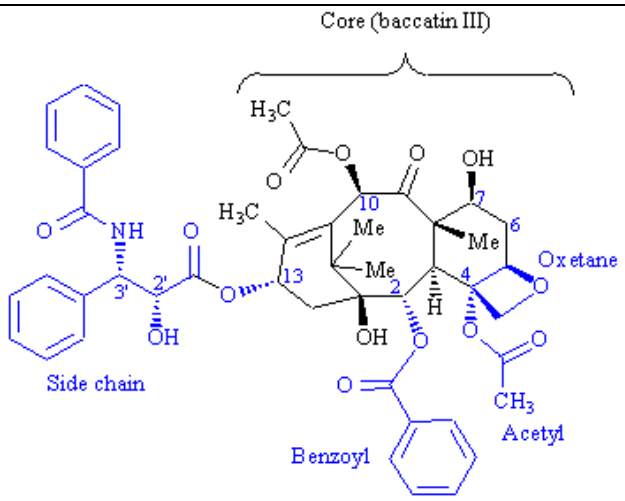
Type: multiple choice		
Title: Chapter 10 - Question 15		
15)	What role is played by valinomycin?	
Feedback: Valinomycin carries potassium ions across the cell membrane in an uncontrollable fashion leading to cell death.		

incorrect	**a.#**	Enzyme inhibitor.
incorrect	**b.#**	Receptor antagonist.
Correct	**c.#**	Ion carrier.
incorrect	**d.#**	Alkylating agent.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 10 - Question 16	
16)	The following diagram is representative of a group of naturally occurring compounds called the Vinca alkaloids that are used in combination therapy to treat a variety of tumours:  What is the target for these compounds?

Feedback: Tubulin is a structural protein which polymerises to produce microtubules in the cell. The Vinca alkaloids prevent this polymerisation.		
incorrect	**a.#**	Aromatase.
Correct	**b.#**	Tubulin.
incorrect	**c.#**	Oestrogen receptor.
incorrect	**d.#**	DNA.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 10 - Question 17	
17)	The following structure is an important anticancer agent used in the treatment of breast and ovarian cancers:



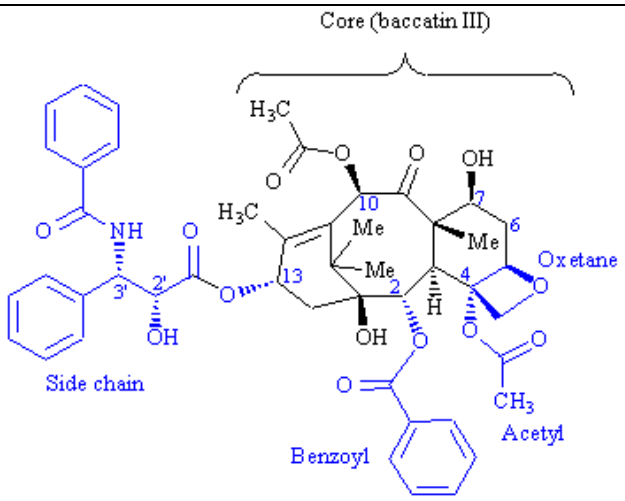
What is the name of the structure?

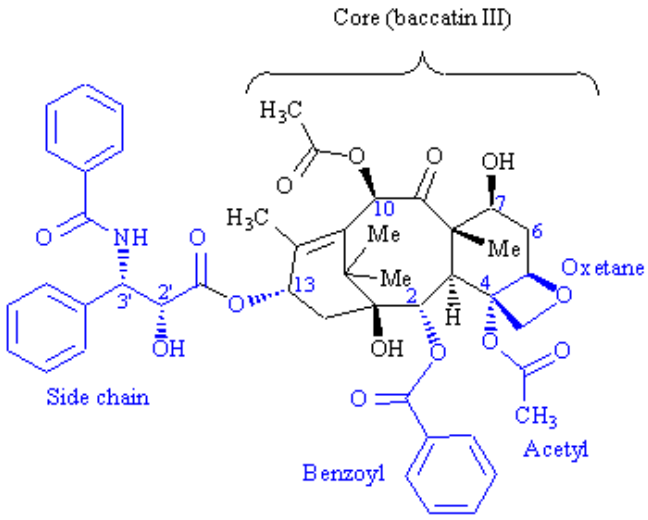
Feedback: The structure is paclitaxel which belongs to a group of compounds called the taxoids. Paclitaxel has a complex tetracyclic ring system which includes an oxetane ring. It was obtained from the bark of yew trees.
Discodermolide is a linear structure containing a six membered lactone. It was isolated from a marine sponge. Epothilone A is a macrocyclic lactone which was isolated from a bacterial source. The combretastatins are relatively simple molecules derived from the African bush willow.

incorrect	**a.#**	Discodermolide.
incorrect	**b.#**	Epothilone A.
incorrect	**c.#**	Combretastatin A-4.
Correct	**d.#**	Paclitaxel.

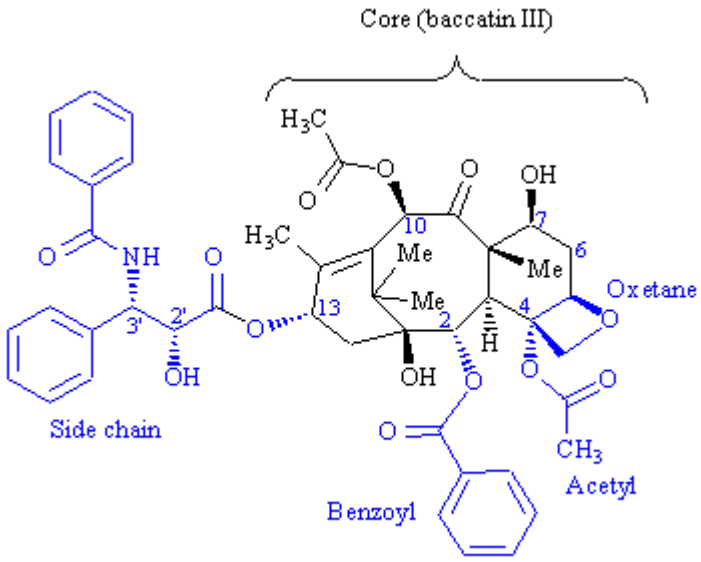
END OF QUESTION

Type: multiple choice	
Title: Chapter 10 - Question 18	
18)	The following structure is called paclitaxel and it is an important anticancer agent used in the treatment of breast and ovarian cancers:

	 What is the natural source for this compound?	
Feedback: Paclitaxel is obtained from the bark of yew trees.		
incorrect	**a.#**	Coral.
Correct	**b.#**	The yew tree.
incorrect	**c.#**	A marine bacterium.
incorrect	**d.#**	The willow tree.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 10 - Question 19	
19)	The following structure is called paclitaxel and it is an important anticancer agent used in the treatment of breast and ovarian cancers:  To which family of compounds does the above structure belong?
Feedback: The taxoids are characterised by a complex tetracyclic ring system which includes a four membered oxetane ring and an eight membered cyclooctane ring.	

incorrect	**a.#**	Combretastatin.
incorrect	**b.#**	Sarcodictyins.
incorrect	**c.#**	Cryptophycins.
Correct	**d.#**	Taxoids.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 10 - Question 20		
20)	The following structure is called paclitaxel and it is an important anticancer agent used in the treatment of breast and ovarian cancers:  How does it work?	
Feedback: Paclitaxel accelerates polymerisation and stabilises the resultant microtubules. As a result depolymerisation is inhibited.		
Correct	**a.#**	It inhibits tubulin depolymerisation.
incorrect	**b.#**	It inhibits tubulin polymerization.
incorrect	**c.#**	It inhibits both processes.
incorrect	**d.#**	It inhibits neither process.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 11 - Question 01		
01)	What is meant by ADME in pharmacokinetics?	
Feedback: Absorption, distribution, metabolism, and excretion are the factors which determine the blood concentration levels and the lifetime of a drug.		
incorrect	**a.#**	Affinity, dosage, marketing, efficacy.
Correct	**b.#**	Absorption, distribution, metabolism, excretion.
incorrect	**c.#**	Agonism, dependence, mobility, efficiency.
incorrect	**d.#**	Antagonism, deficiency, mean, efflux.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 11 - Question 02		
02)	Which of the following does not have a role to play in affecting the blood levels of an orally administered drug?	
Feedback: The gut wall has an important role in determining how much of an orally administered drug is absorbed from the digestive tract. The liver contains enzymes which may metabolise the drug, and the kidney is the main organ for excreting the drug and its metabolites. The lungs are important as far as inhaled drugs are concerned, but have no effect on drugs that are taken orally.		
Correct	**a.#**	Lungs.
incorrect	**b.#**	Liver.
incorrect	**c.#**	Gut wall.
incorrect	**d.#**	Kidneys.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 11 - Question 03		
03)	Which of the following properties is preferred for an orally administered drug?	
Feedback: The drug should have some water solubility to dissolve in the digestive tract and the blood, but it must have some hydrophobic character such that it can pass through the cell membranes of cells lining the digestive tract.		
incorrect	**a.#**	Hydrophilic character.
incorrect	**b.#**	Hydrophobic character.
Correct	**c.#**	A balance of hydrophilic and hydrophobic character.
incorrect	**d.#**	None of the options given are correct.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 11 - Question 04	
04)	What is the pharmacokinetic advantage of drugs having amine functional groups?
Feedback: Amine functional groups are weak bases and so there is equilibrium between the free base or non-ionised form, and the protonated or ionised form. The ionised form shows enhanced water solubility whereas the non-ionised form shows enhanced solubility in fatty cell membranes. As a result, the drug can pass through cell membranes in the non-ionised, hydrophobic form, but still show water solubility as the ionised form.	

incorrect	**a.#**	They are strong bases and are fully ionised.
incorrect	**b.#**	They are very weak bases and are not ionised at all.
Correct	**c.#**	They are weak bases and are in equilibrium between the ionised and free base forms.
incorrect	**d.#**	They are able to form hydrogen bonds.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 11 - Question 05	
05)	Some orally active drugs do not obey the rule of five. For example, some polar drugs with a molecular weight between 200 and 500 are found to be orally active. Which of the following mechanisms is the most likely reason for this?

Feedback: Some polar drugs can be transported across cell membranes by transport proteins if the drug resembles the molecule which is normally carried by that transport protein. Small polar drugs with a molecular weight less than 200 can squeeze through the pores between the cells lining the gut wall. Large polar molecules can be transported across cells by a process known as pinocytosis. Ion channels are not relevant to the absorption process.

Correct	**a.#**	Transport by transport proteins.
incorrect	**b.#**	Passage through pores between the cells of the gut wall.
incorrect	**c.#**	Pinocytosis.
incorrect	**d.#**	Ion channels.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 11 - Question 06	
06)	The blood concentration of a drug falls with time. Which of the following factors does not cause a drop in blood concentration?

Feedback: Distribution of a drug to the tissues will reduce the levels present in the blood supply. Drug metabolism alters the drug to a different compound and so blood concentration levels will fall. Drug excretion removes the drug from the blood supply. Drug absorption will increase blood concentration levels.

incorrect	**a.#**	Distribution to tissues.
incorrect	**b.#**	Drug metabolism.
incorrect	**c.#**	Drug excretion.
Correct	**d.#**	Drug absorption.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 11 - Question 07	
07)	Which of the following statements is true?

Feedback: Once a drug enters the blood supply it is rapidly and evenly distributed round the blood supply. However, the distribution to different tissues and organs is uneven since the level of blood supply to each organ varies.

incorrect	**a.#**	Drugs entering the blood supply are evenly distributed round the blood supply within one minute, resulting in an even distribution to different organs.

incorrect	**b.#**	Drugs entering the blood supply are unevenly distributed round the blood supply within one minute, but are evenly distributed to different organs.
incorrect	**c.#**	Drugs entering the blood supply are unevenly distributed round the blood supply within one minute resulting in an uneven distribution to different organs.
Correct	**d.#**	Drugs entering the blood supply are evenly distributed round the blood supply within one minute, and are unevenly distributed to different organs.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 11 - Question 08	
08)	How can advantage be taken of the blood brain barrier in drug design?
Feedback: Drugs can be made more hydrophilic if their target is in the peripheral system. This prevents the chances of central nervous system side effects. Increasing the hydrophobicity will increase the level of drug entering the brain, but it is not possible to make a drug completely selective for the brain since those drugs can still escape blood vessels in the peripheral regions. Making a drug more hydrophilic will decrease the chances of it entering the brain. Making it more hydrophobic will increase the chances of it entering the brain.	

incorrect	**a.#**	Drugs can be made more hydrophobic such that they act in the brain and not peripherally.
incorrect	**b.#**	Drugs can be made more hydrophilic such that they act in the brain and not peripherally.
incorrect	**c.#**	Drugs can be made more hydrophobic such that they act peripherally and not in the brain.
Correct	**d.#**	Drugs can be made more hydrophilic such that they act peripherally and not in the brain.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 11 - Question 09	
09)	Where is the major site for drug metabolism?
Feedback: The liver is the main organ for drug metabolism and contains metabolic enzymes such as the cytochrome P450 enzymes. However, some metabolic reactions can occur in other organs.	

incorrect	**a.#**	Brain.
Correct	**b.#**	Liver.
incorrect	**c.#**	Gut wall.
incorrect	**d.#**	Kidneys.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 11 - Question 10	
10)	What role does drug metabolism play in making the excretion of drugs more efficient?
Feedback: In general, most metabolic reactions do one of the following: introduce a polar functional group, reveal a polar functional group, or add a second polar molecule to the drug. The more polar the drug, the quicker it is excreted by the kidneys.	

incorrect	**a.#**	It fragments drugs into smaller molecules.
incorrect	**b.#**	It makes drug more hydrophobic.
Correct	**c.#**	It makes drugs more polar.
incorrect	**d.#**	It makes drugs less reactive.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 11 - Question 11		
11)	Which of the following statements is the closest description of Phase I metabolism?	
Feedback: The best description is of reactions which add a polar functional group to a drug. Some phase I metabolic reactions can occur in the blood supply or the gut wall, but they do not take place exclusively at either of these sites. Most of the phase I reactions take place in the liver. Phase II metabolism is where a polar molecule is attached to a polar functional group already present on the drug, or placed there by phase I metabolism.		
incorrect	**a.#**	Reactions which add a polar molecule to a functional group that is already present on a drug or one of its metabolites.
incorrect	**b.#**	Reactions which occur in the blood supply.
Correct	**c.#**	Reactions which add a polar functional group to a drug.
incorrect	**d.#**	Reactions which occur in the gut wall.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 11 - Question 12		
12)	What sort of reactions do cytochrome P450 enzymes catalyse?	
Feedback: Phosphate hydrolysis is catalysed by phosphorylases. Ester hydrolysis is catalysed by esterases. Conjugations are Phase II reactions catalysed by a variety of enzymes. The cytochrome P450 enzymes are oxidases which catalyse phase I metabolic reactions.		
incorrect	**a.#**	Phosphate hydrolysis.
incorrect	**b.#**	Ester hydrolysis.
incorrect	**c.#**	Conjugations.
Correct	**d.#**	Oxidations.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 11 - Question 13		
13)	Which of the following functional groups cannot be formed by a metabolic reaction catalysed by cytochrome P450 enzymes?	
Feedback: Alcohols, ketones, and carboxylic acids can all be formed by oxidation of exposed alkyl groups. Ethers are more likely to be cleaved in a metabolic reaction and are not formed as a result of one.		
Correct	**a.#**	Ethers.
incorrect	**b.#**	Ketones.
incorrect	**c.#**	Alcohols.

incorrect	**d.#**	Carboxylic acids.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 11 - Question 14		
14)	Why may a drug bearing an ester group that is important to target binding have a limited duration of action?	
Feedback: Esters are prone to hydrolysis by esterases. If this group was important to target binding, metabolism would weaken the ability of the drug to bind to its target. Esters are frequently used to protect polar functional groups and to enhance absorption from the gut, or passage through the blood brain barrier, and so it is unlikely that an ester 'fails to be absorbed efficiently from the gut' or 'fails to cross the blood brain barrier'. The ester is not a particularly polar functional group and so much of the drug is likely to be **reabsorbed** in the kidneys.		
incorrect	**a.#**	It fails to be absorbed efficiently from the gut.
Correct	**b.#**	The ester is hydrolysed by esterase enzymes.
incorrect	**c.#**	It fails to cross the blood brain barrier.
incorrect	**d.#**	It is rapidly excreted by the kidneys.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 11 - Question 15		
15)	Benzene is a suspect carcinogen since it is oxidised by cytochrome P450 enzymes to an electrophilic epoxide. As a result, benzene has been largely replaced by toluene as a solvent. Toluene is also oxidised by cytochrome P450 enzymes but the metabolite is less toxic and rapidly excreted. Suggest what the metabolite might be.	
Feedback: The methyl group is 'exposed' to attack by cytochrome P450 enzymes. It is also an activated benzylic carbon and so oxidation through to the carboxylic acid is most likely.		
Correct	**a.#**	Benzoic acid.
incorrect	**b.#**	Benzyl alcohol.
incorrect	**c.#**	Benzaldehyde.
incorrect	**d.#**	*para*-Methylphenol.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 11 - Question 16		
16)	Benzene is a suspect carcinogen since it is oxidised by cytochrome P450 enzymes to an electrophilic epoxide. As a result, benzene has been largely replaced by toluene as a solvent. Toluene is also oxidised by cytochrome P450 enzymes but the metabolite is less toxic and rapidly excreted. Why is the metabolite rapidly excreted?	
Feedback: The metabolite formed is benzoic acid. The carboxylic acid of benzoic acid is highly polar and so benzoic acid is unlikely to be reabsorbed from the kidneys and will be rapidly excreted. Excretion from the lungs is only possible with volatile molecules and gases, and benzoic acid is not volatile.		
incorrect	**a.#**	It is more hydrophobic and is rapidly excreted by the kidneys.

Correct	**b.#**	It is more hydrophilic and is rapidly excreted by the kidneys.
incorrect	**c.#**	It is more hydrophobic and is rapidly excreted by the lungs.
incorrect	**d.#**	It is more hydrophilic and is rapidly excreted by the lungs.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 11 - Question 17		
17)	Which of the following is not an enzyme involved in catalysing Phase II metabolic reactions?	
Feedback: Phase II metabolic reactions are conjugation reactions involving the attachment of a polar molecule to a polar functional group present on the drug or placed there by a phase I metabolic reaction. Adenosyl methionine, glutathione transferase, and sulfotransferase are examples of enzymes which catalyse conjugation reactions. Peptidase catalyses the hydrolysis of a peptide bond to reveal a carboxylic acid and an amine. This is a phase I metabolic reaction.		
incorrect	**a.#**	Adenosyl methionine.
Correct	**b.#**	Peptidase.
incorrect	**c.#**	Glutathione transferase.
incorrect	**d.#**	Sulfotransferase.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 11 - Question 18		
18)	Drugs can be excreted from the body by various routes. Which of the following is generally the most important route?	
Feedback: The kidneys are the most important organs in drug excretion. However, a certain level of drug excretion can take place via sweat, the lungs, or the bile duct. Whether drugs are actually excreted by these alternative routes depends on the particular drug concerned. For example, gaseous general anaesthetics can be excreted from the lungs, whereas non gaseous, non volatile drugs are not.		
incorrect	**a.#**	Sweat.
incorrect	**b.#**	Lungs.
Correct	**c.#**	Kidneys.
incorrect	**d.#**	Bile duct.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 11 -Question 19		
19)	Some drugs are excreted more efficiently than others. Which of the following statements is the best general explanation for this?	
Feedback: The blood is filtered under pressure such that water, drugs, and their metabolites are filtered into the nephrons of the kidney. Water is reabsorbed into the blood supply setting up a concentration gradient where the concentration of drug metabolites in the nephron is higher than in the surrounding blood vessels. Hydrophobic molecules can cross the cell membrane of the nephron down the concentration gradient and re-enter the blood vessels. Polar molecules are trapped in the nephron and excreted. Excretion by the lungs is efficient only for volatile and gaseous hydrophobic molecules.		

incorrect	**a.#**	Hydrophobic drugs are excreted more efficiently by the kidneys.
Correct	**b.#**	Hydrophilic drugs are excreted more efficiently by the kidneys.
incorrect	**c.#**	Hydrophobic drugs are excreted more efficiently by the lungs.
incorrect	**d.#**	Hydrophilic drugs are excreted more efficiently by the lungs.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 11 - Question 20		
20)	Orally taken drugs are absorbed into the blood supply and passed through the liver where a certain percentage of the drug is modified by drug metabolism before the drug is distributed round the rest of the body. What is the term used to describe this process?	
Feedback: The primary effect has no meaning in this context. There is no such thing as an ADME effect. ADME stands for the principle factors to be considered in pharmacokinetics - absorption, distribution, metabolism,and excretion. There is no such thing as the conjugation effect in this context. Conjugation can apply to many things. In drug metabolism, phase II reactions are conjugation reactions.		
incorrect	**a.#**	The primary effect.
incorrect	**b.#**	The ADME effect.
incorrect	**c.#**	The conjugation effect.
Correct	**d.#**	The first pass effect.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 11 - Question 21		
21)	Which of the following methods of administration is the most efficient in achieving high blood concentrations of a drug?	
Feedback: Injection is the most efficient in achieving high blood concentrations of a drug - particularly intravenous injection. Inhalation is also efficient though less so than injection. Topical and oral administrations are less efficient.		
incorrect	**a.#**	Inhalation.
Correct	**b.#**	Injection.
incorrect	**c.#**	Topical.
incorrect	**d.#**	Oral.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 11 - Question 22		
22)	Which of the following methods of drug administration carries the greatest risk?	
Feedback: Since injection is the most efficient method of delivering high blood concentrations of a drug, it is also the most hazardous method of administration. If an excess dose is administered or if unexpected side effects are observed, there is very little time to take remedial action.		
incorrect	**a.#**	Inhalation.

Correct	**b.#**	Injection.
incorrect	**c.#**	Topical absorption.
incorrect	**d.#**	Oral absorption.
END OF QUESTION		

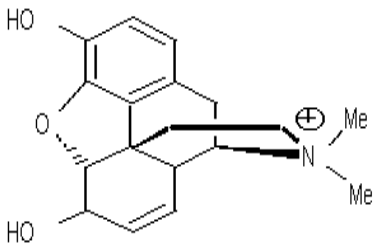
Type: multiple choice		
Title: Chapter 11 - Question 23		
23)	Anti-asthmatic drugs are usually administered by inhalation. Which of the following is a desirable pharmacokinetic property for an anti-asthmatic drug administered in this way?	
Feedback: It is desirable to localise anti-asthmatic drugs in the airways. Polar drugs are less likely to cross the cell membranes of cells lining the airways. This means that they are less likely to enter the blood supply and cause undesirable side effects. Drugs which are hydrophobic are more likely to cross the cell membranes and enter the blood supply. Drugs with a large molecular weight may prove difficult to administer as an aerosol. An aromatic ring is present in several anti-asthmatic agents, but its importance is due to binding interactions with the drug's target and not for pharmacokinetic reasons.		
incorrect	**a.#**	Hydrophobic nature.
incorrect	**b.#**	Large molecular weight.
Correct	**c.#**	Polar character.
incorrect	**d.#**	Presence of an aromatic ring.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 11 - Question 24		
24)	What is meant by topical administration?	
Feedback: Topical administration involves the application of the drug to the skin such that it acts at the surface of the body or is absorbed through the skin.		
incorrect	**a.#**	Injection through the skin.
incorrect	**b.#**	Suppositories.
Correct	**c.#**	Absorption through the skin.
incorrect	**d.#**	Inhalation via the lungs.
END OF QUESTION		

Type: matching			
Title: Chapter 11 - Question 25			
25)	Match the following terms to the descriptions:		
Feedback: An intrathecal injection is injected directly into the spinal cord. An intraperitoneal injection is injected into the abdominal cavity. A subcutaneous injection is injected under the skin. An intramuscular injection is injected into muscle.			
a.#	The injection of a drug under the skin.	=	Subcutaneous
b.#	The injection of a drug into the spinal cord.	=	Intrathecal
c.#	The injection of a drug into muscle.	=	Intramuscular
d.#	The injection of a drug into the abdominal cavity.	=	Intraperitoneal

END OF QUESTION

Type: multiple choice		
Title: Chapter 11 - Question 26		
26)	A drug has a half life of 4 hours. How much of the drug remains after 24 hours?	
Feedback: The level of drug halves every four hours. 50% is left after 4 hours, 25% after 8 hours, 12.5% after 12 hours, 6.25% after 16 hours, 3.125% after 20 hours and 1.5625% after 24 hours.		
incorrect	**a.#**	16.7%
Correct	**b.#**	1.6%
incorrect	**c.#**	0.16%
incorrect	**d.#**	0.016%
END OF QUESTION		

Type: multiple choice		
Title: Chapter 11 - Question 27		
27)	The quaternary salt of morphine binds strongly to CNS opioid receptors *in vitro*. However, the compound is inactive *in vivo* when injected into the blood supply. Explain this apparent contradiction.  	
Feedback: The opioid receptors are located in the central nervous system and so drugs have to cross the blood brain barrier to interact with them. Since the molecule shown has a permanent positive charge, it cannot cross the blood brain barrier and fails to reach the opioid receptors. It has no *in vivo* activity. *In vitro* tests are carried out on isolated cells containing the opioid receptors. There is no blood brain barrier to cross and the drug can interact with the receptors and bind strongly. If the drug is injected directly into the spinal column *in vivo*, it proves to be active.		
incorrect	**a.#**	The drug is rapidly metabolised.
incorrect	**b.#**	The drug is rapidly excreted.
Correct	**c.#**	The drug fails to cross the blood brain barrier.
incorrect	**d.#**	The drug is bound strongly to plasma proteins.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 11 - Question 28	
28)	Which of the following has not been used to assist in drug delivery?
Feedback: Antibodies have been used to target drugs to particular cells such as cancer cells. The drug is linked to an antibody that recognises a specific antigen on target cells. The antibody binds to the antigen on the cell and in doing so delivers the drug to that cell. Thus, the antigen acts as a target and not as a mode of delivery.	

The interior of a liposome or a microsphere can be loaded with a drug such that the latter is protected from metabolic enzymes.		
incorrect	**a.#**	Antibodies.
Correct	**b.#**	Antigens.
incorrect	**c.#**	Liposomes.
incorrect	**d.#**	Microspheres.
END OF QUESTION		

Type: matching			
Title: Chapter 09 - Question 29			
29)	Match the following terms to the descriptions given.		
Feedback: Pharmacodynamics and pharmacokinetics are both relevant to drug design but study different properties of the candidate drug. In pharmacodynamics the focus is on how the candidate drug interacts with its target binding site, whereas in pharmacokinetics one is studying the various factors that might influence how much of the drug reaches the target in vivo. Structure-based drug design is a process whereby drugs can be designed to interact better with their target binding site, based on a knowledge of how they bind. Structure-activity relationships involve the testing of analogues to see which groups or substituents are important for biological activity.			
a.#	The study of how a drug reaches it target in the body and how blood levels are affected by absorption, distribution, metabolism and excretion Pharmacokinetics.	=	Pharmacokinetics.
b.#	The study of how drugs are designed using molecular modelling, based on knowledge of how it interacts with the target binding site.	=	Structure-based drug design
c.#	The study of how a drug interacts with its target binding site Pharmacodynamics.	=	Pharmacodynamics
d.#	The study of which functional groups in a compound are important for binding a drug to its target binding site and/or its activity.	=	Structure-activity relationships
END OF QUESTION			

Type: multiple choice		
Title: Chapter 11 - Question 30		
30)	Which of the following is one of the rules in Lipinski's rule of five?	
Feedback: The molecular weight should be **less** than 500. There should be no mor hydrogen bond acceptor groups. The calculated log*P* value should be **less** than +5		
incorrect	**a.#**	A molecular weight equal to 500.
Correct	**b.#**	No more than five hydrogen bond donor groups.
incorrect	**c.#**	No more than five hydrogen bond acceptor groups.
incorrect	**d.#**	A calculated log*P* value more than +5.

END OF QUESTION

Type: multiple choice

Title: Chapter 11 - Question 31

31) Which of the following descriptions refers to the absorption process known as pinocytosis?

Feedback: Some polar drugs can be transported across cell membranes by transport proteins if the drug resembles the molecule which is normally carried by that transport protein. Small polar drugs with a molecular weight less than 200 can squeeze through the pores between the cells lining the gut wall. Large polar molecules can be transported across cells by a process known as pinocytosis. Ion channels are not relevant to the absorption process.

incorrect	**a.#**	The process by which a drug is 'wrapped up' by a protein in the cell membrane such that it can be carried across the cell membrane.
incorrect	**b.#**	The process by which small molecules can squeeze through the small pores that exist between different cells in the gut wall.
Correct	**c.#**	The process by which drugs of large molecular weight are enveloped by the cell membrane of a cell lining the gut wall, leading to the pinching off of a membane bound vesicle which carries the drug into the cell.
incorrect	**d.#**	The process by which ion channels open to allow the crossing of ions across the cell membrane.

END OF QUESTION

Type: multiple choice

Title: Chapter 11 - Question 32

32) Which type of functional group would be useful to have in a drug designed to treat a gut infection?

Feedback: A carboxylate group is ionised and will make the drug highly polar. A highly polar antibacterial drug will not be able to cross the gut wall, since it will not be able to pass through hydrophobic cell membranes. It will, therefore, remain in the gut and can be used to treat gut infections.

incorrect	**a.#**	An aromatic ring.
incorrect	**b.#**	An alcohol group.
Correct	**c.#**	A carboxylate group.
incorrect	**d.#**	A ketone.

END OF QUESTION

Type: multiple choice

Title: Chapter 11 - Question 33

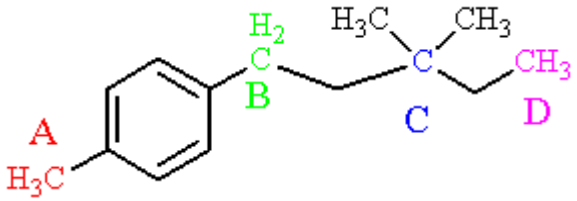
33) Which of the following statements is false regarding the blood brain barrier?

Feedback: The cells making up the walls of capillaries supplying the brain are only one cell thick, but the cells are more tightly packed than those supplying other parts of the body. As a result, drugs have to cross cell membranes to depart the blood vessels and enter the brain. There is also a fatty coating which surrounds the blood vessels and acts as a further barrier. This fatty coating is not part of the capillary wall. For all these reasons, hydrophobic drugs are more likely to cross the blood brain barrier.

incorrect	**a.#**	The walls of the capillaries supplying the brain have tight fitting cells making it difficult for polar drugs to leave the capillaries.

incorrect	**b.#**	The capillaries in the brain have a fatty coating making it more difficult for drugs to enter the brain.
incorrect	**c.#**	The walls of peripheral capillaries are 'leakier' than those in the central nervous system.
Correct	**d.#**	Hydrophilic drugs pass through the blood brain barrier more easily than hydrophobic drugs.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 11- Question 34		
34)	Which of the following statements is not true about cytochrome P450 enzymes?	
Feedback: The cytochrome P450 enzymes belong to a general class of enzymes called monooxygenases. The sulfotransferases are conjugation enzymes.		
incorrect	**a.#**	They contain haem and iron.
Correct	**b.#**	They belong to a general class of enzymes called sulfotransferases.
incorrect	**c.#**	There are over 30 different cytochrome P450 enzymes.
incorrect	**d.#**	Variation in cytochrome P450 enzyme profile between individuals can explain individual variation in drug susceptibility.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 11 - Question 35		
35)	Which of the carbon atoms in the following structure is likely to be least susceptible to cytochrome P450 enzymes? 	
Feedback: Exposed groups such as terminal methyl groups (carbons A and D) are susceptible to oxidation by cytochrome P450 enzymes, as are activated carbons such as allylic and benzylic carbons (carbon B). Quaternary carbon atoms such as carbon C are 'buried' and unlikely to be susceptible to cytochrome P450 enzymes.		
incorrect	**a.#**	Carbon A.
incorrect	**b.#**	Carbon B.
Correct	**c.#**	Carbon C.
incorrect	**d.#**	Carbon D.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 11- Question 36		
36)	Which of the following functional groups can be metabolised by cytochro P450 enzymes?	
Feedback: Cytochrome P450 enzymes catalyse oxidation reactions and can catalyse the		

oxidation of aromatic rings. An epoxide is a possible product from a cytochrome P450 reaction on an alkene or aromatic ring. A carboxylic acid is a possible product from the cytochrome P450 catalysed oxidation of a terminal methyl group or a primary alcohol. An ester is normally hydrolysed by esterase enzymes.		
Correct	**a.#**	Aromatic ring.
incorrect	**b.#**	Epoxide.
incorrect	**c.#**	Carboxylic acid.
incorrect	**d.#**	Ester.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 11 - Question 37		
37)	Which of the following enzymes catalyses a Phase I metabolic reaction?	
Feedback: Monoamine oxidase catalyses the formation of aldehydes from primary amines which is a phase I metabolic reaction. The other enzymes catalyse phase II metabolic reactions.		
incorrect	**a.#**	Sulfotransferases.
Correct	**b.#**	Monoamine oxidases.
incorrect	**c.#**	Glucuronyltransferase.
incorrect	**d.#**	Glutathione transferases.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 11 - Question 38		
38)	Which of the following reactions is not a Phase I metabolic transformation?	
Feedback: Reactions which introduce, modify, or reveal functional groups are phase I metabolic reactions. Conjugation reactions involve the attachment of a polar molecule to a polar functional group and this is a phase II metabolic reaction. Conjugation of a glucuronide to an alcohol is a phase II metabolic reaction.		
incorrect	**a.#**	Oxidation of a methyl group to a carboxylic acid.
Correct	**b.#**	The addition of a glucuronide to an alcohol group.
incorrect	**c.#**	Oxidation of an amine to an *N*-oxide.
incorrect	**d.#**	Epoxidation of an alkene.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 11 - Question 39		
39)	Which of the following is a phase II metabolic reaction?	
Feedback: Phase II metabolic reactions are conjugation reactions involving the attachment of a polar molecule to a polar functional group present on the drug, or placed there by a phase I metabolic reaction. Formation of a sulfate from an alcohol is an example of a conjugation reaction. The other options are phase I metabolic reactions.		
incorrect	**a.#**	Formation of a carboxylic acid from a primary methyl group.
Correct	**b.#**	Formation of a sulfate from an alcohol.
incorrect	**c.#**	Formation of a carboxylic acid and alcohol from an ester.
incorrect	**d.#**	Formation of a ketone at a benzylic carbon.

END OF QUESTION

Type: multiple choice		
Title: Chapter 11 - Question 40		
40)	Which of the following is an enzyme involved in catalysing Phase II metabolic reactions?	
Feedback: Phase II metabolic reactions are conjugation reactions involving the attachment of a polar molecule to a polar functional group present on the drug or which has been placed there by a phase I metabolic reaction. Glutathione transferase is an example of an enzyme which catalyses a conjugation, phase II reaction. The other enzymes catalyse phase I metabolic reactions.		
incorrect	**a.#**	Cytochrome P450.
incorrect	**b.#**	Peptidase.
Correct	**c.#**	Glutathione transferase.
incorrect	**d.#**	Monoamine oxidases.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 12 - Question 01		
01)	Which of the following properties of a drug is most likely to result in a minimum of side effects?	
Feedback: The more selective a drug is for its target, the less it will interact with other targets to produce undesirable side effects. Good oral absorption and fast metabolism are pharmacokinetic issues and do not affect the selectivity of the drug for one target over another. The affinity of a drug may have high affinity for the desired target, but it may also have high affinity for other less desirable targets.		
incorrect	**a.#**	Good oral absorption.
incorrect	**b.#**	Fast metabolism.
Correct	**c.#**	Target selectivity.
incorrect	**d.#**	Target affinity.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 12 - Question 02		
02)	Which of the following is not classed as an *in vivo* test?	
Feedback: *In vivo* tests are carried out on living systems such as animals or human volunteers. *In vitro* tests are carried out on target molecules, whole cells, isolated tissues, or isolated organs. Testing antibacterial agents on microbiological cultures is considered to be *in vitro* testing. The *in vivo* test for an antibacterial agent involves infecting a test animal with a bacterial strain, then administering the antibacterial agent to the animal to see whether it combats the infection.		
incorrect	**a.#**	Testing anticancer drugs on a tumour grown on the back of a rat.
incorrect	**b.#**	Testing analgesics by determining a guinea pig's sensitivity to pain.
Correct	**c.#**	Testing antibacterials by measuring the growth of micro-organisms on a petri dish in their presence.
incorrect	**d.#**	Testing 'smart drugs' by timing how long it takes a rat to find a way through a maze.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 12 - Question 03		
03)	Which of the following is not classed as an *in vitro* test?	
Feedback: *In vivo* tests are carried out on living systems such as animals or human volunteers. *In vitro* tests are carried out on target molecules, whole cells, isolated tissues, or isolated organs. Testing antibacterial agents on microbiological cultures is considered to be *in vitro* testing. The *in vivo* test for an antibacterial agent involves infecting a test animal with a bacterial strain, then administering the antibacterial agent to the animal to see whether it combats the infection.		
incorrect	**a.#**	Receptor binding studies on tissue preparations.
incorrect	**b.#**	Testing inhibitors on isolated enzymes.
incorrect	**c.#**	Testing anticancer drugs by measuring the rate of growth of whole cells.
Correct	**d.#**	Testing antibacterial agents in infected rats.
END OF QUESTION		

Type: multiple choice

Title: Chapter 12 - Question 04		
04)	What is the term used for an animal that has been genetically modified for *in vivo* tests?	
Feedback: The terms described in the other options have no significance in animal testing.		
incorrect	**a.#**	Chimeric animal.
incorrect	**b.#**	Transformed animal.
Correct	**c.#**	Transgenic animal.
incorrect	**d.#**	Hybrid animal.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 12 - Question 05		
05)	Which of the following sources has been particularly fruitful in the discovery of antibacterial agents?	
Feedback: Micro-organisms such as fungi have proved a rich source of antibacterial agents. Fungi have to compete against bacteria for nutrients, and the ability of fungi to produce antibacterial agents give these fungi an advantage in this contest.		
incorrect	**a.#**	Plant extracts.
incorrect	**b.#**	Venoms and toxins.
Correct	**c.#**	Micro-organisms.
incorrect	**d.#**	Animals.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 12 - Question 06		
06)	From which source did one of the lead compounds (teprotide) for the antihypertensive agent captopril come?	
Feedback: Teprotide is a peptide which was isolated from the venom of the Brazilian viper and was one of the lead compounds for the development of the antihypertensive agent captopril.		
incorrect	**a.#**	Marine sources.
Correct	**b.#**	Venoms and toxins.
incorrect	**c.#**	Combinatorial chemistry.
incorrect	**d.#**	Animals.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 12 - Question 07		
07)	What term is used for the chemical which is chiefly responsible for the biological activity of a natural extract?	
Feedback: The other terms are not used in medicinal chemistry.		
incorrect	**a.#**	Active drug.
incorrect	**b.#**	Prime drug.
Correct	**c.#**	Active principle.
incorrect	**d.#**	Prime principle.
END OF QUESTION		

Type: matching			
Title: Chapter 12 – Question 08			
08)	Match the substances on the right to the things they are obtained from, listed on the left.		
Feedback: Tubocurarine is the active principle of the arrow poison ourari. Cocaine is extracted from leaves of the coca bush. Morphine is extracted from opium. Quinine is extracted from cinchona bark.			
a.#	Opium	=	Morphine
b.#	Ourari	=	Tubocurarine
c.#	The cinchona bark	=	Quinine
d.#	The leaves of the coca bush	=	Cocaine
END OF QUESTION			

Type: matching			
Title: Chapter 12 - Question 09			
09)	Match the substances on the right to the things they are obtained from, listed on the left.		
Feedback: Salicin is extracted from the willow tree. Reserpine is extracted from the snakeroot plant. Emetine is extracted from ipecacuanha root. Digitalis is extracted from the foxglove.			
a.#	The willow tree	=	Salicin
b.#	The foxglove	=	Digitalis
c.#	The ipecacuanha root	=	Emetine
d.#	The snakeroot plant	=	Reserpine
END OF QUESTION			

Type: multiple choice		
Title: Chapter 12 - Question 10		
10)	Natural products are often used as lead compounds in the design and synthesis of novel drugs. Which of the following general characteristics of a natural product is most likely to be a disadvantage in synthesising analogues?	
Feedback: Natural products are often complex in structure which makes their synthesis difficult. As a result, the synthesis of analogues can also be difficult.		
incorrect	**a.#**	Novelty of structure.
Correct	**b.#**	Complexity of structure.
incorrect	**c.#**	Level of activity.
incorrect	**d.#**	Availability.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 12 - Question 11		
11)	What is meant when a lead compound is found by serendipity?	
Feedback: There is no pharmaceutical company called serendipity, no serendipity region of the African Veldt and no experimental procedure known as serendipity.		
incorrect	**a.#**	It was found by a pharmaceutical company by the name of serendipity.

incorrect	**b.#**	It was found in the serendipity region of the African Veldt.
Correct	**c.#**	It was discovered by chance.
incorrect	**d.#**	It was found by an experimental procedure known as serendipity.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 12 – Question 12		
12)	Which of the following analytical techniques provides the greatest structural information on a lead compound?	
Feedback: Nuclear magnetic resonance spectroscopy provides the most information about structure since it indicates the type of atoms that are present as well as the way in which these atoms are linked together. Ultra-violet spectroscopy is limited to providing information about the conjugation of a molecule. Infra-red spectroscopy is limited to providing information about the sort of functional groups that are present. Elemental analysis only provides the relative percentage of elements that are present.		
Correct	**a.#**	Nuclear magnetic resonance spectroscopy.
incorrect	**b.#**	Ultra-violet spectroscopy.
incorrect	**c.#**	Infra-red spectroscopy.
incorrect	**d.#**	Elemental analysis.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 12 - Question 13		
13)	Which of the following defines a structure which can be used as the starting point for drug design and development?	
Feedback: A lead compound is a compound that acts as the starting point for drug design and development. It may have disadvantages such as low activity or the presence of side effects, but it acts as a starting point for the development of a better agent. An active principle is the active constituent of a natural extract. A pharmacophore defines the essential groups required for some kind of pharmacological activity. An orphan drug is a compound that is effective against a rare disease.		
incorrect	**a.#**	Active principle.
incorrect	**b.#**	Pharmacophore.
Correct	**c.#**	Lead compound.
incorrect	**d.#**	Orphan drug.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 12 – Question 14		
14)	Which of the following is not an endogenous compound?	
Feedback: Endogenous compounds are any compounds occurring naturally in the body. Desipramine is a synthetic drug and is not an endogenous compound.		
incorrect	**a.#**	Dopamine.
incorrect	**b.#**	Lactate dehydrogenase.
incorrect	**c.#**	Noradrenaline.
Correct	**d.#**	Desipramine.

END OF QUESTION

Type: multiple choice		
Title: Chapter 12 - Question 15		
15)	What term refers to a compound that is chiefly responsible for the biological activity of a natural extract?	
Feedback: A lead compound is a compound that acts as the starting point for drug design and development. It may have disadvantages such as low activity or the presence of side effects, but it acts as a starting point for the development of a better agent. An active principle is the active constituent of a natural extract. A pharmacophore defines the essential groups required for some kind of pharmacological activity. An orphan drug is a compound that is effective against a rare disease.		
Correct	**a.#**	Active principle.
incorrect	**b.#**	Pharmacophore.
incorrect	**c.#**	Lead compound.
incorrect	**d.#**	Orphan drug.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 12 – Question 16		
16)	Which of the following reflects the order in which various stages of the drug discovery and development take place?	
Feedback: It is necessary to decide what the molecular target is before a bioassay can be established. The bioassay is required in order to detect a structure that is pharmacologically active and which can act as the lead compound. Structure-activity relationships can then be carried out on the lead compound.		
Correct	**a.#**	Determining a target, establishing a bioassay, finding a lead compound, structure activity relationships.
incorrect	**b.#**	Establishing a bioassay, determining a target, finding a lead compound, structure activity relationships.
incorrect	**c.#**	Determining a target, establishing a bioassay, structure activity relationships, finding a lead compound.
incorrect	**d.#**	Determining a target, finding a lead compound, structure activity relationships, establishing a bioassay.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 12 - Question 17		
17)	Which of the following statements about high throughput screening is false?	
Feedback: It is an *in vitro* process rather than *in vivo* process. Usually genetically modified cells are used.		
incorrect	**a.#**	It is an automated process.
Correct	**b.#**	It is an *in vivo* process.
incorrect	**c.#**	It is carried out on a small scale.
incorrect	**d.#**	It can be carried out on large numbers of compounds in a short time.
END OF QUESTION		

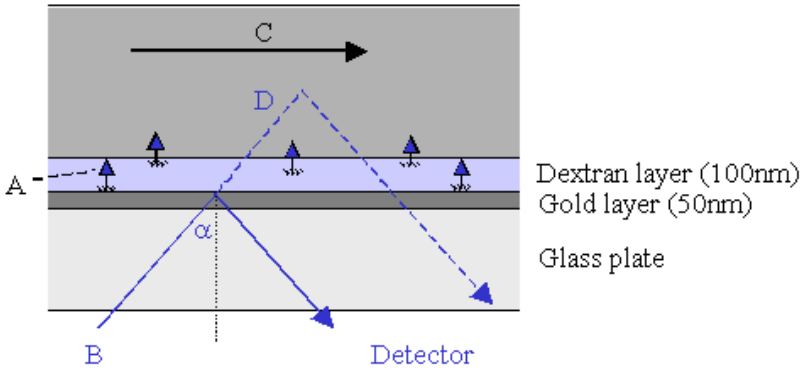
Type: multiple choice
Title: Chapter 12 – Question 18

18)	Which of the following statements is false with respect to NMR screening to detect drug-target interactions?	
Feedback: The method can detect weak bonding, which is one of its advantages over more traditional methods of detecting drug-target interactions.		
incorrect	**a.#**	The procedure relies on small molecules (drugs) having longer relaxation times than large molecules (targets).
incorrect	**b.#**	The procedure can be used on mixtures of compounds.
Correct	**c.#**	The method cannot detect weak binding
incorrect	**d.#**	The method can identify small molecules binding to different regions of the same binding site.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 12 - Question 19		
19)	What is the name of the detection system patented by Pharmacia Biosensor that can detect a ligand binding to a target using the phenomenon known as surface plasmon resonance?	
Feedback: BIAcore is the name of the detection system. The other options are ficticious.		
incorrect	**a.#**	BIOsystems.
incorrect	**b.#**	Bioplasma.
Correct	**c.#**	BIAcore.
incorrect	**d.#**	BIAdetection.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 12 – Question 20		
20)	A scintillation proximate assay makes use of a specific radioisotope which interacts with a scintillant to emit light. Which isotope is involved?	
Feedback: Scintillation proximate assay involves the immobilisation of a target molecule on beads coated with a scintillant. A ligand labelled with ^{125}I binds to the target and the radioisotope interacts with the scintillant to emit light. ^{3}H and ^{14}C are both radioisotopes used in drug metabolism studies. ^{13}C is a stable isotope of carbon which can be detected using nuclear magnetic resonance spectroscopy.		
Correct	**a.#**	^{125}I.
incorrect	**b.#**	^{3}H.
incorrect	**c.#**	^{13}C.
incorrect	**d.#**	^{14}C.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 12- Question 21	
21)	The following diagram illustrates the detection system known as surface plasmon resonance. What is represented by A?

Feedback: Symbol A represents a receptor which is covalently linked to the dextran matrix and which will bind ligands present in the buffer flow.
B is monochromatic plane-polarised light.
C is the flow of buffer solution.
D is the evanescent wave which interacts with free oscillating electrons called plasmons in the metal film when the angle of incidence (α) is right.

incorrect	**a.#**	The evanescent wave.
incorrect	**b.#**	Monochromatic plane-polarised light.
Correct	**c.#**	The receptor.
incorrect	**d.#**	The flow of buffer solution.

END OF QUESTION

Type: multiple choice

Title: Chapter 12- Question 22

22)	There are several sources and methods of discovering new compounds. Which of the following is an *in silico* method?

Feedback: *In silico* methods involve the use of computer software programmes. Database mining involves the computerised search of known compounds for particular pharmacophores.
Screening plant extracts involves a search for active principles that can act as lead compounds. Some natural products are quite bizarre in structure!
Combinatorial chemistry involves the synthesis of a large number of compounds using a standard synthetic route and so the types of structures that will be synthesised are predictable and belong to a similar family of compounds. Me too drugs are analogues of an established drug.

incorrect	**a.#**	Combinatorial chemistry.
Correct	**b.#**	Database mining.
incorrect	**c.#**	Screening plant extracts.
incorrect	**d.#**	Me too drugs.

END OF QUESTION

Type: multiple choice

Title: Chapter 12 - Question 23

23)	From which source did the lead compound for the antiulcer agent cimetidine come from?

Feedback: The lead compound for cimetidine was histamine which is an endogenous compound i.e. one of the body's own chemicals.

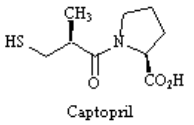
incorrect	**a.#**	Marine sources.
incorrect	**b.#**	Venoms and toxins.
incorrect	**c.#**	Combinatorial chemistry.

Correct	**d.#**	An endogenous compound.
END OF QUESTION		

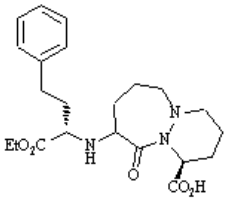
Type: multiple choice	
Title: Chapter 12 – Question 24	
24)	Which of the following drugs is isolated from a natural source?
Feedback: Morphine is isolated from the poppy plant. Aspirin and diamorphine are semi synthetic drugs prepared from compounds isolated from a natural source. Isoniazid is synthesised in the laboratory.	

incorrect	**a.#**	Aspirin.
Correct	**b.#**	Morphine.
incorrect	**c.#**	Isoniazid.
incorrect	**d.#**	Diamorphine (Heroin).
END OF QUESTION		

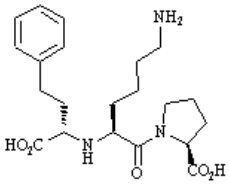
Type: multiple choice	
Title: Chapter 12 - Question 25	
25)	What is the term used for drugs such as cilazapril, lisinopril, and enalapril, all of which were developed as antihypertensives following the discovery of captopril?
Feedback: Orphan drugs are those drugs that are effective against particularly rare ailments. An active principle is a natural product that is chiefly responsible for the pharmacological activity of a natural extract. The drugs described can be described as analogues of captopril, but the term used to describe active analogues that have reached the market is 'me too' drugs.	



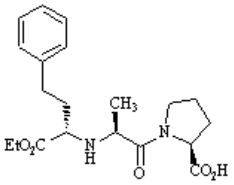
Captopril



Cilazapril
(Hoffmann-LaRoche)



Lisinopril
(Merck)



Enalapril
(Merck)

incorrect	**a.#**	Orphan drugs.
Correct	**b.#**	Me too drugs.
incorrect	**c.#**	Active principles.
incorrect	**d.#**	Analogues.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 12- Question 26	
26)	Natural products are often used as lead compounds in the design and synthesis of novel drugs. Which of the following general characteristics of a natural product is most likely to be a disadvantage?

Feedback: It can often be tedious and expensive to extract an active compound from its natural source. Natural products are also often complex in structure which makes their synthesis difficult. Novelty is usually an advantage. Side effects and poor activity are obviously disadvantages, but if it is possible to prepare analogues by modifying the lead compound or synthesising analogues, it is often possible to increase activity and lower side effects.		
incorrect	**a.#**	Novelty of structure.
Correct	**b.#**	Its availability.
incorrect	**c.#**	Level of activity.
incorrect	**d.#**	Side effects.
END OF QUESTION		

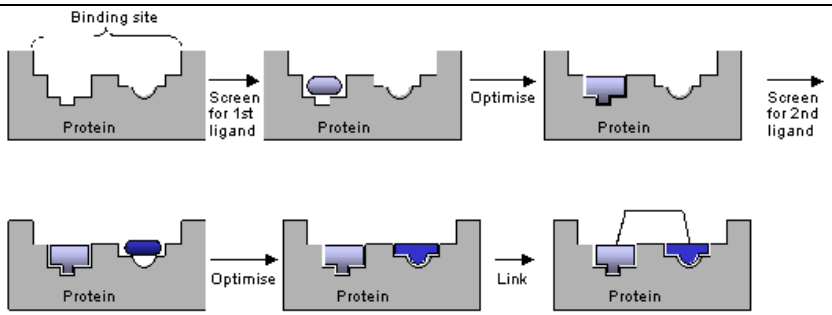
Type: multiple choice	
Title: Chapter 12- Question 27	
27)	Which of the following terms is used to indicate that a lead compound is one of the body's own chemicals?
Feedback: Endogenous compounds are any compounds occurring naturally in the body. These are also biochemicals, but this is not the term that is used in medicinal chemistry. A hormone and a neurotransmitter are examples of endogenous compounds, but not all endogenous compounds are hormones or neurotransmitters.	

incorrect	**a.#**	A neurotransmitter.
incorrect	**b.#**	A biochemical.
incorrect	**c.#**	A hormone.
Correct	**d.#**	An endogenous compound.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 12- Question 28	
28)	Which of the following is not an endogenous compound?
Feedback: Endogenous compounds are any compounds occurring naturally in the body. Fluoxetine is a synthetic drug and is not an endogenous compound.	

incorrect	**a.#**	Adrenaline.
Correct	**b.#**	Fluoxetine.
incorrect	**c.#**	Serotonin.
incorrect	**d.#**	Testosterone.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 12- Question 29	
29)	The following diagram shows a process by which a lead compound can be designed with the aid of nuclear magnetic resonance spectroscopy. What is the process called?

Feedback: Epitopes are small molecules capable of binding to small regions of the overall binding site. By optimising these and linking them, it should be possible to design an active compound. This is known as fragment-based lead discovery.
The process involves NMR screening, but NMR screening could also involve general screening of compounds with no design strategy involved.
De novo drug design and structure-based drug design are design processes involving the use of molecular modelling software.

incorrect	**a.#**	NMR screening.
Correct	**b.#**	Fragment-based lead discovery.
incorrect	**c.#**	*De novo* drug design.
incorrect	**d.#**	Structure-based drug design.

END OF QUESTION

Type: multiple choice

Title: Chapter 12- Question 30

30)	Which of the following statements is not true of herbal medicines?

Feedback: It is quite likely that herbal medicines may have a number of active principles responsible for one or more pharmacological effects.

Correct	**a.#**	Herbal medicines contain a single active principle that is responsible for its action.
incorrect	**b.#**	Herbal medicines can have side effects.
incorrect	**c.#**	It is more difficult to receive an overdose of an active drug if it is present in a herbal medicine rather than as a pure medicine.
incorrect	**d.#**	Certain compounds in herbal medicines may enhance the activity of the active principle.

END OF QUESTION

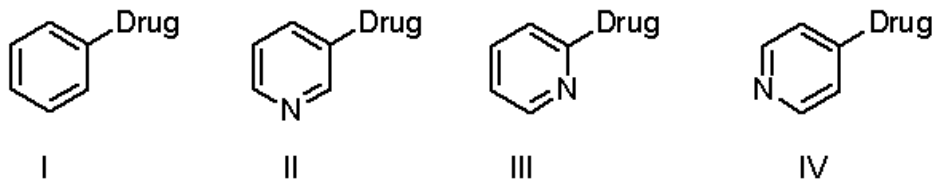
Type: multiple choice		
Title: Chapter 13 - Question 01		
01)	Which of the following statements is true regarding an alcohol?	
Feedback: The alcohol has an OH group where the oxygen can act as a hydrogen bond acceptor and the hydrogen can act as a hydrogen bond donor.		
incorrect	**a.#**	It can only participate in hydrogen bonding as a hydrogen bond donor.
incorrect	**b.#**	It can only participate in hydrogen bonding as a hydrogen bond acceptor.
Correct	**c.#**	It can participate in hydrogen bonding both as a hydrogen bond donor and a hydrogen bond acceptor.
incorrect	**d.#**	It cannot participate in hydrogen bonding at all.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 13 - Question 02		
02)	Which of the following statements is true regarding a tertiary amide?	
Feedback: The tertiary amide (R_2NCOR) has a carbonyl oxygen that can act as a hydrogen bond acceptor. There is no hydrogen attached to nitrogen and so the group cannot act as a hydrogen bond donor. The nitrogen itself does not act as a hydrogen bond acceptor since its lone pair interacts with the carbonyl group.		
incorrect	**a.#**	It can only participate in hydrogen bonding as a hydrogen bond donor.
Correct	**b.#**	It can only participate in hydrogen bonding as a hydrogen bond acceptor.
incorrect	**c.#**	It can participate in hydrogen bonding both as a hydrogen bond donor and a hydrogen bond acceptor.
incorrect	**d.#**	It cannot participate in hydrogen bonding at all.
END OF QUESTION		

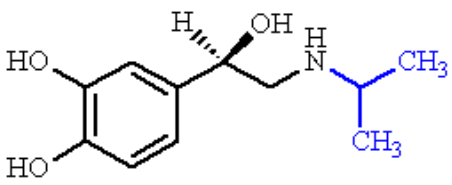
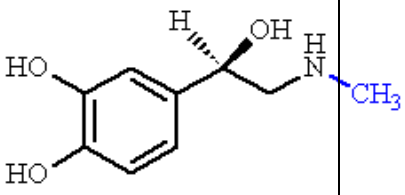
Type: multiple choice		
Title: Chapter 13 - Question 03		
03)	A secondary amide group in a lead compound was reduced to an amine functional group. *In vitro* tests showed that the lead compound was active and that the product was inactive. However, *in vivo* tests showed that both the amide and amine were inactive. Which of the following statements is not plausible?	
Feedback: There is no evidence to suggest that the amine is an important binding group. The most likely scenario is that the amide is acting as a hydrogen bond acceptor through its carbonyl oxygen. This would fit in with the observation that the amine is inactive *in vitro*. The negative result for the amide in the *in vivo* test could be due to hydrolysis of the amide bond by metabolic enzymes to give inactive compounds.		
incorrect	**a.#**	The amide is an important binding group but the amine is not.
Correct	**b.#**	Both the amide and the amine are important binding groups.
incorrect	**c.#**	The carboxyl group of the amide may be an important hydrogen bond acceptor group.
incorrect	**d.#**	In the *in vivo* bioassay, the amide is converted to the amine by metabolic or digestive enzymes.
END OF QUESTION		

Type: multiple choice
Title: Chapter 13 - Question 04

04)	A lead compound contains a carboxylic acid functional group. This was esterified to give an analogue with an ester group. *In vitro* tests showed that the carboxylic acid was active whereas the ester was inactive. On the other hand, *in vivo* tests, where the test compounds were administered orally, showed that the ester was active, and the carboxylic acid was inactive. Which of the following explanations is not plausible?	
Feedback: The *in vitro* tests are at odds with the ester being an important binding group. These tests indicate that the carboxylic acid is an important binding group and not the ester. The most likely explanation is that the ester acts as a prodrug and is metabolised during the *in vivo* studies to give the active carboxylic acid. The carboxylic acid shows no activity *in vivo*, since it is too polar to get absorbed into the blood supply.		
incorrect	**a.#**	The carboxylic acid is not absorbed in the *in vivo* studies.
Correct	**b.#**	The ester is an important binding group but the carboxylic acid group is not.
incorrect	**c.#**	The ester is absorbed in the *in vivo* studies.
incorrect	**d.#**	The ester is metabolised to the carboxylic acid in the *in vivo* studies.
END OF QUESTION		

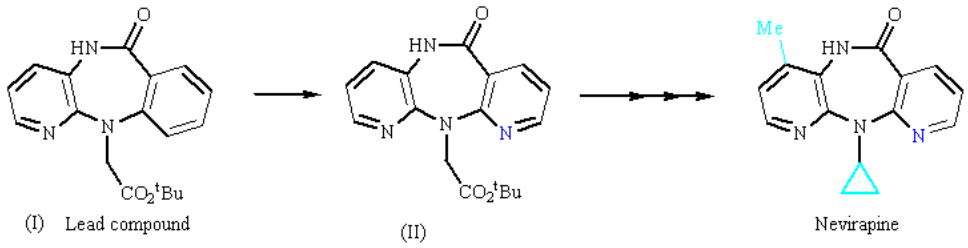
Type: multiple choice		
Title: Chapter 13 - Question 05		
05)	Structures (II-IV) are analogues of a lead compound containing an aromatic ring (structure I). Structures II and III had similar activity to the lead compound whereas structure IV showed a marked increase in activity. Which of the following explanations best fits the facts?  I II III IV	
Feedback: It is highly unlikely that increased activity is due to increasing the basicity, polarity, or water solubility of the analogues, since the activity should have been improved for all three analogues II-IV. An aromatic hydrogen is unlikely to have any significant steric effect, although the possibility cannot be ruled out. The heterocyclic nitrogen can act as a hydrogen bond acceptor with an extra hydrogen bonding region in the binding site. The fact that structure IV has increased activity and not II or III emphasises the importance of having the nitrogen at the correct position of the ring if it is to interact with the extra binding region. This adds weight to the theory.		
incorrect	**a.#**	Introducing a nitrogen increases basicity and so increased basicity is good for activity.
incorrect	**b.#**	Introducing a nitrogen increases the polarity and water solubility of the analogues, and so increased polarity is good for activity.
Correct	**c.#**	Introducing a nitrogen means that an additional hydrogen bonding interaction is possible with an extra binding region in the binding site.
incorrect	**d.#**	Introducing a nitrogen removes an aromatic hydrogen. The aromatic hydrogen removed may have been bad for activity for steric reasons.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 13 - Question 06		
06)	What expression is used to describe the shape that a molecule adopts when it binds to its target binding site?	
Feedback: Conformation is a stereochemical description of the various shapes that a molecule can adopt through single bond rotation. The active conformation is the shape that a drug adopts when it binds to its target. Configuration and constitution are stereochemical terms that are used to describe isomers that have the same molecular formula, but are different molecules having different chemical and physical properties. The pharmacophore indicates the important binding groups of an active drug and their relative positions.		
incorrect	**a.#**	Active configuration.
Correct	**b.#**	Active conformation.
incorrect	**c.#**	Active constitution.
incorrect	**d.#**	Pharmacophore.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 13 - Question 07		
07)	Isoprenaline shows selectivity for β-adrenoceptors over α-adrenoceptors, whereas adrenaline does not. Which of the following statements is the correct explanation for this?  Isoprenaline  Adrenaline	
Feedback: Isoprenaline is certainly more hydrophobic and may or may not be more easily absorbed, but this does not explain the target selectivity of isoprenaline. It is wrong to state that 'β-Adrenoceptors are concentrated in the brain requiring drugs to cross the blood brain barrier'. β-Adrenoceptors are in the periphery, and drugs do not need to cross the blood brain barrier to reach them. It is the **size** of the *N*-substituents that is important for selectivity rather than the basicity of the amine groups.		
incorrect	**a.#**	Isoprenaline is more hydrophobic and is more easily absorbed.
Correct	**b.#**	The amine binding region in the binding site of the β-adrenoceptor is more spacious than the one for the α-adrenoceptor, and can accommodate an isopropyl group more easily.
incorrect	**c.#**	β-Adrenoceptors are concentrated in the brain, requiring drugs to cross the blood brain barrier.
incorrect	**d.#**	The different basicity of the two structures affects target selectivity.
END OF QUESTION		

Type: multiple choice

Title: Chapter 13 - Question 08		
08)	What strategy of drug design is frequently used on complex lead compounds derived from natural products?	
Feedback: Natural products are often complex structures which are difficult to synthesise. Simplification removes excess functional groups and superfluous regions of the molecule to produce simpler molecules which are easier to synthesise and which retain activity.		
incorrect	**a.#**	Extension.
Correct	**b.#**	Simplification.
incorrect	**c.#**	Rigidification.
incorrect	**d.#**	Conformational blocking.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 13 - Question 09		
09)	In the development of the antifungal agent, nevirapine, structure (II) was found to bind more strongly to the target enzyme than the lead compound (I). Which of the following statements is correct?  	
Feedback: Taking the options in turn, the strategy used was one of **ring variation** rather than ring expansion. The extra nitrogen allows an extra **hydrogen bonding** interaction to take place rather than a van der Waals interaction. The extra nitrogen can act as a **hydrogen bond acceptor** and not as a hydrogen bond donor. The extra binding interaction means that an extension strategy can be said to have been used.		
incorrect	**a.#**	The strategy used was one of ring expansion.
incorrect	**b.#**	The extra nitrogen in blue allows an extra binding interaction to take place through van der Waals interactions.
incorrect	**c.#**	The extra nitrogen in blue can act as a hydrogen bond donor.
Correct	**d.#**	The strategy used could be described as extension since a further binding interaction has occurred.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 13 - Question 10		
10)	Hydrogen is often replaced in drugs with a fluorine atom for a variety of reasons. What term is used to describe the fluorine atom when it is used to replace hydrogen?	
Feedback: An isostere is a group having a similar size, shape, and valency to the original group. The other terms have no meaning in medicinal chemistry. An isotope is an element having a different number of neutrons from another element, but the same number of protons and electrons. An isomer is a molecule having the same molecular formula as another, but having a different arrangement of atoms and bonds. An isobar is used to represent atmospheric pressure in meteorological maps.		
incorrect	**a.#**	Isotope.
Correct	**b.#**	Isostere.

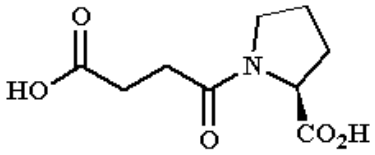
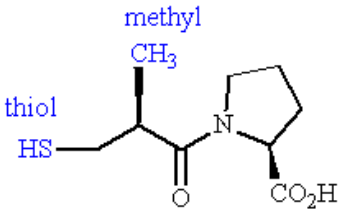
incorrect	**c.#**	Isomer.
incorrect	**d.#**	Isobar.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 13 - Question 11		
11)	Which of the following statements best describes a bio-isostere?	
Feedback: A bio-isostere is a group which can be used in place of another without affecting the activity of a drug. It includes classical and non classical isosteres, and it does not need to have the same valency as the original group. Usually, the bio-isostere would be expected to have a similar size, shape, and chemical property to the original group.		
incorrect	**a.#**	A group having the same valency as another group.
incorrect	**b.#**	A group that is the same as a non-classical isostere.
Correct	**c.#**	A group that can be used in place of another group whilst retaining the important biological activity of the drug.
incorrect	**d.#**	A group having the same size as another group.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 13 - Question 12		
12)	The simplification strategy is commonly used in drug design. What must be retained in a molecule during the simplification process?	
Feedback: The pharmacophore defines the important binding groups and their relative positions. The other factors play a role in drug activity but need not be retained during simplification.		
incorrect	**a.#**	The molecular weight.
incorrect	**b.#**	The stereochemistry.
Correct	**c.#**	The pharmacophore.
incorrect	**d.#**	The hydrophobic character.
END OF QUESTION		

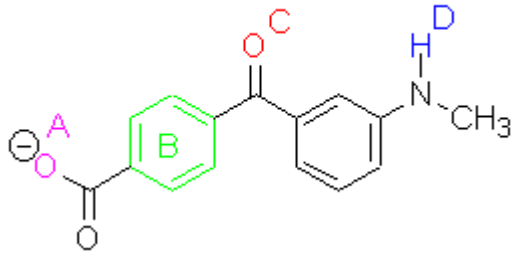
Type: multiple choice		
Title: Chapter 13 - Question 13		
13)	There are three general methods by which molecules can be simplified. Which of the following is not one of these methods?	
Feedback: The three main methods of simplifying a molecule are to remove excess rings, asymmetric centres, and excess functional groups. Simplification normally results in more flexible molecules and not less flexible molecules.		
incorrect	**a.#**	Removal of excess rings.
Correct	**b.#**	Removal of excess flexibility.
incorrect	**c.#**	Removal of excess functional groups.
incorrect	**d.#**	Removal of excess asymmetric centres.
END OF QUESTION		

Type: multiple choice
Title: Chapter 13 - Question 14

14)	Succinyl proline was the lead compound for captopril which acts as an inhibitor of the angiotensin-converting enzyme.  Succinyl proline  Captopril What is the relevance of the methyl group in captopril?	
Feedback: It is wrong to state that the methyl group is an exposed group and is easily oxidised by metabolic enzymes to a carboxylic acid which binds by ionic interactions to a hydrophilic binding region. The methyl group is certainly exposed and could conceivably be oxidised to a carboxylic acid by metabolic enzymes. However, the binding pocket in the active site is hydrophobic and accepts a methyl group rather than a carboxylic acid. There is nothing logically wrong with the two other incorrect options but the real relevance of the methyl group is that it represents an extension strategy where the methyl group binds to an extra hydrophobic binding region.		
Correct	**a.#**	It represents an extension strategy where the methyl group binds to an extra hydrophobic binding region.
incorrect	**b.#**	It acts as a conformational blocker and correctly orientates the important binding groups with respect to each other.
incorrect	**c.#**	It is an exposed group and is easily oxidised by metabolic enzymes to a carboxylic acid which binds by ionic interactions to a hydrophilic binding region.
incorrect	**d.#**	It introduces an asymmetric centre.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 13 - Question 15		
15)	Which of the following terms refers to the study of which structural features of a drug are important to its biological activity?	
Feedback: Structure-activity relationships involves the synthesis of a series of analogues of a lead compound to discover how changes in structure affect the biological activity. In this way, important binding groups and the drug's pharmacophore can be identified.		
incorrect	**a.#**	Pharmacodynamics.
incorrect	**b.#**	Pharmacokinetics.
Correct	**c.#**	Structure-activity relationships.
incorrect	**d.#**	Pharmacological activity.
END OF QUESTION		

Type: matching	
Title: Chapter 13 - Question 16	
16)	Identify which of the bonding interactions shown below are possible for the functional groups highlighted.

Feedback: The oxygens of the carboxylate group could act as hydrogen bond acceptors, as well as participating in ionic bonding. The aromatic ring will interact by van der Waals interactions. The carbonyl oxygen of the ketone is a hydrogen bond acceptor. The NH proton is a hydrogen bond donor. The nitrogen itself is unlikely to be a hydrogen bond acceptor since the lone pair will interact with the π system of the aromatic ring.

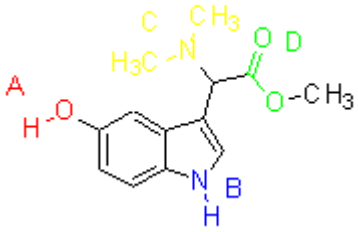
a.#	Ionic bonding and H-bond acceptor	=	The carboxylate group (A)
b.#	Van der Waals interactions	=	The aromatic ring (B)
c.#	Hydrogen bonding as a hydrogen bond acceptor only	=	The ketone (C)
d.#	Hydrogen bonding as a hydrogen bond donor only	=	The NH proton (D)

END OF QUESTION

Type: multiple choice

Title: Chapter 13 - Question 17

17) Which of the groups in the following molecule can act both as a hydrogen bond donor and a hydrogen bond acceptor?



Feedback: Group A is a phenol. The hydrogen can act as a hydrogen bond donor and the oxygen as a hydrogen bond acceptor.
Group B is part of the heterocyclic ring system. The proton can act as a hydrogen bond donor, but the nitrogen is unlikely to act as a hydrogen bond acceptor since the lone pair interacts with the π electrons in the heterocyclic system.
Group C is a tertiary amine and the nitrogen can act as a hydrogen bond acceptor
Group D is an ester where both oxygens can act as hydrogen bond acceptors.

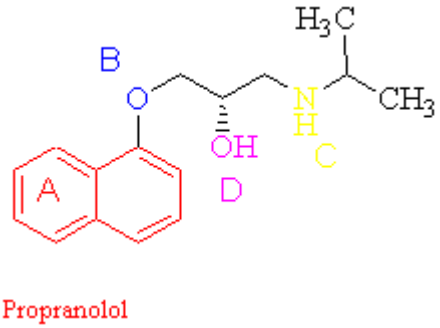
Correct	**a.#**	Group A.
incorrect	**b.#**	Group B.
incorrect	**c.#**	Group C.
incorrect	**d.#**	Group D.

END OF QUESTION

Type: multiple choice

Title: Chapter 13 - Question 18

18) Which of the groups in propranolol can only act as a hydrogen bond

	acceptor?  Propranolol

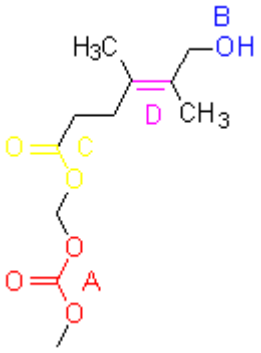
Feedback: Group A is a naphthalene ring. It can only interact by van der Waals interactions.
Group B is an ether and can only act as a hydrogen bond acceptor.
Group C is a secondary amine where the hydrogen can act as a hydrogen bond donor and the nitrogen can act as a hydrogen bond acceptor.
Group D is an alcohol where the hydrogen can act as a hydrogen bond donor and the oxygen as a hydrogen bond acceptor.

incorrect	**a.#**	Group A.
Correct	**b.#**	Group B.
incorrect	**c.#**	Group C.
incorrect	**d.#**	Group D.

END OF QUESTION

Type: multiple choice

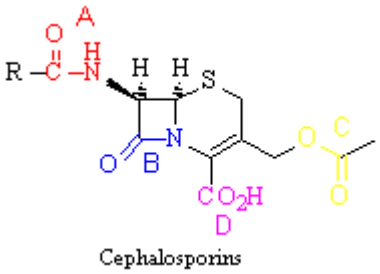
Title: Chapter 13 - Question 19

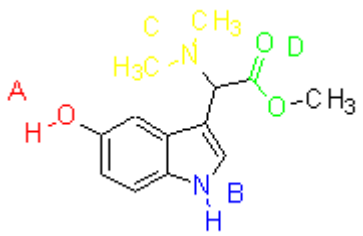
19)	Which of the groups in the following structure can only interact through van der Waals interactions? 

Feedback: Group A is a carbonate group. Each of the three oxygens can act as a hydrogen bond acceptor.
Group B is an alcohol where the hydrogen can act as a hydrogen bond donor and the oxygen as a hydrogen bond acceptor.
Group C is an ester where both oxygens can act as hydrogen bond acceptors.
Group D is an alkene which can only interact by van der Waals interactions.

incorrect	**a.#**	Group A.
incorrect	**b.#**	Group B.
incorrect	**c.#**	Group C.
Correct	**d.#**	Group D.

END OF QUESTION

Type: multiple choice		
Title: Chapter 13 - Question 20		
20)	Which of the groups in the following structure has the potential to form ionic interactions?  Cephalosporins	
Feedback: Group A is a secondary amide. The oxygen can act as a hydrogen bond acceptor and the NH proton can act as a hydrogen bond donor. The nitrogen itself is unlikely to act as a hydrogen bond acceptor since the lone pair of electrons interacts with the carbonyl group. Group B is part of the β-lactam ring. The carbonyl oxygen could potentially act as a hydrogen bond acceptor Group C is an ester where both oxygens can act as hydrogen bond acceptors. Group D is a carboxylic acid which can act as a hydrogen bond donor and a hydrogen bond acceptor in the un-ionised form. However, this group could lose its acidic proton to become ionised and form a stronger ionic bond.		
incorrect	**a.#**	Group A.
incorrect	**b.#**	Group B.
incorrect	**c.#**	Group C.
Correct	**d.#**	Group D.
END OF QUESTION		

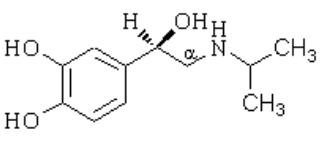
Type: multiple choice		
Title: Chapter 13 - Question 21		
21)	Which of the groups in the following structure can only act as a hydrogen bond donor?  	
Feedback: Group A is a phenol. The hydrogen can act as a hydrogen bond donor and the oxygen as a hydrogen bond acceptor. Group B is part of the heterocyclic ring system. The proton can act as a hydrogen bond donor, but the nitrogen is unlikely to act as a hydrogen bond acceptor since the lone pair interacts with the π electrons in the heterocyclic system. Group C is a tertiary amine and the nitrogen can act as a hydrogen bond acceptor. Group D is an ester where both oxygens can act as hydrogen bond acceptors.		

incorrect	**a.#**	Group A.
Correct	**b.#**	Group B.
incorrect	**c.#**	Group C.
incorrect	**d.#**	Group D.
END OF QUESTION		

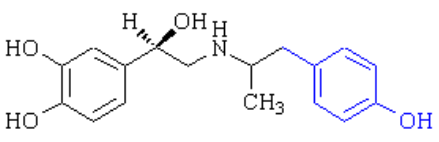
Type: matching			
Title: Chapter 13 - Question 22			
22)	Match the functional groups to the possible intermolecular interactions:		
Feedback: A carboxylate ion has the potential to be a strong binding group since it can interact using both ionic interactions and strong hydrogen bonding interactions. The hydroxyl group contains oxygen which could act as an HBA, and hydrogen which can acts as an HBD. The alkylammonium ion has a permanent positive charge which allows the possibility of a strong ionic interaction. It also has hydrogens which can act as strong HBDs. Therefore, the group is a particularly strong binding group. The oxygen of a ketone can act as an HBA, but there is no hydrogen present to act as an HBD.			
a.#	Carboxylate ion (RCO_2^-)	=	Hydrogen bond acceptor and/or ionic interactions
b.#	Hydroxyl group R-OH	=	Hydrogen bond donor and/or hydrogen bond acceptor
c.#	Alkylammonium group (RNH_3^+)	=	Hydrogen bond donor and/or ionic interactions
d.#	Ketone $R_2C=O$	=	Hydrogen bond acceptor only
END OF QUESTION			

Type: multiple choice		
Title: Chapter 13 - Question 23		
23)	Which of the following aims in drug design is related to the pharmacodynamics of a drug?	
Feedback: The reduction of side effects can be achieved by improving the affinity and selectivity of the drug for its target. The other aims are related to pharmacokinetics.		
Correct	**a.#**	The reduction of side effects.
incorrect	**b.#**	The maximisation of metabolic stability.
incorrect	**c.#**	The minimisation of elimination by the kidneys.
incorrect	**d.#**	The maximisation of oral bioavailability.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 13 - Question 24	
24)	Isoprenaline is an adrenergic agonist. The isoprenaline analogue shown below demonstrates a dramatic rise in activity in *in vitro* studies. What is the most likely conclusion that can be drawn?



Isoprenaline



Analogue

Feedback: The strategy used here is an extension strategy where the rationale is to add extra functional groups to the lead compound with the aim of finding additional binding interactions with the binding site. In this case, the extra aromatic ring could be providing extra van der Waals interactions, while the phenol group provides extra hydrogen bonding interactions. Since *in vitro* studies are involved, drug metabolism is not possible. It is possible that an altered pKa or increased molecular weight may have some influence on activity, but are unlikely to result in a dramatic increase in activity.		
incorrect	**a.#**	The analogue has an altered pKa.
Correct	**b.#**	The extra phenol ring in the analogues is interacting with an additional binding region in the binding site.
incorrect	**c.#**	Activity has increased due to increased molecular weight.
incorrect	**d.#**	The extra substituent is protecting the terminal methyl groups of isoprenaline from metabolic reactions.
END OF QUESTION		

Type: matching			
Title: Chapter 13 - Question 25			
25)	Match the term to the definition:		
Feedback: The pharmacophore defines the important binding groups in a molecule and their relative positions. It is useful in drug design and drug discovery. Molecules with different structures can be shown to have similar pharmacophores. Pharmacodynamics and pharmacokinetics are both relevant to drug design but study different properties of the candidate drug. In pharmacodynamics the focus is on how the candidate drug interacts with its target binding site, whereas in pharmacokinetics one is studying the various factors that might influence how much of the drug reaches the target in vivo. A chromophore relates to a conjugated unsaturated region of a molecule that is responsible for its colour.			
a.#	The important binding groups in a molecule and their relative positions.	=	Pharmacophore
b.#	The study of how a drug interacts with its target binding site.	=	Pharmacodynamics
c.#	The study of how a drug is absorbed, distributed, metabolised and excreted in a living system.	=	Pharmacokinetics
d.#	The conjugated, unsaturated system in a coloured compound.	=	Chromophore
END OF QUESTION			

Type: multiple choice		
Title: Chapter 13 - Question 26		
26)	It is common practice to vary the length and size of alkyl groups when making analogues of a lead compound. Which of the following statements is true?	
Feedback: Activity would increase due to increasing **van der Waals** interactions. There is no obvious reason why an increase in the length or size of an alkyl group should stabilise a molecule, or why increased stability should lead to higher activity and selectivity. An increase in activity will result from better van der Waals interactions with a **hydrophobic** region of the binding site.		
incorrect	**a.#**	Replacing a straight chain alkyl group with a branched alkyl group may increase activity by filling up a hydrophobic pocket

		and increasing hydrogen bonding interactions.
Correct	**b.#**	Increasing the chain length or size of an alkyl group may increase target selectivity if a hydrophobic binding region is more spacious in one binding site than another.
incorrect	**c.#**	Increasing the chain length or size of an alkyl group increases activity and selectivity by stabilising the analogue.
incorrect	**d.#**	Increasing the chain length of an alkyl group may increase activity by leading to better van der Waals interactions with a hydrophilic region of the binding site.
END OF QUESTION		

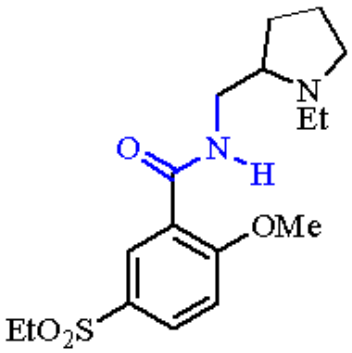
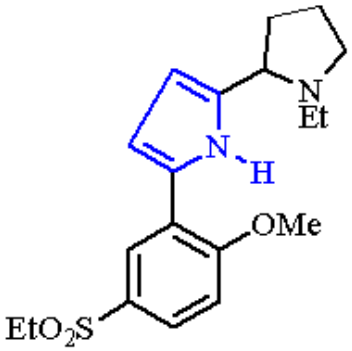
Type: multiple choice	
Title: Chapter 13 - Question 27	
27)	Which of the following is not true of analogues produced by a rigidification strategy?
Feedback: More rigid molecules have less rotatable bonds and a smaller number of possible conformations. Assuming that the active conformation is retained, it is likely that activity and selectivity will increase.	

Correct	**a.#**	They have a greater number of conformations.
incorrect	**b.#**	They are more difficult to synthesise.
incorrect	**c.#**	They are likely to be more selective.
incorrect	**d.#**	They are likely to be more active.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 13 - Question 28	
28)	Which of the following is not necessarily true about analogues produced by the simplification strategy?
Feedback: In general, analogues produced by a simplification strategy are likely to be more flexible. As a result, there is less chance of the active conformation being adopted at any one time.	

Correct	**a.#**	They are more likely to adopt the active conformation.
incorrect	**b.#**	They are easier to synthesise.
incorrect	**c.#**	They are cheaper to synthesise.
incorrect	**d.#**	They have more side effects.
END OF QUESTION		

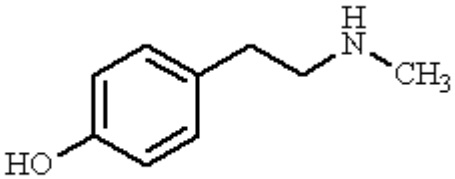
Type: multiple choice	
Title: Chapter 13 - Question 29	
29)	DU 122290 is an analogue of sultopride. The former shows greater selectivity for the D3 receptor over the D2 receptor. Which of the following statements is correct?

	 Sultopride  DU 122290

Feedback: The introduction of the pyrrole ring prevents a single bond rotation and **decreases** the number of possible conformations for the side chain.
The introduction of a pyrrole ring certainly removes carbonyl oxygen and prevents it acting as a hydrogen bond acceptor in DU 122290. However, if this is an important binding interaction for the D3 receptor, DU 122290 should show selectivity for the D2 receptor and not the D3 receptor as observed.
If the active conformation for the D2 receptor was favoured, the analogue should show selectivity for the D2 receptor over the D3 receptor.

incorrect	**a.#**	The introduction of a pyrrole ring increases the number of possible conformations for the side chain.
incorrect	**b.#**	The introduction of a pyrrole ring removes a carbonyl oxygen which could act as a hydrogen bond acceptor to a relevant binding region in the D3 binding site.
incorrect	**c.#**	The introduction of the pyrrole ring introduces increased rigidity and favours the active conformation for the D2 receptor over the active conformation for the D3 receptor.
Correct	**d.#**	The pyrrole ring in DU 122290 represents a bio-isosteric group for the amide bond in sultopride.

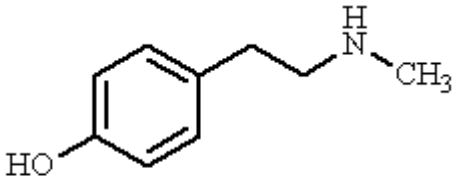
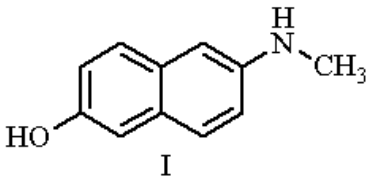
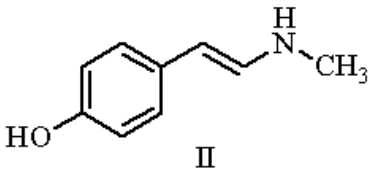
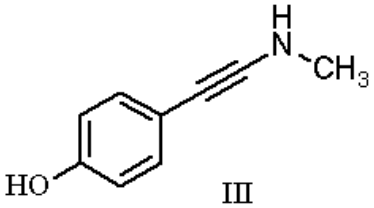
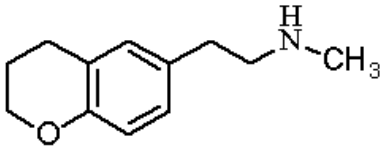
END OF QUESTION

Type: multiple choice	
Title: Chapter 13 - Question 30	
30)	Structure (I) has a highly flexible side chain.  I Which of the following statements is false?

Feedback: A flexible molecule may be able to bind in different binding modes to the one target, but this is more likely to result in decreased activity since some of these binding modes may fail to activate the receptor.

incorrect	**a.#**	A flexible molecule is less likely to be in its active conformation when it approaches its target binding site. This results in decreased activity.

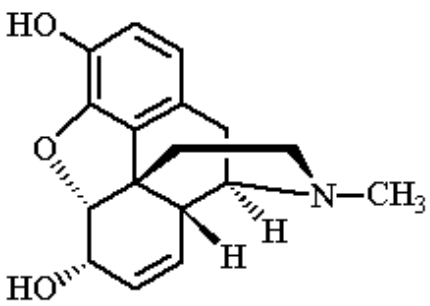
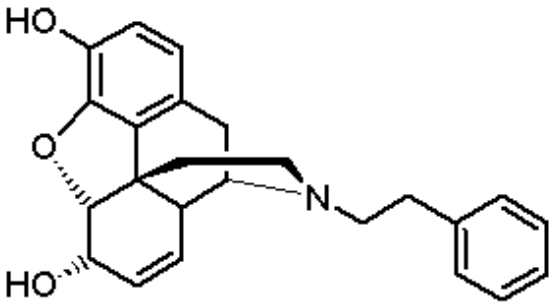
incorrect	**b.#**	A flexible molecule is more likely to adopt conformations that will bind to different targets, resulting in side effects.
Correct	**c.#**	A flexible molecule may be able to bind to its target binding site in different binding modes resulting in an increase in activity.
incorrect	**d.#**	A flexible molecule is less likely to show target selectivity.
END OF QUESTION		

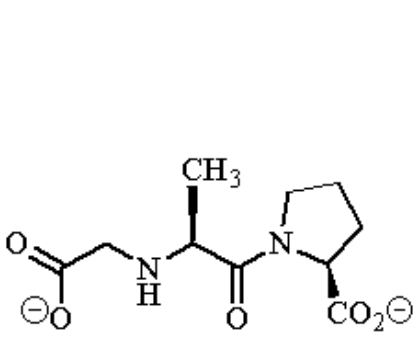
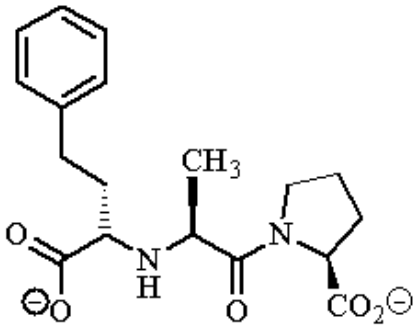
Type: multiple choice	
Title: Chapter 13 - Question 31	
31)	Structure (I) has a highly flexible side chain.  I Which of the following analogues does not represent a suitable rigidification strategy to lower the number of possible conformations?  I  II  III  IV

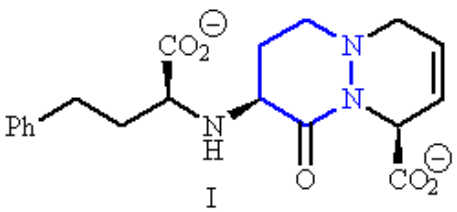
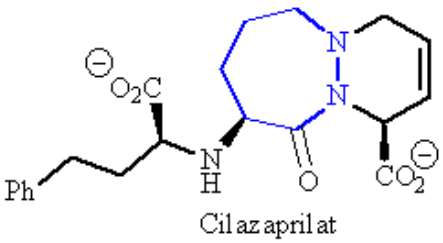
Feedback: In structure IV, none of the rotatable bonds in the side chain are constrained and so the side chain has as many different conformations as the original structure. Structure I has two of the single bonds in the side chain constrained, while structures II and III have one bond constrained.		
incorrect	**a.#**	Structure I.
incorrect	**b.#**	Structure II.
incorrect	**c.#**	Structure III.
Correct	**d.#**	Structure IV.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 13 - Question 32	
32)	A lead compound containing a methyl ester functional group was hydrolysed to give an analogue with a carboxylic acid functional group. An *in vivo* bioassay which involved oral administration

	suggested that the ester was active and that the acid was inactive. However, an *in vitro* bioassay suggested that the ester was inactive and that the acid was active. What is the likely explanation for these results?	
Feedback: The most likely explanation is that the lead compound acts as a prodrug and is metabolised to the active compound once it is absorbed. This implies that the carboxylic acid is an important binding group and that the ester is not. The *in vivo* bioassay for the carboxylic acid would be negative since the compound is too polar to get absorbed into the blood supply. The *in vitro* assay for the ester would be negative since it is not metabolised to the active carboxylic acid. The carboxylic acid is more polar than the ester and should be less easily absorbed. Moreover, esters are more likely to be metabolised to carboxylic acids than the other way round. There is no reason why a target should prefer one functional group under *in vivo* conditions and a different functional group under *in vitro* conditions. It is a possibility that the ester and the carboxylic acid are both metabolised to the active compound *in vivo,* but this is less likely than the ester acting as a prodrug in the *in vivo* studies.		
incorrect	**a.#**	The lead compound fails to get absorbed in the *in vivo* studies whereas the carboxylic acid is absorbed and is metabolised to the ester.
Correct	**b.#**	The ester is acting as a prodrug in the *in vivo* studies.
incorrect	**c.#**	The binding site of the target prefers to bind a carboxylic acid under *in vitro* conditions and an ester under *in vivo* conditions.
incorrect	**d.#**	The ester and the carboxylic acid are both metabolised to the active compound *in vivo*.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 13 - Question 33		
33)	*N*-Phenethylmorphine was synthesised from morphine and found to have enhanced activity.  Morphine  *N*-Phenethylmorphine Why does *N*-phenethylmorphine have higher activity than morphine?	
Feedback: The aromatic ring is binding to an extra hydrophobic binding region by van der Waals interactions.		
Correct	**a.#**	Increased binding to an extra binding region by van der Waals interactions.
incorrect	**b.#**	Increased binding to an extra binding region by hydrogen bonding.
incorrect	**c.#**	Increased binding to an extra binding region by dipole-dipole interactions.
incorrect	**d.#**	Increased binding to an extra binding region by ionic bonding.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 13 - Question 34		
34)	Structure I is an antihypertensive agent which acts by inhibiting the enzyme ACE. Structure II was designed to bind to the binding site more strongly and to have greater activity.  I  II The chain connecting the aromatic ring was varied in length to find the optimum chain length of 2 carbons. Why is a 2-carbon chain length optimum for antihypertensive activity in this situation?	
Feedback: The correct answer is that varying the chain length allows one to determine the best chain length to position the aromatic ring in the best position to interact with a hydrophobic binding region It is not illogical to propose that varying the chain length 'gives the molecule the optimum hydrophobic character for binding' or 'gives sufficient flexibility to the arylalkyl chain to allow a full exploration of the available conformational space'. However, neither is the correct explanation in this case. It makes no sense to say that varying the chain length results in 'the optimum length for molecular stability '. There is no reason why the stability of a molecule should be affected by the length of the carbon chain.		
incorrect	**a.#**	It gives the molecule the optimum hydrophobic character for binding.
incorrect	**b.#**	It gives sufficient flexibility to the arylalkyl chain to allow a full exploration of the available conformational space.
Correct	**c.#**	It places the aromatic ring in the optimum position to bind with the extra binding region.
incorrect	**d.#**	It is the optimum length for molecular stability.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 13 - Question 35	
35)	Cilazaprilat is an antihypertensive agent derived from the lead compound (I) by a ring expansion strategy.  I  Cilazaprilat

	Why was this strategy successful in this situation?

Feedback:
The correct answer is that varying the ring size changes the relative positions of the important binding groups such that they can all interact effectively with corresponding binding regions at the same time. Increasing the size of the ring by a methylene unit would certainly be expected to increase hydrophobicity, but the increase is unlikely to be significant enough to explain the increased activity. It is not very likely that it locked the molecule into the active conformation since this suggests that the molecule has become more rigid. If anything, it is more flexible although this, in itself, does not explain the success of the strategy.

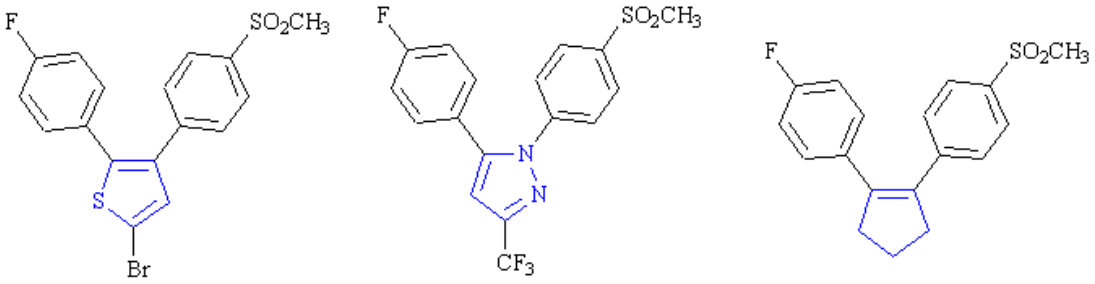
incorrect	**a.#**	It increased the hydrophobicity of the molecule.
incorrect	**b.#**	It locked the molecule into the active conformation.
incorrect	**c.#**	It increased the flexibility of the molecule.
Correct	**d.#**	It placed important binding groups in the optimum position for binding to their respective binding regions.

END OF QUESTION

Type: multiple choice

Title: Chapter 13 - Question 36

36) The following structures are various non-steroidal anti-inflammatory agents developed by a ring variation strategy.



Which of the following terms has been used for drugs of this sort?

Feedback: 'Me too' drugs are often synthesised on a trial and error basis to find structures having similar biological activity which are not covered by patents. Occasionally, such drugs may have improved properties over the parent structure.

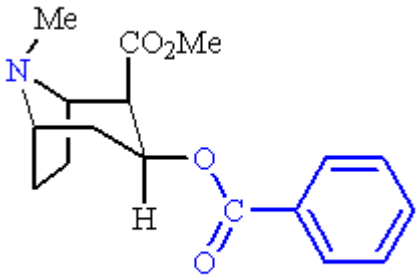
incorrect	**a.#**	Family drugs.
incorrect	**b.#**	Analogue drugs.
Correct	**c.#**	Me-too drugs.
incorrect	**d.#**	Isosteric drugs.

END OF QUESTION

Type: multiple choice

Title: Chapter 13 - Question 37

37) Cocaine was isolated as the active principle from coca leaves. Studies into structure activity relationships showed that the important functional groups for binding are those shown in blue.



Cocaine

What binding interactions are likely for the amine?

Feedback: The amine can act as a hydrogen bond acceptor as the free base. If it is protonated, it can act as a hydrogen bond donor or it can take part in an ionic bond.

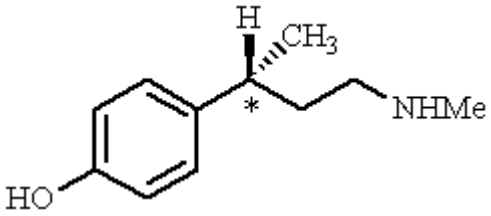
incorrect	**a.#**	Hydrogen bonding only.
incorrect	**b.#**	Ionic bonding only.
incorrect	**c.#**	Van der Waals interactions.
Correct	**d.#**	Ionic bonding and / or hydrogen bonding.

END OF QUESTION

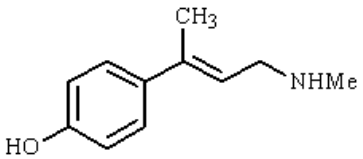
Type: multiple choice

Title: Chapter 13 - Question 38

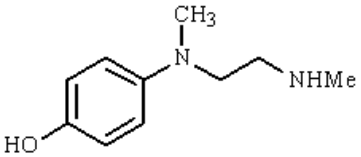
38) The following hypothetical structure has an asymmetric centre as shown by the asterisk.



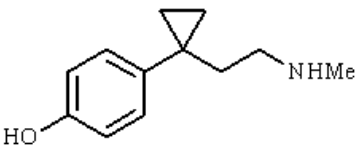
Which of the following structures might be worth synthesising as an analogue of the above lead compound in order to achieve an active compound lacking an asymmetric centre?



A



B

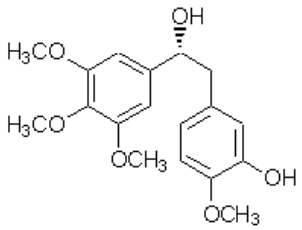
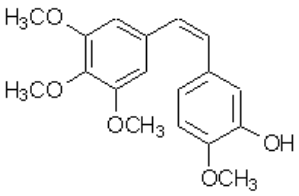
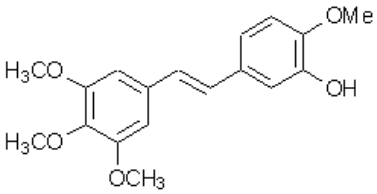


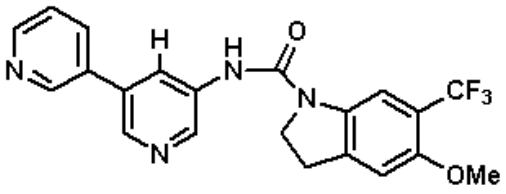
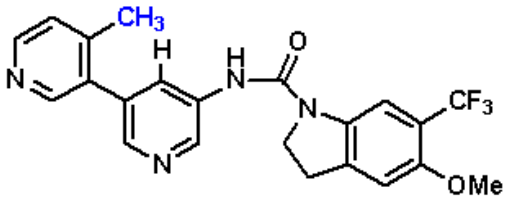
C

Feedback: All of the structures shown are achiral molecules and are all worth considering as analogues.

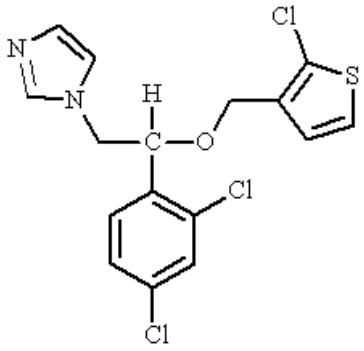
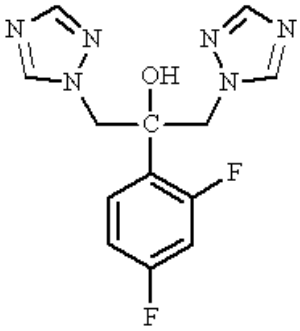
incorrect	**a.#**	Structure A.
incorrect	**b.#**	Structure B.
incorrect	**c.#**	Structure C.
Correct	**d.#**	All of them.

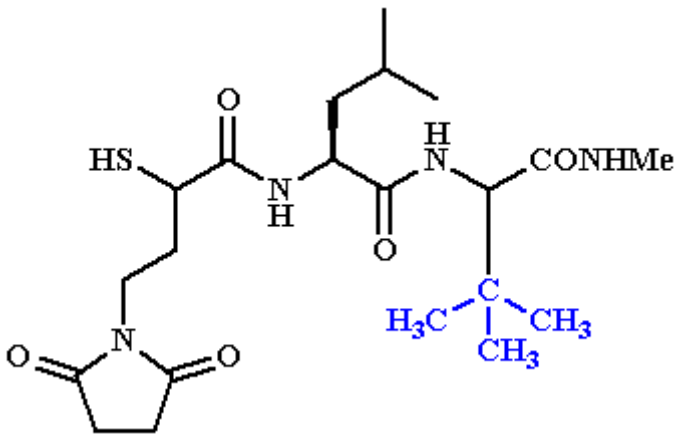
END OF QUESTION

Type: multiple choice		
Title: Chapter 13 - Question 39		
39)	Combretastatin is an anti cancer agent discovered from an African plant. Analogue (I) is more active than combretastatin whereas analogue II is less active.  Combretastatin  Analogue I  Analogue II Which of the following statements is most likely to explain the difference in activity between analogues I and II?	
Feedback: The active conformation (I) has both alkene substituents *cis* with respect to each other. In this configuration they are able to interact with their corresponding binding regions at the same time. In the *trans* isomer, the substituents cannot bind to their corresponding binding regions simultaneously.		
Correct	**a.#**	Rigidification has locked analogue I into the active conformation and locked analogue II into an inactive conformation.
incorrect	**b.#**	Rigidification has locked analogue II into the active conformation and locked analogue I into the inactive conformation.
incorrect	**c.#**	Rigidification has locked both analogues I and II into the active conformation.
incorrect	**d.#**	Rigidification has locked both analogues I and II into the inactive conformation.
END OF QUESTION		

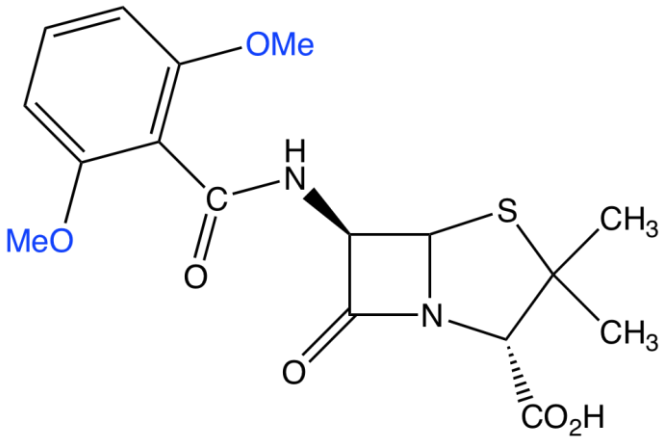
Type: multiple choice		
Title: Chapter 13 - Question 40		
40)	Structure (III) is a serotonin antagonist. A methyl group has been introduced into analogue (IV) resulting in increased activity.  III  IV Which of the following statements is a reasonable explanation for the increase in activity observed for structure IV?	
Feedback: The methyl group acts here as a conformational blocker to force the molecule into the active conformation. Activity should **decrease** if the molecule was prevented from adopting the active conformation. The methyl group should interact with a **hydrophobic** pocket and not a hydrophilic pocket. This is certainly a possibility that would have to be considered. Increasing the basicity of the ring nitrogen should make it a better hydrogen bond **acceptor**. This again would be a possibility worth considering.		
incorrect	**a.#**	The methyl group interacts with an extra hydrophilic binding region through van der Waals interactions.
incorrect	**b.#**	The methyl group increases the basicity of the ring nitrogen, making it a better

		hydrogen bond donor.
incorrect	**c.#**	The methyl group prevents the pyridine rings from being coplanar and prevents the molecule adopting the active conformation.
Correct	**d.#**	The methyl group prevents the pyridine rings from being coplanar and forces the molecule into the active conformation.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 14 - Question 01		
01)	Tioconazole is a non polar antifungal agent which is used topically, whereas fluconazole is a polar drug which is used systemically. Which of the following statements is correct?  Tioconazole  Fluconazole	
Feedback: Fluconazole is more polar and more water soluble than tioconazole due to the extra nitrogen atoms and the alcohol group. The other options are incorrect. The nitrogen atoms make the drug more polar and not less polar. Increased polarity increases water solubility. The fluorine substituents have little effect on water solubility.		
incorrect	**a.#**	The heterocyclic groups in fluconazole contain more nitrogen atoms making the drug less polar.
incorrect	**b.#**	Increased polarity decreases water solubility.
incorrect	**c.#**	The fluorine substituents in fluconazole increase water solubility.
Correct	**d.#**	The alcohol group in fluconazole increases polarity.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 14 - Question 02		
02)	D1927 contains a branched alkyl group shown in blue which protects the terminal peptide bond from hydrolysis. What is the general term given for a group such as this? 	
Feedback: The branched alkyl group is acting purely as a steric shield. There are no electronic effects.		
incorrect	**a.#**	Electronic shield.

incorrect	**b.#**	Stereoelectronic shield.
Correct	**c.#**	Steric shield.
incorrect	**d.#**	Alkyl shield.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 14 - Question 03	
03)	Methicillin was an important penicillin which was effective in the 1960's against penicillin G resistant strains of *Staphylococcus aureus*. Although the drug is more effective than penicillin G against resistant strains, it is not as active against strains which are susceptible to penicillin G. The methoxy substituents of the aromatic ring play an important role in the drug's effectiveness against penicillin G resistant strains. Which of the following statements is the most likely explanation?  Methicillin

Feedback: The β-lactam ring is susceptible to β-lactamase enzymes produced by some bacteria. These enzymes open up the β-lactam ring and deactivate the penicillin. The methoxy groups on the aromatic ring of methicillin shield the lactam ring from attack.		
incorrect	**a.#**	The methoxy groups act as steric shields to protect the aromatic ring from oxidation by metabolic enzymes in the body.
Correct	**b.#**	The methoxy groups act as steric shields to protect the β-lactam ring from hydrolysis by enzymes produced by resistant bacteria.
incorrect	**c.#**	The methoxy groups act as conformational blockers to orientate the aromatic ring out of the plane of the side chain amide group, allowing better binding interactions with the target enzyme.
incorrect	**d.#**	The methoxy groups increase the electron density of the aromatic ring allowing better binding interactions with the target enzyme.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 14 - Question 04	
04)	How does adrenaline prolong the local anaesthetic activity of procaine?

Feedback: Adrenaline constricts blood vessels and helps to localise procaine at the site of injection. The other options are wrong. There is no complex formed between adrenaline and procaine. Adrenaline does not block the metabolic enzymes attacking procaine and adrenaline does not sensitise the receptors for procaine.		
incorrect	**a.#**	It forms a complex with procaine which acts as a more powerful local anaesthetic.
incorrect	**b.#**	It blocks metabolic enzymes that would normally inactivate procaine.
Correct	**c.#**	It constricts blood vessels and helps to localise procaine at the site of injection.
incorrect	**d.#**	It sensitises the receptors normally occupied by procaine.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 14 - Question 05	
05)	Why do many drugs containing ester functional groups have a limited duration of action?

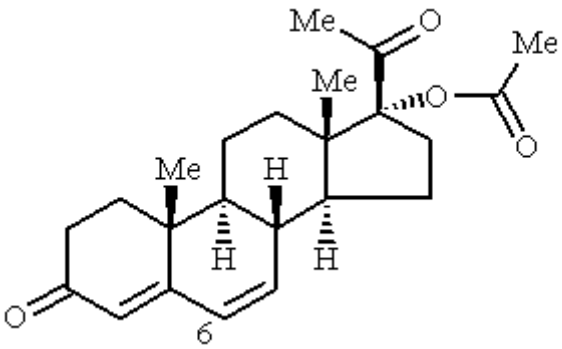
Feedback: Esters are fairly easily hydrolysed by enzymes called esterases in the blood supply. If the ester is important for a binding interaction, the drug will have a limited duration of action. The other options are wrong. Ester groups do not increase polarity as much as a free carboxylic acid or alcohol would. Drugs containing ester functional groups can cross cell membranes relatively easily and some ester prodrugs are used to enhance oral absorption. If the ester was hindering access to the target binding site, the drug would have poor activity rather than a poor duration of activity. Esters are not as polar as the constituent carboxylic acid and alcohol. As a result, they should be excreted more slowly.		
incorrect	**a.#**	The ester groups make the drug more polar and reduce absorption.
Correct	**b.#**	The ester groups are susceptible to esterase enzymes.
incorrect	**c.#**	The ester groups act as steric shields and prevent drugs entering binding sites.
incorrect	**d.#**	Drugs with ester groups are rapidly excreted by the kidneys.
END OF QUESTION		

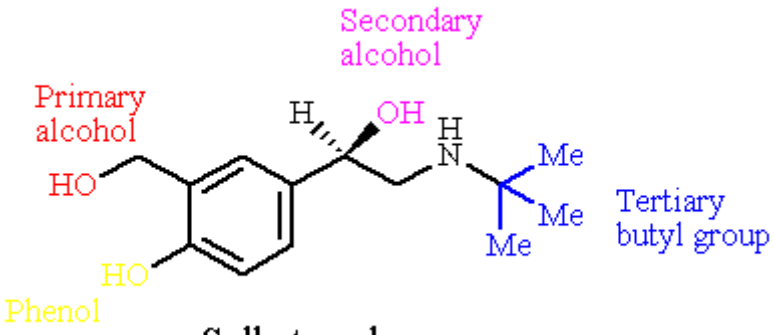
Type: multiple choice	
Title: Chapter 14 - Question 06	
06)	Many drugs containing ester functional groups have a limited duration of action. There are several strategies which can be used to avoid this problem. Which of the following is not a suitable strategy?

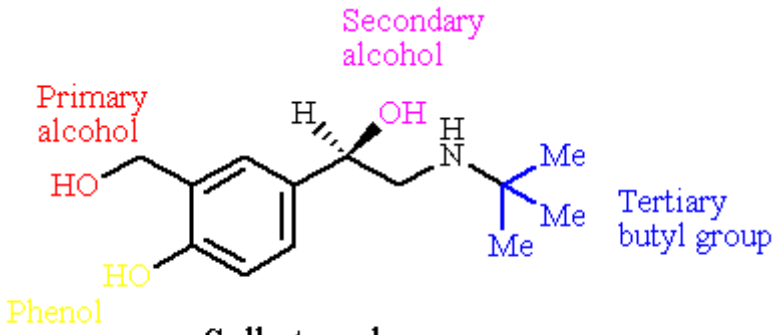
Feedback: Adding an electron withdrawing group to the alkoxy moiety of the ester is more likely to increase its susceptibility to hydrolysis and not decrease it. This is because the alkoxide leaving group would be stabilised by an electron withdrawing group, making it a better leaving group. Replacing the ester group with an amide or replacing the ester group with a urethane will stabilise the functional group since both groups contain a nitrogen with a lone pair of electrons which will interact with the neighbouring carbonyl group to make it less reactive to nucleophiles. Adding a steric shield close to the ester will stabilise the group since the steric shield will hinder the approach of attacking nucleophiles.		
incorrect	**a.#**	Replacing the ester group with an amide.
incorrect	**b.#**	Adding a steric shield close to the ester.
Correct	**c.#**	Adding an electron withdrawing group to the alkoxy moiety of the ester.
incorrect	**d.#**	Replacing the ester group with a urethane.

END OF QUESTION

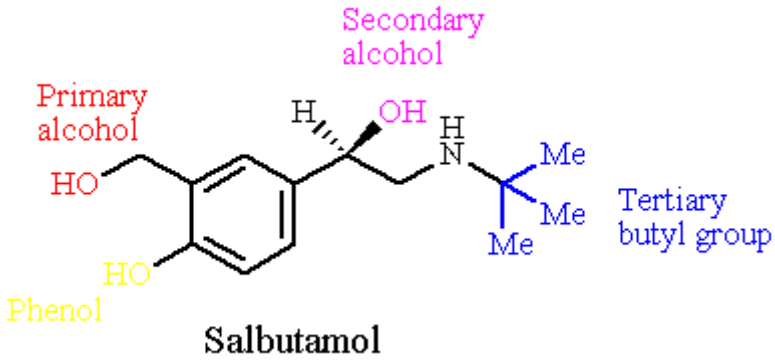
Type: multiple choice		
Title: Chapter 14 - Question 07		
07)	What is the most plausible explanation for the fact that some drugs containing an ester group are inactive *in vitro*, but are active once the drug has been absorbed *in vivo*?	
Feedback: These drugs are described as prodrugs. They do not bind to a target binding site *in vitro* since an important binding group has been masked by the ester. *In vivo*, the ester group is hydrolysed to reveal the binding group. The other proposals are not impossible, but are more unlikely.		
incorrect	**a.#**	The ester can only bind to its target when the target is present in the body. If the target is isolated for *in vitro* studies, the ester cannot bind to it since an important cofactor is missing.
Correct	**b.#**	The ester group is hydrolysed by esterases in the body to produce the active drug.
incorrect	**c.#**	The targets used for *in vitro* tests undergo conformational changes which disguise their binding sites and prevent the ester from binding *in vitro*. The ester binds to the target *in vivo.*
incorrect	**d.#**	The ester forms a complex with an endogenous molecule before binding to the target binding site.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 14 - Question 08		
08)	Megestrol acetate has a short duration of activity since it is metabolised at position 6. Placing a methyl group at the 6-position increases the duration of action. What role does the methyl group play?  	
Feedback: The methyl group acts as a blocker and prevents the metabolic enzyme attacking the 6 position. The other options are all incorrect: it would be rather pointless adding a group to prevent one metabolic reaction if it was susceptible to another. The steroid skeleton is rigid and does not have a variety of conformations available to it. It is true to say that the methyl group is electron donating, but the electron donating effect of the group is not the reason it blocks metabolism. It is a steric effect.		
incorrect	**a.#**	It is a susceptible metabolic group.
Correct	**b.#**	It is a metabolic blocker.
incorrect	**c.#**	It is a conformational blocker.
incorrect	**d.#**	It is an electron donating group.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 14 - Question 09		
09)	Salbutamol is an anti-asthmatic agent which shows an improved activity over previous catechol-like compounds  Salbutamol What feature of salbutamol is crucial to this increased duration of action?	
Feedback: The primary alcohol group represents a group shift where a functional group is retained, but shifted to a different part of the skeleton. In this case, a phenol of the original catechol ring was shifted one carbon unit away from the aromatic ring. This makes it less susceptible to metabolism.		
incorrect	**a.#**	The tertiary butyl group.
incorrect	**b.#**	The secondary alcohol.
incorrect	**c.#**	The phenol.
Correct	**d.#**	The primary alcohol.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 14 - Question 10		
10)	Salbutamol is an anti-asthmatic agent, which shows an improved activity over previous catechol-like compounds  Salbutamol What strategy was used in order to attain the improved duration of action?	
Feedback: A group shift is where a functional group is retained, but shifted to a different part of the skeleton. In this case, a phenol of the original catechol ring was shifted one carbon unit away from the aromatic ring. It could also be argued that the strategy was a chain extension since a methylene group has been inserted between the OH group and the aromatic ring.		

incorrect	**a.#**	Substituent variation.
Correct	**b.#**	Group shift.
incorrect	**c.#**	Chain contraction.
incorrect	**d.#**	Simplification.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 14 - Question 11	
11)	Salbutamol is an anti-asthmatic agent, which shows an improved activity over previous catechol-like compounds.  Salbutamol What was the rationale behind the strategy to attain the improved duration of action?

Feedback: The phenol group that was shifted was susceptible to metabolism but it was also an important binding group for the target binding site. Shifting it one carbon unit away from the ring made it unrecognisable to the metabolic enzyme, but it was still recognisable to the target binding site.
The hydrogen bonding group was already there and so the introduction of a hydrogen bonding group was not the rationale behind this strategy.
The other options are not irrational, but this was not the reason for carrying out this strategy.

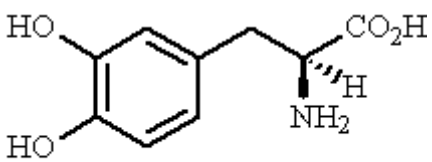
Correct	**a.#**	To shift the position of an important binding group such that it is not recognised by metabolic enzymes.
incorrect	**b.#**	To introduce a group which more closely fits a binding region.
incorrect	**c.#**	To move an important binding group such that it interacts more strongly with a binding region.
incorrect	**d.#**	To introduce a group capable of H-bonding.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 14 - Question 12	
12)	Why should the introduction of an ester group into the main skeleton of a drug decrease the drug's duration of action?

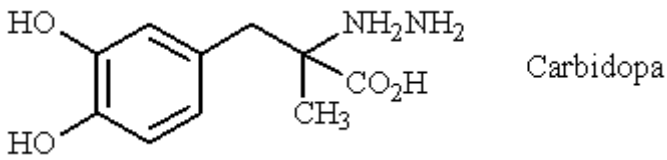
Feedback: Esters are prone to hydrolysis by esterases to generate more polar carboxylic acids and alcohols which are rapidly excreted. If the ester is within the main skeleton of the drug, it is likely that that hydrolysis will split the molecule in two and result in the loss of important binding groups from one half or the other.
The other options are wrong. The introduction of an ester will not result in that large an increase in polarity.
The molecular weight may not be altered very much by the introduction of an ester. In any case, the level of absorption will not affect the lifetime of the drug once it is in the

blood supply. Esters are generally not sufficiently reactive to react with proteins.		
incorrect	**a.#**	It makes the drug more polar and increases the rate of excretion.
incorrect	**b.#**	It increases the molecular weight of the drug and reduces the amount of drug absorbed.
incorrect	**c.#**	It reacts with proteins in the body such that the drug is irreversibly linked to the proteins by a covalent bond.
Correct	**d.#**	It is hydrolysed by esterases to an alcohol and a carboxylic acid which are quickly excreted due to increased polarity.
END OF QUESTION		

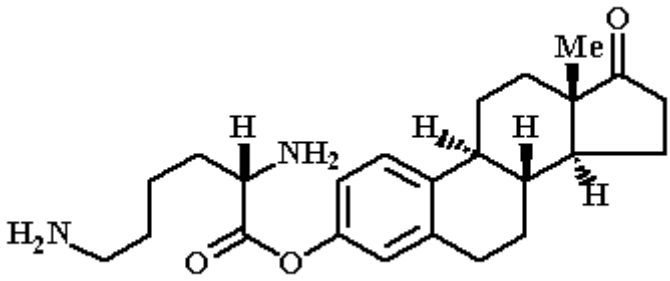
Type: multiple choice		
Title: Chapter 14 - Question 13		
13)	Why should the addition of an alcohol or phenol group to a drug decrease the drug's duration of action?	
Feedback: An alcohol or phenol group is commonly subject to phase II metabolic reactions where polar molecules are linked to the group. The increase in polarity results in rapid excretion and a shorter duration of action. It is true to state that the addition of either group increases the polarity of the drug and reduces the amount of drug absorbed, but a reduction in the amount of drug absorbed is not relevant to the lifetime of the drug in the blood supply once the drug **has** been absorbed. It would be true to say that the increase in polarity will also increase the rate of excretion. It is wrong to state that the modified drug reacts with proteins in the body such that the drug is irreversibly linked to the proteins by a covalent bond. An alcohol or phenol functional group is not a highly reactive group, and drugs containing it should not form irreversible covalent bonds with proteins. It may be true to say that the alcohol or phenol acts as an electron withdrawing group and it is possible that such a group may affect the binding strength of important binding groups depending on the structure of the drug and its target, but this is a pharmacodynamic issue and has nothing to do with the pharmacokinetics of the drug and how long it survives in the body.		
Correct	**a.#**	It acts as a 'polar handle' for conjugation reactions. The conjugates are excreted more quickly.
incorrect	**b.#**	It increases the polarity of the drug and reduces the amount of drug absorbed.
incorrect	**c.#**	It reacts with proteins in the body such that the drug is irreversibly linked to the proteins by a covalent bond.
incorrect	**d.#**	It acts as an electron withdrawing group and affects the binding strength of important binding groups.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 14 - Question 14	
14)	Levodopa is used as a prodrug for dopamine since it enters the brain more efficiently than dopamine. Why?  Levodopa
Feedback: Since the prodrug mimics an amino acid, it can be carried across the blood	

brain barrier by transport proteins that recognise amino acids. The prodrug is **more** polar due to the extra carboxylic acid group, and so it crosses the blood brain barrier **less** effectively. Levodopa still has the catechol system present, which makes it as susceptible to metabolism as dopamine.		
incorrect	**a.#**	It is more polar and crosses the blood brain barrier more effectively.
incorrect	**b.#**	It is less polar and crosses the blood brain barrier more effectively.
Correct	**c.#**	It mimics an amino acid and takes advantage of transport proteins in order to cross the blood brain barrier.
incorrect	**d.#**	It is less susceptible to metabolism.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 14 - Question 15	
15)	What role does carbidopa have in the treatment of Parkinson's disease?  Carbidopa
Feedback: Carbidopa inhibits a decarboxylase enzyme in the periphery such that L-Dopa remains intact and is not converted to dopamine. Carbidopa is too polar to cross the blood brain barrier and does not inhibit the enzyme in the brain. As a result, L-Dopa is only converted to dopamine in the brain where it is needed.	

incorrect	**a.#**	It binds to dopamine receptors in the brain and acts as an agonist.
incorrect	**b.#**	It binds to dopamine receptors in the brain and acts as an antagonist.
incorrect	**c.#**	It inhibits a decarboxylase enzyme in the brain.
Correct	**d.#**	It inhibits a decarboxylase enzyme in the periphery.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 14 - Question 16	
16)	The following structure (III) is used as a prodrug for estrone. What structures are formed following ester hydrolysis?  (III)
Feedback: Lysine is a natural amino acid which is released after hydrolysis of the ester. Since it is a natural amino acid, no toxic by products are formed.	

Serine, arginine, and histidine contain different functional groups on their side chain from the primary amino group present on lysine.		
incorrect	**a.#**	Estrone and serine.
incorrect	**b.#**	Estrone and arginine.
Correct	**c.#**	Estrone and lysine.
incorrect	**d.#**	Estrone and histidine.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 14 - Question 17	
17)	What expression is used to describe drugs which have been designed from peptide lead compounds, but which lack peptide character?
Feedback: Peptidomimetics are drugs that have been developed from peptide lead compounds but which lack peptide character. Antipeptides is a figment of the imagination. Peptoids are peptides where *N*-substituted glycine units are linked together. Antibodies are proteins which are part of the immune response, and which recognise antigens present in foreign cells.	

incorrect	**a.#**	Antipeptides.
Correct	**b.#**	Peptidomimetics.
incorrect	**c.#**	Peptoids.
incorrect	**d.#**	Antibodies.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 14 - Question 18	
18)	Which polymer has been attached to proteins to increase a protein's solubility and stability, as well as decreasing the likelihood of an immune response?
Feedback: Polyethylene glycol (PEG) has been used to increase the solubility and stability of proteins, and to decrease the chances of an immune response. The PEG molecules can be viewed as hydrophilic polymeric shields. The other polymers mentioned have no such applications.	

incorrect	**a.#**	Polythene.
incorrect	**b.#**	Polyvinylchloride.
Correct	**c.#**	Polyethylene glycol.
incorrect	**d.#**	Polyurethane.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 14 - Question 19	
19)	There are several strategies for stabilising susceptible bonds in peptides. Which of the following is not used in this respect?
Feedback: The statement given is wrong since it should be the other way round. L-Amino acids are the natural enantiomers in human proteins, and so replacing an L-amino acid with a D-amino acid is often used to stabilise a susceptible peptide bond since it alters the stereochemistry of the alpha-carbon. This results in the side chain being unable to fit a binding pocket in the active site of metabolic enzymes.	

incorrect	**a.#**	Replacing a peptide link with a stable functional group.
incorrect	**b.#**	Adding an *N*-methyl group to the peptide link.

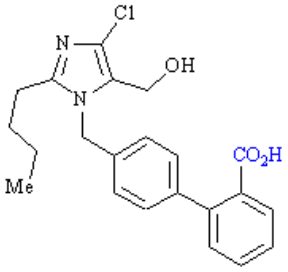
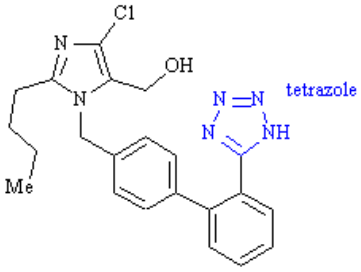
incorrect	**c.#**	Using an unnatural amino acid.
Correct	**d.#**	Replacing a D-amino acid with an L-amino acid.
END OF QUESTION		

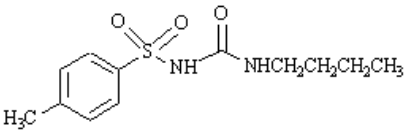
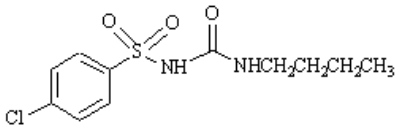
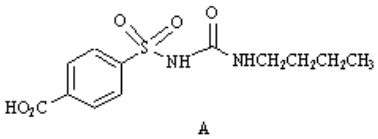
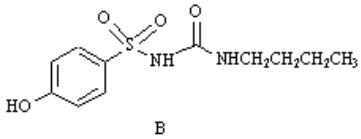
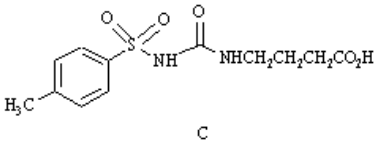
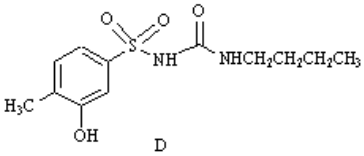
Type: multiple choice		
Title: Chapter 14 - Question 20		
20)	Oligonucleotides suffer disadvantages as drugs since they are susceptible to enzyme-catalysed degradation. Which of the following enzymes is mainly responsible for this?	
Feedback: Nucleases catalyse the hydrolysis of phosphate links. Peptidases catalyse the hydrolysis of peptide bonds. Lipases catalyse the breakdown of fats, and oxidases catalyse oxidation reactions.		
incorrect	**a.#**	Peptidases.
Correct	**b.#**	Nucleases.
incorrect	**c.#**	Lipases.
incorrect	**d.#**	Oxidases.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 14 - Question 21		
21)	What kinds of modifications have been carried out on oligonucleotides to make them more stable?	
Feedback: All of the above strategies have been tried. Modifications to the phosphate group have resulted in agents which have been introduced to the clinic.		
incorrect	**a.#**	Modifications to the phosphate group.
incorrect	**b.#**	Modifications to the sugar group.
incorrect	**c.#**	Modifications to the nucleic acid base.
Correct	**d.#**	All of the answers are correct.
END OF QUESTION		

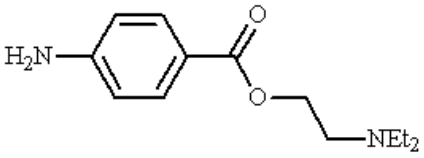
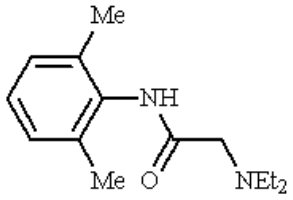
Type: multiple choice		
Title: Chapter 14 - Question 22		
22)	Which of the following strategies will not increase the polarity and water solubility of a drug?	
Feedback: Removing polar functional groups will decrease polarity and water solubility. The other options all increase the polar character of the drug and will increase water solubility.		
Correct	**a.#**	Removing polar functional groups.
incorrect	**b.#**	Removing extra alkyl groups.
incorrect	**c.#**	Replacing an aromatic ring with a nitrogen-containing heterocyclic ring.
incorrect	**d.#**	Replacing an alkyl group with a smaller alkyl group.
END OF QUESTION		

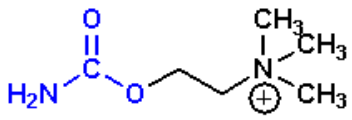
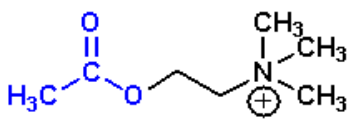
Type: multiple choice		
Title: Chapter 14 - Question 23		
23)	Losartin was developed from structure (I) as an antihypertensive agent by replacing a carboxylic acid group with a tetrazole ring. Which	

	of the following statements is incorrect?  (I)  Losartan	
Feedback: The tetrazole ring is acidic which is one of the reasons why it can be used as a bio-isostere for a carboxylic acid group.		
incorrect	**a.#**	The tetrazole ring represents a bio-isostere.
incorrect	**b.#**	The tetrazole ring mimics a carboxylic acid in being planar.
Correct	**c.#**	The tetrazole ring is basic rather than acidic and so cannot mimic the acidic nature of a carboxylic acid.
incorrect	**d.#**	The tetrazole ring is less polar than a carboxylic acid.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 14 - Question 24		
24)	Chlorpropamide has a longer antibiotic activity than tolbutamide.  Tolbutamide  Chlorpropamide Which of the following structures is the most likely metabolite of tolbutamide?  A  B  C  D	
Feedback: Exposed methyl groups are susceptible to metabolic oxidation which suggests that metabolites A and C are likely. However, the fact that a chloro substituent in chlorpropamide prolongs activity suggests that it is structure A that is the major metabolite. The chloro substituent in chlorpropamide is resistant to metabolic oxidation.		
Correct	**a.#**	Structure A.
incorrect	**b.#**	Structure B.
incorrect	**c.#**	Structure C.

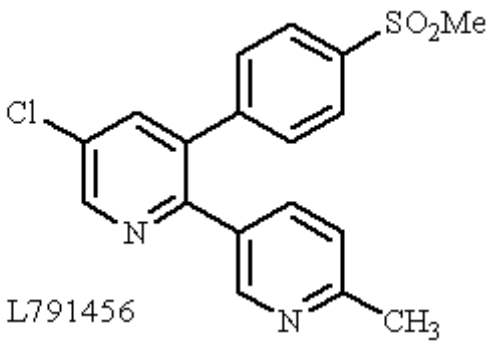
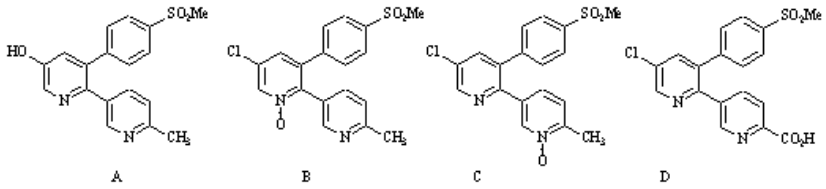
incorrect	**d.#**	Structure D.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 14 - Question 25		
25)	Lidocaine is a longer lasting local anaesthetic than procaine. Which of the following statements is true?  Procaine  Lidocaine	
Feedback: Lidocaine has an amide group which is **more** resistant to hydrolysis and which is also protected by the two methyl steric shields on the aromatic ring. As a result, it is wrong to state that 'lidocaine is less stable to hydrolysis' or that 'the methyl substituents in lidocaine decrease stability'. The amino group plays no role in the stability of the molecule. It is present as an extra binding group for the target binding site.		
incorrect	**a.#**	Lidocaine is less stable to hydrolysis.
incorrect	**b.#**	The methyl substituents in lidocaine decrease stability.
Correct	**c.#**	The amide group in lidocaine increases stability.
incorrect	**d.#**	The NH_2 group in procaine makes procaine less stable than lidocaine.
END OF QUESTION		

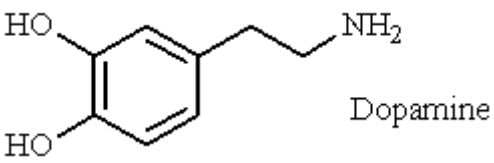
Type: multiple choice		
Title: Chapter 14 - Question 26		
26)	Carbachol is more stable than acetylcholine to hydrolysis. Which of the following statements is true?  Carbachol  Acetylcholine	
Feedback: Acetylcholine is metabolised by esterases which hydrolyse the ester group. It is not metabolised by cytochrome P450 enzymes. Replacing the acyl methyl group of acetylcholine with an amino group alters the **ester** functional group to a urethane functional group. Urethanes are more resistant to hydrolysis since the nitrogen's lone pair interacts with the neighbouring carbonyl group and makes it less reactive to nucleophiles such as water. The urethane NH_2 occupies the same binding region as the original methyl group and there are no additional binding interactions.		
incorrect	**a.#**	The blue coloured methyl group of acetylcholine is susceptible to oxidation by cytochrome P450 enzymes.
incorrect	**b.#**	The blue coloured nitrogen of carbachol acts as an extra binding group.
incorrect	**c.#**	Carbachol contains a urethane group while acetylcholine contains an ether group.
Correct	**d.#**	The blue coloured NH_2 group of carbachol is a bio-isostere for

		the blue coloured CH_3 group of acetylcholine.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 14 - Question 27		
27)	Which of the following definitions best describes a prodrug?	
Feedback: Prodrugs normally undergo a metabolic reaction in the body to produce an active compound. A drug that activates a receptor rather than deactivating it is called an agonist.		
incorrect	**a.#**	A drug that activates a receptor rather than deactivating it.
incorrect	**b.#**	A naturally occurring compound that can be converted synthetically into an active compound.
Correct	**c.#**	A compound that is inactive, but which is converted to an active compound in the body.
incorrect	**d.#**	A drug which has been approved for clinical purposes by a regulatory authority.
END OF QUESTION		

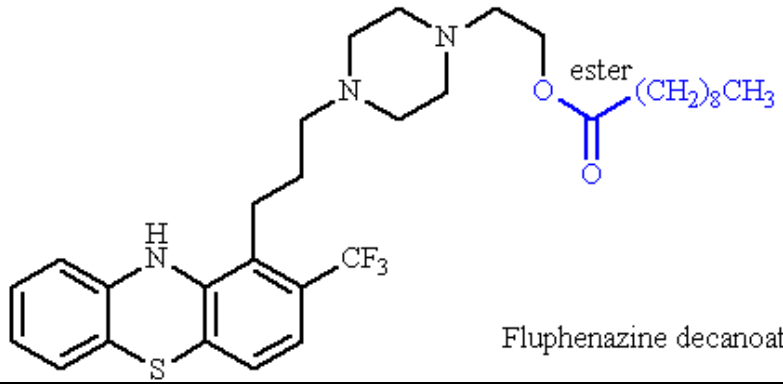
Type: multiple choice		
Title: Chapter 14 - Question 28		
28)	L791456 is an anti-arthritic drug.  L791456 Which of the following structures is not a likely metabolite?  A B C D	
Feedback: A chloro group is resistant to metabolism and is often incorporated into a molecule to block metabolism at a particular part of the molecule.		
Correct	**a.#**	Structure A.
incorrect	**b.#**	Structure B.
incorrect	**c.#**	Structure C.
incorrect	**d.#**	Structure D.
END OF QUESTION		

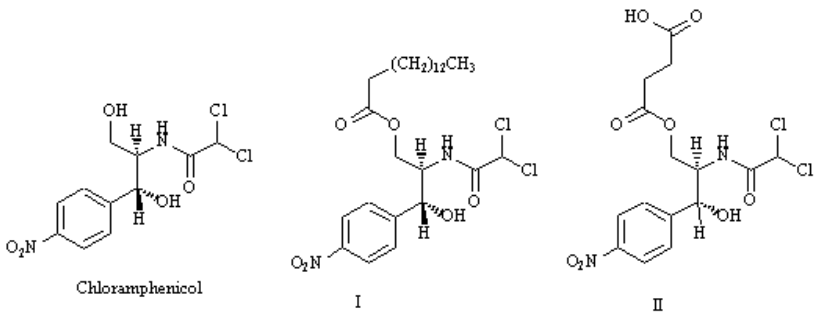
Type: multiple choice

Title: Chapter 14 - Question 29		
29)	In Parkinson's disease, there is a lack of the neurotransmitter dopamine in the brain. Adding dopamine itself is not very effective for a variety of reasons. Which of the following statements is a valid explanation?  Dopamine	
Feedback: Dopamine is metabolised to inactive metabolites, and this is one of the reasons why dopamine itself is not a very effective treatment for Parkinson's disease. The other options are incorrect. Dopamine cannot cross the blood brain barrier, because it is because it is **too** polar rather than insufficiently polar. Dopamine is chemically stable and it shows no selectivity between the dopamine receptors in the brain and the periphery. This is also a reason for it being an unsatisfactory treatment.		
incorrect	**a.#**	Dopamine is not sufficiently polar to cross the blood brain barrier
incorrect	**b.#**	Dopamine is chemically unstable
Correct	**c.#**	Dopamine is metabolised to inactive metabolites
incorrect	**d.#**	Dopamine shows selectivity for peripheral dopamine receptors over dopamine receptors in the brain
END OF QUESTION		

Type: multiple choice		
Title: Chapter 14 - Question 30		
30)	Esters are frequently used as prodrugs. Which of the following statements is false?	
Feedback: The answer is correct to the extent that the preferred leaving group should be a natural compound. However, it is wrong to state that a volatile leaving group is preferred. Formaldehyde is a toxic compound and, although it is released by some prodrugs, a natural alternative would be preferable.		
incorrect	**a.#**	Ester prodrugs are more easily absorbed from the gut than the parent drug.
incorrect	**b.#**	Esters are more susceptible to hydrolysis if the alcohol moiety has an electron withdrawing group.
incorrect	**c.#**	Esters can be used to mask a polar alcohol, phenol, or carboxylic acid group.
Correct	**d.#**	It is preferable if the leaving group from ester hydrolysis is a natural chemical such as an amino acid or a volatile compound such as formaldehyde.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 14 - Question 31		
31)	Fluphenazine decanoate is an ester prodrug for the antipsychotic drug fluphenazine and is administered by intramuscular injection. Which of the following statements best explains its effectiveness?	

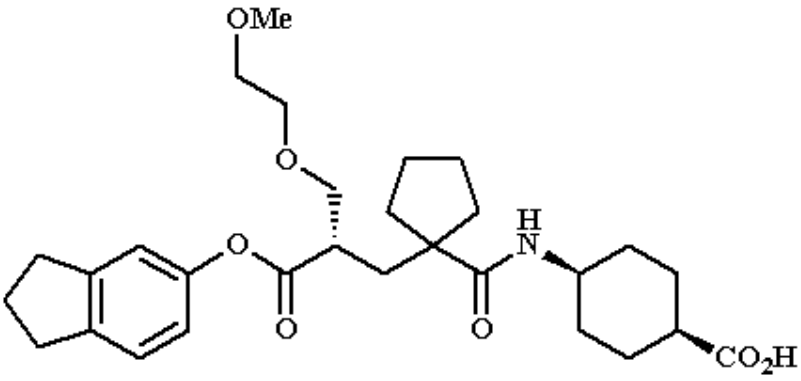
		
Feedback: Fluphenazine decanoate is a prodrug where an important binding group (the alcohol) is masked as an ester. Hydrolysis of the ester reveals the active drug.		
Correct	**a.#**	Hydrolysis of the ester reveals an alcohol group on fluphenazine.
incorrect	**b.#**	The ester is hydrophobic which means that it crosses the blood brain barrier more swiftly.
incorrect	**c.#**	The ester is hydrophilic and rapidly enters the blood supply.
incorrect	**d.#**	The ester is rapidly hydrolysed in blood since the pH of blood is slightly alkaline.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 14 - Question 32	
32)	Structures (I) and (II) are prodrugs of the antibiotic chloramphenicol. Which of the following statements is true? 
Feedback: Structure I has a hydrophobic ester which lowers water solubility. As a result, it reduces the bitter taste of the drug when it is taken by mouth. Structure II has a polar ester due to the carboxylic acid, and so it has good water solubility. It can be used to achieve high concentrations of the drug for injection.	
Correct	**a.#** Structure I is less water soluble than chloramphenicol.
incorrect	**b.#** Structure II is less water soluble than chloramphenicol.
incorrect	**c.#** Structure I is used to achieve higher concentrations for injections.
incorrect	**d.#** Structure II is used to reduce the bitter taste of chloramphenicol.
END OF QUESTION	

Type: multiple choice	
Title: Chapter 14 - Question 33	
33)	Which of the following statements is true with respect to phosphate prodrugs?

Feedback: Phosphate esters are **more** polar and **more** water soluble than the parent drug. As a result, they are less likely to cross cell membranes. They are susceptible to metabolism. Otherwise they would not be prodrugs!		
incorrect	**a.#**	Phosphate esters are less polar in nature than the parent drug.
incorrect	**b.#**	Phosphate esters are less water soluble than the parent drug.
Correct	**c.#**	Phosphate esters are less likely to cross cell membranes than the parent drug.
incorrect	**d.#**	Phosphate esters are resistant to drug metabolism.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 14 - Question 34	
34)	Candoxatril is an ester prodrug for candoxatrilat which inhibits protease enzymes. Which of the following statements is incorrect?



Candoxatril

Feedback: The bicyclic system certainly speeds up the rate of ester hydrolysis, but it is because it is electron **withdrawing** and not electron donating.		
incorrect	**a.#**	Hydrolysis of the ester reveals a carboxylic acid group.
incorrect	**b.#**	The parent drug cannot be administered orally, whereas the ester can.
incorrect	**c.#**	The bicyclic leaving group is non-toxic.
Correct	**d.#**	The bicyclic system is electron donating and speeds up the rate of ester hydrolysis.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 14 - Question 35	
35)	Some peptides and proteins have been used as drugs. Which of the following statements is untrue?

Feedback: One of the problems with peptides and proteins as drugs is the fact that they generally show poor bioavailability. The other statements are true. Proteins and polypeptides are large molecules and can induce an immune response. They are poorly absorbed when taken orally due to their polar nature. The peptide bonds are also prone to hydrolysis by digestive and metabolic enzymes.		
incorrect	**a.#**	Protein drugs suffer a disadvantage in that they could produce an immune response.
Correct	**b.#**	Peptides and proteins generally show good bioavailability.
incorrect	**c.#**	Peptide drugs are susceptible to peptidase enzymes.
incorrect	**d.#**	Peptide drugs are susceptible to metabolic and digestive

		enzymes.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 14 - Question 36		
36)	There are several strategies for stabilising susceptible bonds in peptides. Which of the following is not used in this respect?	
Feedback: An ester bond is just as likely to be susceptible to hydrolysis as a peptide bond due to the presence of esterases.		
Correct	**a.#**	Replacing a peptide link with an ester.
incorrect	**b.#**	Adding an *N*-methyl group to the peptide link.
incorrect	**c.#**	Using an unnatural amino acid.
incorrect	**d.#**	Replacing an L-amino acid with a D-amino acid.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 14 - Question 37		
37)	Which of the following statements is not true?	
Feedback: Monoclonal antibodies are identical, including the variable region.		
incorrect	**a.#**	Drugs with names ending in 'mab' are antibodies.
Correct	**b.#**	Each antibody in a preparation of monoclonal antibodies is identical, except for the variable regions at the *N*-termini of the heavy and light chains.
incorrect	**c.#**	Chimeric antibodies are part human, part mouse in origin.
incorrect	**d.#**	The HAMA response is an immune response triggered in the body by non-human like antibodies used in medicine.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 14 - Question 38		
38)	Peptides and proteins have several disadvantages as drugs. Which of the following is not a disadvantage?	
Feedback: Peptides and proteins are relatively simple to synthesise using automated synthetic techniques or by genetic engineering techniques.		
incorrect	**a.#**	Rapid metabolism.
Correct	**b.#**	Difficult synthesis.
incorrect	**c.#**	Poor absorption from the gut.
incorrect	**d.#**	The possibility of an immune response.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 14 - Question 39	
39)	Which of the following statements is not true?
Feedback: The HAMA response is an immune response triggered by non-human like antibodies that are recognised by the body as being foreign. The body's immune system is activated to destroy them and this can prove harmful rather than beneficial.	

incorrect	**a.#**	Drugs with names ending in 'mab' are antibodies.
incorrect	**b.#**	Each antibody in a preparation of monoclonal antibodies is identical in structure.
incorrect	**c.#**	Chimeric antibodies are part human, part mouse in origin.
Correct	**d.#**	The HAMA response is the mechanism by which antibodies have a beneficial effect in medicine.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 15 - Question 01		
01)	What is a major disadvantage of the therapeutic ratio?	
Feedback: The therapeutic ratio is an indication of the relative dose levels of a drug which will be effective, versus those which will be lethal. It gives no indication of chronic or non-lethal toxicity.		
incorrect	**a.#**	It gives no indication of the efficacy of the drug.
incorrect	**b.#**	It gives no indication of the lethality of the drug.
incorrect	**c.#**	It gives no indication of the dose levels that are more likely to produce the desired effect rather than death.
Correct	**d.#**	It gives no indication of the risk of non-fatal and chronic toxicity.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 15 - Question 02		
02)	Which of the following detection methods is not commonly used to detect isotopically labelled drug metabolites?	
Feedback: Nuclear magnetic resonance can be used to detect metabolites labelled with stable isotopes such as ^{13}C. Scintillation counting is used to detect metabolites labelled with radioactive isotopes such as ^{14}C or ^{3}H. Mass spectrometry can be used to detect metabolites labelled with isotopes such as deuterium.		
Correct	**a.#**	Infra red spectroscopy.
incorrect	**b.#**	Nuclear magnetic resonance spectroscopy.
incorrect	**c.#**	Scintillation counting (detection of radioactivity).
incorrect	**d.#**	Mass spectrometry.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 15 - Question 03		
03)	What term is used in formulation studies to describe the ratio of active drug to the total contents of a dose?	
Feedback: The drug load is the ratio of active drug to the total contents of the dose. A low drug load may cause homogeneity problems. A high drug load may pose flow problems or require large capsules if the compound has a low bulk density		
incorrect	**a.#**	Dose level.
incorrect	**b.#**	Drug level.
Correct	**c.#**	Drug load.
incorrect	**d.#**	Dose load.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 15 - Question 04		
04)	What term is used to signify a preparation that appears identical to the preparation of an active drug, but which has no biological activity?	
Feedback: Placebos are important as a comparison to the active drug in clinical trials. The terms dummy drug and gazebo are not used in medicinal chemistry. The term peptidomimetic refers to a drug that has been derived from a peptide lead compound, but which lacks peptide character.		
incorrect	**a.#**	Dummy drug.

incorrect	**b.#**	Peptidomimetic.
Correct	**c.#**	Placebo.
incorrect	**d.#**	Gazebo.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 15 - Question 05		
05)	What is meant by a double blind clinical trial?	
Feedback: In a double blind clinical trial, it is important that the doctors treating the two groups of patients should not know which group is receiving the drug, and which group is receiving the placebo since they may unwittingly 'give the game away'.		
incorrect	**a.#**	Two patients in a control study are given a placebo rather than the active drug.
incorrect	**b.#**	One batch of patients receives the drug while two batches of patients receive the placebo.
incorrect	**c.#**	One batch of patients receives the drug and one batch receives the placebo. The doctors determine which patients should have priority in receiving the experimental drug.
Correct	**d.#**	One batch of patients receives the drug and one batch receives the placebo. Neither the patient nor the doctor knows whether the drug or the placebo is being administered.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 15 - Question 06		
06)	Which of the following can be patented?	
Feedback: A patent should cover the medicinal use of a particular compound and its possible analogues. Therefore, it makes sense for a pharmaceutical company to patent a series of compounds rather than a specific drug since it makes it more difficult for competitors to discover and patent a similar compound. Similarly, patenting a synthetic route is advantageous since any competitor who finds an alternative drug will have to use a different synthetic route which might be more expensive and make the product less profitable. Consequently, it is best to patent the series of compounds and the synthetic route.		
incorrect	**a.#**	A series of compounds.
incorrect	**b.#**	The medicinal use of a series of compounds.
incorrect	**c.#**	The synthesis of a series of compounds.
Correct	**d.#**	All of the answers are correct.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 15 - Question 07		
07)	What is meant by GLP?	
Feedback: The other terms are not used in medicinal chemistry.		
Correct	**a.#**	Good laboratory practice.
incorrect	**b.#**	Good licensing practice.
incorrect	**c.#**	General licensing procedures.
incorrect	**d.#**	General laboratory principles.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 15 - Question 08		
08)	What is meant by GMP?	
Feedback: The other terms are not used in medicinal chemistry.		
incorrect	**a.#**	Good medicinal practice.
Correct	**b.#**	Good manufacturing practice.
incorrect	**c.#**	General manufacturing procedures.
incorrect	**d.#**	General medicinal principles.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 15 - Question 09		
09)	What is meant by GCP?	
Feedback: The other terms are not used in medicinal chemistry.		
incorrect	**a.#**	Good chemical practice.
incorrect	**b.#**	General chemical practice.
incorrect	**c.#**	General chemical procedures.
Correct	**d.#**	Good clinical practice.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 15 - Question 10		
10)	What is meant by chemical development?	
Feedback: The development of an effective drug is drug development. The development of a production process is called process development. The development of a large scale synthesis is chemical development. The development of new synthetic methods is part of organic chemistry research.		
incorrect	**a.#**	The development of an effective drug.
incorrect	**b.#**	The development of a production process for a drug.
Correct	**c.#**	The development of a large scale synthesis for a drug.
incorrect	**d.#**	The development of new synthetic methods.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 15 - Question 11		
11)	Which of the following is not a priority in chemical development?	
Feedback: In chemical development, the yield, cost, and purity of the final product is crucial. Since the work is being carried out on a single product, it is not possible to optimise the activity. That is fixed for the product concerned.		
incorrect	**a.#**	Optimising the overall yield of a drug.
Correct	**b.#**	Optimising the activity of a drug.
incorrect	**c.#**	Developing a cheap synthetic route.
incorrect	**d.#**	Optimising the purity of a drug.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 15 - Question 12		
12)	What term is used to describe the process by which a synthetic procedure is developed such that it is suitable for a production plant?	
Feedback: The other terms have no significant meaning in this context.		
incorrect	**a.#**	Plant development.
incorrect	**b.#**	Product development.
Correct	**c.#**	Process development.
incorrect	**d.#**	Production development.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 15 - Question 13		
13)	Which of the following is not a priority in process development?	
Feedback: The number of side effects for a particular drug is fixed and unchangeable, so it cannot be altered by how the drug is produced. The other options are features of the process by which the drug is produced and these features can be varied.		
incorrect	**a.#**	Minimising cost.
Correct	**b.#**	Minimising side effects.
incorrect	**c.#**	Minimising the number of reactions.
incorrect	**d.#**	Minimising the number of operations.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 15 - Question 14		
14)	What is meant by the optimum temperature of a reaction?	
Feedback: The temperature at which the maximum yield is obtained is not necessarily the one at which the optimum purity or minimum level of impurities is obtained. The optimum temperature is the one at which the best yield can be obtained with acceptable purity standards.		
incorrect	**a.#**	The temperature that results in the maximum possible yield.
incorrect	**b.#**	The temperature that results in the minimum amount of impurities.
Correct	**c.#**	The temperature that results in an acceptably high level of yield and an acceptably low level of impurities.
incorrect	**d.#**	None of the answers are correct.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 15 - Question 15		
15)	What are the consequences of increasing the temperature for a particular reaction?	
Feedback: It is not possible to come up with a rule that holds for all reactions. For many reactions, increasing the temperature will increase the yield but increase the side products. For others, the yield will decrease as the level of side products increase. It is necessary to monitor both the yield and level of impurities for a particular reaction as the temperature is varied.		
incorrect	**a.#**	Increased yield and increased side products.

incorrect	**b.#**	Increased yield and decreased side products.
incorrect	**c.#**	Decreased yield and increased side products.
Correct	**d.#**	Any of the consequences described.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 15 - Question 16		
16)	Which of the following procedures is not suitable for scale up?	
Feedback: Rotary evaporation is possible in a research laboratory but it is not feasible on a large scale. Distillation would have to be carried out instead.		
incorrect	**a.#**	Liquid-liquid extraction.
incorrect	**b.#**	Crystallisation.
incorrect	**c.#**	Distillation.
Correct	**d.#**	Rotary evaporation.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 15 - Question 17		
17)	What is meant by a drug's 'specifications'?	
Feedback: A drug's specifications are the purity tests that have to be carried out and the purity standards that have to be met. These specifications have to be met by each batch of the drug that is produced in the production plant.		
incorrect	**a.#**	The molecular dimensions of a molecule.
incorrect	**b.#**	The physical properties of a drug.
Correct	**c.#**	The purity tests and purity standards required of a drug.
incorrect	**d.#**	The functional groups on a drug that is important to its activity.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 15 - Question 18		
18)	Which of the following areas of study is part of preclinical trials?	
Feedback: Preclinical trials refer to the various tests and processes which are carried out on a compound prior to clinical trials. The compound concerned is a potential clinical agent and is the result of the drug discovery and development process of which the other options are part.		
incorrect	**a.#**	Drug design.
incorrect	**b.#**	Lead discovery.
Correct	**c.#**	Formulation.
incorrect	**d.#**	Structure-activity relationships.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 15 - Question 19		
19)	Which of the following definitions best describes the ED_{50} of a drug?	
Feedback: The ED_{50} is the dose of drug required to produce an effect in 50% of the animals tested. The LD_{50} value represents the dose of drug which kills 50 % of the		

animals tested. The therapeutic index is the ratio of LD_{50} to ED_{50}.		
incorrect	**a.#**	The dose of a drug required to produce 50% of a maximum effect.
Correct	**b.#**	The dose of a drug required to produce a measurable effect in 50% of the animals tested.
incorrect	**c.#**	The ratio of LD_{50} to ED_{50}.
incorrect	**d.#**	The dose of a drug required to kill 50% of a group of animals.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 15 - Question 20	
20)	Which of the following definitions best describes the LD_{50} of a drug?
Feedback: The ED_{50} is the dose of drug required to produce an effect in 50% of the animals tested. The LD_{50} value represents the dose of drug which kills 50 % of the animals tested. The therapeutic index is the ratio of LD_{50} to ED_{50}	

incorrect	**a.#**	The dose of a drug required to produce 50% of a maximum effect.
incorrect	**b.#**	The dose of a drug required to produce a measurable effect in 50% of the animals tested.
incorrect	**c.#**	The ratio of LD_{50} to ED_{50}.
Correct	**d.#**	The dose of a drug required to kill 50% of a group of animals.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 15 - Question 21	
21)	Which of the following definitions best describes the therapeutic index of a drug?
Feedback: The ED_{50} is the dose of drug required to produce an effect in 50% of the animals tested. The LD_{50} value represents the dose of drug which kills 50 % of the animals tested. The therapeutic index is the ratio of LD_{50} to ED_{50}.	

incorrect	**a.#**	The dose of a drug required to produce 50% of a maximum effect.
incorrect	**b.#**	The dose of a drug required to produce a measurable effect in 50% of the animals tested.
Correct	**c.#**	The ratio of LD_{50} to ED_{50}.
incorrect	**d.#**	The dose of a drug required to kill 50% of a group of animals.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 15 - Question 22	
22)	Which of the following detection processes could be carried out using infra red spectroscopy?
Feedback: Nuclear magnetic resonance can be used to detect metabolites labelled with stable isotopes such as ^{13}C. Scintillation counting is used to detect metabolites labelled with radioactive isotopes such as ^{14}C or ^{3}H. Mass spectrometry can be used to detect metabolites labelled with isotopes such as deuterium. Infra red spectroscopy is useful in identifying specific functional groups within molecules.	

incorrect	**a.#**	Detection of a metabolite labelled with ^{13}C.
Correct	**b.#**	Detection of a metabolite with a ketone functional group.
incorrect	**c.#**	Detection of a metabolite labelled with ^{2}H.
incorrect	**d.#**	Detection of a metabolite labelled with ^{14}C.

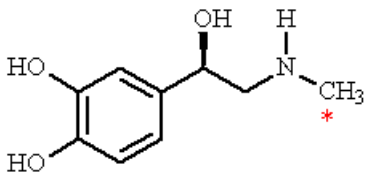
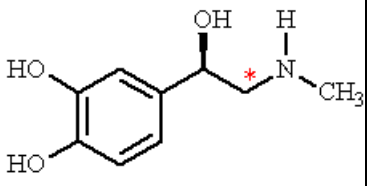
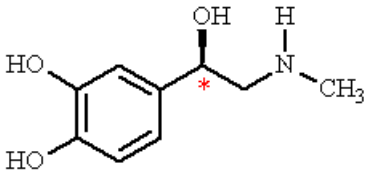
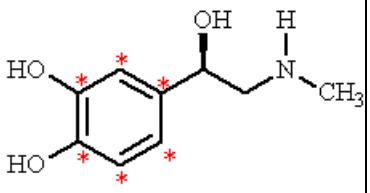
END OF QUESTION

Type: multiple choice		
Title: Chapter 15 - Question 23		
23)	Which of the following detection processes could best be carried out using nmr spectroscopy?	
Feedback: Nuclear magnetic resonance can be used to detect metabolites labelled with stable isotopes such as ^{13}C. Scintillation counting is used to detect metabolites labelled with radioactive isotopes such as ^{14}C or ^{3}H. Mass spectrometry can be used to detect metabolites labelled with isotopes such as deuterium. Infra red spectroscopy is useful in identifying specific functional groups within molecules.		
Correct	**a.#**	Detection of a metabolite labelled with ^{13}C.
incorrect	**b.#**	Detection of a metabolite with a ketone functional group.
incorrect	**c.#**	Detection of a metabolite labelled with ^{2}H.
incorrect	**d.#**	Detection of a metabolite labelled with ^{14}C.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 15 - Question 24		
24)	Which of the following detection processes could best be carried out using scintillation counting?	
Feedback: Nuclear magnetic resonance can be used to detect metabolites labelled with stable isotopes such as ^{13}C. Scintillation counting is used to detect metabolites labelled with radioactive isotopes such as ^{14}C or ^{3}H. Mass spectrometry can be used to detect metabolites labelled with isotopes such as deuterium. Infra red spectroscopy is useful in identifying specific functional groups within molecules.		
incorrect	**a.#**	Detection of a metabolite labelled with ^{13}C.
incorrect	**b.#**	Detection of a metabolite with a ketone functional group.
incorrect	**c.#**	Detection of a metabolite labelled with ^{2}H.
Correct	**d.#**	Detection of a metabolite labelled with ^{14}C.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 15 - Question 25		
25)	Which of the following detection processes could best be carried out using mass spectrometry?	
Feedback: Nuclear magnetic resonance can be used to detect metabolites labelled with stable isotopes such as ^{13}C. Scintillation counting is used to detect metabolites labelled with radioactive isotopes such as ^{14}C or ^{3}H. Mass spectrometry can be used to detect metabolites labelled with isotopes such as deuterium. Infra red spectroscopy is useful in identifying specific functional groups within molecules.		
incorrect	**a.#**	Detection of a metabolite labelled with ^{13}C.
incorrect	**b.#**	Detection of a metabolite with a ketone functional group.
Correct	**c.#**	Detection of a metabolite labelled with ^{2}H.
incorrect	**d.#**	Detection of a metabolite labelled with ^{14}C.
END OF QUESTION		

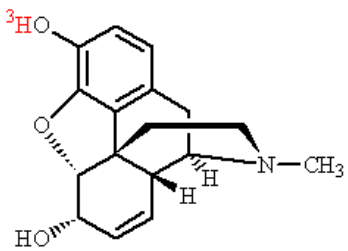
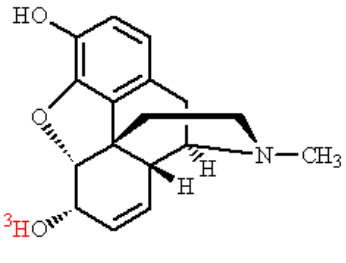
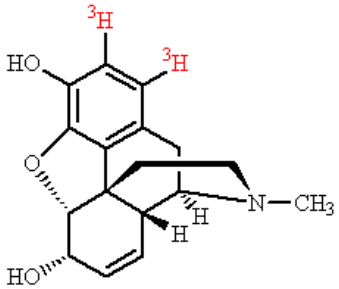
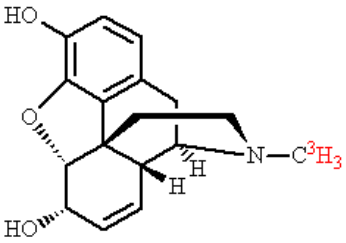
Type: multiple choice
Title: Chapter 15 - Question 26

26)	Which of the following molecules of adrenaline, labelled with the radioisotope ^{14}C would be unsuitable for metabolic studies? (The position of the radioisotope is indicated by the red star.)  A  B  C  D

Feedback: Structure A would be unsuitable since the label is on an *N*-methyl group. An *N*-methyl group is easily lost as a result of a metabolic reaction and so the label would be lost from the molecule.

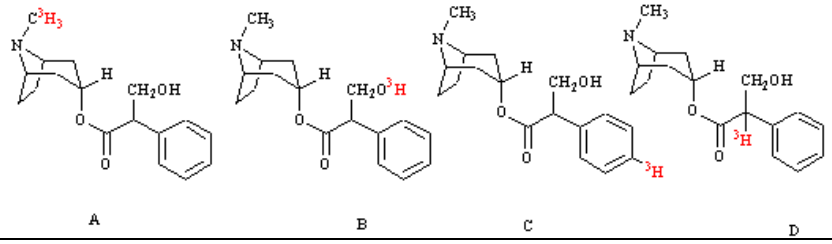
Correct	**a.#**	Structure A.
incorrect	**b.#**	Structure B.
incorrect	**c.#**	Structure C.
Incorrect	**d.#**	Structure D.

END OF QUESTION

Type: multiple choice	
Title: Chapter 15 - Question 27	
27)	Which of the following labelled structures of morphine would you choose to carry out a study on drug metabolism?  A  B  C  D

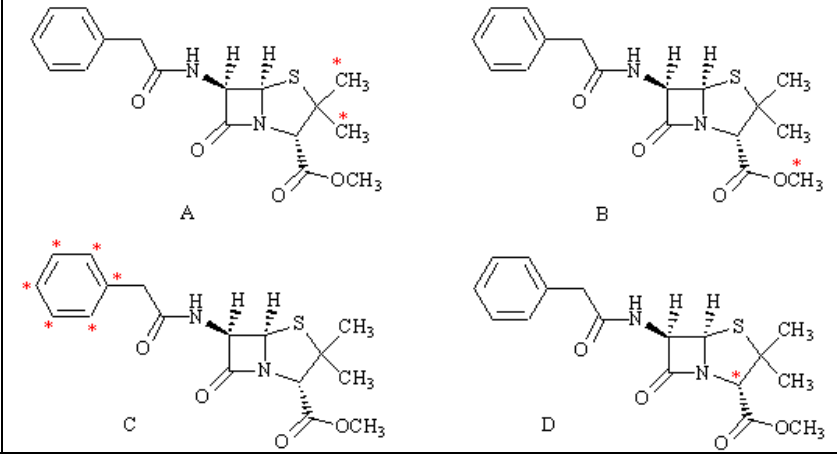
Feedback: Structure C is the structure least likely to lose its tritium label. Structures A and B would be unsuitable since the label is on a phenol or an alcohol. Both of these functional groups easily exchange their proton with water and so a tritium label would be quickly exchanged as well. Structure D would be unsuitable since the label is on an *N*-methyl group. An *N*-methyl group is easily lost as a result of a metabolic reaction and so the label would be lost from the molecule.		
incorrect	**a.#**	Structure A.
incorrect	**b.#**	Structure B.
Correct	**c.#**	Structure C.
incorrect	**d.#**	Structure D.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 15 - Question 28	
28)	Which of the following labelled molecules of atropine would you choose to carry out a metabolic study?



Feedback: Structure C is the structure least likely to lose its tritium label. Structure B would be unsuitable since the label is on an alcohol group. This functional group easily exchanges its proton with water and so a tritium label would be quickly exchanged as well. Structure A would be unsuitable since the label is on an *N*-methyl group. An *N*-methyl group is easily lost as a result of a metabolic reaction and so the label would be lost from the molecule. Structure D is unsuitable since the label is on a carbon next to a carbonyl group. The electron withdrawing effect of the carbonyl group makes the label acidic and it can be easily lost from the molecule.		
incorrect	**a.#**	Structure A.
incorrect	**b.#**	Structure B.
Correct	**c.#**	Structure C.
incorrect	**d.#**	Structure D.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 15 - Question 29	
29)	The following structures are of penicillin G methyl ester, labelled with ^{14}C (indicated by the red star). Which of these would you not use for a metabolic study?

Feedback: In structure B, the labelled carbon is on the alkoxy moiety of an ester. Esters are susceptible to enzyme catalysed hydrolysis which would result in the loss of the isotope from the drug.

incorrect	**a.#**	Structure A.
Correct	**b.#**	Structure B.
incorrect	**c.#**	Structure C.
incorrect	**d.#**	Structure D.

END OF QUESTION

Type: multiple choice	
Title: Chapter 15 - Question 30	
30)	Which of the following is not part of formulation studies?

Feedback: The effect of temperature on the synthesis of a drug is part of chemical development and is not part of formulation studies. Formulation studies relate to how a drug is prepared for administration, as well as the stability and lifetime of the drug in that preparation.

incorrect	**a.#**	The effect of temperature on the stability of a drug.
incorrect	**b.#**	The shelf life of a drug.
Correct	**c.#**	The effect of temperature on the synthesis of a drug.
incorrect	**d.#**	The suitability of containers for storing a drug.

END OF QUESTION

Type: multiple choice	
Title: Chapter 15 - Question 31	
31)	What is meant by chiral switching?

Feedback: Chiral switching is carried out on racemic drugs which were patented before it was fully appreciated that the different enantiomers could have differing activities. It would not be possible to do a chiral switch on a racemic drug that is introduced to the market now.

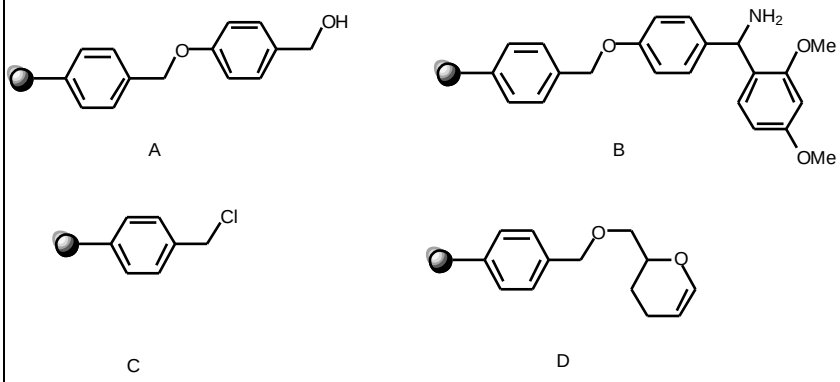
incorrect	**a.#**	The ability of a drug target to bind both possible enantiomers of a chiral drug in two different binding modes.
incorrect	**b.#**	The epimerisation of a chiral centre.
incorrect	**c.#**	The patenting of the opposite enantiomer of a chiral drug that is already on the market.
Correct	**d.#**	The patenting of the active enantiomer of a racemic drug which is already on the market.

END OF QUESTION

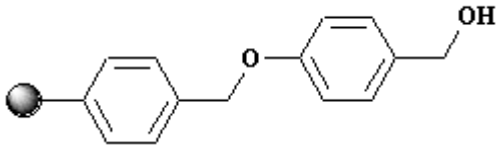
Type: multiple choice		
Title: Chapter 15 - Question 32		
32)	What is meant by an orphan drug?	
Feedback: A novel pharmaceutical agent is known as a New Chemical Entity. A compound that has a useful biological activity and is the starting point for the design of a new drug is known as a lead compound. An established drug that acts as the reference point for related drugs is known as a parent drug.		
incorrect	**a.#**	A novel pharmaceutical agent.
Correct	**b.#**	A drug which is effective against a relatively rare medical problem.
incorrect	**c.#**	A compound that has a useful biological activity and is the starting point for the design of a new drug.
incorrect	**d.#**	An established drug that acts as the reference point for related drugs.
END OF QUESTION		

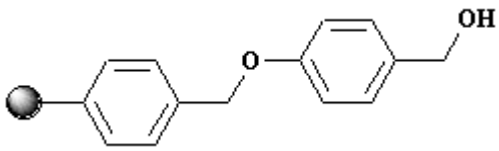
Type: multiple choice		
Title: Chapter 16 - Question 01		
01)	Solid phase synthesis is frequently used in combinatorial chemistry. What is meant by solid phase synthesis?	
Feedback: In solid phase synthesis, molecules are constructed on a solid phase support. Reagents are added in solution.		
incorrect	**a.#**	Reactions are carried out without solvent.
incorrect	**b.#**	Reagents and reactants are attached to a solid phase support.
incorrect	**c.#**	Reagents are used in the solid phase.
Correct	**d.#**	Molecules are constructed on a solid phase support.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 16 - Question 02		
02)	There are several advantages of solid phase synthesis over conventional synthesis. Which of the following statements is **not** an advantage?	
Feedback: This is not an advantage since it is necessary to remove the product from the solid support at the end of the synthesis. The structures should, however, remain attached to the solid support under the various reaction conditions used in the synthesis.		
incorrect	**a.#**	Excess reagents can be used to force the equilibrium of a reaction to products.
incorrect	**b.#**	Impurities and by products are easily removed.
incorrect	**c.#**	Intermediates do not need to be isolated and purified.
Correct	**d.#**	Structures can be strongly linked to the solid support such that they cannot be removed.
END OF QUESTION		

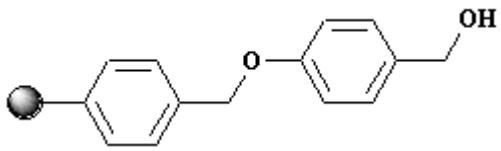
Type: matching	
Title: Chapter 16 - Question 03	
03)	Identify the following resins from the linkers shown: 
Feedback: The Wang resin has an alcohol group which can be used to attach a molecule. The Rink resin has a primary amine for the same purpose. The Merrifield has an alkyl chloride, and the dihydropyran resin has a dihydropyran ring. The molecule that is to be linked to the resin must have a functional group that can react with the functional group on the linker	

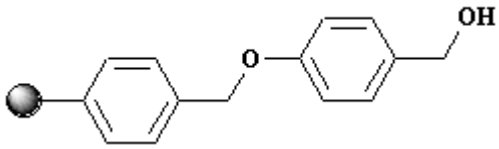
a.#	A	=	Wang resin
b.#	B	=	Rink resin
c.#	C	=	Merrifield resin
d.#	D	=	Dihydropyran-linked resin
END OF QUESTION			

Type: multiple choice		
Title: Chapter 16 - Question 04		
04)	The following is an important resin used in combinatorial chemistry.  Where does a starting material get bound to this resin?	
Feedback: This is a Wang resin. The primary alcohol is the point of attachment. This is not a phenol group since it is not directly attached to the aromatic ring. The ether and the aromatic ring are unreactive under the conditions used.		
incorrect	**a.#**	The phenol group.
incorrect	**b.#**	The aromatic ring.
Correct	**c.#**	The alcohol.
incorrect	**d.#**	The ether.
END OF QUESTION		

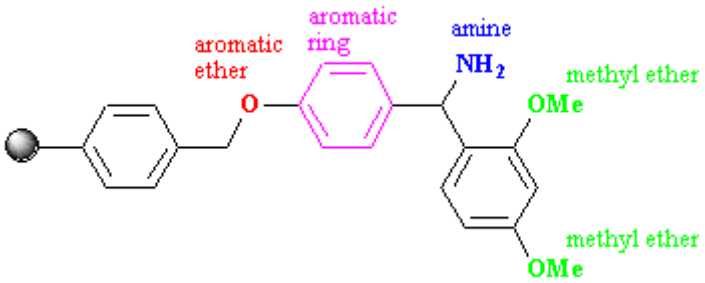
Type: multiple choice		
Title: Chapter 16 - Question 05		
05)	The following is an important resin used in combinatorial chemistry.  What functional group needs to be present on a molecule if it is to be attached to this resin?	
Feedback: This is a Wang resin. The alcohol on the anchor reacts with a carboxylic acid on the first reactant such that an ester bond links the reactant to the anchor.		
incorrect	**a.#**	Phenol.
incorrect	**b.#**	Alcohol.
incorrect	**c.#**	Alkyl halide.
Correct	**d.#**	Carboxylic acid.
END OF QUESTION		

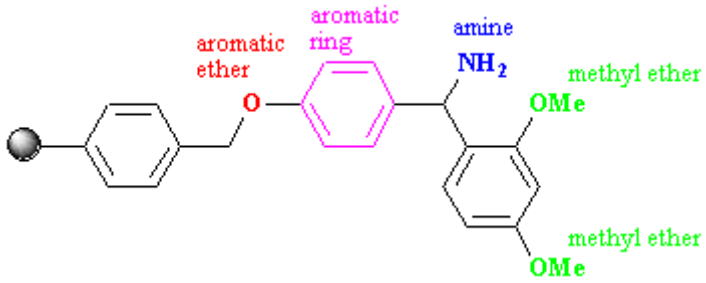
Type: multiple choice

Title: Chapter 16 - Question 06		
06)	The following is an important resin used in combinatorial chemistry.  What conditions are used to cleave a molecule from this resin?	
Feedback: This is a Wang resin. Trifluoroacetic acid is used to cleave the ester bond linking a molecule to the Wang resin, the amide bond linking a molecule to a Rink resin, and the ketal group linking a molecule to a dihydropyran-derivatised resin. It is also used to remove several protecting groups. Hydrofluoric acid is used to cleave the ester bond linking peptides to the Merrifield resin. Piperidine is used to deprotect the Fmoc protecting group from primary amines. Hydrogen and a palladium charcoal catalyst is used to remove benzyl or benzyloxycarbonyl protecting groups		
incorrect	**a.#**	Hydrofluoric acid.
incorrect	**b.#**	Piperidine.
incorrect	**c.#**	Hydrogen and a palladium charcoal catalyst.
Correct	**d.#**	Trifluoroacetic acid.
END OF QUESTION		

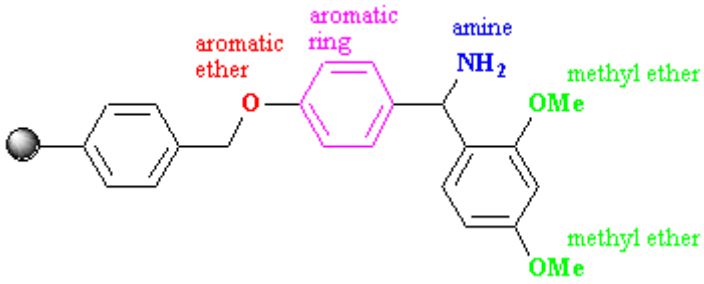
Type: multiple choice		
Title: Chapter 16 - Question 07		
07)	The following is an important resin used in combinatorial chemistry.  What functional group is present on a molecule once it is released from this resin?	
Feedback: This is a Wang resin. The same functional group that was present initially on the molecule attached is regenerated.		
incorrect	**a.#**	Carboxamide.
Correct	**b.#**	Carboxylic acid.
incorrect	**c.#**	Ester.
incorrect	**d.#**	Alcohol.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 16 - Question 08	
08)	The following is an important resin used in combinatorial chemistry.

	 Where does a starting material get bound to this resin?	
Feedback: This is a Rink resin. The aromatic rings and the ethers are unreactive under the reaction conditions used.		
Correct	**a.#**	The amine group.
incorrect	**b.#**	The aromatic ring.
incorrect	**c.#**	The aromatic ether.
incorrect	**d.#**	The methyl ether.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 16 - Question 09		
09)	The following is an important resin used in combinatorial chemistry.  What functional group needs to be present on a molecule if it is to be attached to this resin?	
Feedback: This is a Rink resin. Reaction of a carboxylic acid with the primary amine present on the anchor results in an amide bond linking the first reactant to the anchor.		
incorrect	**a.#**	Phenol.
incorrect	**b.#**	Alcohol.
incorrect	**c.#**	Alkyl halide.
Correct	**d.#**	Carboxylic acid.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 16 - Question 10	
10)	The following is an important resin used in combinatorial chemistry.

	 What conditions are used to cleave a molecule from this resin?

Feedback: This is a Rink resin. Trifluoroacetic acid is used to cleave attached molecules from it.
Hydrofluoric acid is used to cleave the ester bond linking peptides to the Merrifield resin. Piperidine is used to deprotect the Fmoc protecting group from primary amines. Hydrogen and a palladium charcoal catalyst is used to remove benzyl or benzyloxycarbonyl protecting groups. Trifluoroacetic acid is used to cleave the ester bond linking a molecule to the Wang resin, the amide bond linking a molecule to a Rink resin, and the ketal group linking a molecule to a dihydropyran-derivatised resin. It is also used to remove several protecting groups.

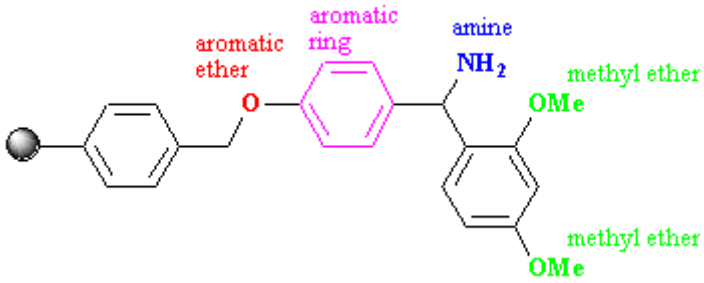
incorrect	**a.#**	Hydrofluoric acid.
incorrect	**b.#**	Piperidine.
incorrect	**c.#**	Hydrogen and a palladium charcoal catalyst.
Correct	**d.#**	Trifluoroacetic acid.

END OF QUESTION

Type: multiple choice

Title: Chapter 16 - Question 11

11) The following is an important resin used in combinatorial chemistry.



What functional group is present on a molecule once it is released from this resin?

Feedback: The amino group that was originally part of the linker is transferred to the final product once it is cleaved from the anchor.

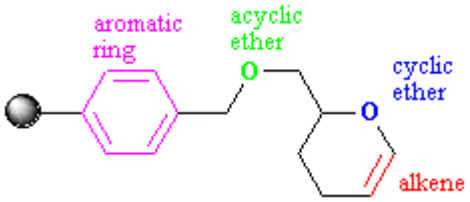
Correct	**a.#**	Carboxamide.
incorrect	**b.#**	Carboxylic acid.
incorrect	**c.#**	Ester.
incorrect	**d.#**	Alcohol.

END OF QUESTION

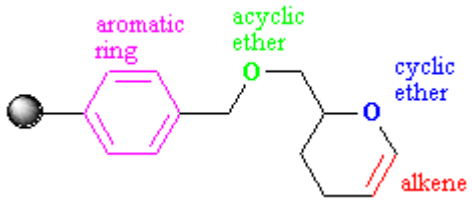
Type: multiple choice

Title: Chapter 16 - Question 12

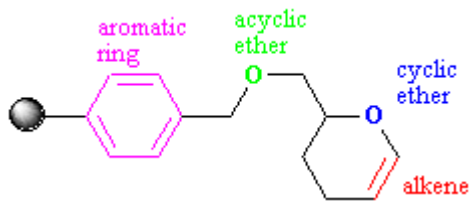
12) The following is an important resin used in combinatorial

	chemistry.  Where does a starting material get bound to this resin?

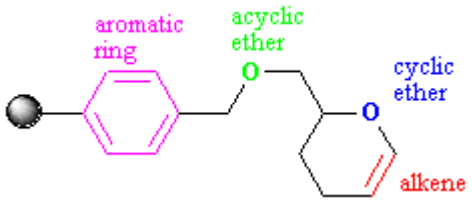
Feedback: This is a dihydropyran-linked resin.		
Correct	**a.#**	The alkene group.
incorrect	**b.#**	The aromatic ring.
incorrect	**c.#**	The acyclic ether.
incorrect	**d.#**	The cyclic ether.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 16 - Question 13	
13)	The following is an important resin used in combinatorial chemistry.  What functional group needs to be present on a molecule if it is to be attached to this resin?

Feedback: This is a dihydropyran-linked resin. The molecule to be attached should have an alcohol group.		
incorrect	**a.#**	Ether.
Correct	**b.#**	Alcohol.
incorrect	**c.#**	Alkyl halide.
incorrect	**d.#**	Nitrile.
END OF QUESTION		

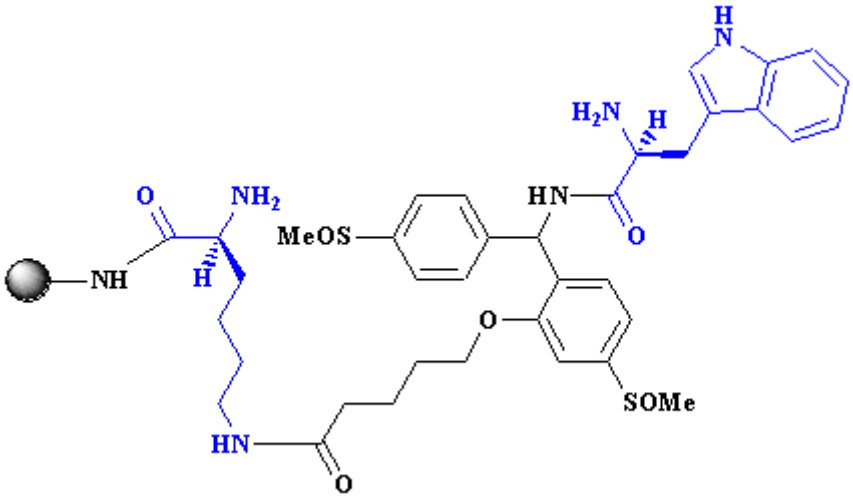
Type: multiple choice	
Title: Chapter 16 - Question 14	
14)	The following is an important resin used in combinatorial chemistry.  What conditions are used to cleave a molecule from this resin?

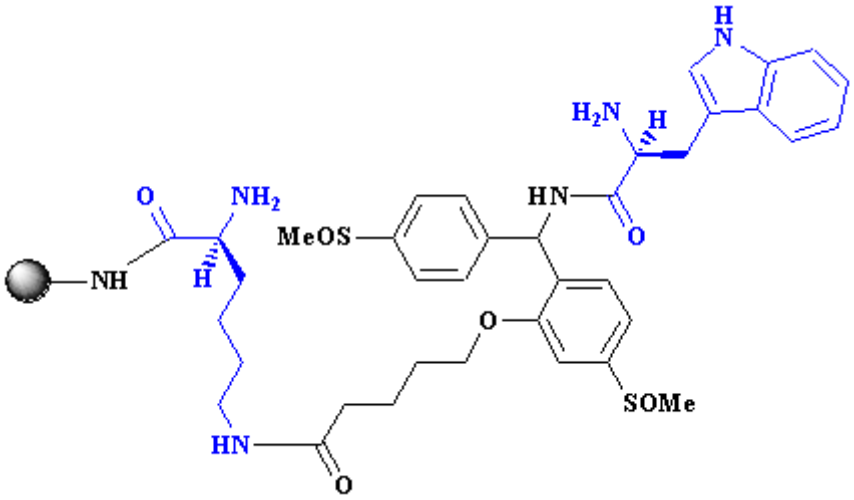
Feedback: This is a dihydropyran-linked resin. Hydrofluoric acid is used to cleave the ester bond linking peptides to the Merrifield resin. Piperidine is used to deprotect the Fmoc protecting group from primary amines. Hydrogen and a palladium charcoal catalyst is used to remove benzyl or benzyloxycarbonyl protecting groups. Trifluoroacetic acid is used to cleave the ester bond linking a molecule to the Wang resin, the amide bond linking a molecule to a Rink resin, and the ketal group linking a molecule to a dihydropyran-derivatised resin. It is also used to remove several protecting groups.		
incorrect	**a.#**	Hydrofluoric acid.
incorrect	**b.#**	Piperidine.
incorrect	**c.#**	Hydrogen and a palladium charcoal catalyst.
Correct	**d.#**	Trifluoroacetic acid.
END OF QUESTION		

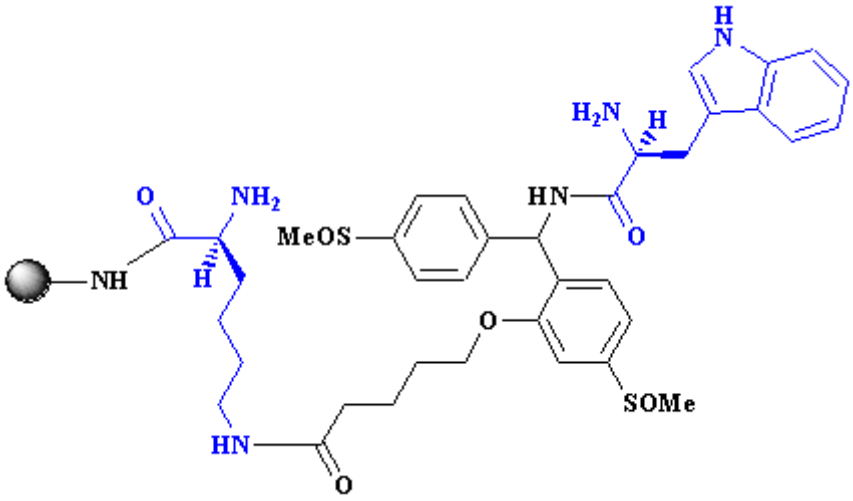
Type: multiple choice		
Title: Chapter 16 - Question 15		
15)	The following is an important resin used in combinatorial chemistry.  What functional group is present on a molecule once it is released from this resin?	
Feedback: This is a dihydropyran-linked resin.The functional group that was originally present on the first reactant is restored.		
incorrect	**a.#**	Ether.
Correct	**b.#**	Alcohol.
incorrect	**c.#**	Alkyl halide.
incorrect	**d.#**	Nitrile.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 16 - Question 16		
16)	A pharmaceutical laboratory wishes to synthesise all the possible dipeptides containing the amino acids (Thr, Ile, Phe, Tyr, Ser). How would you carry out these syntheses using combinatorial techniques?	
incorrect	**a.#**	Mix all five amino acids together and add the mixture to a resin such that the amino acids are attached to the resin, then add another mixture of the five amino acids such that dipeptides are formed on the resin.
Feedback: Incorrect. All the dipeptides are attached to resin beads in the one vessel and there is no easy way of separating them, other than by separating each bead individually.		
Correct	**b.#**	Split the resin between five vessels and add a different amino acid to each vessel such that the amino acid is attached to the resin. Mix the resin beads from all the vessels and resplit the beads back into five vessels, then add a different amino acid

		to each vessel to form dipeptides on each resin.
Feedback: Correct. The procedure will generate five vessels of dipeptides. Each vessel will contain a mixture of dipeptides but no one dipeptide will be in more than one vessel. The identity of the second amino acid will be known for all the dipeptides in any one vessel.		
incorrect	**c.#**	Split the resin between five vessels and add a different amino acid to each vessel such that it is attached to the resin , then add a different amino acid to each vessel to form dipeptides on each resin.
Feedback: Incorrect. The procedure will only produce five dipeptides. Each of the five vessels will contain a single dipeptide.		
incorrect	**d.#**	Split the resin between five vessels and add a different amino acid to each vessel such that it is attached to the resin , then add a mixture of the five amino acids to each vessel to form dipeptides on each resin.
Feedback: Incorrect. The procedure could potentially produce all 25 dipeptides in five different vessels. The first amino acid of the dipeptides in any specific vessel will be identical. However, this procedure suffers from the disadvantage that the different amino acids added at the second stage are competing against each other and some may be more reactive than others. As a result, the five dipeptides expected in a single vessel may not be formed if one of the amino acids is much less reactive than the other four.		
END OF QUESTION		

Type: multiple choice		
Title: Chapter 16 - Question 17		
17)	The following linker is useful in a process called 'tagging'  What are the moieties coloured in blue?	
Feedback: These are amino acid structures. The structures of amino acids are given in Appendix 1. The linker is known as a safety-catch acid-labile linker.		
incorrect	**a.#**	Arginine and tryptophan.
Correct	**b.#**	Lysine and tryptophan.
incorrect	**c.#**	Arginine and lysine.
incorrect	**d.#**	Lysine and phenylalanine.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 16 - Question 18		
18)	The following linker is useful in a process called 'tagging'  What functional groups in the linker molecule are used to attach molecules?	
Feedback: The linker is known as a safety-catch acid-labile linker.		
incorrect	**a.#**	The amide groups.
incorrect	**b.#**	The aromatic rings.
Correct	**c.#**	The primary amino groups.
incorrect	**d.#**	The methanesulfinyl groups.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 16 - Question 19	
19)	The following linker is useful in a process called 'tagging'  What sorts of molecules are usually constructed as tags?
Feedback: Peptides are used since they are easily constructed and easily analysed by automated sequencers. Another option is to use oligonucleotides. The linker is known as a safety-catch acid-labile linker.	

Correct	**a.#**	Peptides
incorrect	**b.#**	Carbohydrates
Incorrec t	**c.#**	Lipids
incorrect	**d.#**	Polyesters
END OF QUESTION		

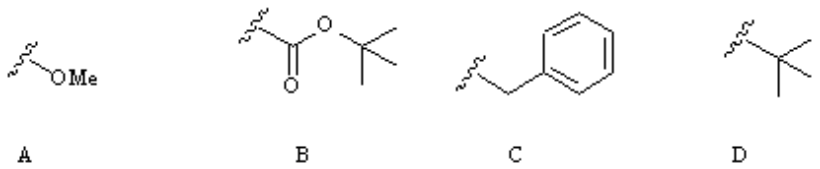
Type: multiple choice		
Title: Chapter 16 - Question 20		
20)	What is the term used to describe the 3-dimensional space around a molecule when it is in a target binding site?	
Feedback: The space accessed by a molecule when it is in the binding site will depend on how flexible the molecule is. This in turn depends on the number of rotatable single bonds that are present. Conformations refer to the different shapes of a molecule that can be adopted by single bond rotation. The other terms are not used in medicinal chemistry.		
incorrect	**a.#**	Stereochemical space.
Correct	**b.#**	Conformational space.
incorrect	**c.#**	Configurational space.
incorrect	**d.#**	Constitutional space.
END OF QUESTION		

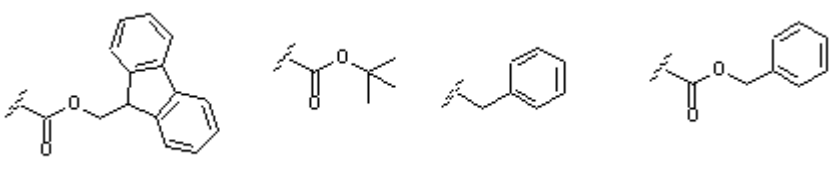
Type: multiple choice		
Title: Chapter 16 - Question 21		
21)	Combinatorial chemistry can be useful at various stages of the drug design / development process. Which of the following is such a stage?	
Feedback: Optimising a lead compound requires the synthesis of a large number of analogues and so combinatorial chemistry can be important at this stage. The other options do not require compounds to be synthesised and so there is no need for combinatorial chemistry.		
incorrect	**a.#**	Purifying a lead compound.
Correct	**b.#**	Optimising a lead compound.
incorrect	**c.#**	Structure determination of the lead compound.
incorrect	**d.#**	Pharmacological testing.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 16 - Question 22		
22)	What term is used for a molecular unit which is attached to the solid support and which contains a reactive functional group that allows attachment of a starting material?	
Feedback: The linker or anchor is the overall moiety which is attached to the solid support and which contains the functional group required to attach the first molecule in the synthetic process.		
incorrect	**a.#**	Resin activator.
incorrect	**b.#**	Hook.
Correct	**c.#**	Linker.

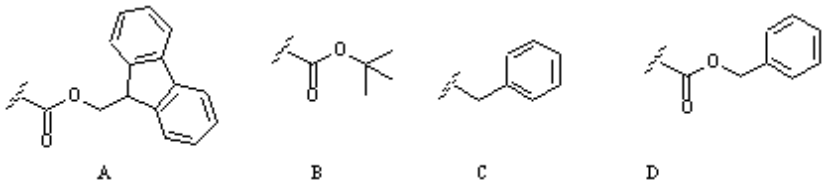
incorrect	**d.#**	Joiner.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 16 - Question 23		
23)	Which of the following statements regarding linkers is wrong?	
Feedback: The statement is true for some linkers but not all. Several linkers will react with a specific functional group and allow the molecule to have the same functional group when it is removed at the end of the synthesis.		
incorrect	**a.#**	The link between the molecule and the solid support must be stable to the reaction conditions used in the synthesis.
incorrect	**b.#**	The link between the molecule and the solid support must be easily cleaved under specific conditions.
Correct	**c.#**	A linker reacts with a specific functional group when the first molecule is attached. At the end of the synthesis the molecule departs with a different functional group.
incorrect	**d.#**	It is important that the resin bead swells to allow molecules to access linkers within the bead.
END OF QUESTION		

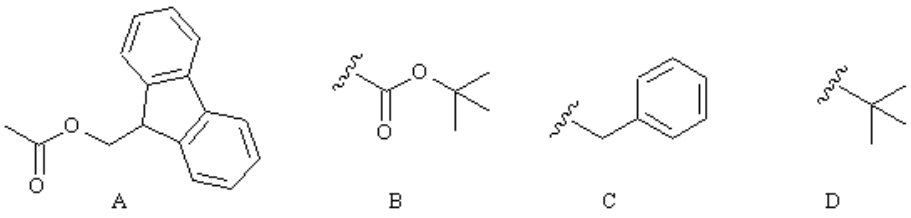
Type: multiple choice		
Title: Chapter 16 - Question 24		
24)	The following structures are protecting groups used in peptide synthesis.  A B C D Which of these would you use to protect an amino group?	
Feedback: The protecting group (B) is a *tertiary*-butyloxycarbonyl protecting group. This protecting group is used to protect the amine functional group of amino acids. The other protecting groups have been used to protect the carboxylic acid group of amino acids.		
incorrect	**a.#**	Structure A.
Correct	**b.#**	Structure B.
incorrect	**c.#**	Structure C.
incorrect	**d.#**	Structure D.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 16 - Question 25		
25)	The following structures are protecting groups used in peptide synthesis.  A B C D	

	Which of these would you use to protect a carboxylic acid group?	
Feedback: Structure C is a benzyl protecting group which can be used to protect carboxylic acids or hydroxyl groups. The other protecting groups have been used to protect the amino group of amino acids.		
incorrect	**a.#**	Structure A.
incorrect	**b.#**	Structure B.
Correct	**c.#**	Structure C.
incorrect	**d.#**	Structure D.
END OF QUESTION		

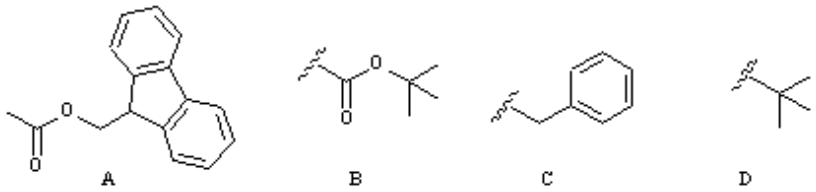
Type: multiple choice	
Title: Chapter 16 - Question 26	
26)	The following structures are protecting groups used in peptide synthesis.  Which of these would you use to protect the side chain of serine?

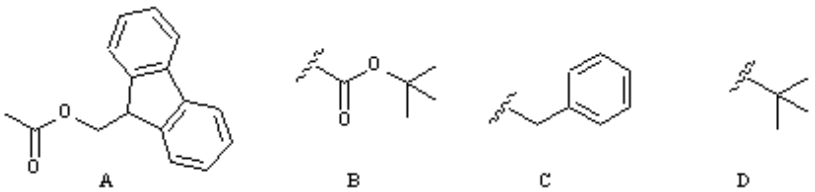
Feedback: Structure C is a benzyl protecting group which can be used to protect carboxylic acids or hydroxyl groups. Serine has a primary alcohol in its side chain and so a benzyl protecting group could be used. The other protecting groups have been used to protect the amino group of amino acids.		
incorrect	**a.#**	Structure A.
incorrect	**b.#**	Structure B.
Correct	**c.#**	Structure C.
incorrect	**d.#**	Structure D.
END OF QUESTION		

Type: matching	
Title: Chapter 16 - Question 27	
27)	The following structures are protecting groups used in peptide synthesis.  Match the structures to the following names:

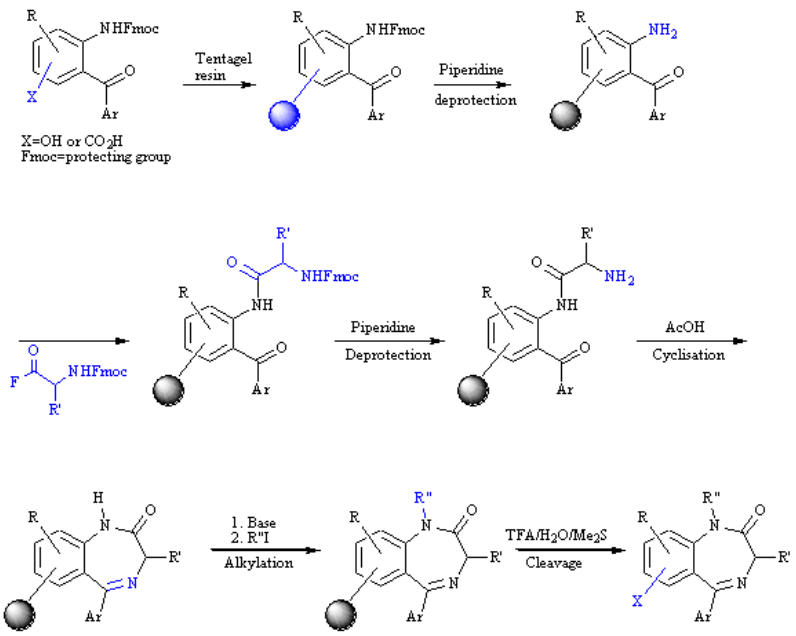
Feedback: Structures A and B are used to protect amino groups. Structures C and D are used to protect carboxylic acids.			
a.#	Structure A	=	9-Fluorenylmethoxycarbonyl

b.#	Structure B	=	*Tertiary*-butyloxycarbonyl
c.#	Structure C	=	Benzyl
d.#	Structure D	=	*Tertiary*-butyl
END OF QUESTION			

Type: multiple choice		
Title: Chapter 16 - Question 28		
28)	The following structures are protecting groups used in peptide synthesis.  Which of these protecting groups can be removed by using piperidine?	
Feedback: Structure A is a 9-fluorenylmethoxycarbonyl group and can be removed under relatively mild conditions compared to the other protecting groups shown.		
Correct	**a.#**	Structure A.
incorrect	**b.#**	Structure B.
incorrect	**c.#**	Structure C.
incorrect	**d.#**	Structure D.
END OF QUESTION		

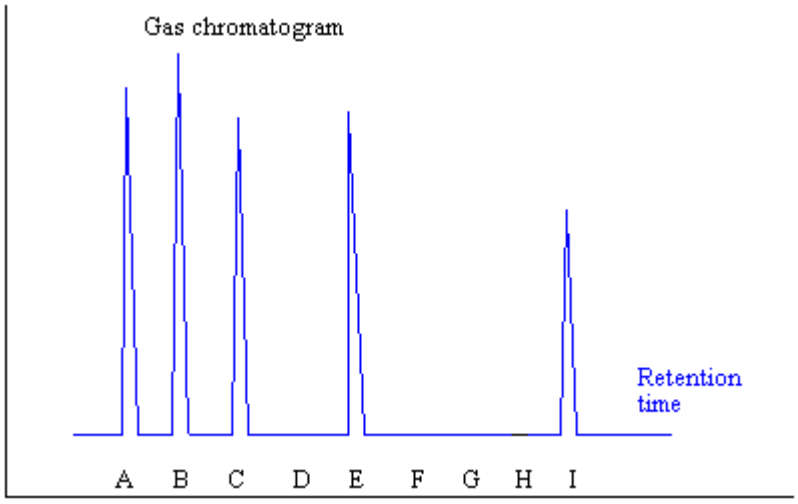
Type: multiple choice		
Title: Chapter 16 - Question 29		
29)	The following structures are protecting groups used in peptide synthesis.  Which of these protecting groups requires the most vigorous conditions for removal?	
Feedback: The benzyl group (structure C) requires vigorous conditions such as hydrofluoric acid for removal. The *tertiary*-butyl and *tertiary*-butyloxycarbonyl groups (structures D and B respectively) can be removed using trifluoroacetic acid. The 9-fluorenylmethoxycarbonyl group (structure A) can be removed using piperidine.		
incorrect	**a.#**	Structure A.
incorrect	**b.#**	Structure B.
Correct	**c.#**	Structure C.
incorrect	**d.#**	Structure D.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 16 - Question 30		
30)	Which of the following statements is true?	
Feedback: Parallel synthesis will not produce a mixture of compounds in each reaction vessel. A single synthetic route is used with different reagents for both parallel and combinatorial syntheses. In a combinatorial synthesis, a large number of compounds are produced with mixtures in each reaction vessel, but a single synthetic route is used with different reagents. In a parallel synthesis, there is only one product per reaction vessel, whereas a combinatorial synthesis generates several products in each reaction vessel.		
incorrect	**a.#**	Parallel synthesis involves the synthesis of a large number of compounds using the same reaction sequence, where there is a mixture of compounds in each reaction vessel.
incorrect	**b.#**	Parallel synthesis involves the synthesis of a large number of compounds using different reaction sequences, where there is a different, single compound formed in each reaction vessel.
incorrect	**c.#**	Combinatorial synthesis involves the synthesis of a large number of compounds using a variety of different synthetic routes to produce a mixture of compounds in each reaction vessel
Correct	**d.#**	A parallel synthesis carried out in a specified number of vessels will produce less compounds than a combinatorial synthesis.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 16 - Question 31	
31)	Consider the following synthesis:  molecular tags (A-I) are constructed on the resin bead as the synthesis is carried out in order to identify R, R' and R". The tagging molecules are used to create an encryption code as follows:

	R	R'	R"
methyl	A	D	G
ethyl	B	E	H
propyl	C	F	I
isopropyl	AB	DE	GH
n-butyl	AC	DF	GI
tert-butyl	BC	EF	GJ
benzyl	ABC	DEF	GHI

On completion of the synthesis, the molecular tags are released from a set of resin beads and identified by gas chromatography as shown below:



What structure has been formed on the bead?

Feedback: The code ABC stands for benzyl, the code E stands for ethyl and the code I stands for propyl.

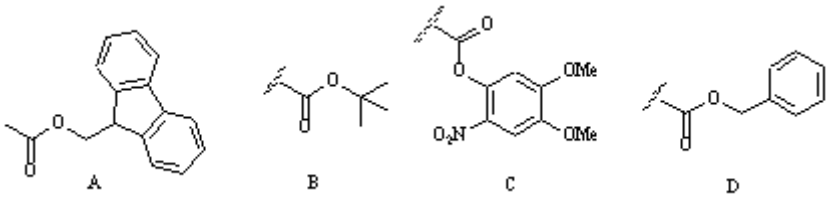
incorrect	**a.#**	R= n-butyl; R'= isopropyl; R" methyl.
incorrect	**b.#**	R= ethyl; R'= propyl; R" benzyl.
Correct	**c.#**	R= benzyl; R'= ethyl; R" propyl.
incorrect	**d.#**	R= n-butyl; R'= ethyl; R" ethyl.

END OF QUESTION

Type: multiple choice

Title: Chapter 16 - Question 32

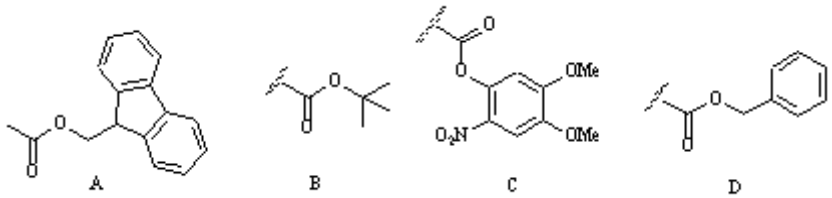
32) The following structures are all protecting groups used in peptide synthesis.

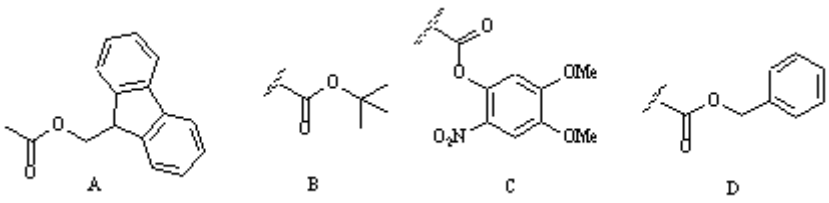


What functional group is normally protected by these groups?

Feedback: All the protecting groups shown can protect the amino group of an amino acid.

incorrect	**a.#**	Carboxylic acid.
Correct	**b.#**	Amine.
incorrect	**c.#**	Alcohol.
incorrect	**d.#**	Phenol.
END OF QUESTION		

Type: matching			
Title: Chapter 16 - Question 33			
33)	The following structures are all protecting groups used in peptide synthesis.  Match the above structures to the following names:		
Feedback: All the protecting groups mentioned here can be used to protect amine functional groups. Once added, urethanes are formed.			
a.#	Tertiary-butyloxycarbonyl	=	Structure B.
b.#	Benzyloxycarbonyl	=	Structure D.
c.#	Nitroveratryloxycarbonyl	=	Structure C.
d.#	9-fluorenylmethoxycarbonyl	=	Structure A.
END OF QUESTION			

Type: multiple choice		
Title: Chapter 16 - Question 34		
34)	The following structures are all protecting groups used in peptide synthesis.  Which of these protecting groups can be removed using light?	
Feedback: Structure C is a nitroveratryloxycarbonyl group.		
incorrect	**a.#**	Structure A.
incorrect	**b.#**	Structure B.
Correct	**c.#**	Structure C.
incorrect	**d.#**	Structure D.

END OF QUESTION

Type: multiple choice		
Title: Chapter 16 - Question 35		
35)	What is the name of the combinatorial technique which involves the use of a photolabile protecting group?	
Feedback: Photolithography is the technique used. Photosensitivity is a term used for chemicals which are susceptible to light. Photofixation is a fictitious term. Photosynthesis is a biological process by which plants capture the energy of light to fuel synthesis.		
incorrect	**a.#**	Photosensitivity.
Correct	**b.#**	Photolithography.
incorrect	**c.#**	Photofixation.
incorrect	**d.#**	Photosynthesis.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 16 - Question 36		
36)	There are several advantages to photolithography. Which of the following statements is untrue regarding these advantages?	
Feedback: The products are tested when they are still bound to the solid support. To do otherwise would defeat the whole point of the exercise; this is to locate specific products at specific locations of the plate.		
incorrect	**a.#**	It allows miniaturisation of the process.
incorrect	**b.#**	It in involves the use of photolabile protecting groups.
incorrect	**c.#**	It allows spatial resolution of the products.
Correct	**d.#**	It requires release of the product from the solid phase before detection of active compounds.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 16 - Question 37		
37)	Dynamic combinatorial chemistry is an alternative method of producing compounds other than the classic mix and split method. Which of the following statements is true about dynamic combinatorial chemistry?	
Feedback: The process depends on the target being present. The process relies on the presence of the target to bind active structures that are formed in the reaction vessel. This in turn leads to an amplification of the active compound. It is necessary to 'freeze' the reaction before identifying active compounds. The process requires the reactions to be in equilibrium such that active compounds are amplified.		
Correct	**a.#**	The target should be present in the reaction flask.
incorrect	**b.#**	There is no scope for amplification.
incorrect	**c.#**	Active compounds can be identified as they are formed.
incorrect	**d.#**	The reactions involved should be irreversible.
END OF QUESTION		

Type: multiple choice
Title: Chapter 16 - Question 38

38)	What name is given to the core structure of a series of related compounds?	
Feedback: The scaffold is the molecular core that is common to a series of compounds.		
incorrect	**a.#**	The lead compound.
incorrect	**b.#**	The skeleton.
Correct	**c.#**	The scaffold.
incorrect	**d.#**	The pharmacophore.
END OF QUESTION		
Type: multiple choice		

Title: Chapter 16 - Question 39		
39)	What term is given to a scaffold that is present in a wide range of drugs with different activities?	
Feedback: A privileged scaffold is a core structure that is present in many different drugs with different activities. Examples of privileged scaffold include steroids, benzodiazepines, and benzenesulfonamides.		
Correct	**a.#**	Privileged.
incorrect	**b.#**	Common.
incorrect	**c.#**	Preferred.
incorrect	**d.#**	Active.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 16 - Question 40		
40)	Which of the following statements is true with respect to scaffolds?	
Feedback: The scaffold should be small. The bigger the scaffold, the less scope there is for adding different substituents since most orally active drugs have a molecular weight less than 500. There is no reason why an important binding group should not be part of the scaffold. It is better to have the substituents spread around the whole molecule since this allows the best exploration of the conformational space around the scaffold. A molecule such as a sugar has various hydroxy groups. If this is to act as a good scaffold, these groups have to be reacted independently of each other such that different substituents can be added to the molecule. This means that protection and deprotection stages have to be introduced to the synthesis which complicates the procedure rather than simplifying it.		
Correct	**a.#**	The scaffold should be small to allow a wide variation of substituents.
incorrect	**b.#**	The scaffold should not be capable of forming any binding interactions with the target.
incorrect	**c.#**	Substituents should be localised at particular regions of the scaffold.
Incorrect	**d.#**	Similar functional groups on the scaffold simplify the synthesis of analogues.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 17 - Question 01		
01)	Which of the following operations or calculations would generally be carried out using quantum mechanics?	
Feedback: Quantum mechanics is used for the calculation of partial atomic charges. The other calculations or procedures are generally done using molecular mechanics		
incorrect	**a.#**	Energy minimisation.
incorrect	**b.#**	Identifying stable conformations.
Correct	**c.#**	Partial atomic charges.
incorrect	**d.#**	Energy calculations for specific conformations.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 17 – Question 02		
02)	Which of the following software programs is not used for drawing 2D chemical structures?	
Feedback: Chem3D is a general molecular modelling software program.		
incorrect	**a.#**	ChemDraw.
incorrect	**b.#**	ChemWindow.
Correct	**c.#**	Chem3D.
incorrect	**d.#**	Isis/Draw.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 17 - Question 03		
03)	Which of the following software programs is not dedicated to the creation of 3D chemical models?	
Feedback: DOCK was one of the early software programs used for docking a ligand to a binding site.		
Correct	**a.#**	DOCK.
incorrect	**b.#**	Alchemy.
incorrect	**c.#**	Hyperchem.
incorrect	**d.#**	Discovery Studio Pro.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 17 – Question 04		
04)	Which of the following is associated with conformational searching?	
Feedback: The Monte Carlo method is used in searching for different conformations of a molecule. LUDI is a software program used in *de novo* drug design. DOCK is used in docking molecules into binding site. CoMFA is used in 3D QSAR.		
incorrect	**a.#**	LUDI.
incorrect	**b.#**	DOCK.
Correct	**c.#**	Monte Carlo method.
incorrect	**d.#**	CoMFA.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 17 – Question 05		
05)	Molecular dynamics can be used to search for different conformations of a molecule. Which of the following statements is false?	
Feedback: Measurements are calculated after each femtosecond of movement.		
incorrect	**a.#**	A variety of different conformations are generated by 'heating' the molecule to 900K.
Correct	**b.#**	The position and velocity of each atom is measured after each nanosecond of movement
incorrect	**c.#**	The programme treats each atom as a moving sphere
incorrect	**d.#**	Each atom is only allowed to move a fraction of a bond length between each cycle of calculations
END OF QUESTION		

Type: multiple choice		
Title: Chapter 17 - Question 06		
06)	Which of the following statements regarding molecular mechanics is untrue?	
Feedback: Molecular mechanics involves the use of equations that follow the laws of classical physics. It treats atoms as spheres and bonds as **springs**.		
incorrect	**a.#**	It treats atoms as spheres.
Correct	**b.#**	It treats bonds as rigid features.
incorrect	**c.#**	It is a molecular modelling computational method.
incorrect	**d.#**	It uses equations that obey the laws of classical physics.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 17- Question 07		
07)	Which of the following statements regarding quantum mechanics is untrue?	
Feedback: In quantum mechanics the assumption is made that nuclei are motionless.		
incorrect	**a.#**	It uses quantum physics to calculate molecular properties.
Correct	**b.#**	It makes the assumption that nuclei are moving independently of each other
incorrect	**c.#**	It makes the assumption that electrons move independently of each other.
incorrect	**d.#**	There are two broad categories - *ab initio* and semi-empirical
END OF QUESTION		

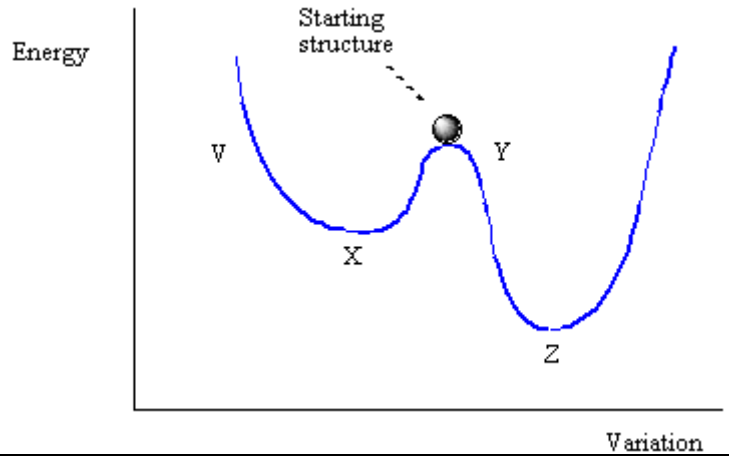
Type: multiple choice		
Title: Chapter 17 - Question 08		
08)	Which of the following statements is true?	
Feedback: Energy minimisation finds a stable conformation for a molecule, but not necessarily the most stable conformation. Bond lengths and bond angles are changed during energy minimisation, but torsion angles are also changed. The energy minimisation stops when there is little change in energy between one structure and the next – corresponding to a stable conformation.		
Correct	**a.#**	Energy minimisation is carried out using molecular mechanics.
incorrect	**b.#**	Energy minimisation is used to find the most stable

		conformation for a molecule.
incorrect	**c.#**	Energy minimisation is carried out by varying only bond angles and bond lengths.
incorrect	**d.#**	Energy minimisation stops when a structure is formed with a much greater stability than the previous one in the process.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 17 - Question 09		
09)	What is meant by a local energy minimum in conformational analysis?	
Feedback: The local energy minimum corresponds to the first stable structure which is formed when energy minimisation is carried out. It may or may not be the most stable conformation. The most stable conformation of a structure is the global energy minimum.		
incorrect	**a.#**	It is a localised region of a molecule which is free of steric strain.
incorrect	**b.#**	It is the most stable conformation of a structure.
incorrect	**c.#**	It is the initial structure that is formed when a 3D model is created, prior to energy minimisation.
Correct	**d.#**	It is the closest stable conformation to the starting structure.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 17 - Question 10		
10)	What is meant by a global energy minimum in conformational analysis?	
Feedback: The global energy minimum corresponds to the most stable conformation of a molecule. The closest stable conformation to the starting structure describes a local energy minimum which corresponds to the first stable structure that is formed when energy minimisation is carried out.		
incorrect	**a.#**	It is a localised region of a molecule which is free of steric strain.
Correct	**b.#**	It is the most stable conformation of a structure.
incorrect	**c.#**	It is the initial structure that is formed when a 3D model is created, prior to energy minimisation.
incorrect	**d.#**	It is the closest stable conformation to the starting structure.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 17 - Question 11	
11)	The following diagram shows the energy of the starting structure prior to energy minimisation. What is the likely result of energy minimisation?

		
Feedback: Energy minimisation would be carried out and the structure would be altered to a stable conformation. The most stable conformation would be Z, but there is no way that the software programme can predict that and so there is an equal likelihood of the structure ending up as X or Z after energy minimisation.		
incorrect	**a.#**	Energy minimisation would have no effect and the structure would remain at position Y.
incorrect	**b.#**	The structure would be energy minimised and give the most stable structure corresponding to position Z.
Correct	**c.#**	The structure would be energy minimised and give a final structure corresponding to either position X or Z.
incorrect	**d.#**	Energy minimisation would be carried out and the final structure would correspond to position X.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 17 - Question 12		
12)	Which of the following statements regarding structural overlays is false?	
Feedback: The calculation continues until the root mean square distance is a **minimum** value.		
incorrect	**a.#**	The two structures to be overlaid should both be in their active conformations.
incorrect	**b.#**	Usually specific pairs of atoms are chosen (one from each molecule) to allow the overlay to take place.
incorrect	**c.#**	The fitting process normally involves both molecules being rigid and not changing conformation during the overlay.
Correct	**d.#**	The overlay is carried out until a maximum value of the root mean square distance between all the atom pairs is achieved.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 17 - Question 13		
13)	Which of the following statements is true?	
Feedback: The active conformation is the conformation that the drug adopts when it binds to the target binding site and this does not necessarily equate with the most stable conformation. Most drugs do not react with their target binding site. Instead, they form intermolecular bonds. Therefore, the active conformation does not need to be a reactive conformation. Conformational analysis will assess the relative stabilities of different conformations, but it cannot predict the active conformation.		
Correct	**a.#**	The most stable conformation of a drug is not necessarily the

		active conformation.
incorrect	**b.#**	The active conformation is the most reactive conformation of a structure.
incorrect	**c.#**	The active conformation is the conformation adopted by a target binding site when it binds a drug.
incorrect	**d.#**	The active conformation can be determined by conformational analysis.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 17 - Question 14		
14)	Which of the following statements is not true of cyclic structures?	
Feedback: Cyclic, rigid structures are generally **more** difficult to synthesise than acyclic simpler molecules.		
incorrect	**a.#**	They are normally more rigid than acyclic structures.
incorrect	**b.#**	They have fewer possible conformations.
incorrect	**c.#**	They are useful in determining the active conformation of a series of related compounds.
Correct	**d.#**	They are easier to synthesise than more flexible acyclic molecules.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 17 - Question 15		
15)	What term is used to describe the process by which drugs are fitted into their target binding sites using molecular modelling?	
Feedback: Docking is where a molecular ligand is fitted into its binding site by molecular modelling.		
incorrect	**a.#**	Energy minimisation.
incorrect	**b.#**	Conformational searching.
Correct	**c.#**	Docking.
incorrect	**d.#**	Overlaying.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 17 - Question 16		
16)	What term is used for the design of a novel drug based on molecular modelling studies of a binding site	
Feedback: *De novo* drug design involves the computer-aided design of a drug based purely on knowledge of a binding site, and without the benefit of a lead compound.		
incorrect	**a.#**	Drug development.
incorrect	**b.#**	Structure-based drug design.
Correct	**c.#**	*De novo* drug design.
incorrect	**d.#**	Drug optimisation.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 17 - Question 17	
17)	Which of the following statements is false in *de novo* drug design?

Feedback: If a molecule is designed to fit snugly, it leaves no margin for error. If the molecule does not bind as predicted, it is unlikely to bind at all. If the molecule is designed such that it does not fill up the available space, an alternative binding mode might be possible.		
incorrect	**a.#**	The design of flexible molecules is superior to rigid ones.
Correct	**b.#**	Molecules should be designed to fit as snugly as possible into the target binding site.
incorrect	**c.#**	Molecules that have to adopt an unstable conformation in order to bind should be rejected.
incorrect	**d.#**	Desolvation energies cannot be ignored.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 17 - Question 18		
18)	What is the software program LUDI used for?	
Feedback: LUDI is a software program used for *de novo* drug design.		
incorrect	**a.#**	Docking molecules into binding sites.
Correct	**b.#**	Automated *de novo* drug design.
incorrect	**c.#**	Creating 3D models of molecules.
incorrect	**d.#**	Conformational analysis.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 17 - Question 19		
19)	Which of the following statements is untrue when using molecular modelling to design a combinatorial library?	
Feedback: It is the other way round. The aim is to synthesise the **minimum** number of structures likely to produce the **maximum** number of pharmacophores.		
incorrect	**a.#**	Pharmacophore triangles can be used to design a library.
Correct	**b.#**	The aim is to synthesise the maximum number of structures likely to produce the minimum number of pharmacophores.
incorrect	**c.#**	Rigid structures should be analysed before flexible ones.
incorrect	**d.#**	Structures should only be included in the library if they represent at least 10% additional new pharmacophores compared to the total represented by structures already present in the library.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 17 - Question 20		
20)	Which of the following operations or calculations would generally be carried out using molecular mechanics?	
Feedback: Generating different conformations is carried out using molecular mechanics. The other three operations or calculations are carried out using quantum mechanics.		
incorrect	**a.#**	Heat of formation for specific conformations.
Correct	**b.#**	Generating different conformations.
incorrect	**c.#**	Partial atomic charges.
incorrect	**d.#**	Transition-state energies.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 17 - Question 21		
21)	Which of the following operations or calculations would generally be carried out using quantum mechanics?	
Feedback: Transition-state energies can be calculated using quantum mechanics. The other three operations can be carried out using molecular mechanics.		
incorrect	**a.#**	Studying molecular motion.
incorrect	**b.#**	Generating different conformations.
incorrect	**c.#**	Energy minimisation.
Correct	**d.#**	Transition-state energies.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 17 - Question 22		
22)	Which of the following software programmes is not used for drawing 2D chemical structures?	
Feedback: LUDI is a software programme used for *de novo* drug design.		
incorrect	**a.#**	ChemDraw.
Correct	**b.#**	LUDI.
incorrect	**c.#**	ChemWindow.
incorrect	**d.#**	Isis/Draw.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 17 - Question 23		
23)	Which of the following software programmes is not dedicated to the creation of 3D chemical models?	
Feedback: CoMFA is a software programme used in 3D-QSAR.		
incorrect	**a.#**	CAChe.
Correct	**b.#**	CoMFA.
incorrect	**c.#**	Hyperchem.
incorrect	**d.#**	Sybil.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 17 - Question 24		
24)	Which of the following statements is false regarding Monte Carlo methods of conformational searching?	
Feedback: The programme can be used to find global energy conformations since a facility is present to avoid the structure being stuck in energy 'wells'. At regular intervals, the programme introduces a large change in conformation to ensure that a structure is moved out of a 'well'.		
incorrect	**a.#**	The Monte Carlo method introduces a bias towards stable conformations.
Correct	**b.#**	The method has the drawback that the structure becomes stuck in a local energy minimum 'well' and fails to reach the global energy minimum.
incorrect	**c.#**	Energies of randomly created conformations are compared,

		then further minor modifications are carried out on the more stable conformations.
incorrect	**d.#**	Random conformations are generated by carrying out bond rotations
END OF QUESTION		

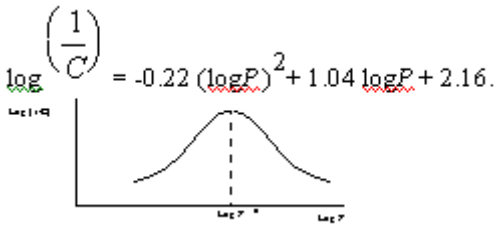
Type: multiple choice		
Title: Chapter 17 - Question 25		
25)	Molecular dynamics can be used to search for different conformations of a molecule. Which of the following statements is false?	
Feedback: Each atom only moves a fraction of a bond length between each round of calculations.		
incorrect	**a.#**	A variety of different conformations are generated by 'heating' the molecule to 900K.
incorrect	**b.#**	The position and velocity of each atom is measured after each femtosecond of movement.
incorrect	**c.#**	The programme treats each atom as a moving sphere.
Correct	**d.#**	Each atom is only allowed to move a bond length between each cycle of calculations.
END OF QUESTION		

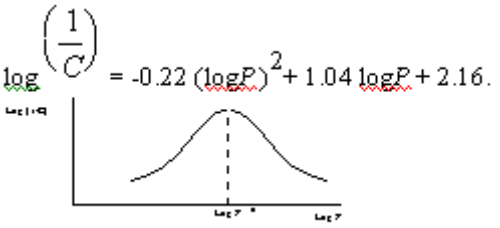
Type: multiple choice		
Title: Chapter 18 - Question 01		
01)	Which of the following physicochemical properties is least commonly considered in QSAR studies?	
Feedback: The most commonly investigated properties are the hydrophobicity of the molecule, the electronic influence of substituents, and the size of substituents. The dipole moment is not often considered, but there are some instances where it is included - see for example cimetidine, chapter 25.		
incorrect	**a.#**	Hydrophobicity.
incorrect	**b.#**	Electronic influence of substituents.
Correct	**c.#**	Dipole moment.
incorrect	**d.#**	Size of substituents.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 18 - Question 02		
02)	What term is used to describe the process by which the best line is fitted through a set of data points on a graph?	
Feedback: In the process, various lines are drawn and the distance of the points from each line is measured as a squared number. The line which has the data points closest to it is the best fitting line.		
incorrect	**a.#**	Linear fitting analysis by the least squares method.
Correct	**b.#**	Linear regression analysis by the least squares method.
incorrect	**c.#**	Linear fitting assessment by the least squares method.
incorrect	**d.#**	Linear regression assessment by the least squares method.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 18 - Question 03		
03)	Which of the following statements is true?	
Feedback: A high level of *P* indicates that more of the drug is taken up by the octanol layer of an octanol / water two phase system. This corresponds to better solubility in a hydrophobic, organic solvent, which indicates that the drug has hydrophobic character. Hydrophilic compounds have a low level of *P*. Finally, the value of *P* does not indicate acidic or basic properties.		
Correct	**a.#**	Hydrophobic compounds have a high value of P.
incorrect	**b.#**	Hydrophilic compounds have a high value of P.
incorrect	**c.#**	Acidic compounds have a high value of P.
incorrect	**d.#**	Basic compounds have a high value of P.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 18 - Question 04	
04)	The QSAR equation relating the general anaesthetic activity of several ethers to their log *P* values is defined by the following equation and graph.

	 Which of the following statements is true?	
Feedback: Hydrophobicity increases as *P* or log *P* increases. The graph shows that activity increases with hydrophobicity at the beginning of the plot. A maximum is reached at log P^0, then activity drops with increasing log *P*.		
incorrect	**a.#**	General anaesthetic activity increases with hydrophobicity.
incorrect	**b.#**	General anaesthetic activity decreases with hydrophobicity.
Correct	**c.#**	General anaesthetic activity increases with hydrophobicity to a maximum value then decreases.
incorrect	**d.#**	General anaesthetic activity decreases with hydrophobicity to a minimum value then decreases.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 18 - Question 05		
05)	The QSAR equation relating the general anaesthetic activity of several ethers to their log *P* values is defined by the following equation and graph.  What is the meaning of log P^0 in the graph?	
Feedback: The level of activity on the y-axis is maximum at the level of hydrophobicity indicated by log P^0 on the x-axis.		
Correct	**a.#**	Log P^0 represents the value of log *P* which is optimum for activity.
incorrect	**b.#**	Log P^0 represents the value of log *P* which is worst for activity.
incorrect	**c.#**	Log P^0 represents the maximum hydrophobicity for the compounds tested.
incorrect	**d.#**	Log P^0 represents the minimum hydrophobicity for the compounds tested.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 18 - Question 06		
06)	How is π measured in QSAR studies?	
Feedback: A positive value of π indicates that a substituent is more hydrophobic than hydrogen. A negative value of π indicates that a substituent is less hydrophobic than hydrogen.		
incorrect	**a.#**	It is measured by calculating molecular dimensions using relevant molecular modelling software.

incorrect	**b.#**	It is measured by comparing the dissociation constants of two weak acids.
incorrect	**c.#**	It is measured by subtracting the log *P* value of an analogue bearing the substituent from the log *P* value of the parent compound lacking the substituent.
Correct	**d.#**	It is measured by subtracting the log *P* value of the parent compound lacking the substituent from the log *P* value of an analogue bearing the substituent.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 18 - Question 07		
07)	Which of the following statements is true?	
Feedback: A positive value of π indicates that a substituent is more hydrophobic than hydrogen. A negative value of π indicates that a substituent is less hydrophobic than hydrogen. π has nothing to do with a substituent's size or electronic character.		
incorrect	**a.#**	A positive value of π indicates that a substituent is larger than H.
incorrect	**b.#**	A positive value of π indicates that a substituent is more electron donating than H.
Correct	**c.#**	A positive value of π indicates that a substituent is more hydrophobic than H.
incorrect	**d.#**	A negative value of π indicates that a substituent is more hydrophobic than H.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 18 - Question 08		
08)	What is the distinction between log *P* and π?	
Feedback: Neither log *P* nor π have anything to do with pH, size, or electronic properties.		
incorrect	**a.#**	Log *P* is a measure of the overall pH of a molecule, whereas π is a measure of the hydrophobicity of a substituent.
Correct	**b.#**	Log *P* is a measure of the overall hydrophobicity of a molecule, whereas π is a measure of the hydrophobicity of a substituent.
incorrect	**c.#**	π is a measure of the overall hydrophobicity of a molecule, whereas log*P* is a measure of the hydrophobicity of a substituent.
incorrect	**d.#**	Log *P* is a measure of the overall pH of a molecule, whereas π is a measure of the electronic effect of a substituent.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 18 - Question 09		
09)	What does E_S represent in a QSAR equation?	
Feedback: The electronic influence of a substituent is measured by σ, the Hammett substituent constant. The measure of a substituent's steric properties is indicated by a variety of methods including Taft's substituent constant E_S. The entropy associated with a substituent is not normally considered in QSAR equations.		
incorrect	**a.#**	The electronic influence of a substituent as an electron withdrawing group.
incorrect	**b.#**	The electronic influence of a substituent as an electron donating group.

incorrect	**c.#**	The entropy associated with a substituent.
Correct	**d.#**	Taft's steric factor.
END OF QUESTION		

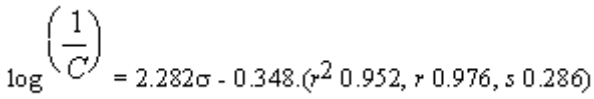
Type: multiple choice		
Title: Chapter 18 - Question 10		
10)	What does *MR* represent in a QSAR equation?	
Feedback: Molar refractivity is a steric factor determined from the index of refraction, the molecular weight, and the density.		
Correct	**a.#**	Molar refractivity as a steric factor.
incorrect	**b.#**	Molar refractivity as an electronic factor.
incorrect	**c.#**	Molar refractivity as a hydrophobic factor.
incorrect	**d.#**	Molar refractivity as a stereoelectronic factor.
END OF QUESTION		

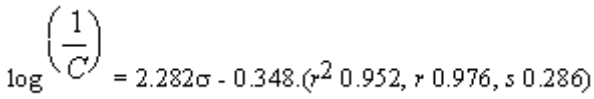
Type: multiple choice		
Title: Chapter 18 - Question 11		
11)	What does the Hammett substituent constant (σ) measure?	
Feedback: The measure of a substituent's steric properties is indicated by a variety of methods including Taft's substituent constant E_S. The electronic effect of a substituent is measured by σ, the Hammett substituent constant. The hydrophobic effect of a substituent is measured by the substituent hydrophobicity constant π.		
incorrect	**a.#**	The steric effect of a substituent.
Correct	**b.#**	The electronic effect of a substituent.
incorrect	**c.#**	The hydrophobic effect of a substituent.
incorrect	**d.#**	The effect on pH of a substituent.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 18 - Question 12		
12)	The σ_m value for a phenol substituent (OH) is 0.12, whereas the σ_p value is -0.37. What do σ_m and σ_p stand for?	
Feedback: σ is a measure of a substituent's electronic effects. The hydrophobic effect of a substituent is measured by the substituent hydrophobicity constant π. The measure of a substituent's steric properties is indicated by a variety of methods including Taft's substituent constant E_S. σ is not a measure of a substituent's effect on acidity.		
Correct	**a.#**	The electronic effect of the phenol substituent at the *meta* and *para* positions respectively.
incorrect	**b.#**	The hydrophobic effect of the phenol substituent at the *meta* and *para* positions respectively.
incorrect	**c.#**	The steric effect of the phenol substituent at the *meta* and *para* positions respectively.
incorrect	**d.#**	The acidic effect of the phenol substituent at the *meta* and *para* positions respectively.
END OF QUESTION		

Type: multiple choice

Title: Chapter 18 - Question 13		
13)	The σ_m value for a phenol substituent (OH) is 0.12, whereas the σ_p value is -0.37. Why are σ_m and σ_p so different?	
Feedback: The Hammett substituent constant has nothing to do with steric or hydrophobic effects. The *meta* substituent is closer to the rest of the molecule and has an electron withdrawing effect, but it does not explain why the *para* substituent has an electron donating effect.		
incorrect	**a.#**	The *meta* substituent is closer to the rest of the molecule than the *para* substituent and has a greater electron withdrawing effect.
incorrect	**b.#**	The *meta* substituent is closer to the rest of the molecule than the *para* substituent and has a greater steric effect.
incorrect	**c.#**	The *meta* substituent is closer to the rest of the molecule than the *para* substituent and has a greater hydrophobic effect.
Correct	**d.#**	The *meta* substituent has an inductive effect which makes it electron withdrawing, whereas the *para* substituent has a resonance effect which makes it electron donating.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 18 - Question 14		
14)	The QSAR equation relating the insecticidal activity of a series of diethyl phenylphosphonates versus σ is shown below.  $\log\left(\frac{1}{C}\right) = 2.282\sigma - 0.348.(r^2\ 0.952, r\ 0.976, s\ 0.286)$ What does log(1/*C*) represent?	
Feedback: The term indicates the insecticidal activity. C is the concentration required to produce a defined activity, and is included in the activity term log(1/*C*). The catalytic activity has no relevance at all since these compounds are not acting as catalysts. The physiochemical property is wrong since the physicochemical properties are on the right hand side of the equation.		
incorrect	**a.#**	The concentration.
incorrect	**b.#**	The catalytic activity.
Correct	**c.#**	The insecticidal activity.
incorrect	**d.#**	The physicochemical property.
END OF QUESTION		

Type: multiple choice		
Title : Chapter 18 – Question 15		
15)	The QSAR equation relating the insecticidal activity of a series of diethyl phenylphosphonates versus σ is shown below.  $\log\left(\frac{1}{C}\right) = 2.282\sigma - 0.348.(r^2\ 0.952, r\ 0.976, s\ 0.286)$ What physicochemical property is beneficial for activity?	
Feedback: A positive value of σ will result in good activity and this corresponds to an electron withdrawing substituent. The Hammett substituent constant has nothing to do with size or hydrophobicity.		

incorrect	**a.#**	An electron donating substituent.
Correct	**b.#**	An electron withdrawing substituent.
Incorrect	**c.#**	A hydrophobic substituent.
Incorrect	**d.#**	A small substituent.
END OF QUESTION		

Type: multiple choice		
Title : Chapter 18 – Question 16		
16)	What do the factors *R* and *F* represent in a QSAR equation?	
Feedback: The factors *R* and *F* are electronic factors which distinguish a substituent's resonance electronic effects from its inductive electronic effects. They have nothing to do with hydrophobicity.		
incorrect	**a.#**	A substituent's steric effect and hydrophobic effect respectively.
Incorrect	**b.#**	A substituent's inductive effect and resonance effect respectively.
Correct	**c.#**	A substituent's resonance effect and inductive effect respectively.
Incorrect	**d.#**	A substituent's electronic effect and hydrophobic effect respectively.
END OF QUESTION		

Type: multiple choice		
Title : Chapter 18 – Question 17		
17)	What is a Craig plot used for?	
Feedback: A Craig plot compares the values of two specified physicochemical properties for a range of substituents. This allows suitable substituents to be chosen for a QSAR study which will demonstrate a range of values for each physicochemical property, but where the values will not be correlated to each other.		
incorrect	**a.#**	To compare the activities of different compounds having different substituents.
Incorrect	**b.#**	To compare the values of a specific physicochemical property for a range of substituents.
Correct	**c.#**	To compare the values of two specified physicochemical properties for a range of substituents.
Incorrect	**d.#**	None of the answers are correct.
END OF QUESTION		

Type: multiple choice		
Title : Chapter 18 – Question 18		
18)	What does CoMFA stand for in 3D QSAR?	
Feedback: CoMFA methodology is based on the assumption that drug–receptor interactions are non-covalent and that changes in biological activity correlate with the changes in the steric and/or electrostatic fields of the drug molecules.		
incorrect	**a.#**	Compound molecular formula assessment.
Incorrect	**b.#**	Compound mass field analysis.
Correct	**c.#**	Comparative molecular field analysis.
Incorrect	**d.#**	Comparative modelling force field analysis.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 18 - Question 19		
19)	What two physical features (or fields) are most important in 3D QSAR studies?	
Feedback: Electrostatic and steric fields are the fields normally used in 3D QSAR studies although it is possible to have a hydrophobic field.		
incorrect	**a.#**	Hydrophobic and steric fields.
Correct	**b.#**	Electrostatic and steric fields.
incorrect	**c.#**	Hydrophobic and electrostatic fields.
incorrect	**d.#**	Steric field and dipole moment.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 18 - Question 20		
20)	Which of the following is not crucial for 3D QSAR studies?	
Feedback: It is crucial that each molecule in a 3D QSAR study is aligned properly. This requires knowledge of the active conformation and the pharmacophore. Knowledge of the target structure and how the molecules bind to it is certainly useful, but it is not essential.		
incorrect	**a.#**	Identification of the active conformation for each molecule.
incorrect	**b.#**	Identification of the pharmacophore for each molecule.
incorrect	**c.#**	Alignment of each molecule.
Correct	**d.#**	Identification of the target structure.
END OF QUESTION		

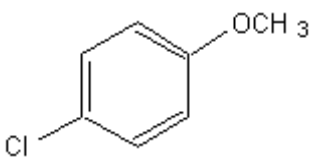
Type: multiple choice		
Title: Chapter 18 - Question 21		
21)	What is the relevance of a probe atom in 3D QSAR?	
Feedback: The probe atom is a positively charged atom such as a carbon or a proton which is placed at each grid point of a lattice surrounding the test molecule. It is used to measure the steric and electronic interactions between it and the test molecule. This in turn allows the calculation of steric and electronic fields for each molecule in the study.		
incorrect	**a.#**	It is used to position a molecule into a lattice of grid points.
incorrect	**b.#**	It is used to determine the position of the various grid points making up the lattice.
incorrect	**c.#**	It is an atom of the test molecule which is used as a spatial reference point.
Correct	**d.#**	It is placed at the grid points of a lattice to determine the important properties of a molecule placed in the lattice.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 18 - Question 22		
22)	Which of the following figures is an acceptable value for the regression coefficient?	
Feedback: A value over 0.90 is an acceptable figure for the regression coefficient.		
incorrect	**a.#**	0.01
incorrect	**b.#**	0.1
incorrect	**c.#**	0.5
Correct	**d.#**	0.95

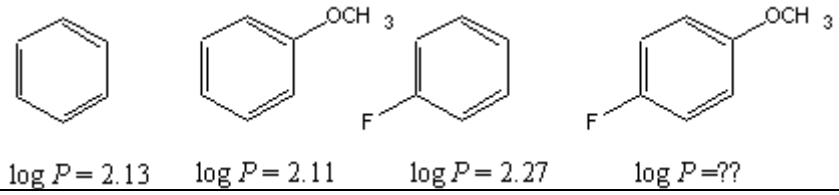
END OF QUESTION

Type: matching			
Title: Chapter 18 - Question 23			
23)	Match the symbols used in QSAR equations to the properties described.		
Feedback: The partition coefficient is a measure of a drug's hydrophobicity and can be measured by comparing the relative concentrations of the drug in a two phase system of water and octanol. pH is a measure of hydrogen ion concentration. The symbol π is used to represent the substituent hydrophobicity constant. The symbol σ is used to represent the Hammett substituent constant.			
a.#	pH.	=	A measure of the acidity of a solution
b.#	π.	=	The hydrophobicity constant for a substituent
c.#	*P.*	=	The partition coefficient
d.#	σ.	=	The Hammett substituent constant
END OF QUESTION			

Type: multiple choice		
Title: Chapter 18 - Question 24		
24)	Which of the following statements is true?	
Feedback: The partition coefficient (*P*) is a measure of a drug's hydrophobicity and is low for hydrophilic compounds. The electronic properties and the size of a substituent are measured by the Hammett substituent constant and a steric constant respectively. The hydrophilic/hydrophobic property of an individual substituent is measured by the substituent hydrophobicity constant and not the partition coefficient. The latter is a measure of the drug's overall hydrophobicity.		
Correct	**a.#**	Hydrophilic compounds have a low value of *P.*
incorrect	**b.#**	An electron withdrawing substituent has a low value of *P.*
incorrect	**c.#**	A small substituent has a low value of *P.*
incorrect	**d.#**	A hydrophilic substituent has a low value of *P.*
END OF QUESTION		

Type: multiple choice		
Title: Chapter 18 - Question 25		
26)	Calculate the log*P* value for structure (I) (log*P* for benzene = 2.13; π(Cl) 0.71; π(OCH$_3$) -0.02)). 	
Feedback: The value is obtained by adding the π values for each substituent to the log*P* value for benzene.		
incorrect	**a.#**	1.40
incorrect	**b.#**	1.44
Correct	**c.#**	2.82

incorrect	**d.#**	2.86
END OF QUESTION		

Type: multiple choice		
Title: Chapter 18 - Question 26		
26)	Given the following log *P* values, what is the predicted log *P* value for the disubstituted structure?  log *P* = 2.13 log *P* = 2.11 log *P* = 2.27 log *P* =??	
Feedback: The log *P* values demonstrate that the hydrophobicity constants for the substituents methoxy and fluoro are -0.02 and 0.14 respectively. This means that the predicted log *P* value of the disubstituted structure will be 2.13 + (-0.02) + 0.14 = 2.25		
incorrect	**a.#**	1.97
incorrect	**b.#**	2.01
Correct	**c.#**	2.25
incorrect	**d.#**	2.29
END OF QUESTION		

Type: matching			
Title: Chapter 18 - Question 27			
27)	Match the symbols used in QSAR equations to the properties described.		
Feedback: *MR* is the symbol used to represent molar refractivity as a steric factor. *R* is used to represent an aromatic substituent's electronic resonance effect. *F* is used to represent an aromatic substituent's electronic inductive effect. E_S is used to represent Taft's substituent constant			
a.#	Molar refractivity	=	*MR*
b.#	The resonance effect of a substituent	=	*R*
c.#	The inductive effect of a substituent	=	*F*
d.#	Taft's steric factor E_S	=	E_S
END OF QUESTION			

Type: multiple choice		
Title: Chapter 18 - Question 28		
28)	What is measured by the software program called Sterimol?	
Feedback: Sterimol is used to measure the Verloop steric parameter. The other constants or factors can be measured experimentally.		
incorrect	**a.#**	The Hammett substituent constant.
incorrect	**b.#**	Taft's steric factor.
Correct	**c.#**	The Verloop steric parameter.
incorrect	**d.#**	The substituent hydrophobicity constant.

END OF QUESTION

Type: multiple choice		
Title: Chapter 18 - Question 29		
29)	What sort of value of σ would signify an electron donating substituent?	
Feedback: A negative value of σ signifies an electron donating substituent whereas a positive value signifies an electron withdrawing substituent. A value close to zero would indicate a neutral electronic effect.		
Correct	**a.#**	Negative value.
incorrect	**b.#**	Positive value.
incorrect	**c.#**	Zero.
incorrect	**d.#**	It is impossible to say.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 18 - Question 30		
30)	What sort of value of σ would signify an electron withdrawing substituent?	
Feedback: A negative value of σ signifies an electron donating substituent whereas a positive value signifies an electron withdrawing substituent. A value close to zero would indicate a neutral electronic effect.		
incorrect	**a.#**	Negative value.
Correct	**b.#**	Positive value.
incorrect	**c.#**	Zero.
incorrect	**d.#**	It is impossible to say.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 18 - Question 31		
31)	What sort of value of σ would signify an aromatic substituent that is neither electron donating nor electron withdrawing?	
Feedback: A negative value of σ signifies an electron donating substituent whereas a positive value signifies an electron withdrawing substituent. A value close to zero would indicate a neutral electronic effect.		
incorrect	**a.#**	Negative value.
incorrect	**b.#**	Positive value.
Correct	**c.#**	Zero.
incorrect	**d.#**	It is impossible to say.
END OF QUESTION		

Type: multiple choice
Title: Chapter 18 - Question 32

32)	The QSAR equation relating the insecticidal activity of a series of diethyl phenylphosphonates versus σ is shown below. What is the value of the correlation coefficient? $\log\left(\frac{1}{C}\right) = 2.282\sigma - 0.348.$ $(r^2\ 0.952, r\ 0.976, s\ 0.286)$

Feedback: *r* represents the regression or correlation coefficient. σ represents the Hammett substituent constant. *s* is the standard deviation.

incorrect	**a.#**	2.282
incorrect	**b.#**	0.348
Correct	**c.#**	0.976
incorrect	**d.#**	0.286

END OF QUESTION

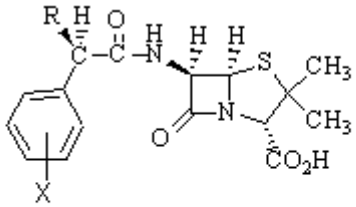
Type: multiple choice	
Title: Chapter 18 - Question 33	
33)	The QSAR equation relating the insecticidal activity of a series of diethyl phenylphosphonates versus σ is shown below. $\log\left(\frac{1}{C}\right) = 2.282\sigma - 0.348.$ $(r^2\ 0.952, r\ 0.976, s\ 0.286)$ What is the value of the standard deviation?

Feedback: *r* represents the regression or correlation coefficient. σ represents the Hammett substituent constant. *s* is the standard deviation.

incorrect	**a.#**	2.282
incorrect	**b.#**	0.348
incorrect	**c.#**	0.976
Correct	**d.#**	0.286

END OF QUESTION

Type: multiple choice	
Title: Chapter 18 - Question 34	
34)	The antibacterial activity of a number of penicillins was found to fit the following QSAR equation:



$$\log\left(\frac{1}{C}\right) = -0.445\pi + 5.673 \quad (n=20, r=0.909)$$

Which of the following statements is true?

Feedback: π is a measure of how hydrophobic the substituent X is. The value of π increases for hydrophobic substituents, but this will have a negative effect in the QSAR equation since it is multiplied by -0.445. As a result, activity drops as the hydrophobic character of X increases.
The value of the regression coefficient is acceptable since it is above 0.9

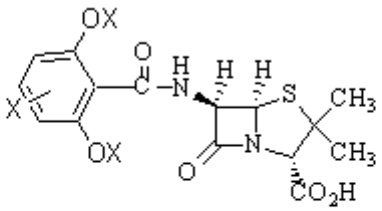
incorrect	**a.#**	The value of the regression coefficient is not acceptable.
incorrect	**b.#**	Activity increases with hydrophobic substituents on the aromatic ring.
Correct	**c.#**	Activity decreases with hydrophobic substituents on the aromatic ring.
incorrect	**d.#**	Activity decreases with electron withdrawing substituents on the aromatic ring.

END OF QUESTION

Type: multiple choice

Title: Chapter 18 - Question 35

35) The antibacterial activity of a number of penicillins was found to fit the following QSAR equation:



$$\log\left(\frac{1}{C}\right) = -0.245\pi + 1.720\sigma + 5.673 \quad (n=8, r=0.929)$$

Which of the following statements is true?

Feedback: Activity will be best for negative values of π and positive values of σ. This corresponds to hydrophilic, electron-withdrawing substituents. The regression coefficient is greater than 0.9 and is acceptable.

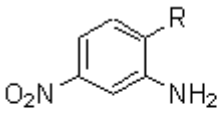
incorrect	**a.#**	The value of the regression coefficient is not acceptable.
incorrect	**b.#**	Activity increases with hydrophobic substituents on the aromatic ring.
incorrect	**c.#**	Activity increases with electron donating substituents on the aromatic ring.
Correct	**d.#**	Activity increases with electron withdrawing substituents on the aromatic ring.

END OF QUESTION

Type: multiple choice

Title: Chapter 18 - Question 36

36) The relative sweetness of a variety of compounds was found to fit the following QSAR equation:



$$\log\left(\frac{1}{C}\right) = 1.610\pi - 1.831\sigma + 1.729 \quad (n=9, r=0.936)$$

Which of the following statements is not true?

Feedback: Activity will be best for positive values of π and negative values of σ. This corresponds to hydrophobic, electron-donating substituents. The regression coefficient is greater than 0.9 and is acceptable. There is no steric factor in the equation and so it is not possible to say whether there is a steric influence or not.

incorrect	**a.#**	The value of the regression coefficient is acceptable.
incorrect	**b.#**	Activity increases with hydrophobic substituents on the aromatic ring.
incorrect	**c.#**	Activity increases with electron donating substituents on the aromatic ring.
Correct	**d.#**	Activity increases with large substituents on the aromatic ring.

END OF QUESTION

Type: multiple choice

Title: Chapter 18 - Question 37

37) A Hansch analysis is being carried out in order to relate biological activity to σ and π. Which of the following substituents would best suit the study?

Feedback: The substituents $CONH_2$, I, NMe_2, OH, NO_2 are from different quadrants of the Craig plot show in fig. 13.12. There is no correlation between the values of σ and π. The substituents also display a range of values for both σ and π.
F, Cl, Br, I is no good since the substituents are from the same quadrant and are all substituents with a positive value of σ and a positive value of π.
SO_2NH_2, $CONH_2$, COOH, F, Me is no good since the values for σ and π are correlated. As π increases, σ decreases.
SO_2NH_2, CH_3SO_2, CN, CF_3, SF_3 is no good since the substituents have similar values of σ.

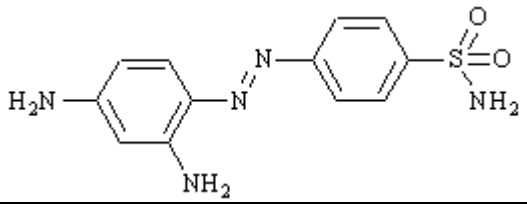
incorrect	**a.#**	F, Cl, Br, I.
incorrect	**b.#**	SO_2NH_2, $CONH_2$, COOH, F, Me.
incorrect	**c.#**	SO_2NH_2, CH_3SO_2, CN, CF_3, SF_3.
Correct	**d.#**	$CONH_2$, I, NMe_2, OH, NO_2.

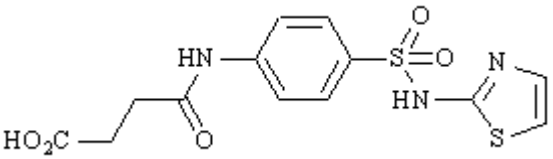
END OF QUESTION

Type: multiple choice

Title: Chapter 18 - Question 38		
38)	What is the name of the plot that is used to compare the values of two specified physicochemical properties for a range of substituents?	
Feedback: A Craig plot compares the values of two specified physicochemical properties for a range of substituents. This allows suitable substituents to be chosen for a QSAR study which will demonstrate a range of values for each physicochemical property, but where the values will not be correlated to each other. The other plots do not exist.		
incorrect	**a.#**	Topliss plot.
incorrect	**b.#**	Hansch plot.
Correct	**c.#**	Craig plot.
incorrect	**d.#**	Free-Wilson plot.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 18 - Question 39		
39)	Which of the following statements is untrue when comparing 3D QSAR with conventional QSAR?	
Feedback: Experimental parameters are not required by 3D-QSAR		
incorrect	**a.#**	Drugs of the different structural classes can be studied by 3D QSAR but not by QSAR.
incorrect	**b.#**	3D-QSAR has a predictive quality which QSAR does not.
Correct	**c.#**	Experimental parameters are required by 3D-QSAR and QSAR.
incorrect	**d.#**	Results can be shown graphically in 3D-QSAR, but not by QSAR.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 19 - Question 01		
01)	Which of these statements is true about the following structure? 	
Feedback: The structure shown is prontosil. Prontosil acts as a prodrug and is converted to sulphanilamide in the body. Since it needs to be metabolised to the active compound, it is inactive when tested *in vitro*. Prontosil itself is inactive.		
incorrect	**a.#**	It is a prodrug which is converted to prontosil in the body.
Correct	**b.#**	It is a prodrug which is converted to sulphanilamide in the body.
incorrect	**c.#**	It was the first active sulphonamide.
incorrect	**d.#**	It shows antibacterial activity *in vitro*.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 19 - Question 02		
02)	What type of infection is succinyl sulphathiazole used for? 	
Feedback: Succinyl sulfathiazole is a prodrug. Since the prodrug is ionised and polar, it fails to cross the gut wall and is concentrated in the gut when taken orally. It is, therefore, effective against gut infections.		
incorrect	**a.#**	Urinary tract infections.
incorrect	**b.#**	Eye infections.
incorrect	**c.#**	Mucous membrane infection.
Correct	**d.#**	Gut infections.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 19 - Question 03		
03)	What structure or structures is/are formed if the acyl side chain of a penicillin is hydrolysed?	
Feedback: Penicillenic acids and penillic acids are formed if the acyl side chain promotes the hydrolysis of the β lactam ring. Removing the acyl side chain gives 6-APA. The amino group is at position 6 of the penicillin nucleus and not position 7.		
incorrect	**a.#**	Penicillenic acids.
incorrect	**b.#**	Penillic acids.
incorrect	**c.#**	7-Aminopenicillanic acid.

Correct	**d.#**	6-Aminopenicillanic acid.
END OF QUESTION		

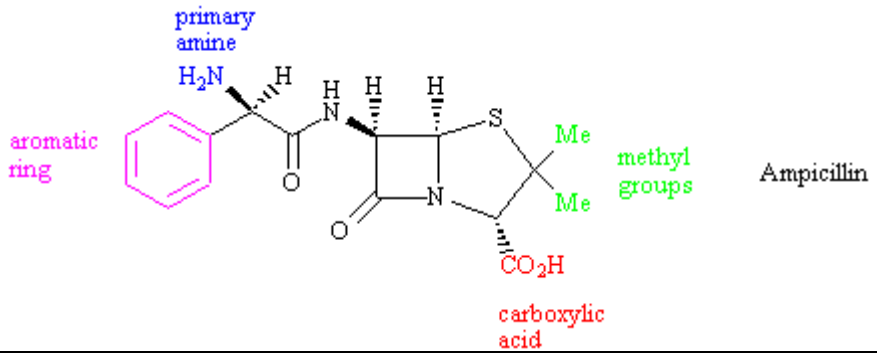
Type: multiple choice		
Title: Chapter 19 - Question 04		
04)	Penicillins act on an important bacterial enzyme called transpeptidase. What role does this enzyme play?	
Feedback: Transpeptidase catalyses the final crosslinking stage in cell wall synthesis. The reaction L-alanine to D-alanine is carried out by L-Ala racemase. The transport of building blocks across the cell membrane is carried out by a carrier lipid and this is unaffected by penicillins.		
incorrect	**a.#**	It racemises L-alanine to D-alanine prior to its incorporation into the bacterial cell wall.
incorrect	**b.#**	It acylates amino acid residues in the bacterial cell wall.
Correct	**c.#**	It catalyses the final crosslinking stage in cell wall synthesis.
incorrect	**d.#**	It catalyses the transport of the building blocks for cell wall synthesis across the cell membrane.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 19 - Question 05		
05)	Which of the following statements best describes the mechanism of action of penicillins?	
Feedback: A serine residue acts as a nucleophile in the active site of the enzyme and not lysine. It reacts with the β-lactam ring and not the carboxylate group. The latter group is ionised and forms an ionic interaction with the active site, but this is not involved in the mechanism leading to irreversible inhibition.		
incorrect	**a.#**	A serine residue in the active site of the enzyme acts as a nucleophile and reacts with the carboxylate group of penicillin to form an ester.
incorrect	**b.#**	A lysine residue in the active site of the enzyme acts as a nucleophile and reacts with the carboxylate group of penicillin to form an ester.
Correct	**c.#**	A serine residue in the active site of the enzyme acts as a nucleophile and reacts with the β-lactam ring.
incorrect	**d.#**	A lysine residue in the active site of the enzyme acts as a nucleophile and reacts with the β-lactam ring.
END OF QUESTION		

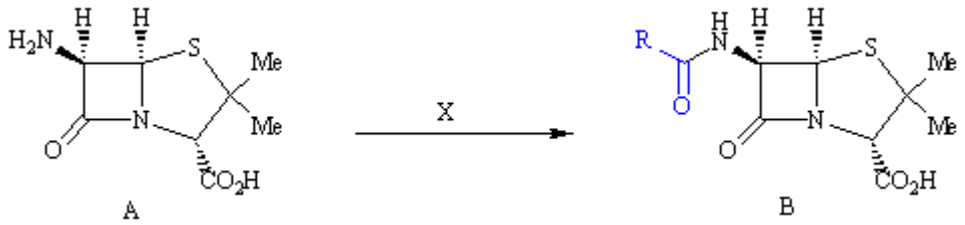
Type: multiple choice		
Title: Chapter 19 - Question 06		
06)	Which of the following statements best describes the importance of the bicyclic system in penicillins to antibacterial activity?	
Feedback: There is some truth in all the statements but the most important influence of the bicyclic system is to increase the reactivity of the β-lactam ring to attacking nucleophiles by increasing bond angle strain.		
incorrect	**a.#**	It acts as a scaffold to hold the carboxylate group and acyl side chain in the correct relative orientations for binding.
incorrect	**b.#**	It is the correct shape and size to fit the binding site.
Correct	**c.#**	It increases the strain of the β-lactam ring and increases reactivity and antibacterial activity.

incorrect	**d.#**	It is a folded ring system and protects the β-lactam ring from hydrolysis.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 19 - Question 07		
07)	What tactic is successfully used to increase the stability of penicillins to acid hydrolysis whilst retaining antibacterial activity?	
Feedback: Adding an electron-withdrawing group to the side chain decreases the nucleophilicity of the carbonyl group in that side chain, and reduces the rate at which it participates in the opening of the β-lactam ring. Expanding a ring to reduce ring strain would increase the stability of the penicillin but would also reduce activity. Adding an electron-withdrawing group to the β-lactam ring would not be easy to do synthetically and is likely to affect activity adversely. Removing the acyl side chain would increase stability since it removes the carbonyl group on the side chain, but biological activity would also be lost since important binding groups are lost as well.		
incorrect	**a.#**	Expanding one or other ring to relieve ring strain.
incorrect	**b.#**	Adding an electron withdrawing group to the β-lactam ring.
Correct	**c.#**	Adding an electron withdrawing group to the acyl side chain.
incorrect	**d.#**	Removing the acyl side chain.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 19 - Question 08		
08)	What feature of ampicillin is important in its acid stability?  	
Feedback: The primary amine acts as an electron withdrawing group and decreases the nucleophilicity of the carbonyl group in the side chain. This reduces the rate at which the carbonyl group participates in opening the β-lactam ring.		
incorrect	**a.#**	The carboxylic acid.
incorrect	**b.#**	The aromatic ring.
Correct	**c.#**	The primary amine.
incorrect	**d.#**	The methyl groups.
END OF QUESTION		

Type: multiple choice
Title: Chapter 19 - Question 09

09)	The following reaction is carried out to synthesise a range of semi-synthetic penicillins.  How is structure A (6-aminopenicillanic acid) obtained?

Feedback: 6-APA can be obtained by the enzymatic removal of the acyl side chain of benzyl penicillin.
A total synthesis is impractical since it would take too long and would be low yielding.
Converting a penicillin structure to a cephalosporin structure can be done but not the other way round.
Benzyl penicillin could be converted to 6-APA by a chemical hydrolysis, but not with sodium hydroxide. This would also open the β-lactam ring.

incorrect	**a.#**	By total synthesis.
incorrect	**b.#**	By ring contraction of a cephalosporin.
Correct	**c.#**	By enzymatic hydrolysis of benzyl penicillin.
incorrect	**d.#**	By chemical hydrolysis of benzyl penicillin with sodium hydroxide.

END OF QUESTION

Type: multiple choice

Title: Chapter 19 - Question 10

10)	What is the relevance of β-lactamase enzymes to penicillins?

Feedback: β-Lactamase enzymes catalyse the opening of β-□lactam rings and result in the deactivation of penicillins. Thus, bacteria that are able to produce these enzymes are resistant to penicillins and are not susceptible. Fungi and mammalian cells do not produce β-lactamase enzymes.

incorrect	**a.#**	Fungi containing β-lactamase enzymes can synthesise penicillins.
Correct	**b.#**	Bacteria containing β-lactamase enzymes show resistance to penicillins.
incorrect	**c.#**	Mammalian cells are unaffected by penicillins because they contain β-lactamase enzymes.
incorrect	**d.#**	Only bacteria containing β-lactamase enzymes are susceptible to penicillins.

END OF QUESTION

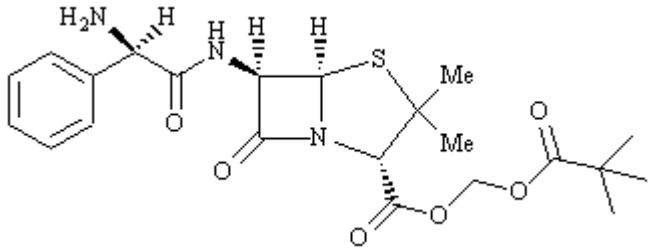
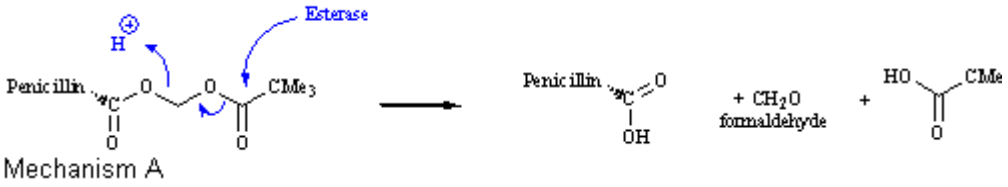
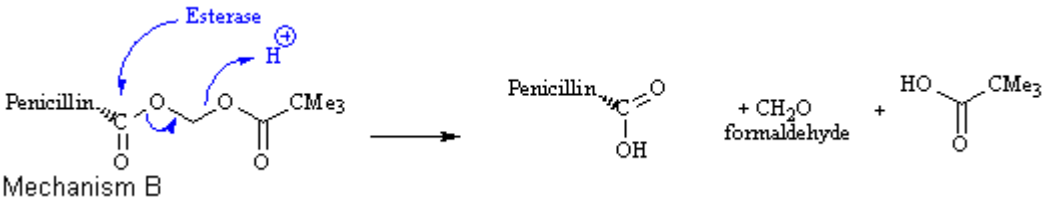
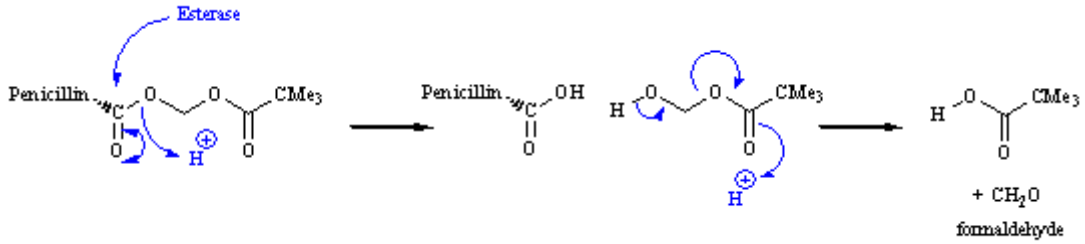
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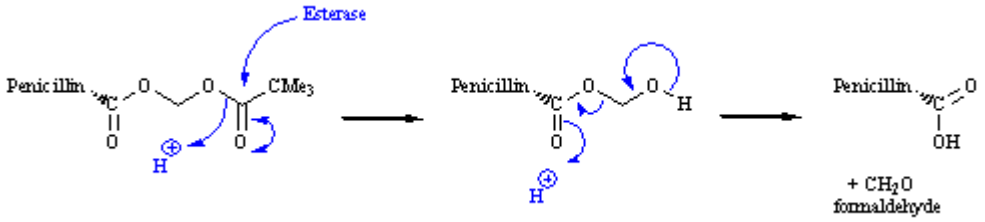
Title: Chapter 19 - Question 11

11)	Why is a methyl ester of a penicillin not used as a prodrug?

Feedback: The methyl ester is too close to the penicillin skeleton. As a result, the latter acts as a steric shield and prevents esterase enzymes from hydrolysing the ester.
Resistance to acid catalysed hydrolysis in the digestive system would be an advantage since it would allow the prodrug to survive the acids of the stomach and to be absorbed more effectively.
Inactivity is not an issue since prodrugs are not meant to be active.
Hydrolysis of the methyl ester to produce methanol is a drawback, but the level of

methanol produced from a typical dose of penicillin would be relatively low and unlikely to be toxic.		
incorrect	**a.#**	The methyl ester is resistant to acid-catalysed hydrolysis.
incorrect	**b.#**	The methyl ester is inactive.
Correct	**c.#**	The methyl ester is resistant to enzymatic hydrolysis in humans.
incorrect	**d.#**	The methyl ester would produce toxic methanol on hydrolysis.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 19 - Question 12	
12)	Pivampicillin (below) is used as a prodrug for ampicillin.  Which of the following mechanisms best describes the conversion of the prodrug to ampicillin?  Mechanism A  Mechanism B  Mechanism C

	 Mechanism D

Feedback: The mechanism shown in D is the most accurate. The mechanisms shown in B and C are both wrong since the esterase enzyme is shown attacking the ester group closest to the penicillin nucleus, and it is known that this ester is shielded from attack. The mechanism in A suggests that ampicillin is released as a direct result of the enzyme action, when in fact a chemically unstable intermediate is formed first as shown in mechanism D.

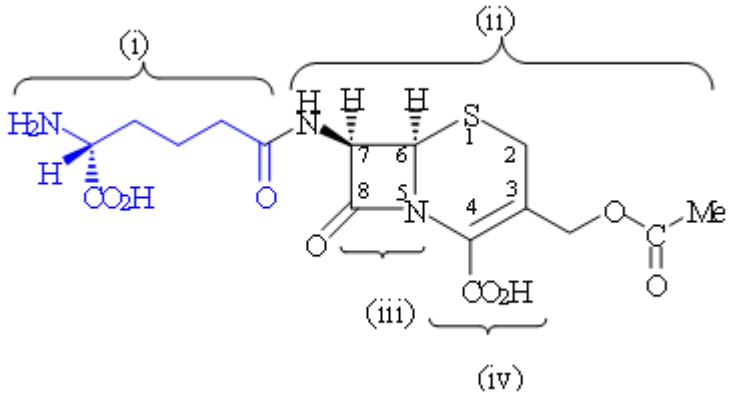
incorrect	**a.#**	Mechanism A.
incorrect	**b.#**	Mechanism B.
incorrect	**c.#**	Mechanism C.
Correct	**d.#**	Mechanism D.

END OF QUESTION

Type: matching

Title: Chapter 19 - Question 13

13) Identify the component parts (i), (ii), (iii), and (iv) of cephalosporin C.



Feedback: The side chain is the 7-aminoadipic group. The bicyclic system including the ring substituents and the 7-amino group is 7-aminocephalosporinic acid. The 4-membered ring is a β-lactam ring. The 6-membered ring is a dihydrothiazine ring.

a.#	(i)	=	7-aminoadipic group
b.#	(ii)	=	7-aminocephalosporinic acid
c.#	(iii)	=	β-Lactam
d.#	(iv)	=	Dihydrothiazine

END OF QUESTION

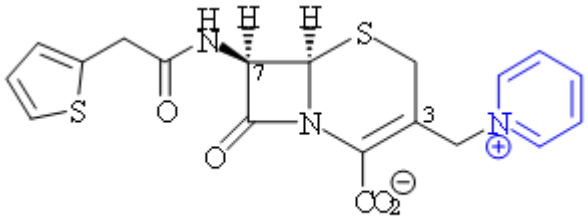
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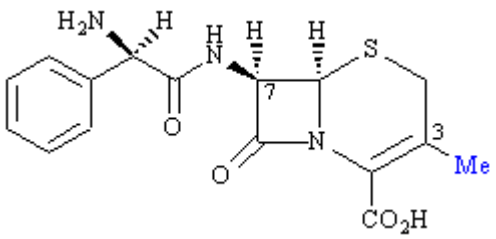
Title: Chapter 19 - Question 14

14) Which of the following statements is true?

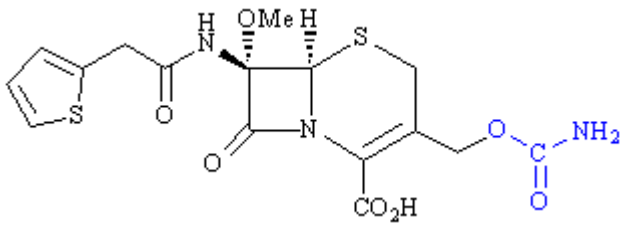
Feedback: Cephalosporin analogues can be prepared from 7-aminocephalosporinic acid (7-ACA). None of the procedures described in the remaining options are feasible for cephalosporins. 7-ACA was prepared from cephalosporins by removing the 7-acyl chain with a chemical procedure.

Correct	**a.#**	Cephalosporin analogues can be prepared by a semi-synthetic synthesis.
incorrect	**b.#**	Cephalosporin analogues can be prepared by fermentation.
incorrect	**c.#**	7-Aminocephalosporinic acid can be prepared by fermentation, then used in the synthesis of cephalosporin analogues.
incorrect	**d.#**	7-Aminocephalosporinic acid can be prepared by the enzymatic cleavage of the 7-acyl side chain of cephalosporin C, then used in the synthesis of cephalosporin analogues.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 19 - Question 15		
15)	Cephaloridine has a pyridinium ring in place of the acetyl group in cephalothin. Which of the following statements is true?  Cephaloridine	
Feedback: Cephaloridine is poorly absorbed through the gut wall and has to be injected. It has a longer duration of action since the acyl group that was replaced is susceptible to enzymatic hydrolysis. The pyridinium ring does act as a good leaving group in the inhibition mechanism. It is also less easily metabolised than the acetyl group of cephalothin.		
incorrect	**a.#**	Cephaloridine is more easily absorbed through the gut wall.
incorrect	**b.#**	Cephaloridine has a shorter duration of action than cephalothin.
Correct	**c.#**	The pyridinium ring acts as a good leaving group in the inhibition mechanism.
incorrect	**d.#**	The pyridinium ring is more easily metabolised than the acetyl group of cephalothin.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 19 - Question 16		
16)	The following structure (cefalexin) is a first-generation cephalosporin.  Which of the following statements is true for the methyl substituent at position 3?	

Feedback: A methyl group at the 3-position is generally **bad** for activity since it is a **poor** leaving group. However, if a suitable 7-acyl side chain is present, the methyl group at position 3 is good for oral activity since it aids oral absorption. It does **not** act as a steric shield.		
incorrect	**a.#**	It is a good leaving group.
incorrect	**b.#**	It is generally good for activity.
Correct	**c.#**	It is good for oral activity.
incorrect	**d.#**	It acts as a steric shield.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 19 - Question 17	
17)	The following structure is an example of a cephamycin.  What is the significance of the urethane group (in blue)?

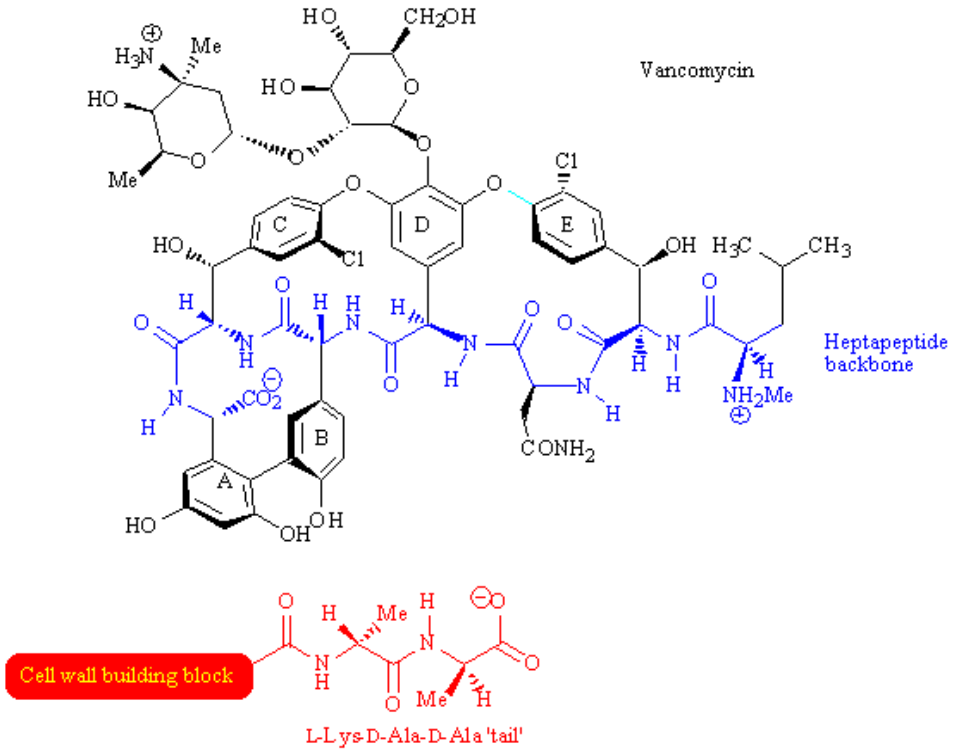
Feedback: Compared to an ester group, a urethane is more stable to acid and enzyme catalysed hydrolysis due to the influence of the nitrogen's lone pair of electrons on the neighbouring carbonyl group. The urethane group has no steric effect.		
incorrect	**a.#**	It acts as a steric shield leading to longer activity.
Correct	**b.#**	It is resistant to hydrolysis by esterases leading to longer activity.
incorrect	**c.#**	It acts as a steric shield for the β-lactam ring leading to lower activity.
incorrect	**d.#**	It acts as a polar group resulting in better water solubility.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 19 - Question 18	
18)	What molecule is mimicked by the antibacterial agent D-cycloserine?

Feedback: D-Cycloserine mimics D-alanine and inhibits L-Ala racemase and D-Ala-D-Ala ligase. The former enzyme catalyses the racemisation of L-Ala to D-Ala. The latter enzyme catalyses the linkage of two D-Ala amino acids to give the dipeptide D-Ala-D-Ala.		
incorrect	**a.#**	L-alanine.
Correct	**b.#**	D-alanine.
incorrect	**c.#**	Glycine.
incorrect	**d.#**	Penicillin.
END OF QUESTION		

Type: multiple choice

Title: Chapter 19 - Question 19		
19)	Vancomycin forms interactions to a 'tail' segment of an important biosynthetic building block. What amino acids are involved in the tail segment?	
Feedback: Lysine has a butyl side chain with a primary amino group at the end. Alanine has a simple methyl side chain. The side chains themselves are not involved in the important interactions with vancomycin.		
incorrect	**a.#**	D-Lys-L-Ala-L-Ala.
incorrect	**b.#**	L-Ala-D-Lys-D-Lys.
incorrect	**c.#**	D-Ala-L-Lys-L-Lys.
Correct	**d.#**	L-Lys-D-Ala-D-Ala.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 19 - Question 20		
20)	Vancomycin forms interactions to a 'tail' segment of an important biosynthetic building block. How many of these interactions are there? 	
Feedback: There are three interactions involving the terminal alanine unit of the tail, one interaction involving the penultimate alanine, and one interaction involving the lysine unit.		
incorrect	**a.#**	2.
incorrect	**b.#**	3.
incorrect	**c.#**	4.
Correct	**d.#**	5.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 19 - Question 21		
21)	Which of the following statements is true regarding vancomycin?	
Feedback: Vancomycin is not an enzyme inhibitor. It 'caps' the building blocks for cell wall synthesis and acts as a steric shield, preventing these substrates from entering the active sites of enzymes.		
incorrect	**a.#**	It inhibits the transglycosidase enzyme.
incorrect	**b.#**	It inhibits the transpeptidase enzyme.
incorrect	**c.#**	It inhibits both enzymes.
Correct	**d.#**	It inhibits neither enzyme.
END OF QUESTION		

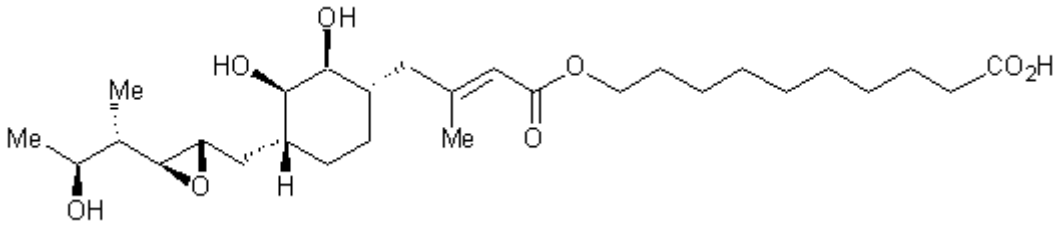
Type: multiple choice		
Title: Chapter 19 - Question 22		
22)	What is meant by VRSA?	
Feedback: The correct answer is 'Vancomycin-*resistant Staphylococcus aureus'*. The terms 'Vigorously resistant *Staphylococcus aureus*' and 'Vancomycin immune resistant *Staphylococcus aureus*' do not exist. The term 'vancomycin-resistant enterococci' (VRE) is used and refers to different microorganisms that are resistant to vancomycin and which can cause life-threatening gut infections.		
incorrect	**a.#**	Vigorously-resistant *Staphylococcus aureus.*
Correct	**b.#**	Vancomycin-resistant *Staphylococcus aureus.*
incorrect	**c.#**	Vancomycin-resistant *Enterococcus faecalis.*
incorrect	**d.#**	Vancomycin-immune resistant *Staphylococcus aureus.*
END OF QUESTION		

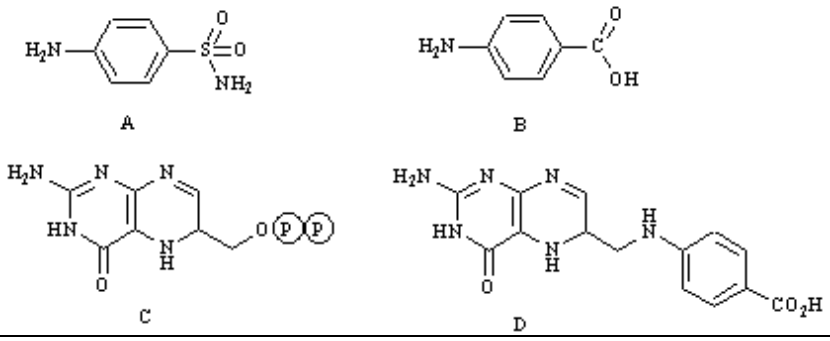
Type: multiple choice		
Title: Chapter 19 - Question 23		
23)	Vancomycin-resistant enterococci have resulted due to a modification of a D-Ala group in a cell wall precursor. What modification has taken place?	
Feedback: The terminal group is lactic acid rather than alanine. Lactic acid has a hydroxyl group instead of the amino group present in alanine. This means that the terminal unit is linked to the 'tail' of the biosynthetic precursor by an ester group rather than an amide group.		
Correct	**a.#**	D-Alanine to D-lactic acid.
incorrect	**b.#**	D-Alanine to L-alanine.
incorrect	**c.#**	D-Alanine to D-lysine.
incorrect	**d.#**	D-Alanine to L-lysine.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 19 - Question 24		
24)	Aminoglycosides are given an enhanced activity in the presence of vancomycin. Why does this occur?	
Feedback: Disruption of the cell membrane structure allows aminoglycosides to enter the bacterial cell more easily. Disruption of the bacterial cell wall has no significant effect on the ease with which aminoglycosides enter the cell, since these		

molecules can pass through cell walls in any case. Vancomycin disrupts RNA synthesis and this would disrupt protein synthesis. Therefore, the disruption of RNA synthesis and reduction of the extent to which other antibiotics are metabolised or pumped back out of the bacterial cell are not inconceivable. However, it is vancomycin's effect on cell membranes that explains why aminoglycoside activity is enhanced.		
Correct	**a.#**	Vancomycin disrupts the bacterial cell membrane making it easier for the above antibiotics to enter the bacterial cell.
incorrect	**b.#**	Vancomycin disrupts the bacterial cell wall making it easier for the above antibiotics to enter the bacterial cell.
incorrect	**c.#**	Vancomycin disrupts RNA synthesis and reduces the extent to which other antibiotics are metabolised.
incorrect	**d.#**	Vancomycin disrupts RNA synthesis and reduces the extent to which other antibiotics are pumped back out of the bacterial cell.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 19 - Question 25		
25)	What cell structure is targeted by drugs such as the macrolides, tetracyclines, and aminoglycosides?	
Feedback: These compounds bind to bacterial ribosomes and disrupt the translation process leading to the synthesis of proteins.		
Correct	**a.#**	Ribosomes.
incorrect	**b.#**	Nucleus.
incorrect	**c.#**	Cell membrane.
incorrect	**d.#**	Cell wall.
END OF QUESTION		

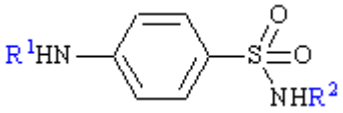
Type: multiple choice		
Title: Chapter 19 - Question 26		
26)	The following structure (mupirocin) has been used as a topical agent for skin infections, and as a means of combating the transmission of *Staphylococcus aureus* in hospitals.  What is the target for the above structure?	
Feedback: The target is isoleucyl tRNA synthetase. Transpeptidase is the target for penicillins and cephalosporins. DNA-dependant RNA-polymerase is the target for rifampicin. Topoisomerases are the target for quinolones and isoquinolones		
incorrect	**a.#**	Transpeptidase.
incorrect	**b.#**	DNA-dependant RNA polymerase.
incorrect	**c.#**	Topoisomerases.
Correct	**d.#**	Isoleucyl tRNA synthetase.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 19 - Question 27		
27)	Which of the following structures do the sulphonamides mimic when acting as enzyme inhibitors?  	
Feedback: Structure B is *para*-aminobenzoic acid, which is a substrate for the enzyme dihydropteroate synthetase. The sulphonamides mimic this feature. Structure A is sulphanilamide which is a sulphonamide. Structure B is the other substrate for dihydropteroate synthetase, and which gets linked to *para*-aminobenzoic acid. Structure D is dihydropteroate which is the product from the enzyme-catalysed reaction catalysed by dihydropteroate synthetase.		
incorrect	**a.#**	Structure A.
Correct	**b.#**	Structure B.
incorrect	**c.#**	Structure C.
incorrect	**d.#**	Structure D.
END OF QUESTION		

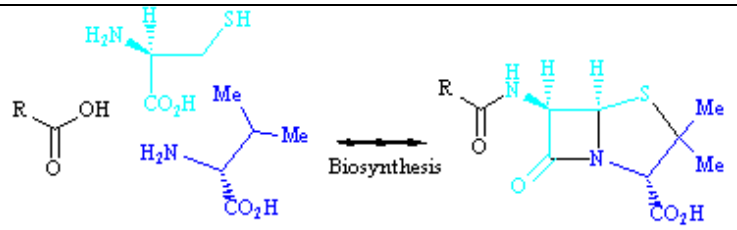
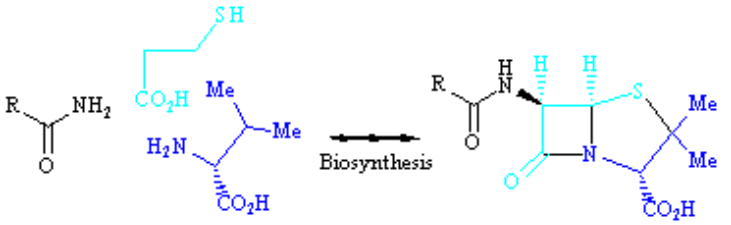
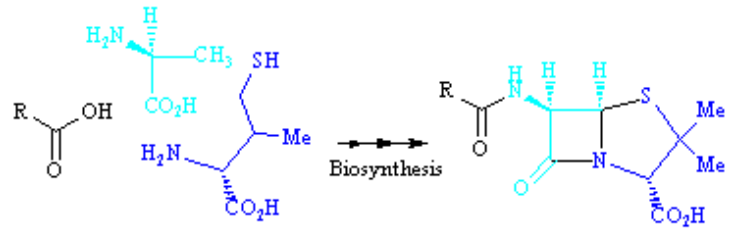
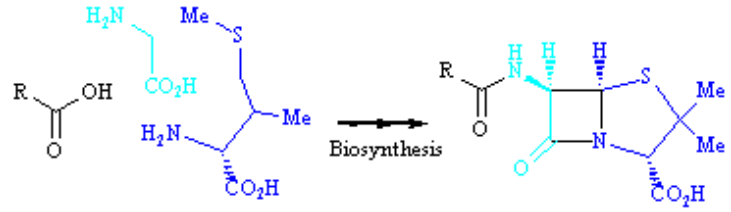
Type: multiple choice		
Title: Chapter 19 - Question 28		
28)	Which of the following statements is untrue?	
Feedback: A bacterial cell is prokaryotic whereas an animal cell is eukaryotic.		
Correct	**a.#**	A bacterial cell is eukaryotic whereas an animal cell is prokaryotic.
incorrect	**b.#**	A bacterial cell has a cell wall whereas an animal cell does not.
incorrect	**c.#**	Bacterial cells do not have a well defined nucleus, whereas animal cells do.
incorrect	**d.#**	Bacteria may contain enzymes that are not present in animal cells.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 19 - Question 29	
29)	What is meant by the term sequential blocking?
Feedback: Sequential blocking is used to inhibit two separate enzymes in the one biosynthetic pathway. It is more effective than inhibiting one or other of the separate enzymes. It allows lower and safer doses of each inhibitor to achieve the same effect as using one inhibitor. 'Administering two different drugs' is true to the extent that two separate drugs are used,	

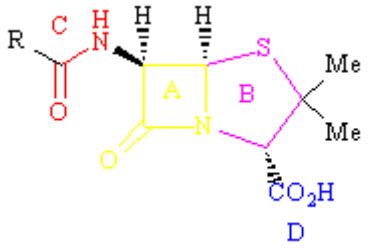
but it is not specified whether the drugs act on targets within the same biosynthetic pathway. There is no truth in the remaining two options.		
incorrect	**a.#**	Administering two different drugs.
Correct	**b.#**	The inhibition of two different enzymes in the same biosynthetic pathway.
incorrect	**c.#**	The order in which different binding regions of a binding site are occupied by a drug structure.
incorrect	**d.#**	The protection of vulnerable functional groups in a drug.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 19 - Question 30	
30)	The following general structure is representative of sulphonamides. Which of the following statements is true for active sulphonamides? 
Feedback: R^1 can be a hydrogen or an acyl group. The latter is removed in the body by enzyme-catalysed hydrolysis to release a primary amine which is essential for activity. A variety of groups can be placed at R^2. The aromatic ring and sulphonamide group are both essential for activity.	
Correct	**a.#** R^1 can be H or an acyl group.
incorrect	**b.#** R^2 must be hydrogen.
incorrect	**c.#** The aromatic ring is not essential.
incorrect	**d.#** The sulphonamide functional group can be replaced with an ester.
END OF QUESTION	

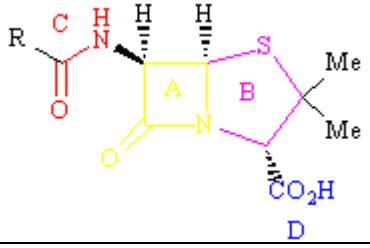
Type: multiple choice	
Title: Chapter 19 - Question 31	
31)	Which of the following diagrams represents the building blocks used in the biosynthesis of penicillins?

	A  B  C  D 	
Feedback: The building blocks are the two amino acids cysteine and valine, as well as a carboxylic acid as the side chain. Cysteine provides the sulfur atom in the penicillin structure, while valine is the source for the geminal dimethyl groups. (Geminal means that the methyl groups are attached to the same carbon).		
Correct	**a.#**	Diagram A.
incorrect	**b.#**	Diagram B.
incorrect	**c.#**	Diagram C.
incorrect	**d.#**	Diagram D.
END OF QUESTION		

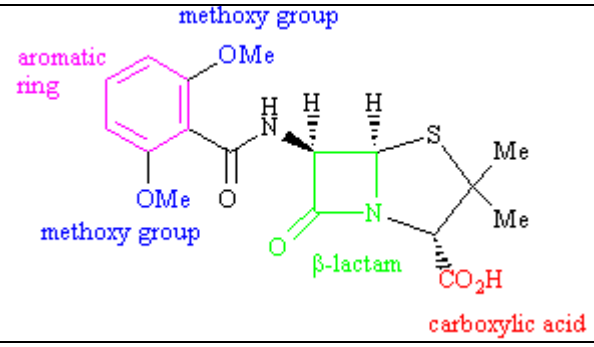
Type: matching	
Title: Chapter 19 - Question 32	
32)	The diagram shows a general structure of penicillins.

	 Match the following functional groups or rings with the regions shown.

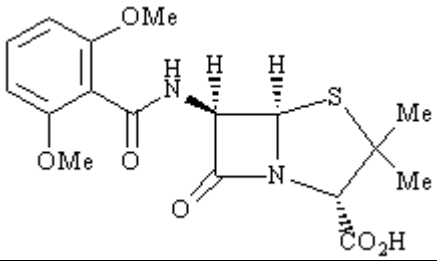
Feedback: All four groups are important to the activity of penicillins			
a.#	Carboxylic acid	=	Region D.
b.#	Seconday amide	=	Region C.
c.#	Thiazolidine	=	Region B.
d.#	β-Lactam	=	Region A.
END OF QUESTION			

Type: multiple choice		
Title: Chapter 19 - Question 33		
33)	What crucial feature of a penicillin is involved in its mechanism of action? 	
Feedback: Region A is the β-lactam ring which is reactive to nucleophiles. The carboxylic acid and acyl side chain are important binding groups in binding the penicillin to the active site prior to reaction. The thiazolidine ring increases the reactivity of the β-lactam ring due to increased ring strain.		
incorrect	**a.#**	Region D.
Correct	**b.#**	Region A.
incorrect	**c.#**	Region C.
incorrect	**d.#**	Region B.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 19 - Question 34	
34)	Methicillin is an important penicillin that was introduced in the 1960s to counter the threat of penicillin-resistant strains of *Staphylococcus aureus*. Which group or groups in methicillin plays an important role in making this penicillin more effective than penicillin G against resistant strains of *S. aureus*?

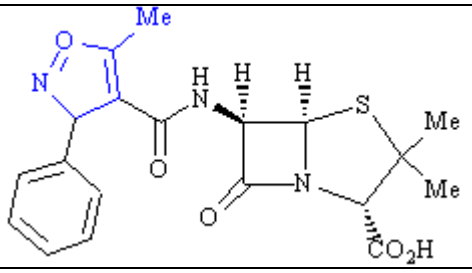
	

Feedback: The methoxy groups act as steric shields to hinder the ability of β-lactamases to hydrolyse the β□-lactam ring.		
incorrect	**a.#**	The aromatic ring.
Correct	**b.#**	The methoxy groups.
incorrect	**c.#**	The β-lactam ring.
incorrect	**d.#**	The carboxylic acid.
END OF QUESTION		

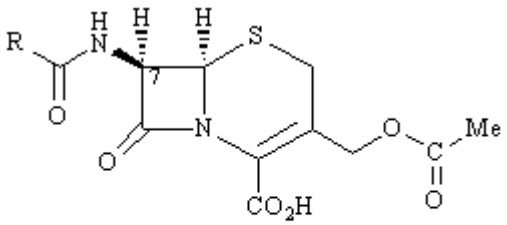
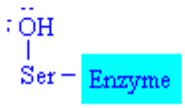
Type: multiple choice	
Title: Chapter 19 - Question 35	
35)	Methicillin is an important penicillin that was introduced in the 1960s to counter the threat of penicillin-resistant strains of *Staphylococcus aureus*. Which of the following statements is false? 

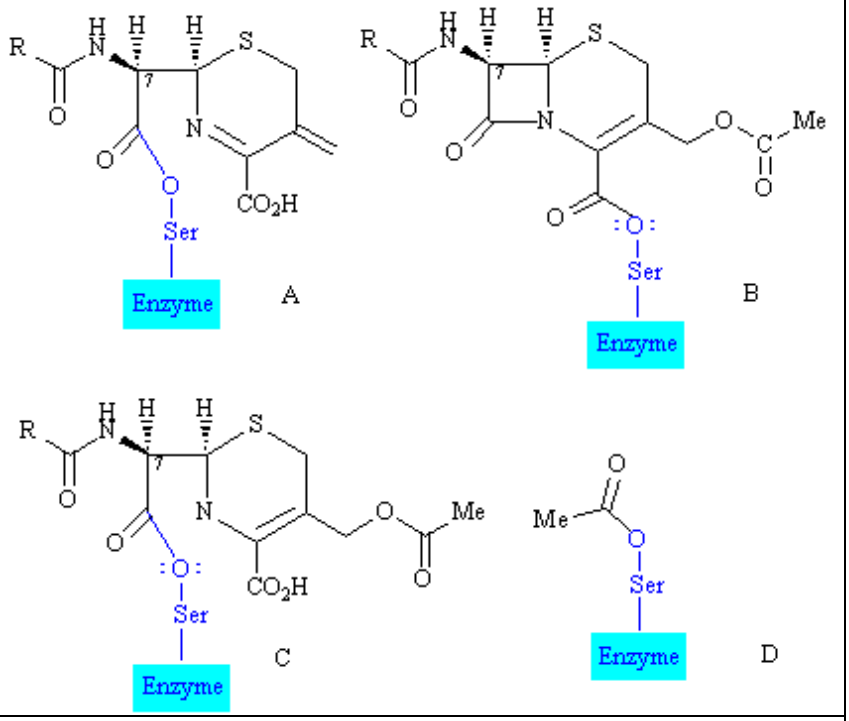
Feedback: The methoxy groups may well be electron withdrawing in an inductive sense, but they can also interact with the ring through resonance. This has the effect of increasing electron density in the ring and making it electron rich. As a result, the aromatic ring cannot act as an electron-withdrawing group with respect to the neighbouring side chain amide, and so the structure is acid sensitive.		
Correct	**a.#**	The methoxy groups on the aromatic ring withdraw electrons from the ring making it electron deficient.
incorrect	**b.#**	The methoxy groups act as steric shields for the β-lactam ring.
incorrect	**c.#**	Replacing ethoxy groups on the aromatic ring would increase acid stability but decrease activity.
incorrect	**d.#**	The methoxy groups are at the *ortho* position of the aromatic ring. If they were at the *meta* position, they would be less effective.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 19 - Question 36	
36)	Oxacillin was introduced as an improvement over methicillin for the treatment of penicillin-resistant strains of *Staphylococcus aureus.* Which of the following statements is false?

Feedback: The blue coloured ring has an electron-**withdrawing** effect on the neighbouring amide group, and it is this effect which stabilises the molecule to acid and allows it to be administered orally.		
Correct	**a.#**	The feature coloured blue has an electron-donating effect on the neighbouring amide.
incorrect	**b.#**	The feature coloured blue has a steric effect which protects the β-lactam ring.
incorrect	**c.#**	The feature coloured blue makes the structure more stable to acid.
incorrect	**d.#**	The feature coloured blue makes the structure orally active.
END OF QUESTION		

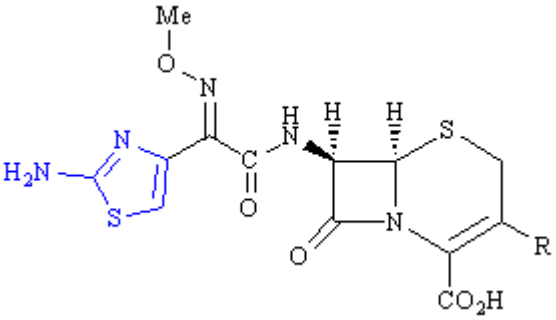
Type: multiple choice	
Title: Chapter 19 - Question 37	
37)	A serine residue in the active site of the transpeptidase enzyme reacts with cephalosporins, resulting in a covalent bond being formed between the enzyme and the cephalosporin.   What is the structure of this enzyme-bound product?

Feedback: The serine residue reacts with the β-lactam ring of cephalosporins and so options B and D are wrong. Option C is wrong because ring opening coincides with a concerted mechanism that sees the expulsion of the acetoxy moiety as an acetate ion. The double bond in the dihydrothiazine ring is important in allowing this mechanism to take place.

Correct	**a.#**	Structure A.
incorrect	**b.#**	Structure B.
incorrect	**c.#**	Structure C.
incorrect	**d.#**	Structure D.

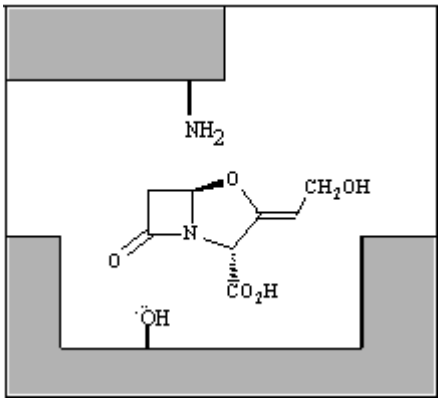
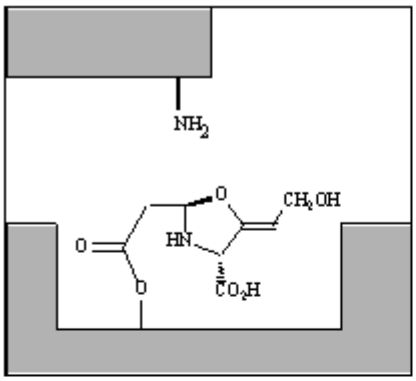
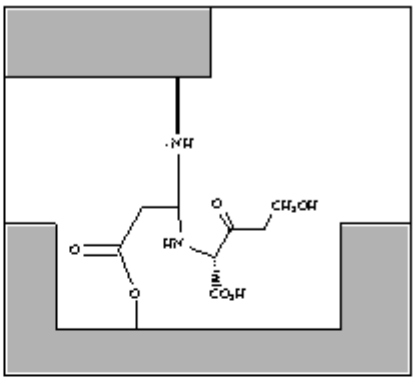
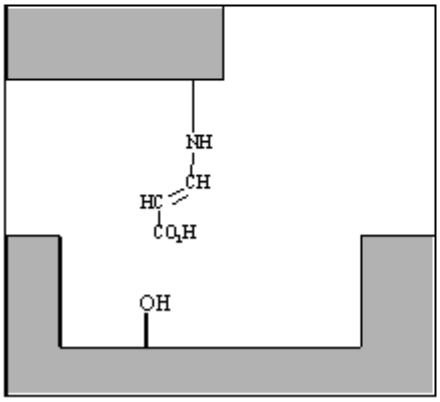
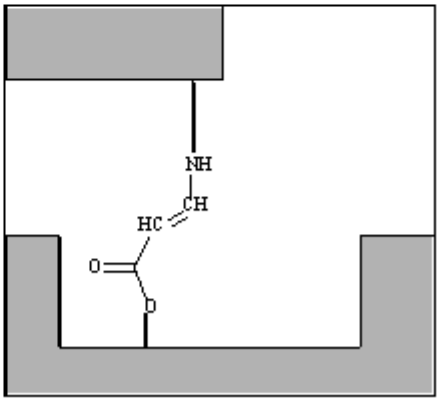
END OF QUESTION

Type: multiple choice	
Title: Chapter 19 - Question 38	
38)	Third and fourth generation cephalosporins contain an aminothiazole ring (shown in blue). Which of the following statements is false? 

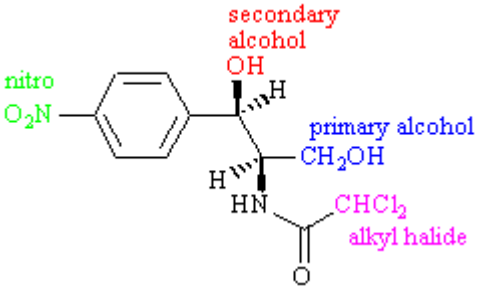
Feedback: The advantages of having an aminothiazole ring present are in increasing binding interactions and penetration through the outer membrane of Gram negative bacteria. It has no influence on the reactivity of the β-lactam ring.

incorrect	**a.#**	Third- and fourth-generation cephalosporins show enhanced activity against Gram negative bacteria.
incorrect	**b.#**	The aminothiazole ring enhances the penetration of

		cephalosporins through the outer membrane of Gram negative bacteria.
incorrect	**c.#**	The aminothiazole ring may increase affinity for the transpeptidase enzyme.
Correct	**d.#**	The aminothiazole ring increases the reactivity of the β-lactam ring.
END OF QUESTION		

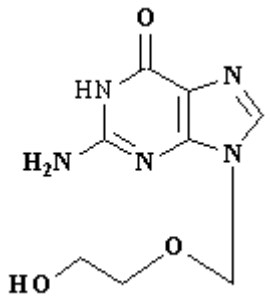
Type: multiple choice	
Title: Chapter 19 - Question 39	
39)	The following diagram shows clavulanic acid in the active site of a β-lactamase enzyme.  What is the final result of the interaction of clavulanic acid with the enzyme?  A  B  C  D

Feedback: Diagrams A and B are formed in the mechanism leading to the final structure. Diagram C would occur if the ester bond was broken, but this reaction is very slow.		
incorrect	**a.#**	Diagram A.
incorrect	**b.#**	Diagram B.
incorrect	**c.#**	Diagram C.
Correct	**d.#**	Diagram D.
END OF QUESTION		

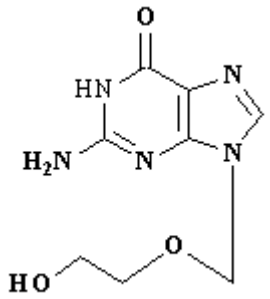
Type: multiple choice		
Title: Chapter 19 - Question 40		
40)	Which functional group is thought to be chiefly responsible for the toxicity of chloramphenicol?  	
Feedback: Aromatic nitro groups are normally avoided if possible since they commonly cause toxic effects.		
incorrect	**a.#**	A secondary alcohol.
Correct	**b.#**	An aromatic nitro group.
incorrect	**c.#**	An alkyl chloride.
incorrect	**d.#**	A primary alcohol.
END OF QUESTION		

Type: matching			
Title: Chapter 20 - Question 01			
01)	Match the following terms and definitions.		
Feedback: The purpose of the capsid is to protect the viral nucleic acid when the virion moves between host cells.			
a.#	The protein coat that contains the nucleic acid of a virus	=	Capsid
b.#	The form a virus takes outside the host cell	=	Virion
c.#	The protein subunits that make up the protein coat of a virus	=	Protomer
d.#	The assembly of viral nucleic acid within a protein coat	=	Nucleocapsid
END OF QUESTION			

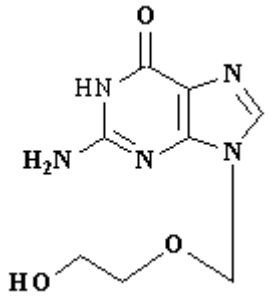
Type: multiple choice		
Title: Chapter 20 - Question 02		
02)	What is meant by (+) ssRNA?	
Feedback: RNA stands for ribonucleic acid. The (+) indicates that the RNA is identical to the messenger RNA used to code for viral proteins. The letters ss stand for single strand.		
incorrect	**a.#**	Single stranded deoxyribonucleic acid that is complementary to the messenger RNA used to code for viral proteins.
incorrect	**b.#**	Double stranded ribonucleic acid.
Correct	**c.#**	Single stranded ribonucleic acid that is identical to the messenger RNA used to code for viral proteins.
incorrect	**d.#**	Single stranded ribonucleic acid that is complementary to the messenger RNA used to code for viral proteins.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 20 - Question 03		
03)	Which of the following is not treated by aciclovir? 	
Feedback: Aciclovir is used against diseases caused by the herpes simplex virus and the varcella-zoster virus. Both of these are DNA viruses, whereas HIV is an RNA virus.		
incorrect	**a.#**	Chickenpox.

Correct	**b.#**	Human immunodeficiency virus.
incorrect	**c.#**	Cold sores.
incorrect	**d.#**	Shingles.
END OF QUESTION		

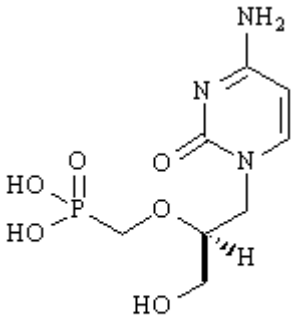
Type: multiple choice	
Title: Chapter 20 - Question 04	
04)	Which enzyme is inhibited by aciclovir? 
Feedback: DNA polymerase is the enzyme which duplicates new DNA using each strand of parent DNA as a template for the construction. The enzymes reverse transcriptase and protease are present in HIV which is an RNA virus. They are not present in the DNA viruses targeted by aciclovir. RNA polymerase is present in the flu virus which is an RNA virus. It is not present in the viruses targeted by aciclovir.	

Correct	**a.#**	DNA polymerase.
incorrect	**b.#**	RNA polymerase.
incorrect	**c.#**	Reverse transcriptase.
incorrect	**d.#**	Protease.
END OF QUESTION		

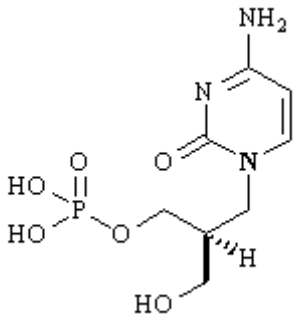
Type: multiple choice	
Title: Chapter 20 - Question 05	
05)	Why is aciclovir selective for virally infected cells? 
Feedback: Aciclovir is phosphorylated by viral thymidine kinase to the monophosphate which is then taken on to the active triphosphate by the action of cellular kinases. Aciclovir is a poor substrate for cellular thymidine kinase and so the drug is activated more slowly in non-infected cells. There is certainly a selective uptake of aciclovir into infected cells, but the drug can still enter non-infected cells. The activated form of aciclovir can inhibit cellular DNA polymerase in normal cells. However, this inhibition is weaker than the inhibition of viral DNA polymerase. The drug is not rapidly metabolised in non-infected cells.	

incorrect	**a.#**	It cannot enter normal cells.
Correct	**b.#**	It can only be activated in the presence of viral thymidine kinase.
incorrect	**c.#**	The activated form fails to inhibit the target in uninfected cells.
incorrect	**d.#**	It is rapidly metabolised in normal cells.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 20 - Question 06		
06)	The activated form of aciclovir has a second mechanism of action other than enzyme inhibition. What is it?	
Feedback: Once the drug is activated to the triphosphate derivative, it can be incorporated into the growing DNA chain. Since the sugar moiety is incomplete, no further nucleotides can be added and chain growth stops.		
incorrect	**a.#**	Antagonism of a receptor.
Correct	**b.#**	Chain terminator for DNA.
incorrect	**c.#**	Alkylating agent for DNA.
incorrect	**d.#**	Blocking a carrier protein in the cell membrane.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 20 - Question 07		
07)	What nucleotide is mimicked by cidofovir?  Cidofovir	
Feedback: Cidofovir contains the nucleic acid base cytosine. It also has a phosphonomethylene group to mimic a monophosphate group.		
incorrect	**a.#**	Deoxyadenosine monophosphate.
incorrect	**b.#**	Deoxythymidine monophosphate.
Correct	**c.#**	Deoxycytidine monophosphate.
incorrect	**d.#**	Deoxyguanosine monophosphate.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 20 - Question 08	
08)	What disadvantage might the following structure have as an

	antiviral agent compared to cidofovir? 

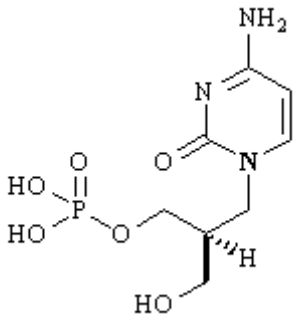
Feedback: Since the structure has a phosphate group, it is susceptible to enzymatic hydrolysis. The other options are not valid. The structure could be phosphorylated to the active triphosphate if it reached infected cells, but most of it will get hydrolysed before it does so. The drug is no more polar or hydrophobic than cidofovir, and so this is not a disadvantage either.

incorrect	**a.#**	It will not be activated to the active drug.
Correct	**b.#**	It is susceptible to hydrolysis.
incorrect	**c.#**	It is too polar.
incorrect	**d.#**	It is too hydrophobic.

END OF QUESTION

Type: multiple choice

Title: Chapter 20 - Question 09

09)	What is the target for the following structure (zidovudine)? 

Feedback: The activated form of zidovudine is a reverse transcriptase inhibitor.

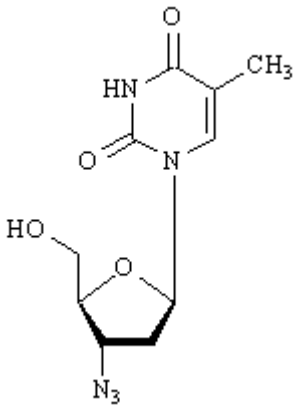
incorrect	**a.#**	Integrase.
incorrect	**b.#**	Protease.
Correct	**c.#**	Reverse transcriptase.
incorrect	**d.#**	DNA polymerase.

END OF QUESTION

Type: multiple choice

Title: Chapter 20 - Question 10

10)	What is the active form of zidovudine?

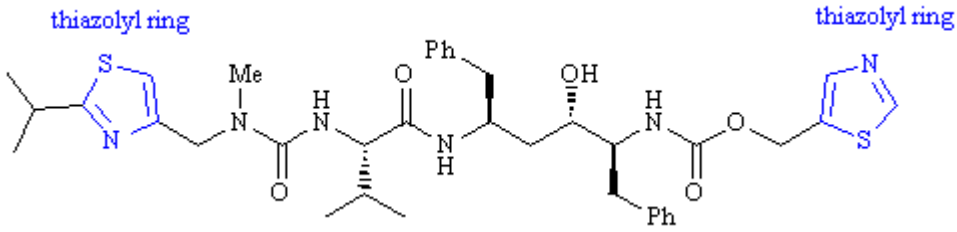
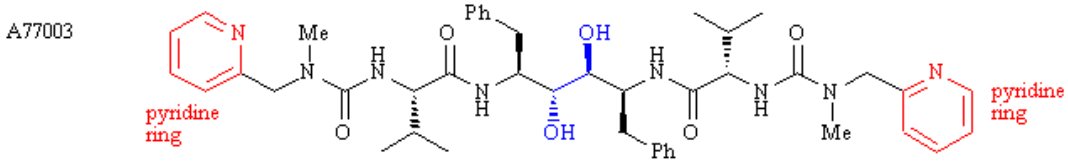
	

Feedback: The triphosphate is the active form. Once the triphosphate is formed it can mimic the natural nucleotide building blocks of DNA. This allows it to be recognised by the target enzyme more readily.		
incorrect	**a.#**	The structure itself.
incorrect	**b.#**	The monophosphate.
incorrect	**c.#**	The diphosphate.
Correct	**d.#**	The triphosphate.
END OF QUESTION		

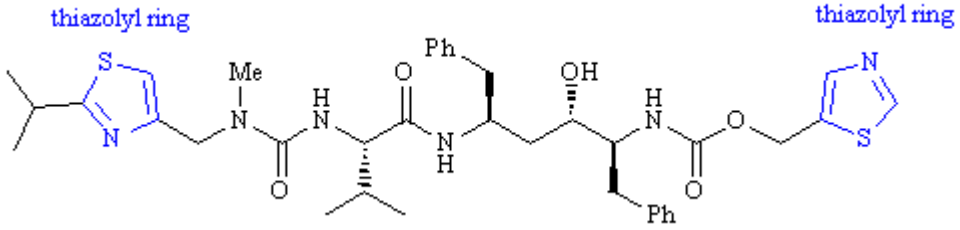
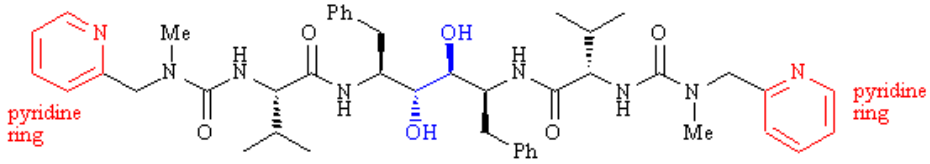
Type: multiple choice		
Title: Chapter 20 - Question 11		
11)	How many protein units make up the HIV protease enzyme?	
Feedback: There are two identical protein subunits making up the protease enzyme. The protease enzyme is symmetrical.		
incorrect	**a.#**	1
Correct	**b.#**	2
incorrect	**c.#**	3
incorrect	**d.#**	4
END OF QUESTION		

Type: multiple choice		
Title: Chapter 20 - Question 12		
12)	The HIV protease enzyme can catalyse the hydrolysis of a particular peptide bond that is not hydrolysed by mammalian proteases. Between which amino acids does this take place?	
Feedback: The protease enzyme catalyses the hydrolysis of peptide bonds between proline and phenylalanine where the *N*-terminus of proline is linked to the *C*-terminus of phenylalanine.		
incorrect	**a.#**	Phenylalanine and lysine.
incorrect	**b.#**	Proline and lysine.
incorrect	**c.#**	Phenylalanine and tyrosine.
Correct	**d.#**	Proline and phenylalanine .
END OF QUESTION		

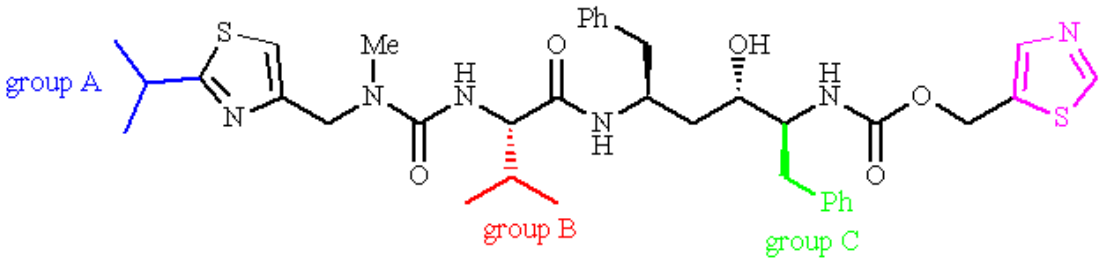
Type: multiple choice		
Title: Chapter 20 - Question 13		
13)	Which substituent is the ideal binding group for the S1 binding pocket of HIV-protease?	
Feedback: The benzyl group is the residue that is present on the amino acid phenylalanine.		
incorrect	**a.#**	Phenyl (-Ph).
incorrect	**b.#**	Phenethyl (CH_2CH_2Ph).
Correct	**c.#**	Benzyl ($-CH_2Ph$).
incorrect	**d.#**	$-CH_2CONH_2$.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 20 - Question 14		
14)	The following structure (ritonavir) is a protease inhibitor.  The structure A77003 (below) was an early compound in the design of ritonavir. This contained a diol shown in blue. One of these hydroxyl groups was subsequently removed. Why?  	
Feedback: Both hydroxy groups are involved in hydrogen bonding interactions, but one of the hydroxy groups is more involved than the other. Removing one of the OH groups results in less binding interactions, but there is a net gain in binding stability since less energy is required to desolvate one hydroxyl group prior to binding than having to desolvate two such groups. The hydroxyl group that was removed was not acting as a steric shield and was not causing the molecule to bind in an inactive mode.		
incorrect	**a.#**	The group that was removed was not taking part in any hydrogen bonding interactions with the active site.
incorrect	**b.#**	The hydroxyl group that was removed was acting as a steric shield and preventing the molecule from binding properly.
Correct	**c.#**	Removing the hydroxyl group reduced the energy penalty associated with desolvation prior to binding.
incorrect	**d.#**	The group that was removed was participating in a hydrogen bonding interaction that caused the molecule to bind in an inactive mode.
END OF QUESTION		

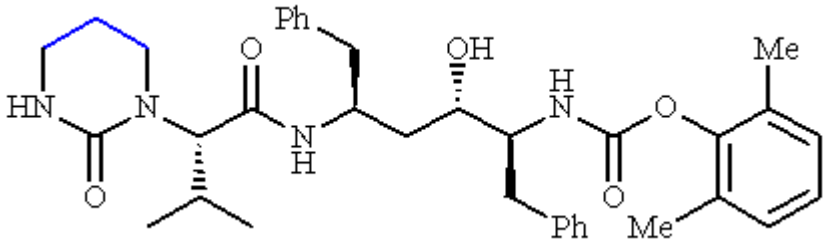
Type: multiple choice	
Title: Chapter 20 - Question 15	
15)	The following structure (ritonavir) is a protease inhibitor.

	 The structure A77003 (below) was an early compound in the design of ritonavir. The pyridine rings in A77003 were replaced with thiazolyl heterocycles. Why? A77003 

Feedback: The pyridine rings were prone to metabolism leading to *N*-oxides which were quickly excreted, mainly in bile. The thiazolyl rings were less susceptible to this form of metabolism.		
Correct	**a.#**	The pyridine rings were susceptible to metabolic oxidation.
incorrect	**b.#**	The thiazolyl rings were smaller and decreased the molecular weight.
incorrect	**c.#**	The nitrogen atoms of the pyridine rings could be protonated and make the drug too polar.
incorrect	**d.#**	The thiazolyl rings fitted better into binding pockets than the pyridine rings.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 20 - Question 16		
16)	Ritonavir is a protease inhibitor.A mutation in the protease enzyme results in resistance to ritonavir. An important interaction between the drug and the active site is lost as a result of the mutation. Which group in ritonavir is involved in this interaction? 	
Feedback: The isopropyl group A forms a hydrophobic interaction with the isopropyl side chain present on valine.		
Correct	**a.#**	Group A.
incorrect	**b.#**	Group B.
incorrect	**c.#**	Group C.
incorrect	**d.#**	Group D.
END OF QUESTION		

Type: multiple choice

Title: Chapter 20 - Question 17		
17)	Lopinavir is a protease inhibitor that was developed from ritonavir. Lopinavir is active against viral strains that are resistant to ritonavir. What is the significance of the region shown in blue in lopinavir?  	
Feedback: The ring imparts conformational restraint on the urea functional group. This enhances H-bonding interactions between the urea group and the S2 subsite.		
incorrect	**a.#**	It is involved in a hydrophobic interaction with the active site.
Correct	**b.#**	It acts as a conformational restraint for the urea group.
incorrect	**c.#**	It acts as a conformational blocker and orientates the rest of the chain.
incorrect	**d.#**	It prevents one of the nitrogens of the urea group acting as a hydrogen bond donor.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 20 - Question 18		
18)	What feature of the protease active site permits a hybridisation strategy to be used in the design of protease inhibitors?	
Feedback: Since the active site is symmetrical, inhibitors can bind 'left to right' or 'right to left'. Since that is the case, it is feasible to take the right hand half of one inhibitor and link it to the left hand half of a different inhibitor. The other statements above are perfectly correct but do not explain why the hybridisation strategy is feasible. For example, one option indicates that there are the same number of subsites in either half, but does not indicate that the subsites in one half are identical to the corresponding subsites in the other half.		
incorrect	**a.#**	There are two aspartate residues in the active site which are involved in the catalytic mechanism of the enzyme.
incorrect	**b.#**	There are two isoleucine residues in the flap region of the active site.
Correct	**c.#**	The active site is symmetrical.
incorrect	**d.#**	There are the same number of subsites in one half of the active site as the other half.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 20 - Question 19		
19)	What advantage is there in replacing the benzyl group at the P1 position of a protease inhibitor with a $PhSCH_2$ group?	
Feedback: Incorporating a sulphur atom extends the side chain allowing simultaneous binding to the S1 and S3 subsites.		
incorrect	**a.#**	The new group fits the S1 subsite more efficiently.
incorrect	**b.#**	The new group binds to the S1 and S2 subsites simultaneously.

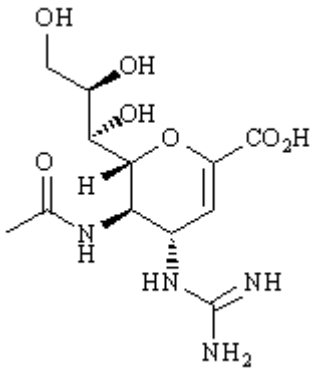
Correct	**c.#**	The new group binds to the S1 and S3 subsites simultaneously.
incorrect	**d.#**	The new group binds to the S3 subsite rather than the S1 subsite.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 20 - Question 20		
20)	Which type of influenza virus is most susceptible to mutations?	
Feedback: Mutations occur almost yearly with influenza A, but much more slowly with influenza B. Mutations do not appear to occur at a significant rate for influenza C.		
Correct	**a.#**	Influenza A.
incorrect	**b.#**	Influenza B.
incorrect	**c.#**	Influenza C.
incorrect	**d.#**	None of the answers are correct. They are all equally susceptible.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 20 - Question 21		
21)	Amantadine was one of the earliest antiviral drugs used clinically against the influenza virus. What is its mechanism of action at low concentration?	
Feedback: At high concentration, amantadine acts as a buffer. At low concentrations, it blocks a viral ion channel. It does not bind to viral nucleic acid or to a viral enzyme.		
incorrect	**a.#**	It acts as a base and buffers the pH of endosomes, preventing the acidic pH required for fusion to take place between the viral membrane and that of the endosome.
incorrect	**b.#**	It binds to viral nucleic acid and prevents replication and transcription.
Correct	**c.#**	It binds to a specific region of a viral ion channel and blocks it.
incorrect	**d.#**	It binds to a viral enzyme and inhibits it.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 20 - Question 22		
22)	Sialic acid binds to the active site of neuraminidase. Which feature of sialic acid has the most significant binding interaction?	
Feedback: All the functional groups mentioned have a binding role, but the carboxylate ion is able to form the strongest interaction since it is ionic.		
incorrect	**a.#**	The triol.
Correct	**b.#**	The carboxylate ion.
incorrect	**c.#**	The amide.
incorrect	**d.#**	The alcohol.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 20 - Question 23	
23)	Zanamivir has been used in the treatment of influenza.



What is the target for the above structure?

Feedback: Hemagglutinin is a glycoprotein on the surface of the flu virus which plays a role in the adsorption process. RNA and DNA polymerases are enzymes that catalyse the synthesis of RNA and DNA respectively. RNA polymerase is carried by the flu virus but not DNA polymerase. Neuraminidase is an enzyme on the surface of the flu virus that catalyses the splitting of sialic acid from glycoproteins or glycolipids. This is the target for zanamivir and oseltamivir.

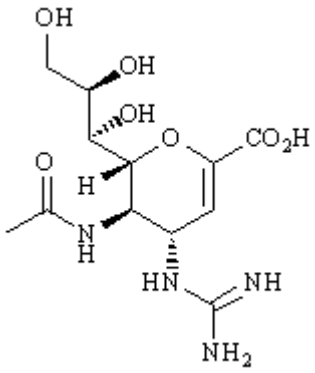
incorrect	**a.#**	Hemagglutinin.
incorrect	**b.#**	RNA polymerase.
incorrect	**c.#**	DNA polymerase.
Correct	**d.#**	Neuraminidase.

END OF QUESTION

Type: multiple choice

Title: Chapter 20 - Question 24

24) Zanamivir has been used in the treatment of influenza.

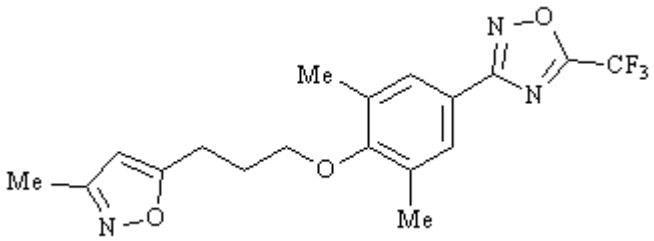


What sort of drug is it?

Feedback: The drug mimics the transition state for the catalytic reaction carried out by neuraminidase, but is stable to the catalytic process.

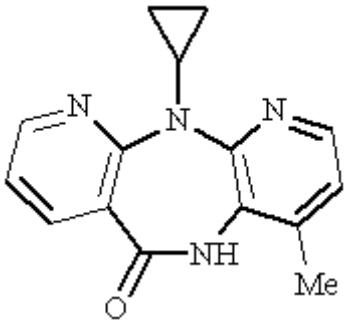
incorrect	**a.#**	Antagonist.
incorrect	**b.#**	Prodrug.
incorrect	**c.#**	Suicide substrate.
Correct	**d.#**	Transition-state inhibitor.

END OF QUESTION

Type: multiple choice		
Title: Chapter 20 - Question 25		
25)	Pleconaril entered phase III clinical trials for the treatment of the cold virus.  How does pleconaril have an antiviral effect?	
Feedback: The drug stabilises the viral protein coat and prevents the uncoating process.		
incorrect	**a.#**	It blocks the replication and transcription of viral nucleic acid.
Correct	**b.#**	It prevents the uncoating process.
incorrect	**c.#**	It inhibits an enzyme-catalysed reaction.
incorrect	**d.#**	It prevents adsorption.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 20 - Question 26		
26)	Which is the normal order of the following events in the life cycle of viruses?	
Feedback: The life cycle starts with the virus adsorbing or attaching to the cell membrane of the host cell. Penetration then takes place with uncoating to free the viral nucleic acid. Replication and transcription leads to new viral protein allowing the assembly of nucleocapsids. Finally, the virus departs the host cell, usually by budding.		
incorrect	**a.#**	Penetration, uncoating, replication and transcription, budding, assembly of nucleopcapsids.
incorrect	**b.#**	Budding, uncoating, penetration, replication and transcription, assembly of nucleocapsids
incorrect	**c.#**	Uncoating, penetration, replication and transcription, assembly of nucleocapsids, budding.
Correct	**d.#**	Uncoating, penetration, replication and transcription, assembly of nucleocapsids, budding.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 20 - Question 27		
27)	Which of the following targets is not a current or potential drug target in anti-HIV therapy?	
Feedback: Neuraminidase is a drug target in tackling the flu virus but not HIV.		
incorrect	**a.#**	Reverse transcriptase.
incorrect	**b.#**	Integrase.
incorrect	**c.#**	Protease.
Correct	**d.#**	Neuraminidase.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 20 - Question 28		
28)	Which of the following statements is true regarding nevirapine? 	
Feedback: Nevirapine is a chiral molecule that binds to an allosteric binding site next to the active site of reverse transcriptase. It bears no resemblance to a nucleoside and it is a first-generation, non-nucleoside reverse transcriptase inhibitor.		
Correct	**a.#**	It is a non-nucleoside reverse transcriptase inhibitor (NNRTI).
incorrect	**b.#**	It binds to the substrate binding site of reverse transcriptase.
incorrect	**c.#**	It is an achiral molecule.
incorrect	**d.#**	It is an example of a second-generation drug of its class.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 20 - Question 29		
29)	Which of the following statements is true for non-nucleoside reverse transcriptase inhibitors?	
Feedback: The non-nucleoside reverse transcriptase inhibitors bind to an allosteric binding site next to the active site of reverse transcriptase. Since they do not occupy the active site, they can be used alongside nucleoside reverse transcriptase inhibitors which **do** occupy the active site. There are significant binding interactions between second- and third-generation NNRTIs and the main protein chain in the binding site. Resistance can arise due to a mutation where lysine is mutated to asparagine.		
incorrect	**a.#**	They cannot be used alongside nucleoside reverse transcriptase inhibitors.
incorrect	**b.#**	They bind to the active site.
Correct	**c.#**	Resistance can arise due to a mutation where a lysine in the target binding site is mutated to asparagine.
incorrect	**d.#**	Binding interactions between NNRTIs and the main protein chain in the binding site are not important.
END OF QUESTION		

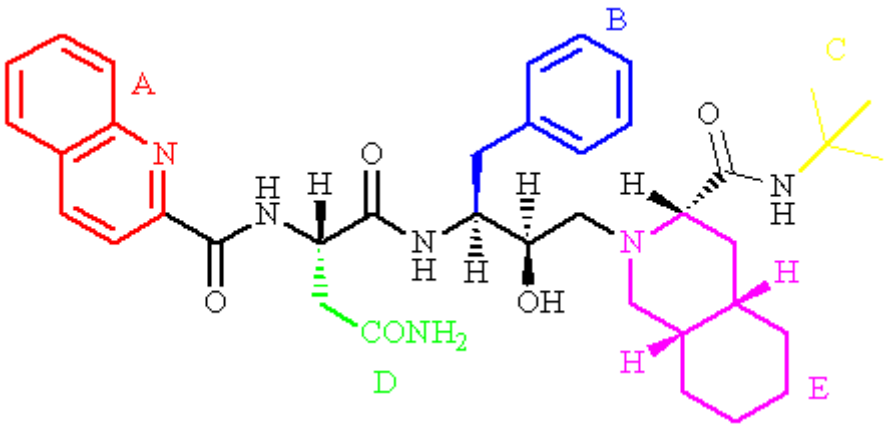
Type: multiple choice		
Title: Chapter 20 - Question 30		
30)	What role do the aspartate residues Asp 25 and Asp 25' play in the HIV protease enzyme?	
Feedback: Aspartic acid 25 and 25' are involved in the catalytic mechanism which hydrolyses the peptide bond of a protein substrate.		
incorrect	**a.#**	They bind directly to the substrate.
Correct	**b.#**	They are involved in the catalytic mechanism which leads to peptide bond hydrolysis in a protein substrate.

incorrect	**c.#**	They are located in the flap region of the protease active site and are involved in binding a water molecule. The latter acts as a hydrogen bonding bridge to a bound protein substrate.
incorrect	**d.#**	They have no significant role.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 20 - Question 31		
31)	What role do the isoleucine residues Ile 50 and Ile 50' play in the HIV protease enzyme?	
Feedback: The amino acids concerned do not bind directly to the substrate and are not involved directly with the mechanism.		
incorrect	**a.#**	They bind directly to the substrate.
incorrect	**b.#**	They are involved in the catalytic mechanism which leads to peptide bond hydrolysis in a protein substrate.
Correct	**c.#**	They are located in the flap region of the protease active site and are involved in binding a water molecule. The latter acts as a hydrogen bonding bridge to a bound protein substrate.
incorrect	**d.#**	They have no significant role.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 20 - Question 32		
32)	Which statement is false?	
Feedback: Only the HIV protease enzyme is symmetrical. Since there is a significant difference between the HIV protease enzyme and mammalian protease enzymes in terms of symmetry, it is feasible to design inhibitors that show selectivity for the viral enzyme over the mammalian enzymes.		
Correct	**a.#**	Both mammalian and HIV protease enzymes are symmetrical.
incorrect	**b.#**	It is possible to design inhibitors that prove selective for the HIV protease over mammalian proteases.
incorrect	**c.#**	The protease enzyme of HIV hydrolyses peptide bonds in protein substrates.
incorrect	**d.#**	The protease enzyme in the HIV enzyme has different substrate selectivity from protease enzymes in humans, despite catalysing the same reaction.
END OF QUESTION		

Type: matching	
Title: Chapter 20 - Question 33	
33)	The following structure is saquinavir. Match the coloured moieties (A-E) to the following binding subsites in the HIV protease active site.



Feedback: The coloured moieties mimic the amino acid residues of protein substrates that would normally fit into these binding subsites.

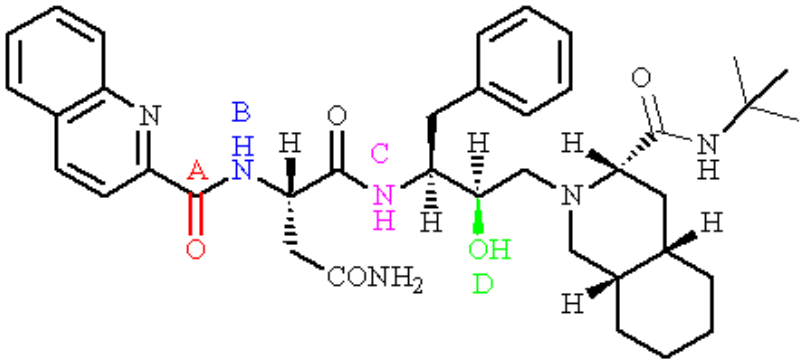
a.#	Subsite S1'	=	Moiety E.
b.#	Subsite S2'	=	Moiety C.
c.#	Subsite S1	=	Moiety B.
d.#	Subsite S2	=	Moiety D.
e.#	Subsite S3	=	Moiety A.

END OF QUESTION

Type: multiple choice

Title: Chapter 20 - Question 34

34) The following structure is saquinavir. Which coloured moiety forms a hydrogen bond to Asp25 and Asp25' when the drug is bound to the active site of the HIV protease enzyme?



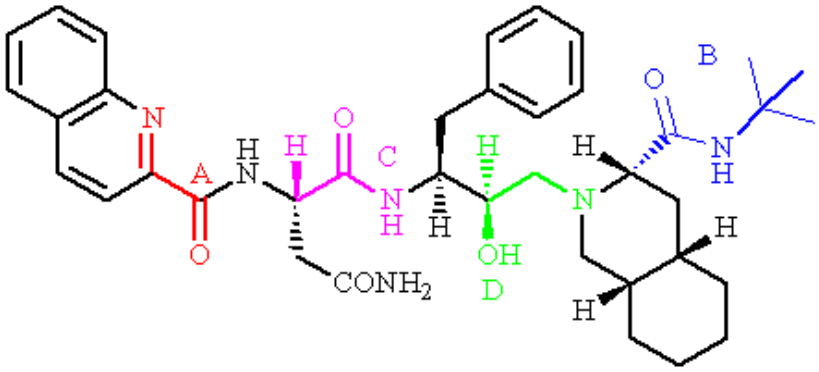
Feedback: The hydroxy group is part of a hydroxyethylamine transition-state isostere that mimics a peptide bond but is not hydrolysed.

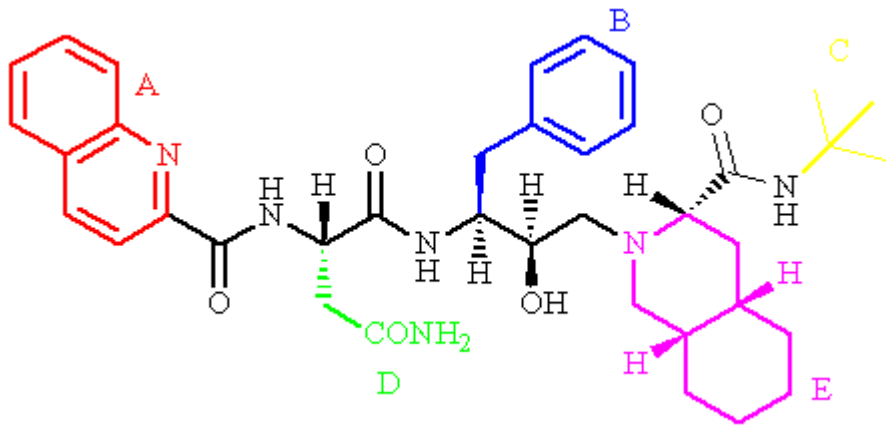
incorrect	**a.#**	Moiety A.
incorrect	**b.#**	Moiety B.
incorrect	**c.#**	Moiety C.
Correct	**d.#**	Moiety D.

END OF QUESTION

Type: multiple choice

Title: Chapter 20 - Question 35

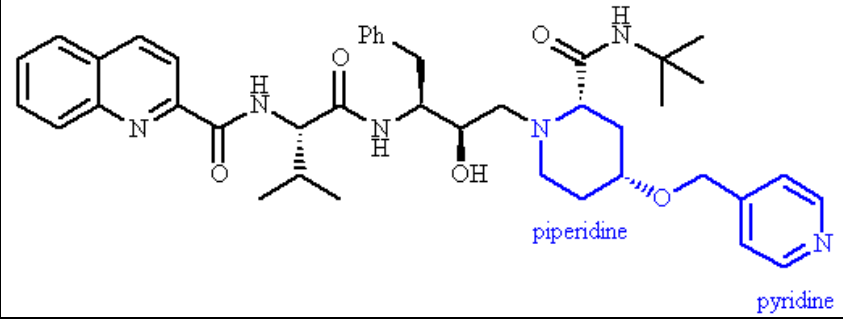
35)	The following structure is saquinavir. Which moiety acts as a transition-state isostere?  		
Feedback: Moiety D is a hydroxylamine moiety which mimics the transition state for the catalytic reaction. It is stable to the catalytic reaction.			
incorrect	**a.#**	Moiety A.	
incorrect	**b.#**	Moiety B.	
incorrect	**c.#**	Moiety C.	
Correct	**d.#**	Moiety D.	
END OF QUESTION			

Type: matching			
Title: Chapter 20 - Question 36			
36)	The following structure is saquinavir. Match the following groups to the coloured moieties shown (A-E).  		
Feedback: The moieties shown are designed to fit the enzyme binding subsites that would normally be occupied by the amino acid residues of protein substrates.			
a.#	Quinoline ring	=	Moiety A.
b.#	Decahydroisoquinoline ring	=	Moiety E.
c.#	Benzyl group	=	Moiety B.
d.#	Tertiary butyl group	=	Moiety C.
e.#	Asparagine residue	=	Moiety D.
END OF QUESTION			

Type: multiple choice		
Title: Chapter 20 - Question 37		
37)	The dipeptide L-phenylalanyl-L-proline was chosen as the lead compound for the development of saquinavir. Which of the following statements is the best explanation for this choice?	
Feedback: The dipeptide was susceptible to hydrolysis. If it had been stable, it would have been a poor choice as a lead compound since it is unlikely that it could bind efficiently to the active site. As a result, analogues would be unlikely to bind as well. There is nothing wrong with the statement 'the residues of the two constituent amino acids fit binding subsites in the active site and so analogues will be easily recognised and bound strongly.' However, that would not be a reason in itself for choosing this dipeptide. If the dipeptide also was bound strongly to mammalian proteases, there would be little chance of achieving inhibitor selectivity. The dipeptides of all the naturally occurring amino acids are easy to synthesise.		
incorrect	**a.#**	The dipeptide is stable to hydrolysis by the HIV protease and so any analogues are likely to act as inhibitors rather than substrates
incorrect	**b.#**	The residues of the two constituent amino acids fit binding subsites in the active site and so analogues will be easily recognised and bound strongly
Correct	**c.#**	The HIV protease catalyses the hydrolysis of the dipeptide whereas mammalian proteases do not. This allows the possibility of target selectivity.
incorrect	**d.#**	The dipeptide was easy to synthesise compared to other dipeptides and so subsequent analogues would also be easily obtainable
END OF QUESTION		

Type: multiple choice		
Title: Chapter 20 - Question 38		
38)	The lead compound and early structures leading to the important protease inhibitors ritonavir and lopinavir were symmetrical molecules. Why were symmetrical molecules chosen?	
Feedback: They were easy to synthesise and could bind left to right or right to left in the symmetrical binding site, increasing the chances of good activity but these were not the principal reasons for choosing symmetrical lead compounds. The molecules may have been symmetrical but they still contained chiral centres.		
incorrect	**a.#**	They were easy to synthesise.
incorrect	**b.#**	They did not have chiral centres.
Correct	**c.#**	The binding site for the HIV protease enzyme is symmetrical whereas the binding sites for mammalian proteases are not
incorrect	**d.#**	They could bind left to right or right to left in the symmetrical binding site, increasing the chances of good activity.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 20 - Question 39	
39)	Palinavir is an HIV protease inhibitor. Which of the following statements is true regarding the group coloured blue?

Feedback: The blue coloured moiety spans two binding subsites and does away with the need for two separate binding groups. As a result, the molecular weight of the molecule is decreased.

incorrect	**a.#**	The moiety in blue was added such that water solubility could be increased with the addition of a pyridine ring.
Correct	**b.#**	The moiety in blue was designed to span the two binding subsites (S1' and S3').
incorrect	**c.#**	The moiety in blue was added such that the pyridine ring could form a hydrogen bonding interaction with a binding region not normally accessible to protein substrates.
incorrect	**d.#**	The blue coloured moiety was added as a steric shield to prevent hydrolysis of the *C*-terminal peptide bond.

END OF QUESTION

Type: multiple choice

Title: Chapter 20 - Question 40

40)	Which of the following statements is untrue regarding the life cycle of the flu virus?

Feedback: The vesicles are called **endosomes**.

incorrect	**a.#**	Some virions are taken into a host cell by a process known as endocytosis which involves the cell membrane wrapping round the virion and being pinched off to produce vesicles.
Correct	**b.#**	Vesicles that have brought virions into the host cell are called exosomes.
incorrect	**c.#**	Membrane bound vesicles that contain virions fuse with cell structures called lysosomes.
incorrect	**d.#**	Host cell enzymes can aid the virus in the uncoating process.

END OF QUESTION

Type: multiple choice

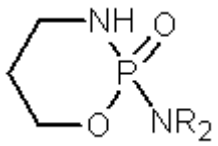
Title: Chapter 20 - Question 41

41)	Which of the following statements is false regarding the characteristics of a good protein target for antiviral drugs?

Feedback: The ideal target should be shared between various viruses in order to design a drug with a broad antiviral activity.

incorrect	**a.#**	It should be important to the life cycle of the virus.
incorrect	**b.#**	It should bear little resemblance to human proteins.
Correct	**c.#**	It should be unique for a specific virus.
incorrect	**d.#**	It should be important in the early stages of the virus life cycle.

END OF QUESTION

Type: multiple choice		
Title: Chapter 21 - Question 01		
01)	The following drug is commonly used in anticancer therapy.  Which of the following statements is not true of the above structure?	
Feedback: The drug is called cyclophosphamide. It can be taken orally and since it is a prodrug, it is not toxic to the gut wall when taken orally.		
incorrect	**a.#**	It is relatively non toxic.
incorrect	**b.#**	It acts as a prodrug.
Correct	**c.#**	It cannot be taken orally.
incorrect	**d.#**	The structure is metabolised to release acrolein.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 21 - Question 02		
02)	The Ras protein plays an important part in cell signalling pathways. Abnormal Ras protein is constantly active and is observed in many cancers. Which of the following statements best explains this constant activity?	
Feedback: The Ras protein is active when it binds GTP and inactive when it binds GDP. The inability to autocatalyse bound GTP means that abnormal Ras is constantly active. None of the other options account for this.		
incorrect	**a.#**	The abnormal Ras protein is unable to autocatalyse the hydrolysis of bound GDP.
incorrect	**b.#**	The abnormal Ras protein is unable to bind GTP.
Correct	**c.#**	The abnormal Ras protein is unable to autocatalyse the hydrolysis of bound GTP.
incorrect	**d.#**	The abnormal Ras protein is unable to bind GDP.
END OF QUESTION		

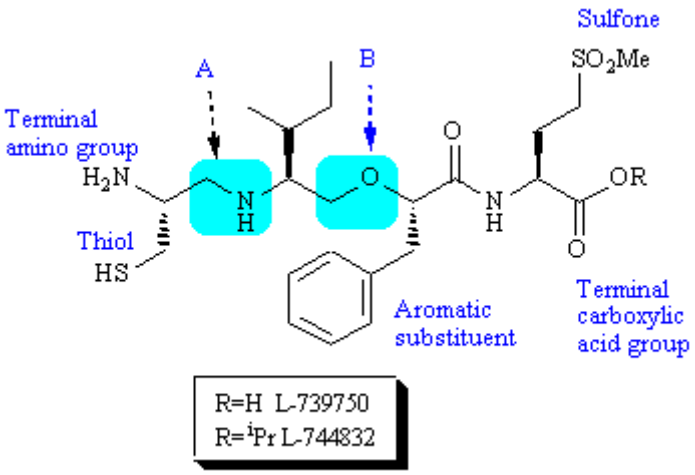
Type: multiple choice		
Title: Chapter 21 - Question 03		
03)	When the Ras protein is present in the cytoplasm, it can undergo an enzyme-catalysed reaction that leads to the attachment of a farnesyl hydrocarbon chain to the protein. What role does the farnesyl chain play once it is linked to the Ras protein?	
Feedback: The farnesyl chain is hydrophobic and so it anchors the Ras chain to the inner surface of the cell membrane. Activation or deactivation of Ras depends on whether Ras binds GTP or GDP respectively. The farnesyl chain plays no part in protecting the Ras protein from degradative enzymes.		
incorrect	**a.#**	It activates the Ras protein.
incorrect	**b.#**	It deactivates the Ras protein.
Correct	**c.#**	It anchors the Ras protein to the cell membrane.
incorrect	**d.#**	It protects the Ras protein from degradative enzymes.

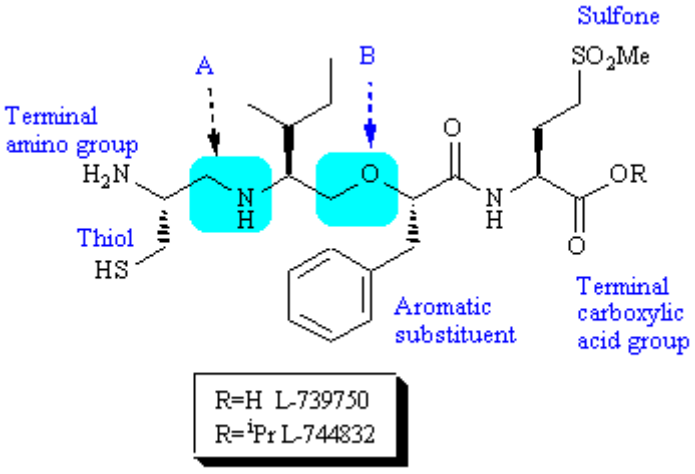
END OF QUESTION

Type: multiple choice		
Title: Chapter 21 - Question 04		
04)	The CaaX tetrapeptide is a substrate for the enzyme involved in attaching the farnesyl chain to the Ras protein. However, placing a specific amino acid at the underlined position Ca<u>a</u>X results in an enzyme inhibitor. Which amino acid can be placed at this position to produce an inhibitor?	
Feedback: The a in CaaX is normally valine, isoleucine or leucine. Changing the amino acid (a) next to X from one of these amino acids to phenylalanine results in an inhibitor rather than a substrate.		
incorrect	**a.#**	Serine.
incorrect	**b.#**	Glycine.
incorrect	**c.#**	Valine.
Correct	**d.#**	Phenylalanine.
END OF QUESTION		

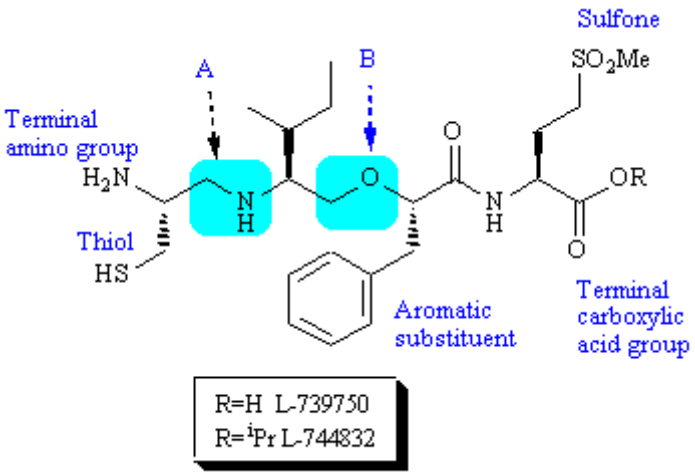
Type: multiple choice		
Title: Chapter 21 - Question 05		
05)	Specific CaaX tetrapeptides have been shown to be inhibitors of the enzyme involved in attaching the farnesyl chain to the Ras protein. However, such structures are susceptible to hydrolysis by aminopeptidases. Which peptide link in CaaX analogues is particularly susceptible to aminopeptidases?	
Feedback: The bond between C and a is susceptible to aminopeptidases. These enzymes catalyse the cleavage of the amino acid at the *N*-terminus of a peptide chain.		
incorrect	**a.#**	Between C and the rest of the Ras protein.
Correct	**b.#**	Between C and a.
incorrect	**c.#**	Between a and a.
incorrect	**d.#**	Between a and X.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 21 - Question 06	
06)	The following structures inhibit the enzyme responsible for linking farnesyl chains to the Ras protein:

	 What is the rationale for moiety A?	
Feedback: There is nothing illogical in any of the options presented, but the group was introduced to remove a peptide bond that was susceptible to aminopeptidases.		
incorrect	**a.#**	It replaces a peptide bond such that the N-H group can act both as a hydrogen bond donor and a hydrogen bond acceptor.
incorrect	**b.#**	It replaces a peptide bond such that chain flexibility is increased, allowing the active conformation to be adopted more easily.
Correct	**c.#**	It replaces a metabolically susceptible peptide bond with a metabolically stable group.
incorrect	**d.#**	It replaces a peptide bond such that the carbonyl oxygen can no longer act as a hydrogen bond acceptor.
END OF QUESTION		

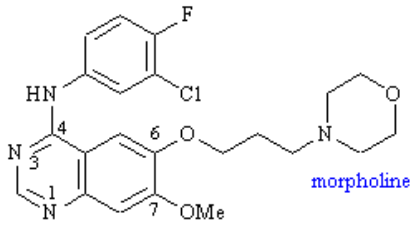
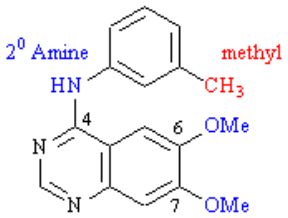
Type: multiple choice	
Title: Chapter 21 - Question 07	
07)	The following structures inhibit the enzyme responsible for linking farnesyl chains to the Ras protein.  What is the rationale for the peptide bond isostere - B?
Feedback: The carbonyl has been removed and the NH has been replaced with O. The nitrogen of an amine is more nucleophilic than the nitrogen of an amide and can take part in an intramolecular cyclisation with a terminal ester.	

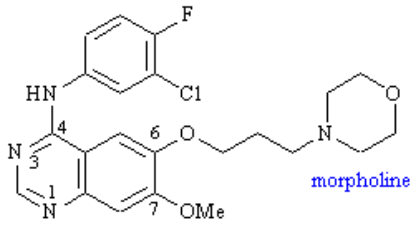
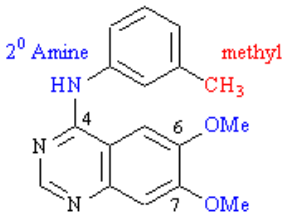
incorrect	**a.#**	It replaces a metabolically susceptible peptide bond with a metabolically stable group.
Correct	**b.#**	It is used instead of an amine isostere since the latter can result in an intramolecular cyclisation
incorrect	**c.#**	The oxygen acts as a hydrogen bond acceptor whereas the NH group of the original peptide group acted as a hydrogen bond donor.
incorrect	**d.#**	It allows greater conformational freedom for the phenylalanine residue.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 21 - Question 08	
08)	The following structures inhibit the enzyme responsible for linking farnesyl chains to the Ras protein:  Which feature of the molecule forms an interaction with a zinc ion cofactor?
Feedback: The thiol functional group is known to be a good ligand for a zinc ion cofactor. It is present in several drugs that interact with zinc metalloenzymes. Captopril is one such example.	

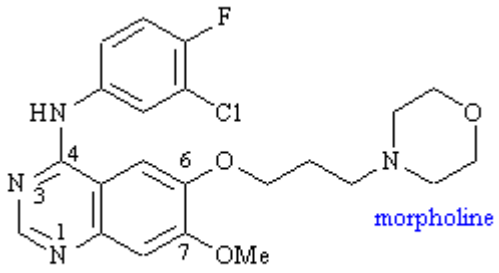
incorrect	**a.#**	The terminal amino group.
incorrect	**b.#**	The sulfone.
incorrect	**c.#**	The terminal carboxylic acid group.
Correct	**d.#**	The thiol.
END OF QUESTION		

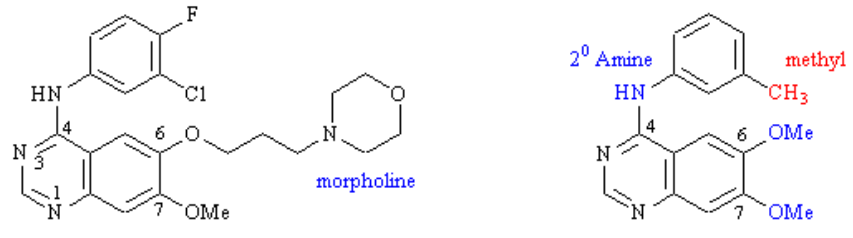
Type: multiple choice	
Title: Chapter 21 - Question 09	
09)	Gefitinib (I) has been used for the treatment of refractory lung cancers.

	 I  II Structure II was the lead compound for structure I. Why was the methyl group replaced with a chloro group?	
Feedback: The methyl group is susceptible to metabolic oxidation from cytochrome P450 enzymes to give a hydroxymethylene group. The chloro substituent does not get oxidised. It is true that the chloro substituent is an inductive electron-withdrawing group whereas the methyl substituent is electron donating, but this was not the reason for the change. A chloro group is often used to replace a methyl group because it is much the same size and lipophilicity.		
incorrect	**a.#**	To replace an inductive electron-donating group with an inductive electron-withdrawing group.
incorrect	**b.#**	To replace the methyl group with a smaller chloro group.
incorrect	**c.#**	To replace a lipophilic methyl group with a more polar chloro group.
Correct	**d.#**	To replace a metabolically susceptible methyl group with a metabolically stable chloro group.
END OF QUESTION		

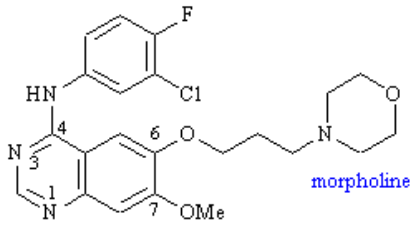
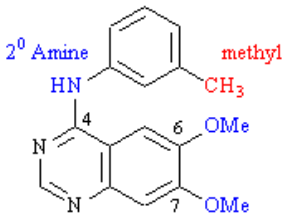
Type: multiple choice		
Title: Chapter 21 - Question 10		
10)	Gefitinib (I) has been used for the treatment of refractory lung cancers.  I  II Structure II was the lead compound for structure I. Why was a fluoro group introduced into gefitinib and what role does it play?	
Feedback: Cytochrome P450 enzymes oxidise the *para* position of the aromatic ring to introduce a phenol group. Placing a fluoro substituent at the *para* position blocks this reaction. The fluoro substituent acts as an electron-withdrawing group but this is not the reason for its inclusion. Fluorine is a very poor hydrogen bond acceptor. The fluorine atom is a similar size to hydrogen and so it could not fill a hydrophobic pocket better than a hydrogen atom.		
Correct	**a.#**	It blocks metabolism at that position of the aromatic ring.
incorrect	**b.#**	It is electron withdrawing and makes the aromatic ring less electron rich.

incorrect	**c.#**	It acts as a hydrogen bond acceptor.
incorrect	**d.#**	It fills a hydrophobic pocket better than a hydrogen atom.
END OF QUESTION		

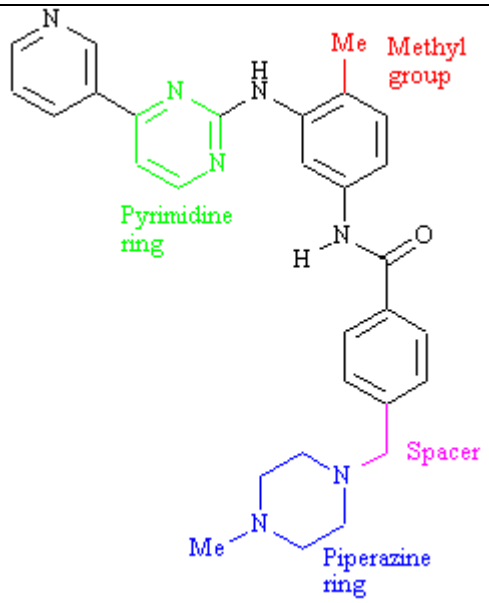
Type: multiple choice		
Title: Chapter 21 - Question 11		
11)	Gefitinib has been used for the treatment of refractory lung cancers.  I What role does the morpholine ring play in gefitinib?	
Feedback: The addition of a morpholine ring is a common tactic to increase the water solubility of a drug. In theory, the oxygen and the nitrogen could act as hydrogen bond acceptor groups, but not in this case. Similarly, it is feasible that the nitrogen could act as a hydrogen bond donor if it was protonated, but that is not the case here.		
incorrect	**a.#**	It acts as a hydrogen bond acceptor at the nitrogen atom.
incorrect	**b.#**	It acts as a hydrogen bond acceptor at the oxygen atom.
incorrect	**c.#**	It is protonated at blood pH and acts as a hydrogen bond donor.
Correct	**d.#**	It increases water solubility.
END OF QUESTION		

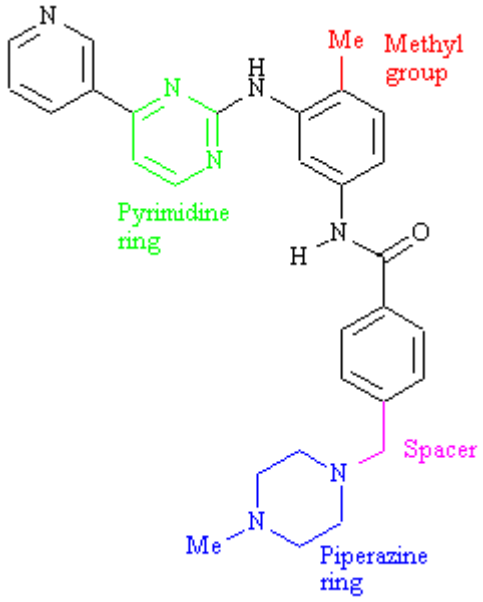
Type: multiple choice		
Title: Chapter 21 - Question 12		
12)	Gefitinib has been used for the treatment of refractory lung cancers.  I II One of the nitrogen atoms in gefitinib is an important hydrogen bond acceptor to the amino acid methionine in the active site. Which one?	
Feedback: This nitrogen acts as a hydrogen bond acceptor.		
incorrect	**a.#**	The nitrogen at position 3.
Correct	**b.#**	The nitrogen at position 1.

incorrect	**c.#**	The aniline nitrogen.
incorrect	**d.#**	The morpholine nitrogen.
END OF QUESTION		

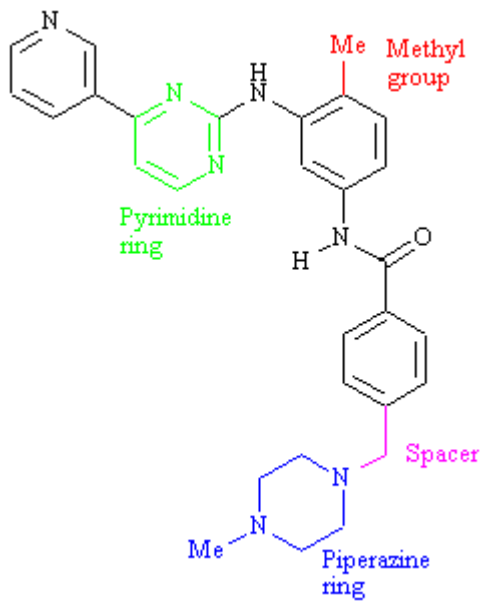
Type: multiple choice		
Title: Chapter 21 - Question 13		
13)	Gefitinib has been used for the treatment of refractory lung cancers.  I  II One of the nitrogen atoms in structure I forms an important hydrogen bond to a water molecule. The latter acts as a bridge between the drug and the amino acid threonine in the active site. Which nitrogen atom is involved in this interaction?	
Feedback: This nitrogen acts as a hydrogen bond acceptor to water.		
Correct	**a.#**	The nitrogen at position 3.
incorrect	**b.#**	The nitrogen at position 1.
incorrect	**c.#**	The aniline nitrogen.
incorrect	**d.#**	The morpholine nitrogen.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 21 - Question 14	
14)	Imatinib is an important anticancer drug which targets a protein kinase.

	 What is the role of the methyl group in the structure?	
Feedback: The methyl group restricts the number of conformations available to the molecule. The other options are feasible reasons for adding a methyl group, but are not relevant to this case.		
Correct	**a.#**	It acts as a conformational blocker.
incorrect	**b.#**	It acts as an accessible group for drug metabolism.
incorrect	**c.#**	It has an electron-donating effect on the aromatic ring.
incorrect	**d.#**	All of the options given are correct.
END OF QUESTION		

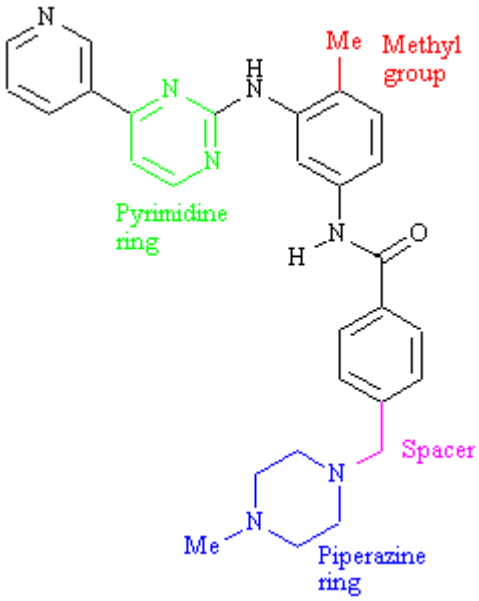
Type: multiple choice	
Title: Chapter 21 - Question 15	
15)	Imatinib is an important anticancer drug which targets a protein kinase.  What effect does the methyl group have on activity?

Feedback: The methyl group prevents the molecule from adopting the active conformation for serine-threonine kinases but allows the active conformation for tyrosine kinases. The methyl group was not introduced to accelerate metabolism or to increase the binding interactions of the aromatic ring.		
Correct	**a.#**	It introduces specificity for tyrosine kinases over serine-threonine kinases.
incorrect	**b.#**	It introduces specificity for serine-threonine kinases over tyrosine kinases.
incorrect	**c.#**	It allows the drug to be metabolised in a controlled fashion, thus reducing side effects from alternative metabolic reactions.
incorrect	**d.#**	It increases the activity of the drug by increasing binding interactions between the aromatic ring and the binding site.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 21 - Question 16	
16)	Imatinib is an important anticancer drug which targets a protein kinase.  What is the principle reason for the spacer unit?

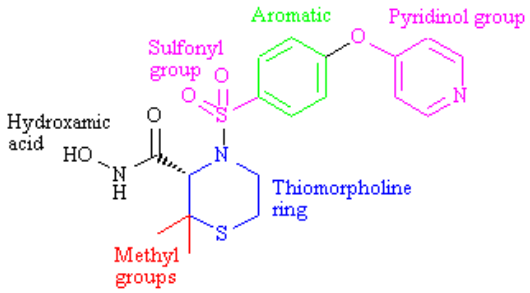
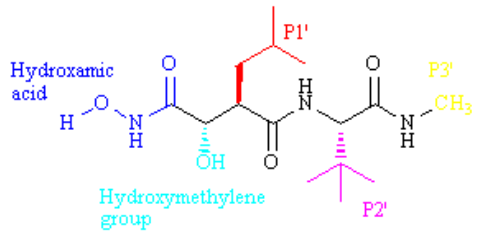
Feedback: A nitrogen next to an aromatic ring is defined as being aniline like. Such groups are potentially toxic and so a spacer was introduced such that the nitrogen was not directly linked to the aromatic ring.		
incorrect	**a.#**	It positions the piperazine ring in the correct position for binding.
incorrect	**b.#**	It allows increased flexibility such that the piperazine ring can access greater conformational space.
Correct	**c.#**	It avoids the potential toxicity of an aniline-like nitrogen.
incorrect	**d.#**	It increases the hydrophobicity of the molecule.
END OF QUESTION		

Type: multiple choice
Title: Chapter 21 - Question 17

17)	Imatinib is an important anticancer drug which targets a protein kinase:  What role does the piperazine ring have?	
Feedback: The terminal nitrogen of the piperazine ring is protonated. This increases water solubility and also allows an extra ionic bonding interaction to take place with the active site. This increases target binding affinity and selectivity.		
incorrect	**a.#**	It increases water solubility.
incorrect	**b.#**	It increases target selectivity.
incorrect	**c.#**	It increases target binding.
Correct	**d.#**	All of the options given are correct.
END OF QUESTION		

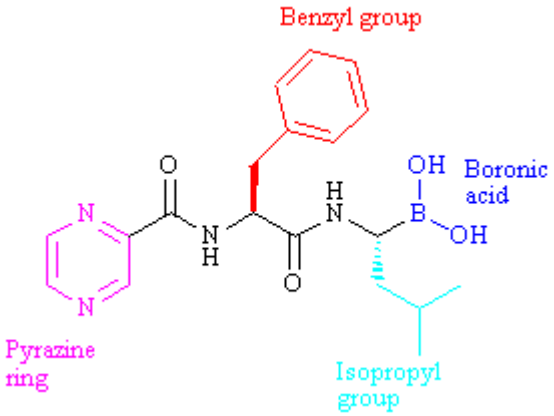
Type: multiple choice		
Title: Chapter 21 - Question 18		
18)	Matrix metalloproteinases are enzymes which are targets for potential anticancer agents. The collagenase enzymes are examples of matrix metalloproteinases which catalyse the hydrolysis of peptide links in protein substrates. Which of the following features is not present in peptide-based inhibitors of collagenase enzymes?	
Feedback: One or more substituents are required to fit into the enzyme subsites, but they should form van der Waals interactions and not ionic interactions.		
incorrect	**a.#**	Replacement of the susceptible peptide bond with a moiety which is stable to hydrolysis.
Correct	**b.#**	One or more substituents that can fit into the enzyme subsites and form ionic interactions.
incorrect	**c.#**	At least one functional group capable of forming a H-bond to the enzyme backbone.
incorrect	**d.#**	A group capable of strong interactions with the zinc ion cofactor.
END OF QUESTION		

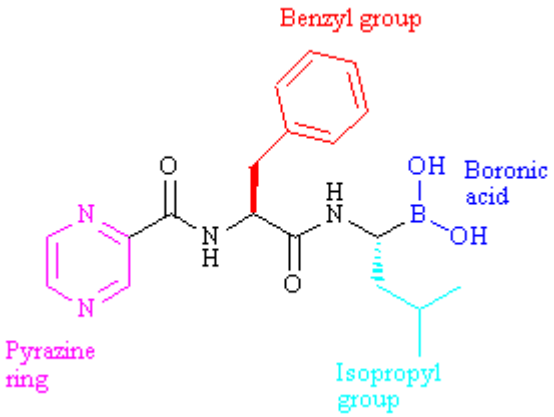
Type: multiple choice

Title: Chapter 21 - Question 19		
19)	The following molecule is a second-generation matrix metalloproteinase inhibitor called prinomastat:  Prinomastat  Marimastat What advantage does this structure have over the first-generation structure called marimastat?	
Feedback: Prinomastat has **decreased** peptide character. As a result, it has improved pharmacokinetic properties. Unfortunately, it suffers from a lack of selectivity.		
incorrect	**a.**#	It has increased peptide character.
incorrect	**b.** #	It has increased selectivity.
Correct	**c.**#	It has improved pharmacokinetic properties.
incorrect	**d.** #	All of the options given are correct.
END OF QUESTION		

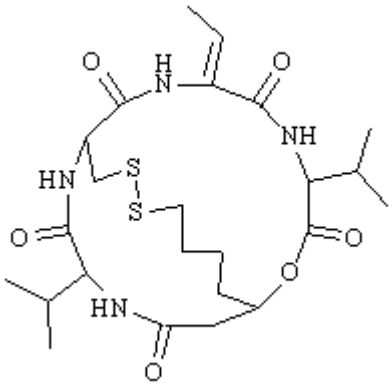
Type: multiple choice
Title: Chapter 21 - Question 20

20)	Bortezomib is a dipeptide that has been approved for the treatment of multiple myeloma.

	 What is the target for the structure?	
Feedback: The target is the proteasome. This is a structure that can destroy redundant proteins in the cell.		
incorrect	**a.#**	Cyclooxygenase 2.
incorrect	**b.#**	Matrix metalloproteinases.
Correct	**c.#**	Proteasomes.
incorrect	**d.#**	Tyrosine hydroxylase.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 21 - Question 21		
21)	Bortezomib is a dipeptide that has been approved for the treatment of multiple myeloma.  The above structure is thought to form an important interaction with a threonine group in the target's active site. Which group is involved in this interaction?	
Feedback: It is thought that a boron-threonine bond is formed at the active site.		
incorrect	**a.#**	The pyrazine ring.
incorrect	**b.#**	The benzyl group.
incorrect	**c.#**	The isopropyl group.
Correct	**d.#**	The boronic acid group.

END OF QUESTION

Type: multiple choice		
Title: Chapter 21 - Question 22		
22)	Depsipeptide is a natural product currently being studied as an anticancer agent:  What is the target for this compound?	
Feedback: Histone deacetylase catalyses the deacetylation of *N*-acetylated lysine residues present in histone. The enzyme is inhibited by depsipeptide.		
incorrect	**a.#**	Methionine aminopeptidase.
Correct	**b.#**	Histone deacetylase.
incorrect	**c.#**	Caspases.
incorrect	**d.#**	Tyrosine hydroxylase.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 21 - Question 23		
23)	What sort of structures are the anticancer agents trastuzumab, rituximab, and cetuximab?	
Feedback: These agents are antibodies which recognise distinctive antigens on cancer cells which have been overexpressed.		
incorrect	**a.#**	Signalling proteins.
Correct	**b.#**	Monoclonal antibodies.
incorrect	**c.#**	Steroids.
incorrect	**d.#**	Carbohydrates.
END OF QUESTION		

Type: multiple choice
Title: Chapter 21 - Question 24
24) Which of the following statements is true regarding antibody-drug conjugates?
Feedback: It is important that the bond between drug and antibody is stable when the complex circulates in the blood supply due to the toxicity of the drugs which have to be used in this therapy. Part of the structure of chimeric antibodies is derived from mice antibodies and so chimeric antibodies can still induce an immune response. The antibody should be humanised to reduce the risks of an immune response.

The antigen that is recognised on the cancer cell should be overexpressed. Antigens on cancer cells are usually likely to be present on normal cells as well, but in smaller quantities. The antibody-drug complex is incapable of dissolving through the cell membrane. The conjugates are taken into the cell by receptor-mediated endocytosis.		
incorrect	**a.#**	Chimeric antibodies should be used since they do not induce an immune response.
incorrect	**b.#**	The antibody is chosen such that it recognises an antigen present on cancer cells but not normal cells.
incorrect	**c.#**	The antibody-drug conjugate should be capable of dissolving through the cell membrane.
Correct	**d.#**	The link between the antibody and the drug should be stable until the conjugate has entered the cell.
END OF QUESTION		

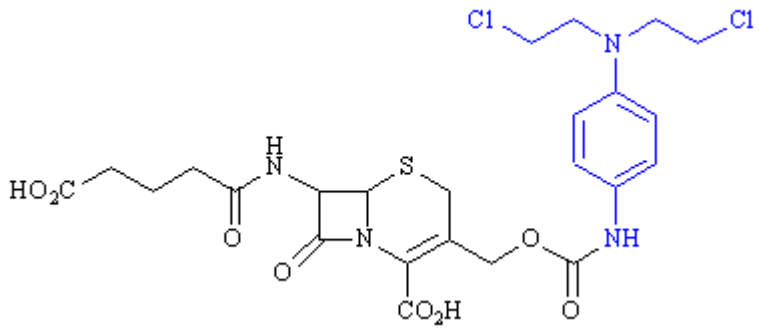
Type: multiple choice	
Title: Chapter 21 - Question 25	
25)	Which of the following groups on antibodies have been used for the attachment of drugs?
Feedback: All three groups can be used and there are advantages and disadvantages for each group. The disulfide group is reduced to thiols in order to attach drugs. The carbohydrate groups are oxidised to produce aldehyde groups to which drugs can be attached. There are lysine groups already present which can be used for the attachment of drugs. However, it is possible that drugs would bind to lysine residues in the arms of the antibody and hinder recognition of the target cell by the antibody.	

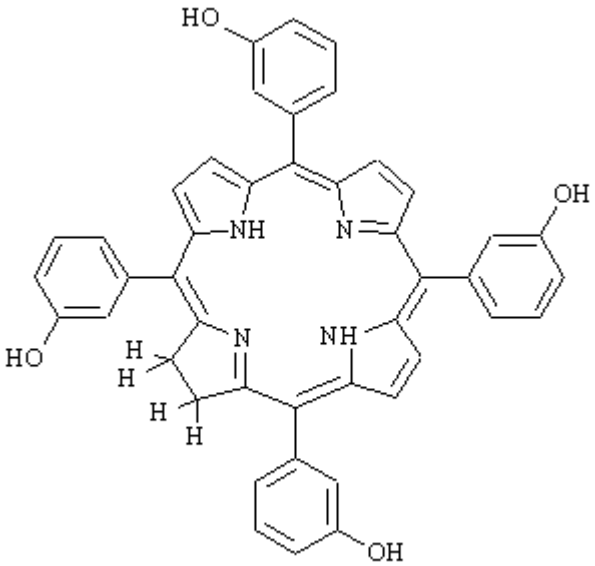
incorrect	**a.#**	Lysine groups.
incorrect	**b.#**	Disulfide links.
incorrect	**c.#**	Carbohydrate groups.
Correct	**d.#**	All of the options given are correct.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 21 - Question 26	
26)	Which of the following drugs is suitable for using in an antibody-drug conjugate?
Feedback: The drug that is used in the antibody-drug conjugate has to be extremely potent since very little of it is delivered to each cancer cell. Agents such as cisplatin, doxorubicin, and 5-fluorouracil are ineffective as the concentrations obtained by this therapy are lower than the therapeutic levels normally required for these agents.	

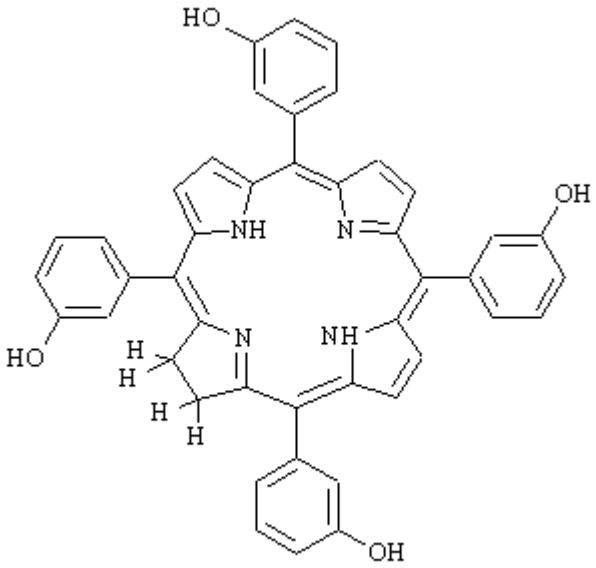
incorrect	**a.#**	Cisplatin.
incorrect	**b.#**	Doxorubicin.
Correct	**c.#**	Calicheamicin.
incorrect	**d.#**	5-fluorouracil.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 21 - Question 27	
27)	The following structure is a cephalosporin prodrug of an alkylating agent. What enzyme can be linked to an antibody to release the alkylating agent?

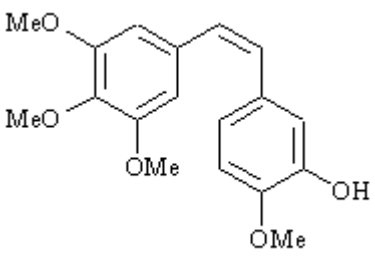
		
Feedback: The β-lactamase catalyses the opening of the β-lactam ring This results in a concerted mechanism which shifts the double bond in the six-membered ring and breaks the bond linking the anticancer drug to the penicillin.		
incorrect	**a.#**	Carboxypeptidase.
Correct	**b.#**	β-Lactamase.
incorrect	**c.#**	Aminopeptidase.
incorrect	**d.#**	Esterase.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 21 - Question 28		
28)	The following structure is undergoing clinical trials for the treatment of various cancers:  What sort of therapy is involved with this structure?	
Feedback: The structure shown is called temoporfin. It is used in photodynamic therapy. The other terms are not used.		
incorrect	**a.#**	Photoactive therapy.
incorrect	**b.#**	Photosynthesis therapy.
Correct	**c.#**	Photodynamic therapy.

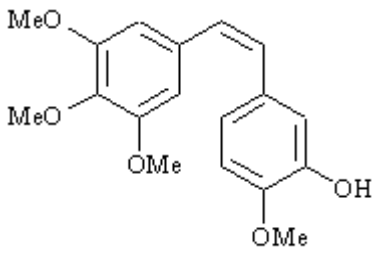
incorrect	**d.** #	Photochemical therapy.
END OF QUESTION		

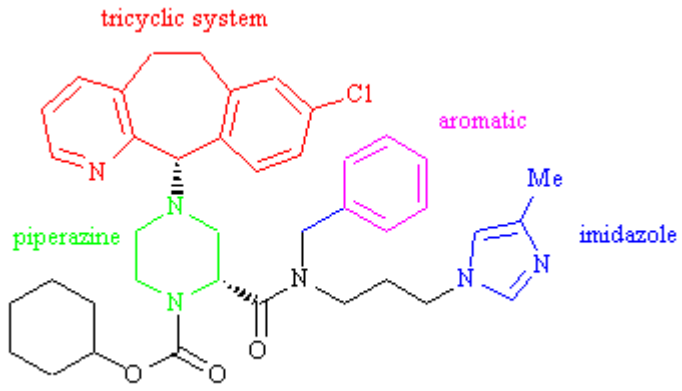
Type: multiple choice	
Title: Chapter 21 - Question 29	
29)	Temoporfin is undergoing clinical trials for the treatment of various cancers.  What is the most serious disadvantage of the above structure?

Feedback: Photosensitivity arises if the drug can accumulate in the eyes and skin where it is activated by the red light in normal sunlight.		
incorrect	**a.#**	Lack of solubility.
Correct	**b.** #	Photosensitivity.
incorrect	**c.#**	Metabolic susceptibility.
incorrect	**d.** #	Rapid excretion.
END OF QUESTION		

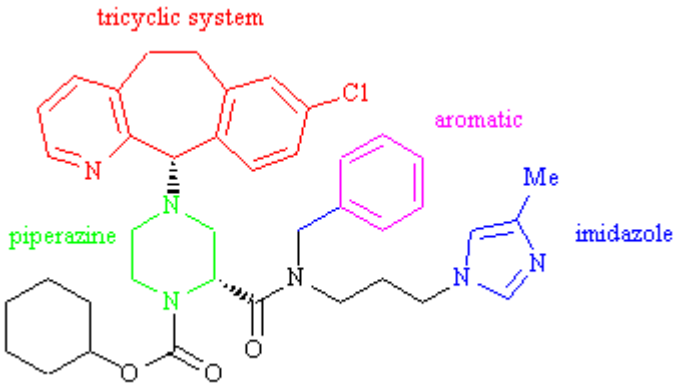
Type: multiple choice	
Title: Chapter 21 - Question 30	
30)	The following structure is combretastatin A-4. What sort of target is affected by this drug? 

Feedback: The drug inhibits tubulin polymerisation by binding to tubulin.		
Correct	**a.#**	Structural protein.
incorrect	**b.#**	Enzyme.
incorrect	**c.#**	Receptor.
incorrect	**d.#**	DNA.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 21 - Question 31		
31)	Which of the following statements is untrue regarding combretastatin A-4  	
Feedback: The drug inhibits angiogenisis rather than promoting apoptosis.		
incorrect	**a.#**	The above structure is being tested as a phosphate prodrug in cancer chemotherapy.
Correct	**b.#**	It promotes apoptosis.
incorrect	**c.#**	The *cis* isomer is more active than the *trans* isomer.
incorrect	**d.#**	It bears a structural similarity to colchicine.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 21 - Question 32		
32)	The following structure is an inhibitor of the enzyme responsible for adding a farnesyl chain to the Ras protein:  What role is played by the imidazole ring system?	
Feedback: The coloured regions shown have the following roles. The imidazole ring is a		

good ligand for zinc ion cofactors and has been used in several drugs that target zinc metalloenzymes. The aromatic ring was added as a steric shield to protect the molecule from metabolism. The tricyclic system and the piperazine ring play a role in binding the drug to the active site of the enzyme.		
incorrect	**a.#**	It is present to increase water solubility.
incorrect	**b.#**	It plays a role in binding the drug to amino acids present in the er site.
incorrect	**c.#**	It is present to protect the molecule from metabolism.
Correct	**d.#**	It binds to a zinc ion cofactor in the enzyme's active site.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 21 - Question 33		
33)	The following structure is an inhibitor of the enzyme responsible for adding a farnesyl chain to the Ras protein:  What role is played by the aromatic ring?	
Feedback: The coloured regions shown have the following roles. The imidazole ring is a good ligand for zinc ion cofactors and has been used in several drugs that target zinc metalloenzymes. The aromatic ring was added as a steric shield to protect the molecule from metabolism. The tricyclic system and the piperazine ring play a role in binding the drug to the active site of the enzyme.		
incorrect	**a.#**	It fits into a h.ydrohobic pocket in the enzyme's active site
incorrect	**b.#**	It plays a role in binding the drug to amino acids present in the enzyme's active site.
Correct	**c.#**	It is present to protect the molecule from metabolism.
incorrect	**d.#**	It binds to a zinc ion cofactor in the enzyme's active site.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 21 – Question 34		
34)	Thalidomide entered phase III clinical trials as a treatment for renal cancer and multiple myeloma. Revamid and actimid are thalidomide analogues which have a primary amino group added to the aromatic ring. What advantage does this group offer?	
Feedback: The presence of a primary amino group may well increase water solubility, but the main advantage in its presence is a reduction in drug toxicity. Using an amino group as a metabolic blocker is not a good idea since the group is susceptible to metabolism itself.		
incorrect	**a.#**	It allows an extra binding interaction and increases activity.

incorrect	**b.#**	It increases water solubility.
Correct	**c.#**	It reduces toxicity.
incorrect	**d.#**	It blocks metabolism at that position.
END OF QUESTION		

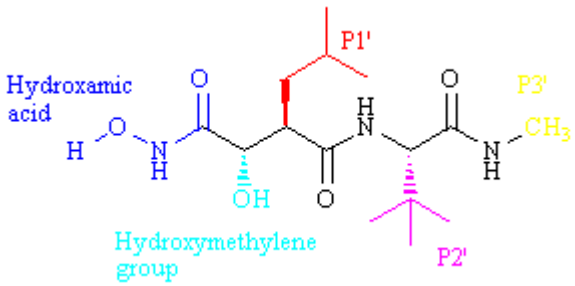
Type: multiple choice		
Title: Chapter 21 – Question 35		
35)	Crisantaspase is used as an anticancer agent in the treatment of leukemia. Which of the following statements is false?	
Feedback: Crisantaspase is a preparation of the asparaginase enzyme which catalyses the breakdown of asparagine such that the cancer cells are starved of this amino acid. However, it is only effective against cancer cells that are unable to synthesise their own asparagine. Therefore, not all leukaemia cells are susceptible.		
incorrect	**a.#**	Crisantaspase is a preparation of the asparaginase enzyme.
Incorrect	**b.#**	The asparaginase enzyme catalyses the breakdown of the amino acid asparagine.
Correct	**c.#**	The agent is effective against all leukaemias.
Incorrect	**d.#**	Leukaemia cells starved of the amino acid asparagine will be killed.
END OF QUESTION		

Type: matching			
Title: Chapter 21 - Question 36			
36)	What terms are used to indicate the following descriptions?		
Feedback: The birth of a cancer is known as carcinogenesis. There are various mechanisms by which this might take place. Apoptosis is the process by which cells commit suicide. If this is inhibited, it increases the chance of carcinogenesis. Metastasis is the process by which tumour cells escape from a primary tumour and set up secondary tumours in other parts of the body. Mutagenesis is the process by which mutations can occur in the cell's DNA which may result in tumour formation.			
a.#	The birth of a cancer Carcinogenesis	=	Carcinogenesis.
b.#	The process by which cells 'commit suicide'	=	Apoptosis.
c.#	The spread of a cancer to different parts of the body and the appearance of secondary tumours	=	Metastasis.
d.#	The promotion of mutations in a cell's DNA	=	Mutagenesis.
END OF QUESTION			

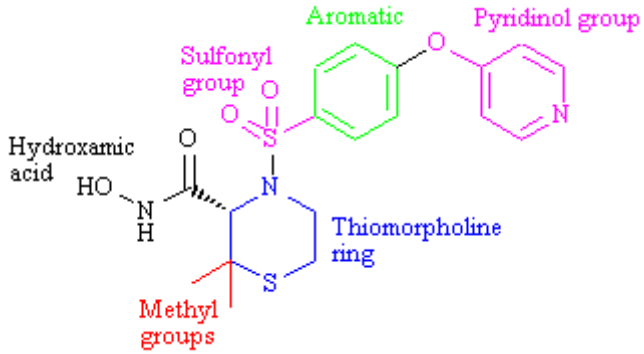
Type: matching			
Title: Chapter 21 - Question 37			
37)	What processes are catalysed by the following enzymes?		
Feedback: All the processes catalysed by the enzymes mentioned play an important role in various tumours. The enzymes can be viewed as potential targets for novel anticancer agents			
a.#	The construction of 'tails' at the end of chromosomes	=	Telomerases.
b.#	Protein degradation during apoptosis	=	Caspases.
c.#	Breaking down the basement membranes around blood vessels to allow angiogenesis	=	Matrix metalloproteinases.

d.#	Control of the cell cycle	=	Cyclin-dependent kinases.
END OF QUESTION			

Type: matching			
Title: Chapter 21 - Question 38			
38)	What type of agents are the following?		
Feedback: The nitrogen mustards contain alkyl halides which act as alkylating agents. Cisplatin is a platinum based drug which metallates DNA. Doxorubicin can intercalate into DNA and act as a topoisomerase II poison. Methotrexate is classed as an antimetabolite since it inhibits the enzyme dihydrofolate reductase, and indirectly blocks the synthesis of the DNA building block dTMP			
a.#	Nitrogen mustards	=	Alkylating agents.
b.#	Cisplatin	=	Metallating agents.
c.#	Methotrexate	=	Antimetabolites.
d.#	Doxorubicin	=	Intercalating agents.
END OF QUESTION			

Type: matching			
Title: Chapter 21 - Question 39			
39)	The following structure is called marimastat and is a matrix metalloproteinase inhibitor that has undergon clinical trials for the treatment of breast and prostate cancers:  What roles are played by the following groups?		
Feedback: A transition-state isostere mimics the binding interactions of the transitions state of an enzyme-catalysed reaction, but is stable to that reaction. The hydroxamic acid group acts as a bidentate ligand for the zinc ion. The side chain P2' acts as a steric shield to protect the terminal amide from hydrolysis. The side chain P1' fits into a hydrophobic pocket and forms van der Waals interactions.			
a.#	Transition-state isostere	=	Hydroxymethylene group.
b.#	Binding group for a zinc ion cofactor	=	Hydroxamic acid group.
c.#	A binding group for a binding sub pocket which prefers van der Waals interactions	=	The side chain P1'.
d.#	A steric shield	=	The side chain P2' .
END OF QUESTION			

Type: matching	
Title: Chapter 21 - Question 40	
40)	Prinomastat is a second-generation matrix metalloproteinase inhibitor:



What roles are played by the following groups?

Feedback: The methyl groups act as a steric shield to protect the hydroxamic acid group from metabolism. Incorporating the steric shield into a thiomorpholine ring restricts the number of possible conformations and increases activity. The aromatic ring fits the S1' binding sub pocket and can form van der Waals interactions.

a.#	Steric shield	=	The methyl groups.
b.#	Binding group for a zinc ion cofactor	=	Hydroxamic acid group.
c.#	Rigidifying group	=	Thiomorpholine ring.
d.#	A group added to increase the number of binding interactions	=	Pyridinol group.
e.#	A group capable of interacting with the S1'subsite of the target active site	=	Aromatic ring.

END OF QUESTION

Type: multiple choice		
Title: Chapter 23 - Question 01		
01)	What are the two main types of adrenergic receptor?	
Feedback: The two main types of adrenergic receptors are the α and β-adrenoceptors. The muscarinic and nicotinic receptors are the two main types of cholinergic receptors. A-receptors are adenosine receptors. Numerals such as 1 and 2 are used to indicate receptor subtypes such as the α_1-adrenoceptor.		
incorrect	**a.#**	Muscarinic and nicotinic.
incorrect	**b.#**	1 and 2.
Correct	**c.#**	α and β.
incorrect	**d.#**	A1 and A2.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 23 - Question 02		
02)	What kind of receptors are the adrenergic receptors?	
Feedback: The adrenergic receptors bind G-proteins when they are activated and split the G-protein into an α-subunit and a β,γ-dimer. These in turn trigger signal transduction pathways.		
Correct	**a.#**	G-Protein coupled receptors.
incorrect	**b.#**	Kinase linked receptors.
incorrect	**c.#**	Intracellular receptors.
incorrect	**d.#**	Ion channels.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 23 - Question 03		
03)	How many subtypes of the β-adrenoceptor are there?	
Feedback: The subtypes are the β_1-adrenoceptor, β_2-adrenoceptor and β_3-adrenoceptor.		
incorrect	**a.#**	2
Correct	**b.#**	3
incorrect	**c.#**	4
incorrect	**d.#**	5
END OF QUESTION		

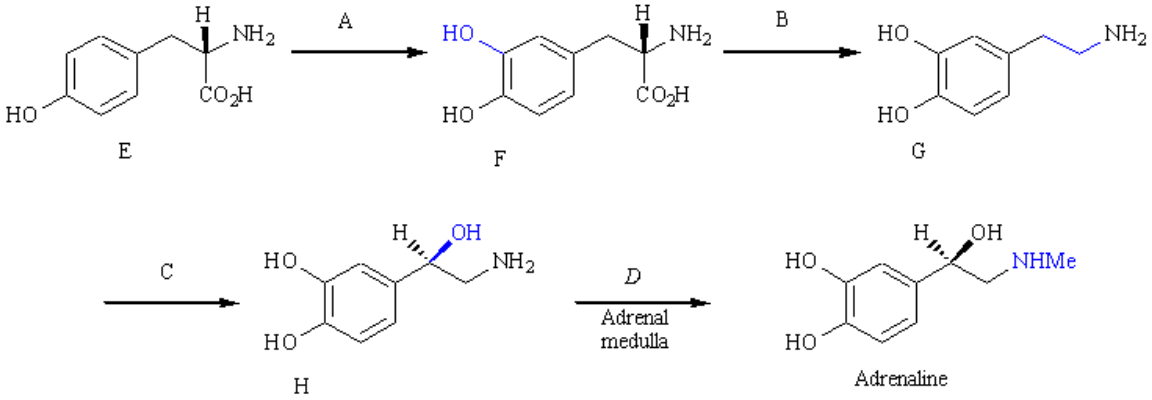
Type: matching	
Title: Chapter 23 - Question 04	
04)	What physiological effect results from the following?
Feedback: The different types and subtypes of adrenoceptors are not uniformly distributed around the body. Certain organs have a predominance of particular receptor types and subtypes. Agents which are designed to be selective for a particular type or subtype will therefore show selectivity for particular tissues or organs. Knowing what the normal effect of receptor activation has on a particular tissue or organ allows one to understand the medicinal uses for drugs that interact with a particular	

receptor. For example, agonists for β_2-adrenoceptors are useful in the treatment of asthma.			
a.#	Activation of β_1-adrenoceptors in heart muscle	=	Muscle contraction.
b.#	Activation of β_2-adrenoceptors in the airways	=	Muscle relaxation.
c.#	Activation of β_3-adrenoceptors	=	Activation of enzymes leading to fat breakdown.
d.#	Activation of β_2-adrenoceptors in smooth muscle of arterioles supplying blood to the skeletal muscles	=	Muscle relaxation.
END OF QUESTION			

Type: matching

Title: Chapter 23 - Question 05

05) The following diagram shows the biosynthetic pathway to adrenaline.

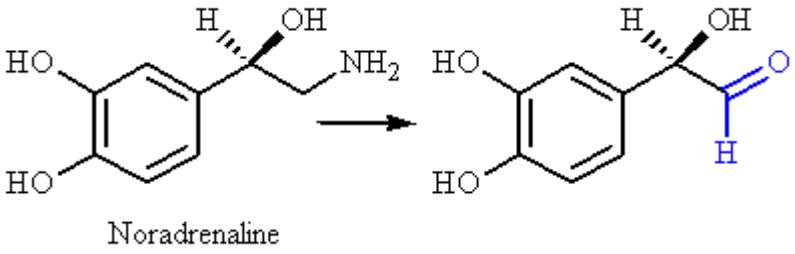


[

What are the enzymes A-D and the structures E-H?

Feedback: Tyrosine hydroxylase catalyses the oxidation of the aromatic ring present in tyrosine to introduce a second phenol group and the formation of levodopa. Dopa decarboxylase catalyses the decarboxylation of levodopa to produce dopamine. Dopamine β-hydroxylase catalyses the oxidation of dopamine in order to introduce an alcohol function leading to the formation of noradrenaline. *N*-Methyl transferase introduces a methyl group onto the nitrogen atom of noradrenaline

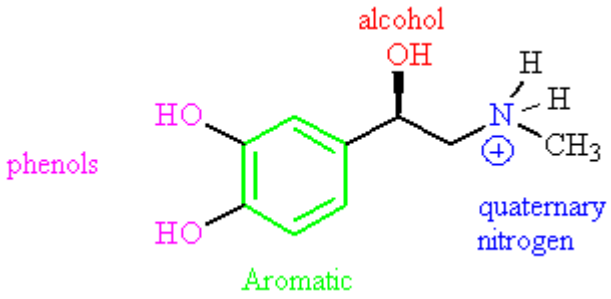
a.#	Tyrosine hydroxylase	=	Enzyme A
b.#	Dopa decarboxylase	=	Enzyme B
c.#	Dopamine β-hydroxylase	=	Enzyme C
d.#	*N*-Methyl transferase	=	Enzyme D
e.#	L-Tyrosine	=	Structure E
f.#	Levodopa	=	Structure F
g.#	Dopamine	=	Structure G
h.#	Noradrenaline	=	Structure H
END OF QUESTION			

Type: multiple choice		
Title: Chapter 23 - Question 06		
06)	Identify the enzyme that is responsible for catalysing the following metabolic reaction.  Noradrenaline	
Feedback: Monoamine oxidase catalyses the oxidation of a primary amino group to an aldehyde. Catechol *O*-methyltransferase catalyses the methylation of one of the catechol phenol groups to a methyl ether. The other two enzymes do not exist.		
incorrect	**a.#**	Monoamine *O*-methyltransferase.
Correct	**b.#**	Monoamine oxidase.
incorrect	**c.#**	Catechol *O*-methyltransferase.
incorrect	**d.#**	Catechol oxidase.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 23 - Question 07		
07)	How is noradrenaline removed from a nerve synapse once it has served its function as a chemical messenger?	
Feedback: Transport proteins in the presynaptic nerve transport noradrenaline back into the nerve such that it can be taken up in vesicles. It is then ready to be released again when the nerve becomes active.		
incorrect	**a.#**	It is metabolised by enzymes present on the outer surface of nerve membranes.
Correct	**b.#**	It is transported into the presynaptic nerve by transport proteins.
incorrect	**c.#**	It is transported into the postsynaptic nerve by transport proteins.
incorrect	**d.#**	It diffuses out of the synapse and enters the blood vessels to be taken to the kidneys.
END OF QUESTION		

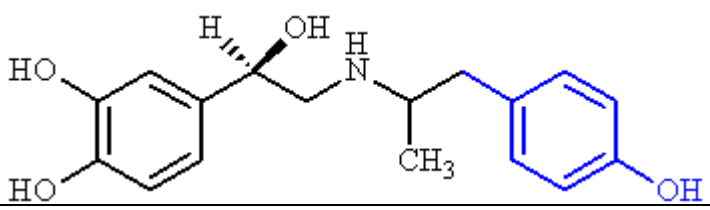
Type: multiple choice		
Title: Chapter 23 - Question 08		
08)	Which of the following agents has a controlling influence on the release of noradrenaline from adrenergic nerves?	
Feedback: Prostaglandins, acetylcholine, and noradrenaline itself all inhibit the release of noradrenaline from presynaptic nerves. Each of these chemical messengers interacts		

with a receptor on the pre-synaptic nerve that is specific for that chemical messenger.		
incorrect	**a.#**	Prostaglandins.
incorrect	**b.#**	Acetylcholine.
incorrect	**c.#**	Noradrenaline.
Correct	**d.#**	All of the answer are correct.
END OF QUESTION		

Type: matching	
Title: Chapter 23 - Question 09	
09)	The structure shown below is adrenaline. What sort of binding interactions do the following functional groups have with the binding site of the adrenergic receptor; i) the phenols, ii) the aromatic ring, iii) the alcohol, and iv) the quaternary nitrogen atom? 

Feedback: The phenol groups and the alcohol group act as hydrogen bond donors. The aromatic ring interacts through van der Waals interactions. The quaternary nitrogen interacts through ionic interactions.

a.#	(i) The phenols	=	Hydrogen bond donor (1).
b.#	(ii) The aromatic ring	=	Van der Waals interactions.
c.#	(iii) The alcohol	=	Hydrogen bond donor (2).
d.#	(iv) The quaternary nitrogen atom	=	Ionic interactions.
END OF QUESTION			

Type: multiple choice	
Title: Chapter 23 - Question 10	
10)	The following structure was found to have an 800-fold increase in agonist activity when the blue coloured group was added. What drug design strategy is illustrated here? 

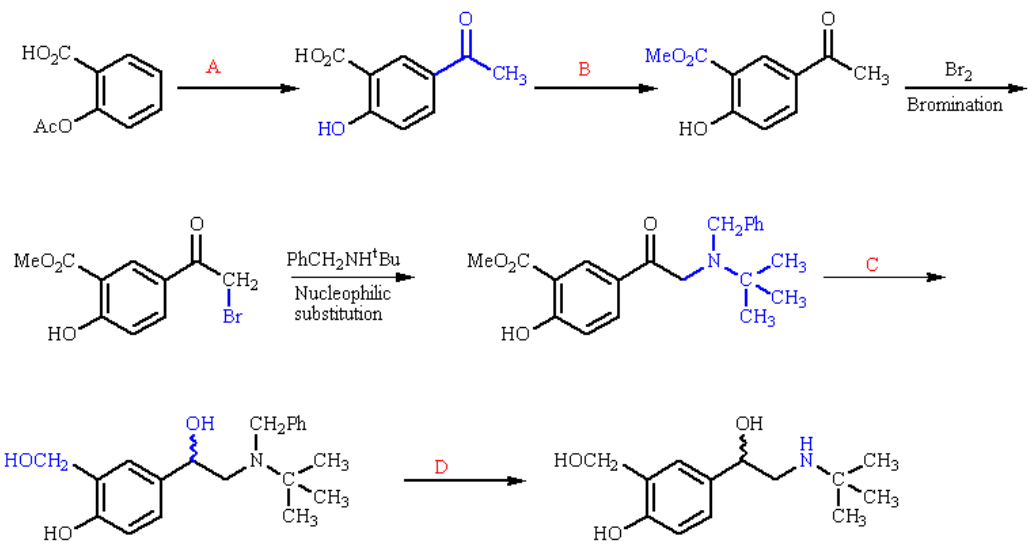
Feedback: The extra phenol is capable of interacting with an extra binding region by means of hydrogen bonding, and so this illustrates the strategy of extension - adding

extra groups to find extra binding regions within the binding site.		
Correct	**a.#**	Extension.
incorrect	**b.#**	Rigidification.
incorrect	**c.#**	Substituent variation.
incorrect	**d.#**	Conformational blocking.
END OF QUESTION		

Type: matching

Title: Chapter 23 – Question 11

11) The following synthetic route has been used to prepare SALBUTAMOL. What are the reagents A – D?



Feedback: Aluminium chloride and nitrobenzene are used to carry out a Fries rearrangement to produce a keto acid from aspirin. Methanol and hydrochloric acid are used to carry out an esterification with the hydrochloric acid acting as an acid catalyst. Lithium aluminium hydride is a reducing agent which converts the ketone present in the structure to an alchohol. Tetrahydrofuran is present as solvent. A hydrogenolysis reaction is carried out using hydrogen gas in the presence of a palladium/charcoal catalyst.

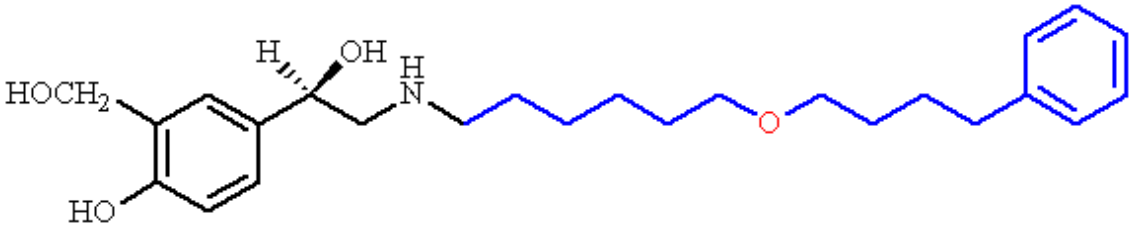
a.#	Aluminium trichloride and nitrobenzene	=	A
b.#	Methanol and hydrochloric acid	=	B
c.#	Lithium aluminium hydride and tetrahydrofuran	=	C
d.#	Hydrogen gas over a palladium charcoal catalyst	=	D

END OF QUESTION

Type: multiple choice

Title: Chapter 23 - Question 12

12) The following structure (salmeterol) has twice the potency of salbutamol, and an extended action of 12 hours.

	 What role does the red coloured oxygen play?

Feedback: It does increase the polarity of the molecule but the group was not introduced in order to increase polarity for its own sake, but to introduce a hydrogen bond acceptor. There is no evidence to indicate that a methylene group at that position is overly susceptible to metabolism. It is impossible for an ether to act as a hydrogen bond donor.

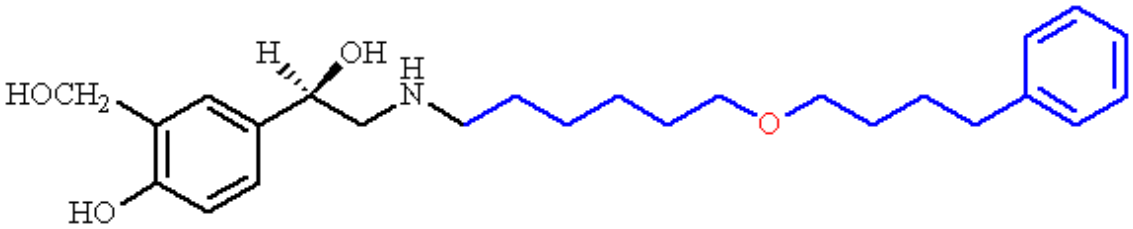
incorrect	**a.#**	It increases the polarity of the molecule.
incorrect	**b.#**	It acts as an isostere for a methylene group that is susceptible to metabolism.
Correct	**c.#**	It acts as a hydrogen bond acceptor to an extra hydrogen bonding region in the binding site.
incorrect	**d.#**	It acts as a hydrogen bond donor to an extra hydrogen bonding region in the binding site.

END OF QUESTION

Type: multiple choice

Title: Chapter 23 - Question 13

13) The following structure (salmeterol) has twice the potency of salbutamol, and an extended action of 12 hours.



What was the principle reason for including the blue coloured regions of the molecule?

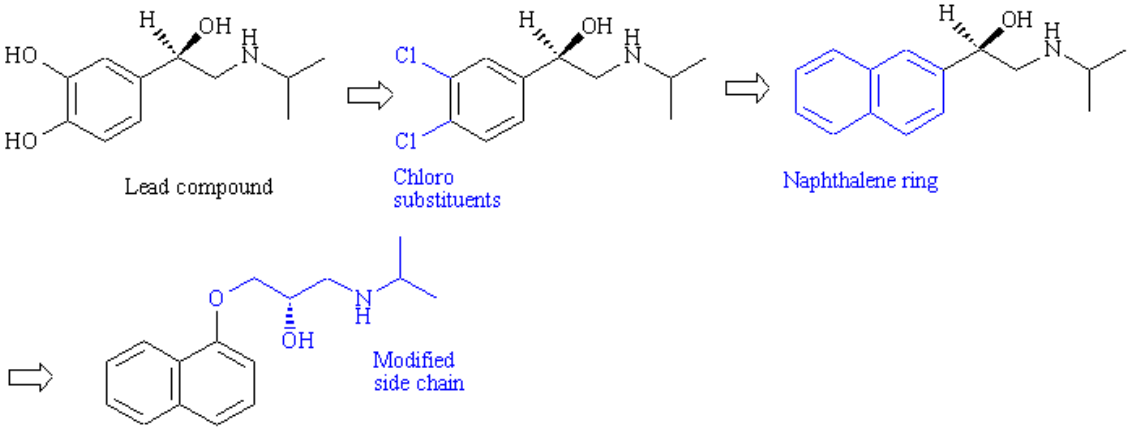
Feedback: The principle reason for adding the extended alkyl chain and aromatic ring is that they increase the hydrophobicity of the molecule such that it binds more strongly and for longer periods to lipophilic tissue in the vicinity of adrenoceptors.. However, it is possible that these groups also interact with hydrophobic binding regions in the binding site of the receptor.

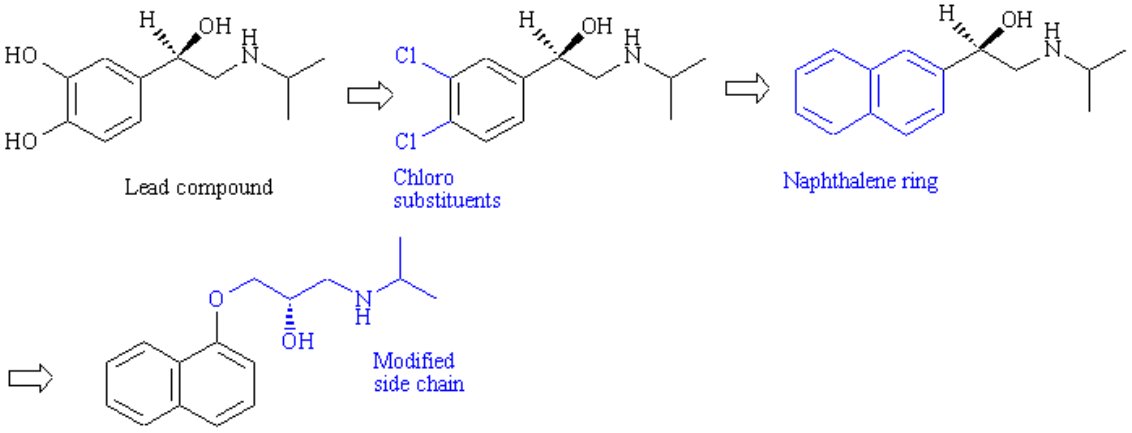
incorrect	**a.#**	They interact with hydrophobic binding regions in the binding site.
incorrect	**b.#**	The alkyl chain acts as a flexible arm positioning the aromatic ring correctly such that it can interact with an extra hydrophobic region by van der Waals interactions.
incorrect	**c.#**	They increase the hydrophobicity of the molecule such that it is less solvated.
Correct	**d.#**	They increase the hydrophobicity of the molecule such that it binds more strongly and for longer periods to lipophilic tissue in the vicinity of adrenoceptors.

END OF QUESTION

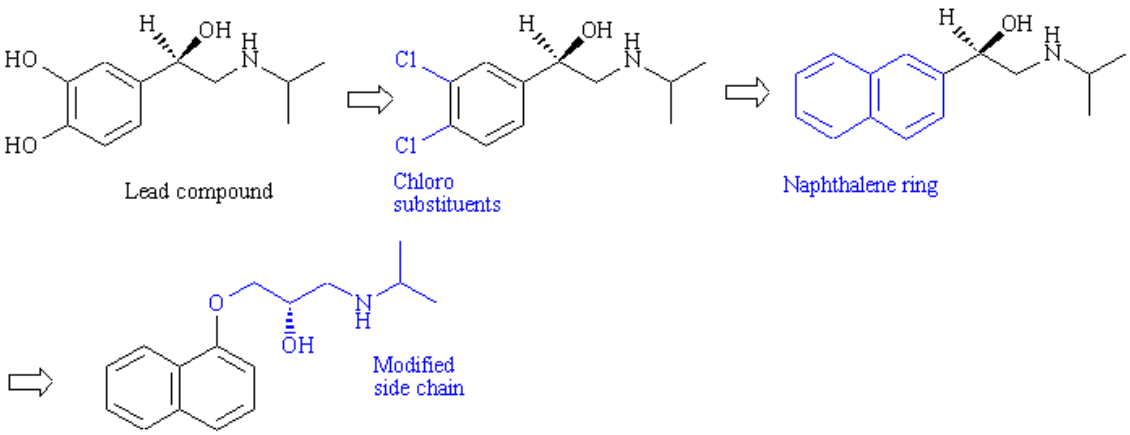
Type: multiple choice

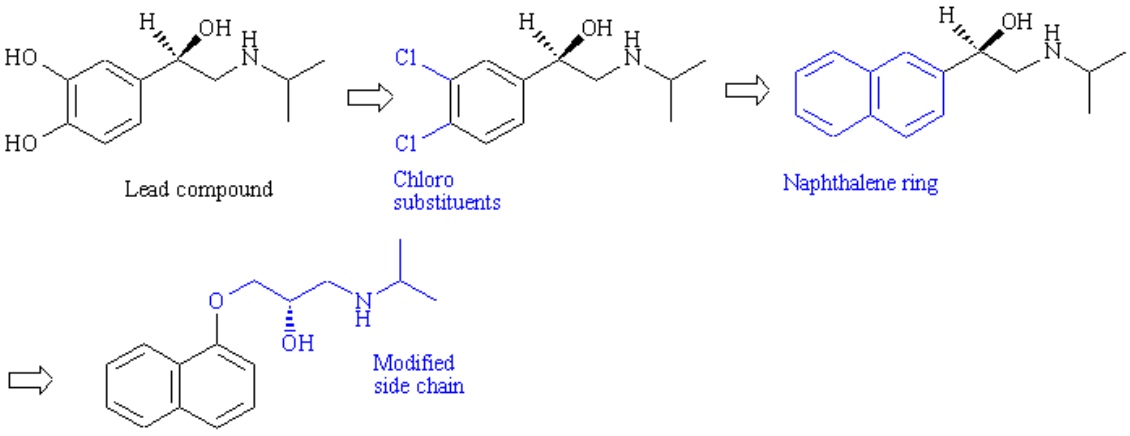
Title: Chapter 23 - Question 14

14)	The following diagram illustrates some of the compounds that were involved in the development of an important clinical agent.  What sort of agent is the final structure?	
Feedback: The agent is propranolol. Propranolol was first introduced for the treatment of angina but is now more commonly used for the treatment of hypertension.		
incorrect	**a.#**	Anti-asthmatic agent.
Correct	**b.#**	Cardiovascular agent.
incorrect	**c.#**	Analgesic.
incorrect	**d.#**	Anti-ulcer agent.
END OF QUESTION		

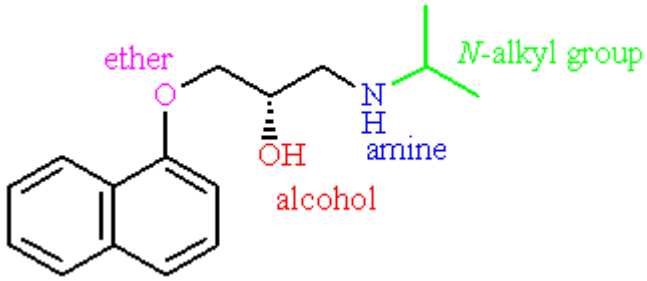
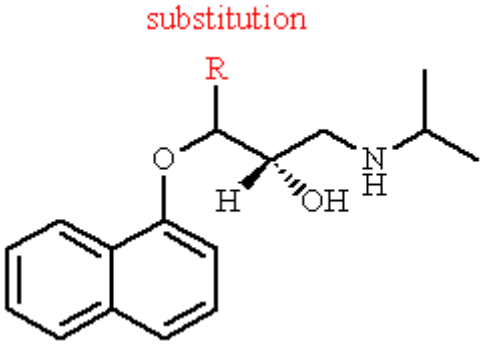
Type: multiple choice		
Title: Chapter 23 - Question 15		
15)	The following diagram illustrates some of the compounds that were involved in the development of an important clinical agent.  What role does the clinical agent play?	
Feedback: The agent is propranolol. Propranolol acts as an antagonist of the β_1-adrenoceptors present in heart muscle. This inhibits the force and frequency of heart muscle contractions and relieves the symptoms of angina.		
incorrect	**a.#**	An agonist at α-adrenoceptors.
incorrect	**b.#**	An antagonist at α-adrenoceptors.
incorrect	**c.#**	An agonist at β-adrenoceptors.

Correct	**d.#**	An antagonist at β-adrenoceptors.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 23 - Question 16		
16)	The following diagram illustrates some of the compounds that were involved in the development of the important clinical agent – propranolol.  What effect does the replacement of phenol groups in the lead compound with chloro groups have?	
Feedback: A partial agonist still activates its receptor, but not to the same extent as a full agonist.		
incorrect	**a.#**	It makes the agent an agonist.
incorrect	**b.#**	It makes the agent an antagonist.
Correct	**c.#**	It makes the agent a partial agonist.
incorrect	**d.#**	It makes the agent inactive.
END OF QUESTION		

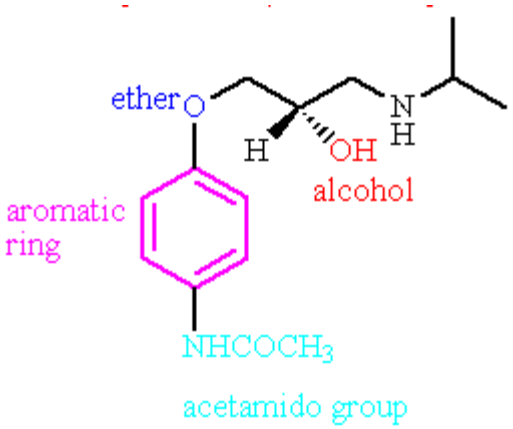
Type: multiple choice		
Title: Chapter 23 - Question 17		
17)	The following diagram illustrates some of the compounds that were involved in the development of the important clinical agent - propranolol.  What effect does modification of the side chain have in the final structure?	

Feedback: Modification of the side chain makes the agent an antagonist. It binds to the receptor but fails to activate it.		
incorrect	**a.#**	It makes the agent an agonist.
Correct	**b.#**	It makes the agent an antagonist.
incorrect	**c.#**	It makes the agent a partial agonist.
incorrect	**d.#**	It makes the agent inactive.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 23 - Question 18	
18)	The diagram below indicates propranolol.  How does substitution on the side chain affect activity? 

Feedback: Substitution on the side chain decreases activity. This may be due to bad steric interactions which make the molecule a poorer 'fit' for the binding site.		
incorrect	**a.#**	It increases activity.
Correct	**b.#**	It decreases activity.
Incorrect	**c.#**	It has no effect.
Incorrect	**d.#**	It is impossible to predict.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 23 - Question 19	
19)	Practolol is a second-generation β-blocker with selectivity for β_2-adrenoceptors over β_1-adrenoceptors.



Which feature of the molecule is responsible for this advantage?

Feedback: The acetamido group is capable of forming an extra binding interaction with the β_1-adrenoceptor.

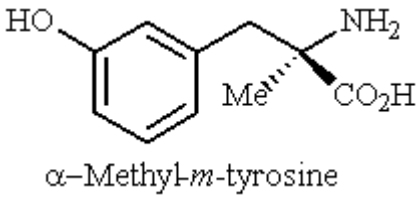
incorrect	**a.#**	The aromatic ring.
Correct	**b.#**	The acetamido group at the *para*-position of the aromatic ring.
incorrect	**c.#**	The ether.
incorrect	**d.#**	The alcohol.

END OF QUESTION

Type: multiple choice

Title: Chapter 23 - Question 20

20) The following structure is a prodrug used in the treatment of shock. It is converted in the body to metaraminol which is the active agent.



α–Methyl-*m*-tyrosine

What function does the active agent serve?

Feedback: Metaraminol is a false transmitter which is taken up into vesicles and released like the natural neurotransmitter. However, it is less effective at activating adrenergic receptors, and so it down regulates the adrenergic system.

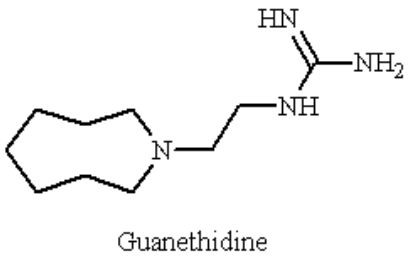
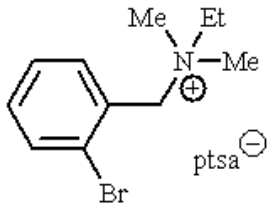
incorrect	**a.#**	It is an adrenergic antagonist.
Correct	**b.#**	It is a false transmitter.
incorrect	**c.#**	It is an enzyme inhibitor.
incorrect	**d.#**	It inhibits noradrenaline uptake by the presynaptic nerve.

END OF QUESTION

Type: multiple choice

Title: Chapter 23 - Question 21

21)	What is the mode of action of the natural product reserpine?	
Feedback: Reserpine binds to the transport proteins responsible for transporting noradrenaline into the storage vesicles within the presynaptic nerve. This prevents the uptake of noradrenaline into storage vesicles.		
incorrect	**a.#**	Adrenergic antagonist.
incorrect	**b.#**	False transmitter.
incorrect	**c.#**	Enzyme inhibitor.
Correct	**d.#**	Transport protein inhibitor.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 23 - Question 22		
22)	What is the mode of action of guanethidine and bretylium?  Guanethidine  Bretylium	
Feedback: These agents are taken up into storage vesicles and displace noradrenaline. They also prevent release of the vesicle's contents into the synaptic gap.		
incorrect	**a.#**	They are adrenergic antagonists.
Correct	**b.#**	They inhibit the release of noradrenaline from storage vesicles.
incorrect	**c.#**	They are enzyme inhibitors.
incorrect	**d.#**	They inhibit noradrenaline uptake into presynaptic nerves.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 23 - Question 23		
23)	What is the mode of action of the tricyclic antidepressants?	
Feedback: The tricyclic antidepressants inhibit the transport proteins that transport noradrenaline from the synaptic gap back into the presynaptic nerve. Noradrenaline levels remain high in the synaptic gap leading to increased adrenergic activity.		
incorrect	**a.#**	They are adrenergic antagonists.
incorrect	**b.#**	They inhibit the release of noradrenaline from storage vesicles.
incorrect	**c.#**	They are enzyme inhibitors.
Correct	**d.#**	They inhibit noradrenaline uptake by presynaptic nerves.

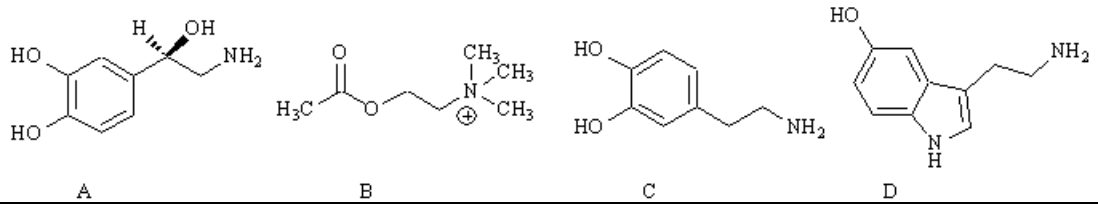
Type: multiple choice

Title: Chapter 23 - Question 24		
24)	What is the clinical use of monoamine oxidase inhibitors?	
Feedback: These agents were used for the treatment of depression. They inhibit the metabolic enzyme monoamine oxidase, and thus reduce the level at which catecholamines such as adrenaline and noradrenaline are metabolised. This results in an increase in adrenergic activity.		
incorrect	**a.#**	Treatment of asthma.
incorrect	**b.#**	Treatment of hypertension.
Correct	**c.#**	Treatment of depression.
incorrect	**d.#**	Treatment for obesity.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 23 - Question 25		
25)	Ripe cheese contains a chemical which results in severe headaches for people who are taking monoamine oxidase inhibitors (MAOIs). What is the chemical in cheese responsible for this effect?	
Feedback: Tyramine has a structure which is similar to noradrenaline. Too much tyramine can enhance the adrenergic system and lead to acute hypertension and severe headaches. This can occur if a patient is taking monoamine oxidase inhibitors.		
incorrect	**a.#**	Tyrosine.
Correct	**b.#**	Tyramine.
incorrect	**c.#**	Threonine.
incorrect	**d.#**	Tryptophan.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 23 - Question 26		
26)	Ripe cheese contains a chemical which results in severe headaches for people who are taking monoamine oxidase inhibitors (MAOIs). Why does this effect take place?	
Feedback: Tyramine has a structure which is similar to noradrenaline. Too much tyramine can enhance the adrenergic system and lead to acute hypertension and severe headaches. This can occur if a patient is taking monoamine oxidase inhibitors since these inhibit the ennzymes which normally metabolise tyramine.		
incorrect	**a.#**	The MAOI and the chemical react with each other to produce a different compound which produces the symptoms.
Correct	**b.#**	The MAOI inhibits the enzyme which normally metabolises the chemical and so there is a build up of the latter in the body. This produces the observed symptoms.
incorrect	**c.#**	The MAOI inhibits the enzyme which normally metabolises the chemical, and so the chemical undergoes an alternative metabolic reaction to produce a different metabolite which is responsible for the symptoms.
incorrect	**d.#**	The chemical binds to monoamine oxidase and blocks the MAOI from binding. Free MAOI produces the observed symptoms.

END OF QUESTION

Type: multiple choice		
Title: Chapter 23 – Question 27		
27)	Which of the following is a natural chemical messenger for the adrenergic receptor?  A B C D	
Feedback: Structure A is noradrenaline. This is the neurotransmitter that activates adrenergic receptors. The other compounds (B-D) are acetylcholine, dopamine, and serotonin. These are neurotransmitters which have their own receptors.		
Correct	**a.#**	Structure A.
incorrect	**b.#**	Structure B.
incorrect	**c.#**	Structure C.
incorrect	**d.#**	Structure D.
END OF QUESTION		

Type: matching			
Title: Chapter 23 - Question 28			
28)	Identify which organ has a predominance of the following receptors.		
Feedback: The β_1-adrenoceptor is predominant in heart muscle. The β_2-adrenoceptor is the predominant type of β-adrenoceptor in bronchial smooth muscle, arteriole smooth muscle, veins, the liver, gastrointestinal tract smooth muscle, and kidney. The β_3-adrenoceptor is predominant in fat cells. The α-adrenoceptors are predominant in arteriole smooth muscle where the arterioles are not supplying muscles.			
a.#	β_1-Adrenoceptors	=	Heart muscle.
b.#	β_2-Adrenoceptors	=	Bronchial smooth muscle.
c.#	β_3-Adrenoceptors	=	Fat cells.
d.#	α-Adrenoceptors	=	Arteriole smooth muscle not supplying muscles.
END OF QUESTION			

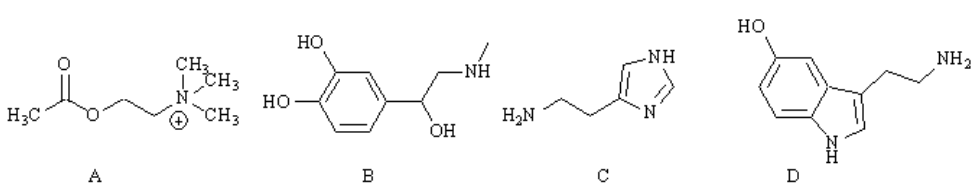
Type: multiple choice	
Title: Chapter 23 - Question 29	
29)	Which of the following has a predominance of α-adrenoceptors?
Feedback: The β_1-adrenoceptor is predominant in heart muscle. The β_2-adrenoceptor is the predominant type of β-adrenoceptor in bronchial smooth muscle, arteriole smooth muscle, veins, the liver, gastrointestinal tract smooth muscle, and kidney. The β_3-adrenoceptor is predominant in fat cells. The α-adrenoceptors are predominant in arteriole smooth muscle where the arterioles are not supplying muscles.	

incorrect	**a.#**	Fat cells.
incorrect	**b.#**	Bronchial smooth muscle.
incorrect	**c.#**	Heart muscle.
Correct	**d.#**	Arteriole smooth muscle not supplying muscles.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 23 – Question 30		
30)	Which of the following adrenoceptors is the main target for anti-asthmatic drugs?	
Feedback: Agonists of the β_2-adrenoceptors in bronchial smooth muscle result in smooth muscle relaxation and dilation of the airways to relieve asthma.		
incorrect	**a.#**	β_1-Adrenoceptor.
Correct	**b.#**	β_2-Adrenoceptor.
incorrect	**c.#**	β_3-Adrenoceptor.
incorrect	**d.#**	α_2-Adrenoceptor.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 23 - Question 31		
31)	Which of the following adrenoceptors is the main target for anti-hypertensive drugs?	
Feedback: Antagonists of the β_1-adrenoceptors were originally designed as β-blockers to inhibit heart muscle contraction and force, and thus treat angina. However, they are more commonly used to treat hypertension (high blood pressure).		
Correct	**a.#**	β_1-Adrenoceptor.
incorrect	**b.#**	β_2-Adrenoceptor.
incorrect	**c.#**	β_3-Adrenoceptor.
incorrect	**d.#**	α_2-Adrenoceptor.

END OF QUESTION

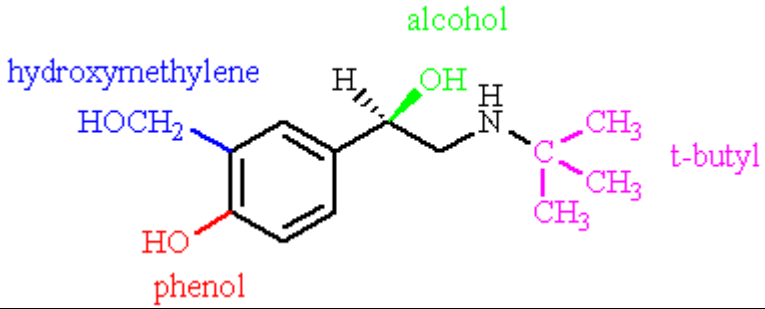
Type: multiple choice	
Title: Chapter 23 – Question 32	
32)	Which of the following structures is a catecholamine?  A B C D

Feedback: Structure B is a catecholamine. The catecholamines all contain an aromatic ring with two neighbouring phenolic groups, as well as an alkyl chain bearing an amino group. Structure B is adrenaline. This is a hormone that activates adrenergic receptors. The other compounds (A, C and D) are acetylcholine, dopamine, and serotonin respectively. These are neurotransmitters which have their own receptors.		
incorrect	**a.#**	Structure A.
Correct	**b.#**	Structure B.
incorrect	**c.#**	Structure C.
incorrect	**d.#**	Structure D.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 23 - Question 33		
33)	Which of the following statements is false regarding analogues of noradrenaline?	
Feedback: The selectivity is the other way round with selectivity for β-receptors over α-receptors.		
incorrect	**a.#**	A naphthalene ring is good for selective β-antagonists.
Correct	**b.#**	A bulky *N*-alkyl substituent introduces selectivity for α-receptors over β-receptors.
incorrect	**c.#**	Removal of one of the catechol phenol groups leads to a drop in activity.
incorrect	**d.#**	Addition of an α-methyl group increases selectivity for the α_2-adrenoceptor.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 23 – Question 34		
34)	Which of the following statements about adrenaline is false?	
Feedback: Adrenaline acts like a hormone when it is released in the body. It is not rapidly metabolised and has a reasonable life time. It also has a good activity. The disadvantage with adrenaline is that it activates all the adrenergic receptors that it reaches and shows no selectivity between different types and sub types. It is a useful medicine, especially in emergency situations of cardiac arrest or anaphylactic reactions.		
incorrect	**a.#**	It acts like a hormone.
Correct	**b.#**	It is rapidly metabolised.
incorrect	**c.#**	It shows poor selectivity between different types of adrenoceptor.
incorrect	**d.#**	It is a useful medicine.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 23 - Question 35	
35)	The following structure is the anti-asthmatic agent salbutamol. Which of the following groups is important in conferring selectivity for β-

	adrenoceptors? 

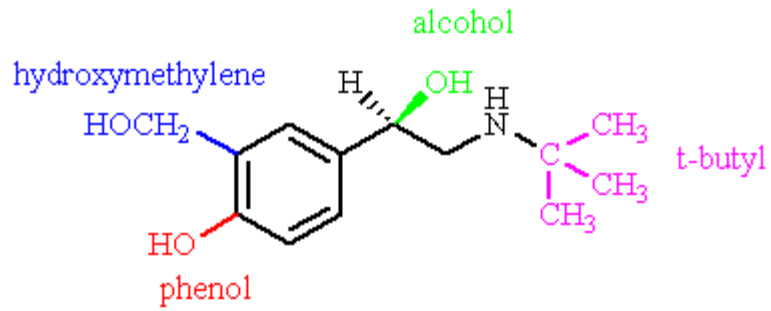
Feedback: The β-adrenoceptor has a larger hydrophobic pocket available to *N*-alkyl substituents than the α-adrenoceptor, and so bulky *N*-alkyl substituents confer selectivity for β-adrenoceptors.

incorrect	**a.#**	The hydroxymethylene group.
incorrect	**b.#**	The phenol.
incorrect	**c.#**	The alcohol.
Correct	**d.#**	The *tertiary* butyl group.

END OF QUESTION

Type: multiple choice

Title: Chapter 23 – Question 36

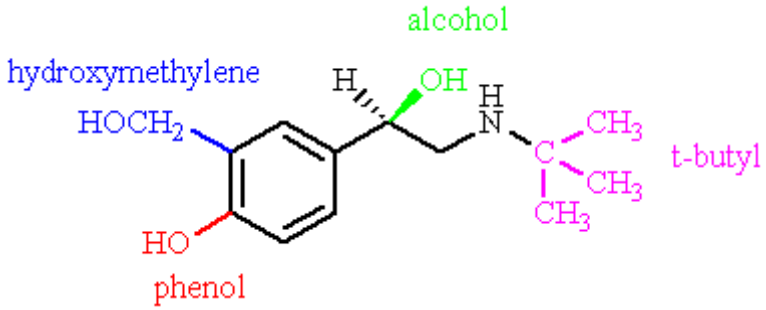
36)	The following structure is the anti-asthmatic agent salbutamol. Which of the coloured groups interacts with a hydrophobic binding region in the adrenergic binding site? 

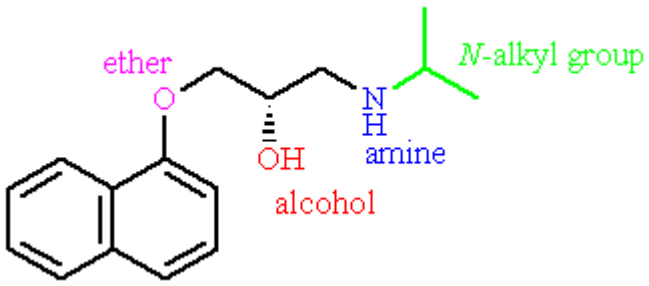
Feedback: The β-adrenoceptor has a larger hydrophobic pocket available to *N*-alkyl substituents than the α-adrenoceptor, and so bulky *N*-alkyl substituents confer selectivity for β-adrenoceptors by interacting with this pocket via van der Waals interactions.

incorrect	**a.#**	The hydroxymethylene group.
Incorrect	**b.#**	The phenol.
incorrect	**c.#**	The alcohol.
Correct	**d.#**	The *tertiary* butyl group.

END OF QUESTION

Type: multiple choice

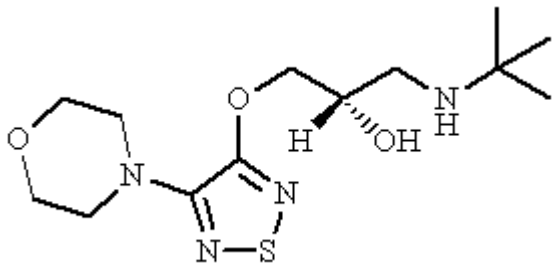
Title: Chapter 23 - Question 37		
37)	The following structure is the anti-asthmatic agent salbutamol. Which of the following groups is important in conferring greater metabolic stability compared to noradrenaline?  	
Feedback: The original phenol is metabolised to a methyl ether by the metabolic enzyme catechol *O*-methyltransferase. Shifting the OH group away from the aromatic ring by one bond makes it unrecognisable to the metabolic enzyme, but the group can still interact by hydrogen bonding with the receptor binding site.		
Correct	**a.#**	The hydroxymethylene group.
incorrect	**b.#**	The phenol.
incorrect	**c.#**	The alcohol.
incorrect	**d.#**	The *tertiary* butyl group.
END OF QUESTION		

Type: matching			
Title: Chapter 23 - Question 38			
38)	Identify the binding interactions that take place to the adrenergic receptor for the following groups in propr  		
Feedback: The amine is protonated and interacts with the binding site by ionic interactions. The ether acts as a hydrogen bond acceptor. The alcohol can act both as a hydrogen bond donor and a hydrogen bond acceptor. The *N*-alkyl group interacts with a hydrophobic region by van der Waals interactions.			
a.#	Hydrogen bonding as a hydrogen bond acceptor	=	Ether.
b.#	Hydrogen bonding as a hydrogen bond donor and/or acceptor	=	Alcohol.
c.#	Ionic interactions	=	Amine.
d.#	Van der Waals interactions	=	*N*-Alkyl group.
END OF QUESTION			

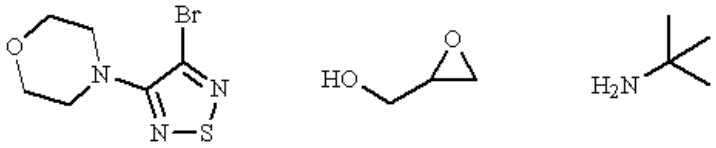
Type: multiple choice

Title: Chapter 23 – Question 39

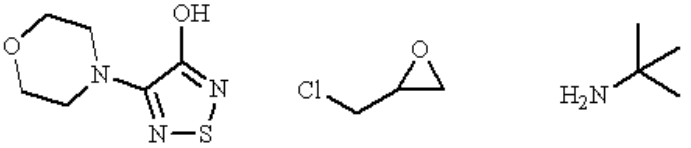
39) What reagents could be used to synthesise timolol?



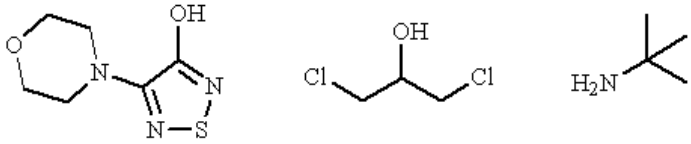
Reagent A



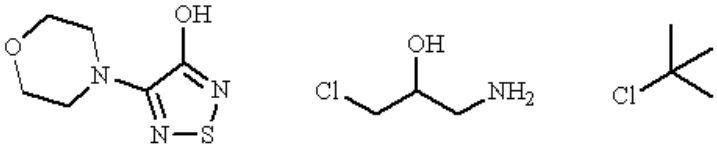
Reagent B



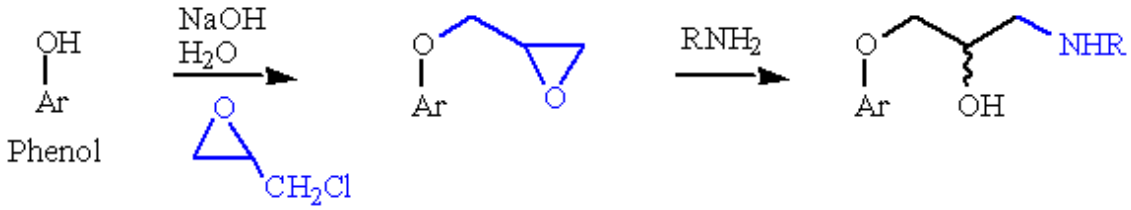
Reagent C



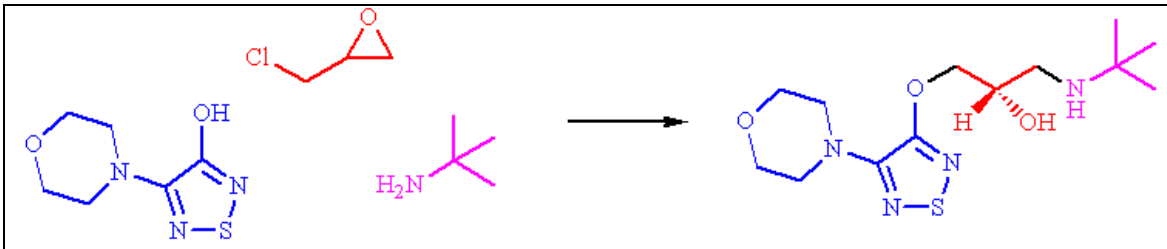
Reagent D



Feedback: The general synthesis for aryloxypropanolamines is shown below:



Therefore the 'building blocks' for timolol are as shown below:



incorrect	**a.#**	Reagent A.
Correct	**b.#**	Reagent B.
incorrect	**c.#**	Reagent C.
incorrect	**d.#**	Reagent D.
END OF QUESTION		

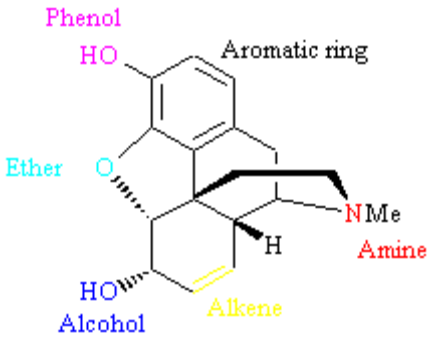
Type: multiple choice		
Title: Chapter 23 – Question 40		
40)	Which of the following agents inhibits the transport proteins that transport noradrenaline from the synaptic gap back into the presynaptic nerve.?	
Feedback: Desipramine is an example of a tricyclic antidepressant. The tricyclic antidepressants inhibit the transport proteins that transport noradrenaline from the synaptic gap back into the presynaptic nerve. Noradrenaline levels remain high in the synaptic gap leading to increased adrenergic activity.		
incorrect	**a.#**	Propranolol.
incorrect	**b.#**	Guanethidine.
Correct	**c.#**	Desipramine.
incorrect	**d.#**	Salbutamol.
END OF QUESTION		

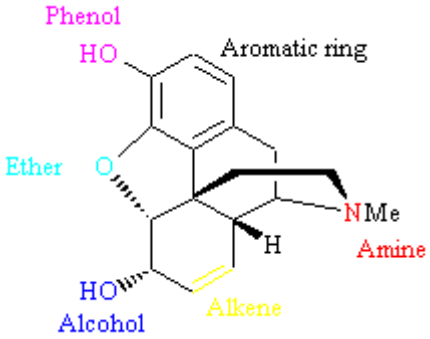
Type: multiple choice		
Title: Chapter 23 - Question 41		
41)	What is the mode of action of phenelzine?	
Feedback: Phenelzine inhibits the metabolism of noradrenaline leading to increased levels of the neurotransmitter and increased adrenergic activity.		
Correct	**a.#**	Monoamine oxidase inhibitor.
incorrect	**b.#**	Inhibitor of noradrenaline uptake.
incorrect	**c.#**	Adrenergic agonist.
incorrect	**d.#**	Adrenergic antagonist.
END OF QUESTION		

Type: multiple choice
Title: Chapter 23 – Question 42

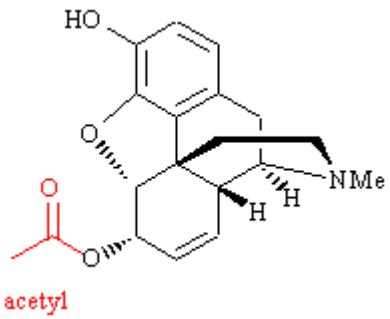
42)	What is the mode of action of methylphenidate?	
Feedback: Methylphenidate inhibits noradrenaline uptake and acts as a stimulant.		
incorrect	**a.#**	Monoamine oxidase inhibitor.
Correct	**b.#**	Inhibitor of noradrenaline uptake.
incorrect	**c.#**	Adrenergic agonist.
incorrect	**d.#**	Adrenergic antagonist.
END OF QUESTION		

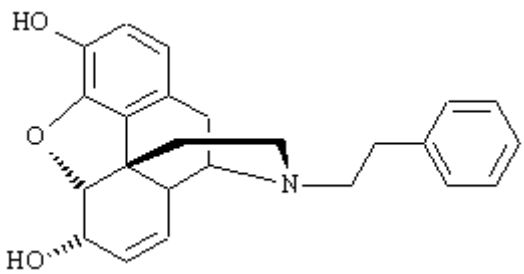
Type: multiple choice		
Title: Chapter 23 - Question 43		
43)	What side effect can affect patients taking monoamine oxidase inhibitors (MAOI's) if they eat ripe cheese?	
Feedback: Severe headache can arise. Ripe cheese contains tyramine which is normally metabolised by monoamine oxidases. Inhibition of these enzymes allows tyramine levels to increase, resulting in the effects observed.		
incorrect	**a.#**	Dizziness.
incorrect	**b.#**	Nausea.
Correct	**c.#**	Severe headache.
incorrect	**d.#**	Constipation.
END OF QUESTION		

Type: matching			
Title: Chapter 24 - Question 01			
01)	What interactions are involved in binding the following groups to an opioid receptor b  		
Feedback: The important binding groups for morphine are the aromatic ring, phenol and amine. The amine is protonated and ionised when it interacts with the target binding site.			
a.#	The group has no binding role.	=	Alkene
b.#	Hydrogen bonding.	=	Phenol
c.#	Ionic bonding.	=	Amine
d.#	Van der Waals interactions.	=	Aromatic ring
END OF QUESTION			

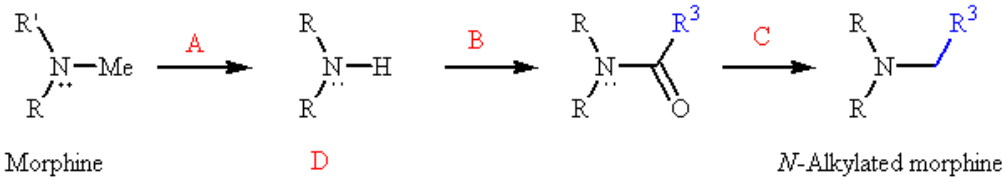
Type: multiple choice		
Title: Chapter 24 - Question 02		
02)	Morphine is an important analgesic.  Which is the most serious short term side effect morphine?	
Feedback: An overdose of morphine can suppress the respiratory centre in the central nervous system leading to cessation of breathing and death.		
incorrect	**a.#**	Tolerance and depression.
incorrect	**b.#**	Constipation.
Correct	**c.#**	Depression of the respiratory centre.
incorrect	**d.#**	Pupil constriction.
END OF QUESTION		

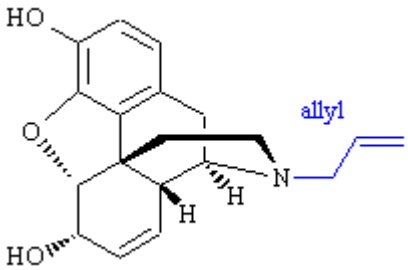
Type: multiple choice

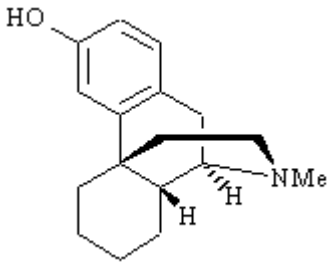
Title: Chapter 24 - Question 03		
03)	The following structure is more active than morphine as an analgesic.  What is the principle reason for this?	
Feedback: The extra acetyl group masks a polar alcohol group such that the drug can cross the blood brain barrier more easily. Once in the brain, it can interact immediately with analgesic receptors since the important phenol group is free. There is no extra binding region for a group of this sort. A polar group is masked, but the drug interacts with receptors on the surface of the cells and does not need to cross cell membranes to reach its target. The phenol is the major site for metabolic conjugation, not the alcohol. Therefore, masking the alcohol does little to reduce metabolism and excretion.		
incorrect	**a.#**	The extra acetyl group interacts with an extra binding region within the binding site by acting as a hydrogen bond acceptor.
Correct	**b.#**	The extra acetyl group masks a polar group allowing the structure to cross the blood brain barrier more efficiently.
incorrect	**c.#**	The extra acetyl group masks a polar group allowing the structure to cross cell membranes more efficiently.
incorrect	**d.#**	The extra acetyl group masks an alcohol group which would otherwise undergo a phase II metabolic conjugation reaction leading to rapid excretion of the drug.
END OF QUESTION		

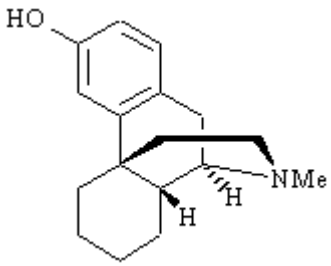
Type: multiple choice		
Title: Chapter 24 - Question 04		
04)	The *N*-phenethyl analogue of morphine has an analgesic activity which is 14 times greater than that of morphine. What is the principle reason for this?  	
Feedback: The aromatic ring interacts with an extra hydrophobic binding region by van der Waals interactions. This is the main reason for the increased activity although the increased hydrophobicity should also increase the levels of drug crossing the blood brain barrier. Aromatic rings are hydrophobic and will not interact well with hydrophilic binding regions. The major metabolic reaction for opiates is a phase II conjugation reaction carried out on the phenol group.		
incorrect	**a.#**	The aromatic ring of the phenethyl group interacts with an extra hydrophilic region of the binding site.
Correct	**b.#**	The aromatic ring of the phenethyl group interacts with an extra hydrophobic region of the binding site.

incorrect	**c.#**	The phenethyl group increases the hydrophobicity of the molecule allowing a greater percentage to cross the blood brain barrier.
incorrect	**d.#**	The phenethyl group acts as a steric shield to protect the amine group from drug metabolism.
END OF QUESTION		

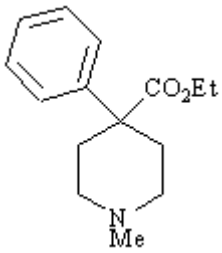
Type: matching			
Title: Chapter 24 - Question 05			
05)	The following synthetic route can be used to prepare *N*-alkylated analogues of morphine.  Morphine D *N*-Alkylated morphine What reagents are used in step A , step B, and step C, and what is structure D?		
Feedback: Vinyloxycarbonyl chloride in methanol is a mild reagent that demethylates the tertiary amine of morphine to form a secondary amine. Reaction with an acid chloride acylates the secondary amine to give a tertiary amide. Reduction with Lithium aluminium hydride reduces the amide to an amine.			
a.#	Vinyloxycarbonyl chloride and methanol	=	Reagent A
b.#	An acid chloride	=	Reagent B
c.#	Lithium aluminium hydride	=	Reagent C
d.#	Noradrenaline	=	Structure D
END OF QUESTION			

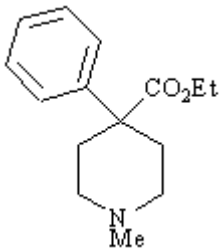
Type: multiple choice		
Title: Chapter 24 - Question 06		
06)	The following structure (nalorphine) is the *N*-allyl analogue of morphine.  What sort of property does this molecule have?	
Feedback: The opioids interact with receptors. Opioids such as morphine act as agonists, but the presence of an *N*-allyl group results in antagonist properties.		
incorrect	**a.#**	It is an agonist.
Correct	**b.#**	It is an antagonist.
incorrect	**c.#**	It is an enzyme inhibitor.
incorrect	**d.#**	It is an inverse agonist.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 24 - Question 07		
07)	Levorphanol is an analgesic which is five times more active than morphine.  What drug design strategy has been used in developing this structure from the lead compound morphine?	
Feedback: The ring containing the ether group has been removed from the morphine structure, as have non-essential functional groups. Therefore, this is a simplification.		
Correct	**a.#**	Simplification.
incorrect	**b.#**	Rigidification.
incorrect	**c.#**	Extension.
incorrect	**d.#**	Substituent variation.
END OF QUESTION		

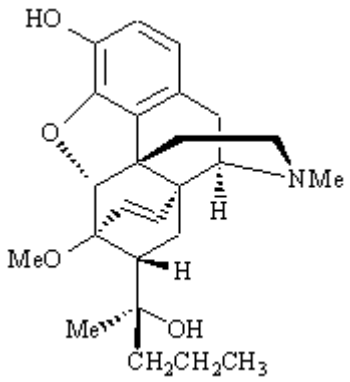
Type: multiple choice		
Title: Chapter 24 - Question 08		
08)	Levorphanol is an analgesic which is five times more active than morphine.  What happens when the *N*-methyl group is replaced with an *N*-phenethyl group (NCH_2CH_2Ph)?	
Feedback: The extra aromatic ring interacts with an extra hydrophobic binding region in the binding site by means of van der Waals interactions. This leads to an increase in agonist activity.		
Correct	**a.#**	There is an increase in activity.
incorrect	**b.#**	There is a decrease in activity.
incorrect	**c.#**	There is a loss of all activity.
incorrect	**d.#**	The compound becomes an antagonist.
END OF QUESTION		

Type: multiple choice

Title: Chapter 24 - Question 09		
09)	The following agent has been used as an analgesic during child birth.  To which class of compounds does this structure belong?	
Feedback: The structure is called pethidine and is an example of a group of compounds called the 4-phenylpiperidines.The 4-phenylpiperidines contain two of the rings originally present in morphine. Benzomorphans and morphinans are more complex structures containing more rings. Enkephalins are endogenous pentapeptides produced in the brain.		
incorrect	**a.#**	Benzomorphans.
Correct	**b.#**	4-phenylpiperidines.
incorrect	**c.#**	Morphinans.
incorrect	**d.#**	Enkephalins.
END OF QUESTION		

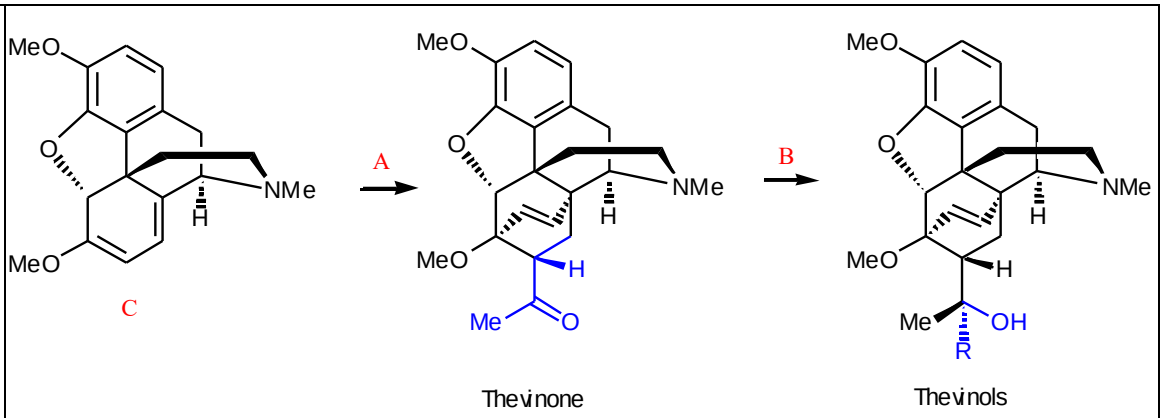
Type: multiple choice		
Title: Chapter 24 - Question 10		
10)	Pethidine has been used as an analgesic during child birth.  Which of the following statements is true regarding the above structure?	
Feedback: The *N*-substituents that lead to antagonism or increased activity in more complex opiates, such as the morphine analogues, the morphinans, and the benzomorphans, do not have these effects in the 4-phenylpiperidine series. This indicates that the 4-phenylpiperidines are binding to the binding site in a different manner.		
incorrect	**a.#**	Replacing the *N*-methyl group with an *N*-allyl group results in an antagonist.
incorrect	**b.#**	Replacing the *N*-methyl group with an *N*-allyl group results in a partial agonist.
incorrect	**c.#**	Replacing the *N*-methyl group with an *N*-phenethyl group increases activity.
Correct	**d.#**	The molecule has a different binding mode with the receptor binding site compared to morphine.
END OF QUESTION		

Type: multiple choice
Title: Chapter 24 - Question 11

11)	Etorphine is used in veterinary medicine.  To which class of compounds does the above structure belong	
Feedback: Benzomorphans are structures lacking two of the rings originally present in morphine. Endorphins are polypeptide structures and endomorphins are tetrapeptides.		
incorrect	**a.#**	Benzomorphans.
incorrect	**b.#**	Endorphins.
Correct	**c.#**	Oripavines.
incorrect	**d.#**	Endomorphins.
END OF QUESTION		

Type: matching			
Title: Chapter 24 - Question 12			
12)	The following synthetic route is used to produce compounds called thevinols. Same diagram as below in Q13 What are the reagents A and B and structure C?		
Feedback: Reaction of thebaine with methyl vinyl ketone results in a Diels Alder reaction which produces an extra six-membered ring. The product also has a ketone group which can undergo reaction with Grignard reagents.			
a.#	Methyl vinyl ketone	=	Reagent A
b.#	Grignard reagent	=	Reagent B
c.#	Thebaine	=	Structure C
END OF QUESTION			

Type: multiple choice	
Title: Chapter 24 - Question 13	
13)	The following synthetic route is used to produce compounds called thevinols.



The starting material has no analgesic activity. Which of the following explanations is the most likely reason for this?

Feedback: The phenol group is an essential part of the analgesic pharmacophore. It is masked in thebaine and so the molecule is inactive. The alkene and the alcohol groups are not part of the pharmacophore, and so the masking or absence of these groups does not explain the lack of activity. The principle reason for inactivity is the masked phenol group.

incorrect	**a.#**	The alcohol group is masked as an ether and cannot act as a binding group.
Correct	**b.#**	The phenol group is masked as an ether and cannot act as a binding group.
incorrect	**c.#**	The alkene group present in morphine is no longer present as a binding group.
incorrect	**d.#**	The diene group distorts the molecule such that it does not fit the binding site.

END OF QUESTION

Type: multiple choice

Title: Chapter 24 - Question 14

14) What pharmacophore is shared by the opioid analgesics?

Feedback: The charged nitrogen interacts with the binding site by ionic bonding. The phenol interacts by hydrogen bonding. The aromatic ring interacts by van der Waals interactions.

Correct	**a.#**	A charged nitrogen, a phenol, and an aromatic ring.
incorrect	**b.#**	A charged nitrogen, an alcohol, and an aromatic ring.
incorrect	**c.#**	A charged nitrogen, a phenol, an aromatic ring, and an ethylene hydrocarbon bridge.
incorrect	**d.#**	A charged nitrogen, a phenol, an aromatic ring, and an *N*-methyl substituent.

END OF QUESTION

Type: multiple choice

Title: Chapter 24 - Question 15

15) What sort of receptors are the opioid receptors?

Feedback: The opioid receptors are G-protein-coupled receptors that activate Gi or Go signal proteins.

incorrect	**a.#**	Calcium ion channel.
Correct	**b.#**	G-protein linked receptor.
incorrect	**c.#**	Potassium ion channel.
incorrect	**d.#**	Kinase-linked receptor.

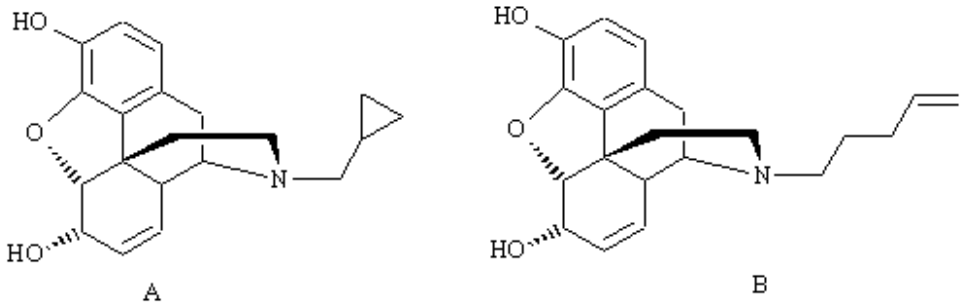
END OF QUESTION

Type: multiple choice		
Title: Chapter 24 - Question 16		
16)	Which of the following receptors is considered the safest target for opioids?	
Feedback: The κ receptor has the least serious side effects. The μ receptor has the most serious side effects. The σ receptor is not an opioid receptor and has no analgesic activity associated with it. Some opioids interact with it and produce hallucinogenic side effects.		
incorrect	**a.#**	μ.
Correct	**b.#**	κ.
incorrect	**c.#**	δ.
incorrect	**d.#**	σ.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 24 -Question 17		
17)	Which of the following describes the properties of nalorphine?	
Feedback: Nalorphine is a safer analgesic than morphine since it acts as an agonist at the κ receptor and as an antagonist at the μ receptor. The κ receptor has less serious side effects associated with it than the μ receptor.		
incorrect	**a.#**	It is an agonist at the μ receptor and the κ receptor.
incorrect	**b.#**	It is an antagonist at the μ receptor and the κ receptor.
Correct	**c.#**	It is an antagonist at the μ receptor and an agonist at the κ receptor.
incorrect	**d.#**	It is an antagonist at the κ receptor and an agonist at the μ receptor.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 24 - Question 18		
18)	Which of the following describes the properties of buprenorphine?	
Feedback: Buprenorphine is a safer analgesic than might be expected for a partial agonist of the μ receptor. This is related to the rate at which the drug binds and dissociates from the binding site.		
incorrect	**a.#**	It is an agonist at the μ receptor, the κ receptor and the δ receptor.
incorrect	**b.#**	It is an antagonist at the μ receptor, the κ receptor and the δ receptor.
Correct	**c.#**	It is an antagonist at the κ and δ receptors, and a partial agonist at the μ receptor.
incorrect	**d.#**	It is an antagonist at the μ receptor and the δ receptor, and a partial agonist at the κ receptor.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 24 - Question 19	
19)	What properties would you expect from the following structures?



Feedback: Structure A has a cyclopropylmethylene substituent which can interact with the hydrophobic antagonist region close to the ionic binding region. Structure B has a hydrophobic alkene group at the end of a lengthy alkyl chain. It is more likely to bind to the agonist binding region than the antagonist binding region since the former region is further from the ionic binding region. Therefore, this compound should act as an agonist.

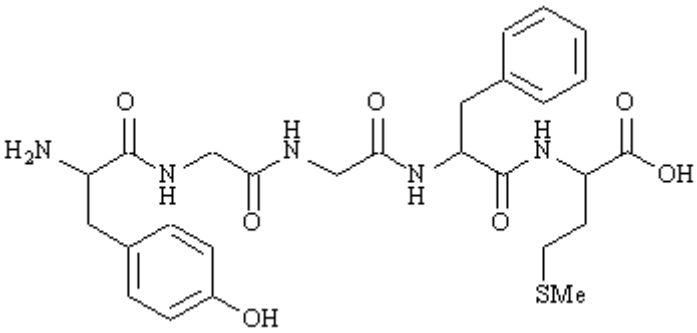
incorrect	**a.#**	Both structures would be agonists.
incorrect	**b.#**	Both structures would be antagonists.
incorrect	**c.#**	Structure A would be an agonist and structure B would be an antagonist.
Correct	**d.#**	Structure A would be an antagonist and structure B would be an agonist.

END OF QUESTION

Type: multiple choice

Title: Chapter 24 - Question 20

20) Consider the following analgesic, Met-enkephalin:



which amino acid in the above structure is crucial to activity?

Feedback: The tyrosine is thought to have the same phenol and amine groups that are present in morphine.

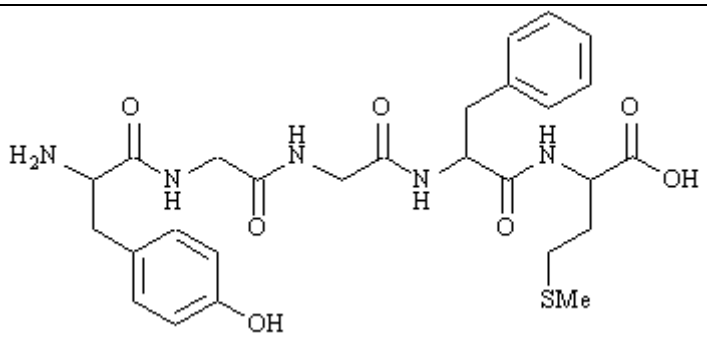
Correct	**a.#**	Tyrosine.
incorrect	**b.#**	Glycine.
incorrect	**c.#**	Phenylalanine.
incorrect	**d.#**	Cysteine.

END OF QUESTION

Type: multiple choice

Title: Chapter 24 - Question 21

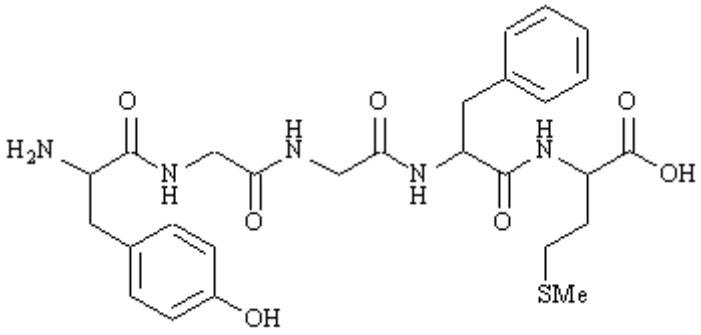
21) Consider the following analgesic, Met-enkephalin:

	 which peptide bond in the above structure is susceptible to enzymatic hydrolysis?

Feedback: The peptide bond between tyrosine and glycine is susceptible to enzymatic hydrolysis. Various enkephalin analogues have been made to try and prevent this hydrolysis – for example replacing glycine with an unnatural amino acid.

incorrect	**a.#**	gly-gly.
incorrect	**b.#**	phe-gly.
Correct	**c.#**	tyr-gly.
incorrect	**d.#**	phe-met.

END OF QUESTION

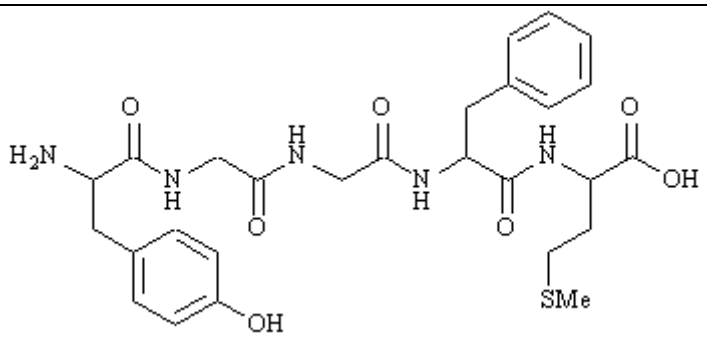
Type: multiple choice	
Title: Chapter 24 - Question 22	
22)	Consider the following analgesic, Met-enkephalin:  which of the following strategies have been tried to design metabolically stable analogues of the above structure?

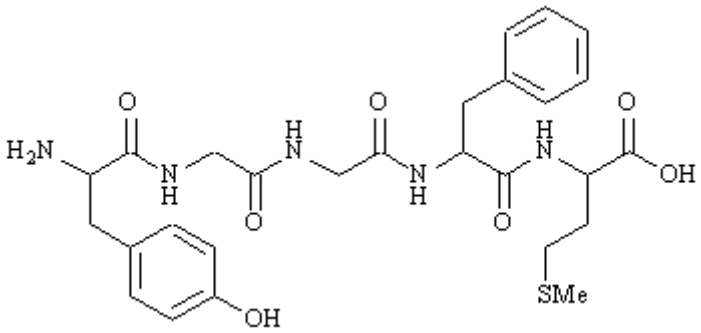
Feedback: All the strategies have been tried, but it is important to retain L-tyrosine since this contains important binding groups.

incorrect	**a.#**	*N*-Methylation of the peptide bond.
incorrect	**b.#**	Introduction of a D-amino acid.
incorrect	**c.#**	Introduction of an unnatural amino acid.
Correct	**d.#**	All of the strategies have been tried.

END OF QUESTION

Type: multiple choice	
Title: Chapter 24 - Question 23	
23)	Consider the following analgesic, Met-enkephalin:

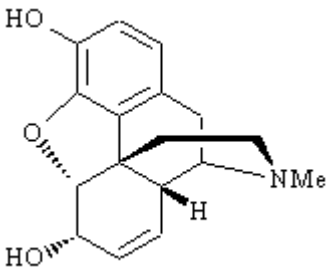
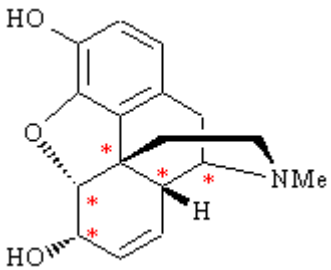
	 which receptor shows some selectivity for this structure?	
Feedback: The δ receptor is a G_i-protein-linked receptor.		
incorrect	**a.#**	μ.
incorrect	**b.#**	κ.
Correct	**c.#**	δ.
incorrect	**d.#**	σ.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 24 - Question 24	
24)	Consider the following analgesic, Met-enkephalin:  what compound has been used as an enzyme inhibitor to reduce the level of enkephalin hydrolysis?
Feedback: Thiorphan inhibits peptidase enzymes, resulting in a longer lifetime for enkephalins and enkephalin analogues. Pro-opiomelanocortin is a protein precursor for endorphins and enkephalins. Phencyclidine is a drug of abuse that is thought to interact with σ-receptors. γ-Aminobutyric acid is a neurotransmitter in the brain.	

incorrect	**a.#**	Pro-opiomelanocortin.
incorrect	**b.#**	Phencyclidine.
Correct	**c.#**	Thiorphan.
incorrect	**d.#**	γ-Aminobutyric acid.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 24 - Question 25	
25)	What is the name for the recently discovered tetrapeptides which have analgesic activity?
Feedback: Enkephalins are pentapeptides with analgesic activity. Endorphins and dynorphins are larger polypeptides with analgesic activity. Endomorphins are	

tetrapeptides with analgesic activity.		
incorrect	**a.#**	Enkephalins.
incorrect	**b.#**	Endorphins.
incorrect	**c.#**	Dynorphins.
Correct	**d.#**	Endomorphins.
END OF QUESTION		

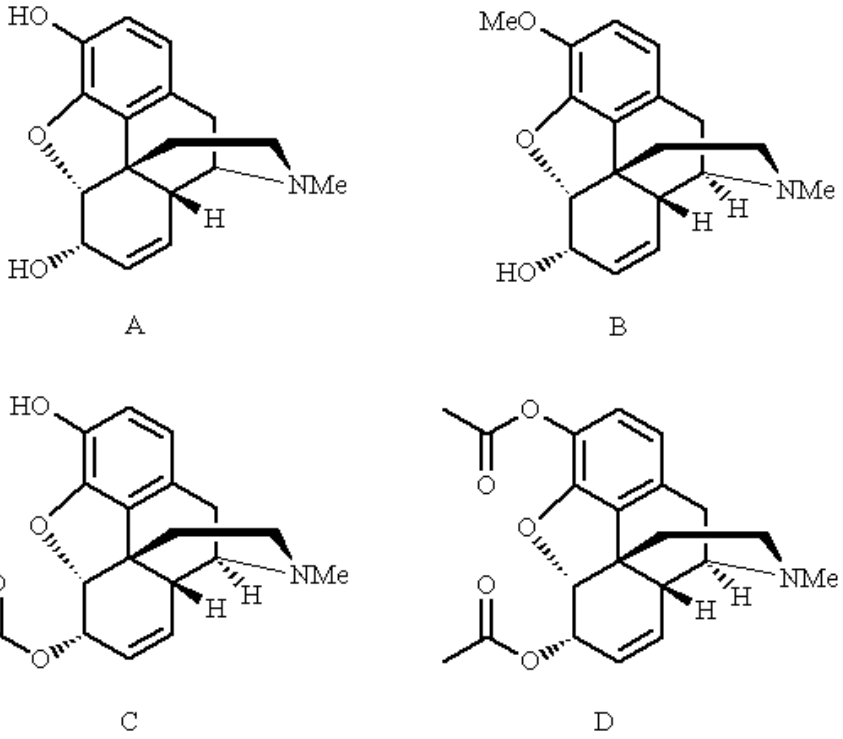
Type: multiple choice	
Title: Chapter 24 - Question 26	
26)	How many asymmetric centres are present in morphine? 
Feedback: The asymmetric centres are represented by red stars below  	

incorrect	**a.#**	2
incorrect	**b.#**	3
incorrect	**c.#**	4
Correct	**d.#**	5
END OF QUESTION		

Type: multiple choice		
Title: Chapter 24 - Question 27		
27)	The quaternary methiodide salt of morphine has no analgesic activity when injected into the blood supply, but it is active when injected directly into the brain. Explain these results.	
Feedback: Since the salt has a permanent positive charge, it cannot cross the blood brain barrier, and so it shows no activity when injected into the blood supply. When it is injected directly into the brain, it can bind to analgesic receptors since it has the necessary positive charge for an ionic interaction.		
incorrect	**a.#**	The salt is rapidly metabolised in the blood supply before it has the chance to reach the brain.
incorrect	**b.#**	The salt is rapidly excreted in the blood supply before it has a chance to reach the brain.

Correct	**c.#**	The salt is unable to cross the blood brain barrier.
incorrect	**d.#**	The salt is metabolised to an active compound by enzymes which are only present in the brain.
END OF QUESTION		

Type: matching	
Title: Chapter 24 – Question 28	
28)	Identify the following agents.

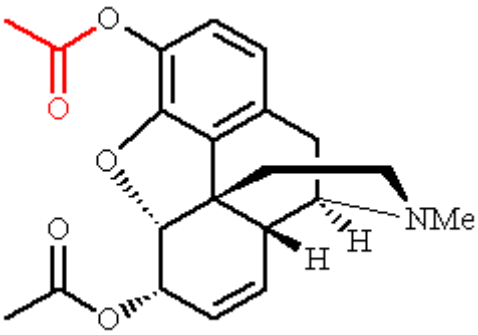


Feedback: Morphine is the main active principle of opium. Codeine is an analogue of morphine where the phenol group is methylated. 6-Acetylmorphine is an analogue of morphine where the alcohol group at position 6 is acetylated. Diamorphine is an analogue of morphine where both the phenol and alcohol functions are acetylated.

a.#	Codeine	=	Agent B
b.#	Diamorphine	=	Agent D
c.#	6-Acetylmorphine	=	Agent C
d.#	Morphine	=	Agent A
END OF QUESTION			

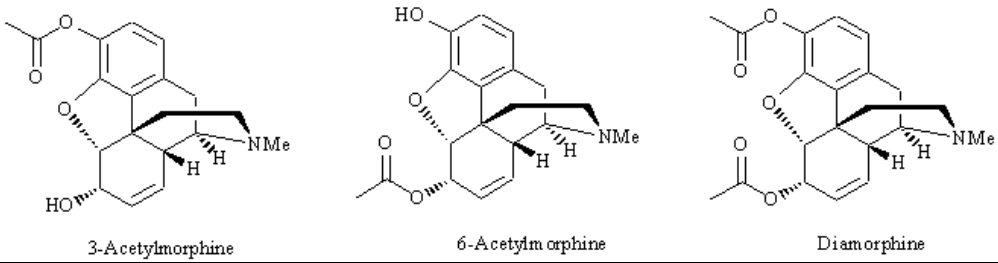
Type: multiple choice		
Title: Chapter 24 – Question 29		
29)	Codeine is used as an analgesic. Which of the following statements is false?	
Feedback: Codeine acts as a prodrug and is converted to morphine by enzymes in the liver which remove the methyl ether. This reveals an important phenol binding group. It is a weaker analgesic than morphine since only 20% of it is converted to morphine.		
incorrect	**a.#**	The structure is a weaker analgesic than morphine.

incorrect	**b.#**	The structure acts as a prodrug.
incorrect	**c.#**	The structure is converted to morphine in the liver.
Correct	**d.#**	It contains a methyl ether that acts as an important binding group.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 24 - Question 30	
30)	Diamorphine is used as an analgesic.  Which of the following statements is false?

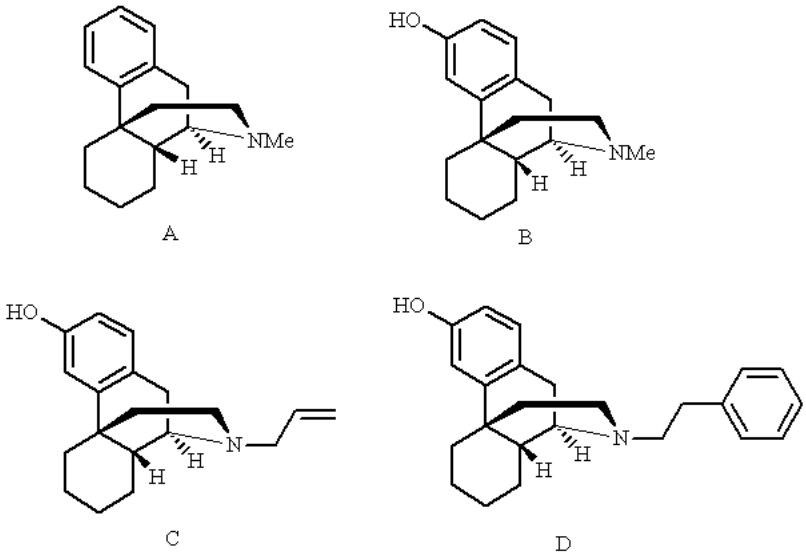
Feedback: Diamorphine is a prodrug which is converted to **morphine** in the brain. The acetyl group in red is hydrolysed by esterases to reveal the phenol group: an important binding group. Diamorphine produces a stronger analgesic effect than morphine. This is because it can cross the blood brain barrier more easily and lead to higher levels of morphine in the brain. The acetyl masks an important binding group and has to be hydrolysed before the drug is active.

incorrect	**a.#**	The structure is a stronger analgesic than morphine.
incorrect	**b.#**	The structure acts as a prodrug.
Correct	**c.#**	The structure is converted to codeine in the brain.
incorrect	**d.#**	The coloured acetyl group is not an important binding group.
END OF QUESTION		

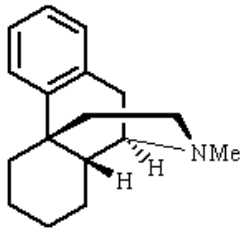
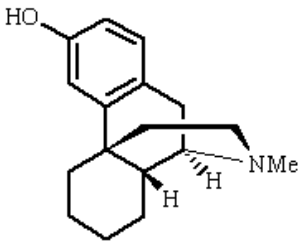
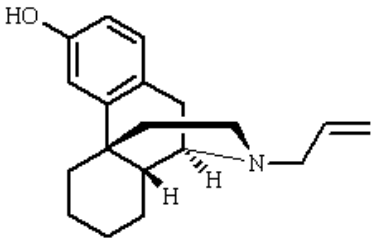
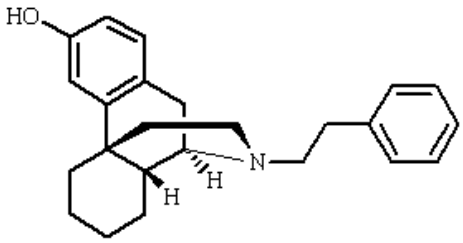
Type: multiple choice	
Title: Chapter 24 – Question 31	
31)	Which of the following statements is true? 

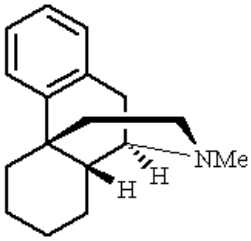
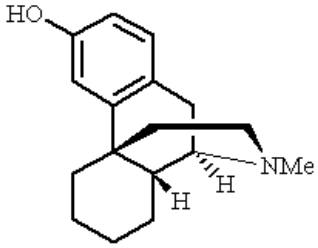
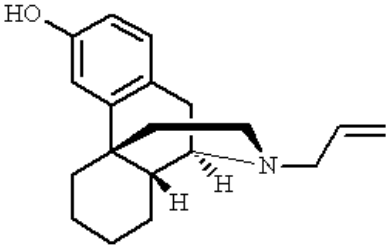
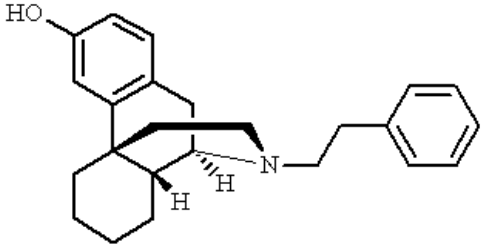
Feedback: 3-Acetylmorphine should be less active than 6-acetylmorphine. Both structures should cross the blood brain barrier equally well since each has a polar group masked by an acetyl group. However, esterases have to remove the acetyl group of 3-acetylmorphine before it can be active, whereas 6-acetylmorphine is inherently active since the phenol group is free.
3-Acetylmorphine should be less active than diamorphine. The latter structure has two polar groups masked by acetyl groups and should be able to cross the blood brain barrier more effectively leading to higher brain concentrations of the drug. Both agents are likely to be similarly activated by esterases, but

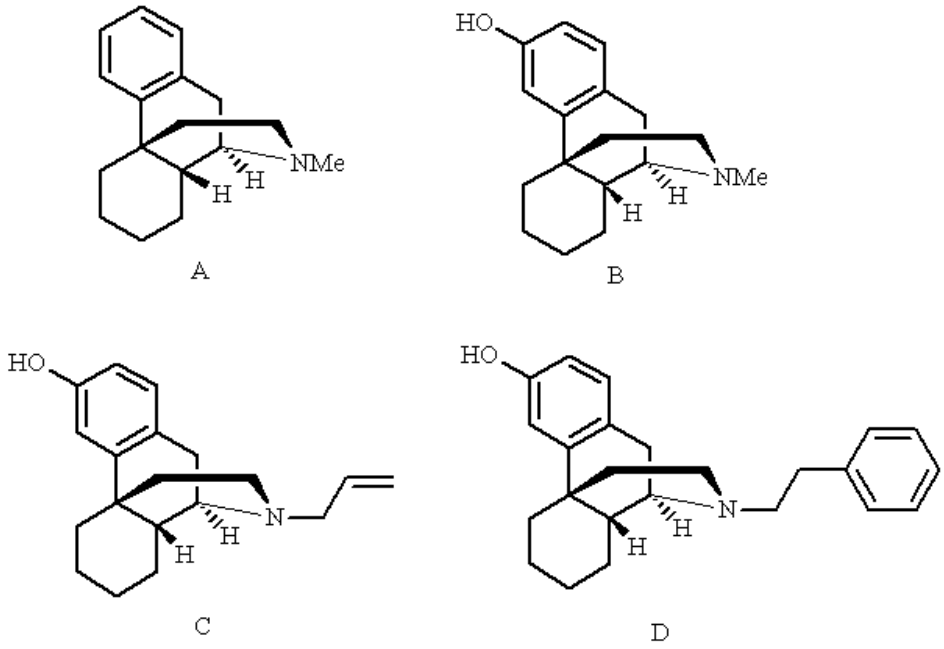
the diamorphine should be more effective since it is present at higher levels.		
incorrect	**a.#**	3-Acetylmorphine should be more active than 6-acetylmorphine since the alcohol group is free to bind.
Correct	**b.#**	3-Acetylmorphine should be less active than 6-acetylmorphine since the phenol group is masked.
incorrect	**c.#**	3-Acetylmorphine should be more active than diamorphine since it should cross the blood brain barrier more easily.
incorrect	**d.#**	3-Acetylmorphine should be less active than diamorphine since it lacks an acetyl group at the 6-position which is important for binding.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 24 – Question 32	
32)	To which class of compounds do the following structures belong? 
Feedback: Morphinans lack the ring containing the ether oxygen. Benzomorphans and 4-phenylpiperidines are simpler structures with fewer rings. Enkephalins are endogenous pentapeptides that are produced in the brain. The structures shown (A-D) are *N*-methylmorphinan, levorphanol, levallorphan, and *N*-phenethyllevorphanol respectively.	
incorrect	**a.#** Benzomorphans.
incorrect	**b.#** 4-Phenylpiperidines.
Correct	**c.#** Morphinans.
incorrect	**d.#** Enkephalins.
END OF QUESTION	

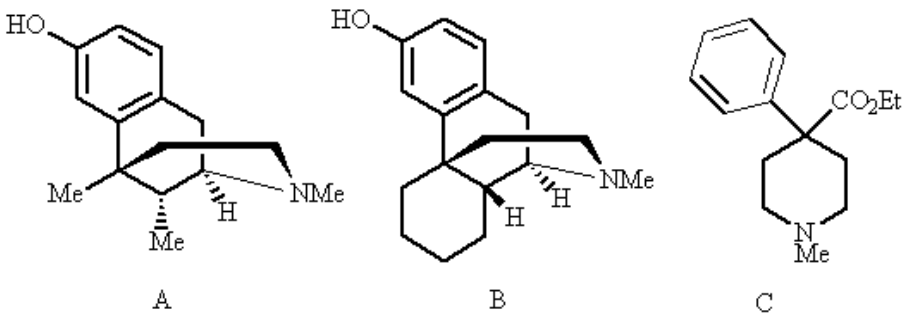
Type: multiple choice	
Title: Chapter 24 - Question 33	
33)	Which of the following structures is an opioid antagonist?

	 A  B  C  D	
Feedback: The structures shown (A-D) are *N*-methylmorphinan, levorphanol, levallorphan, and *N*-phenethyllevorphanol respectively. The presence of an *N*-Allyl or *N*-cyclopropyl substituent on an opioid structure is known to change it from an agonist into an antagonist.		
incorrect	**a.#**	Structure A.
incorrect	**b.#**	Structure B.
Correct	**c.#**	Structure C.
incorrect	**d.#**	Structure D.
END OF QUESTION		

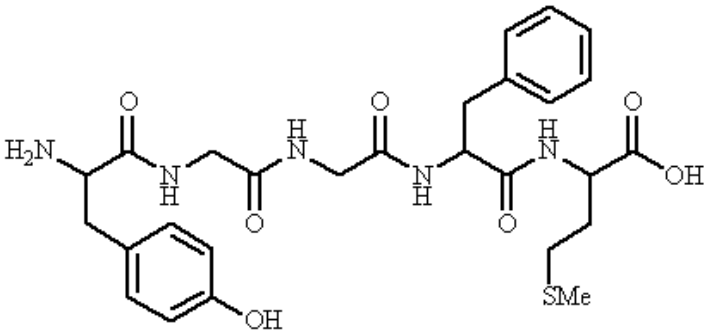
Type: multiple choice	
Title: Chapter 24 – Question 34	
34)	Which of the following structures has the greatest analgesic activity?  A  B  C  D
Feedback: The structures shown (A-D) are *N*-methylmorphinan, levorphanol, levallorphan, and *N*-phenethyllevorphanol respectively. Levallorphan is an antagonist and not an agonist. *N*-Methylmorphinan is	

the weakest agonist since it lacks the important phenol group. *N*-Phenethyllevorphanol is the strongest agonist. The extra aromatic ring interacts with an extra hydrophobic binding region in the binding site by means of van der Waals interactions. This leads to an increase in agonist activity.		
incorrect	**a.#**	Structure A.
incorrect	**b.#**	Structure B.
incorrect	**c.#**	Structure C.
Correct	**d.#**	Structure D.
END OF QUESTION		
Type: multiple choice		
Title: Chapter 24 – Question 35		
35)	Which of the following structures has the weakest analgesic activity? 	
Feedback: The structures shown (A-D) are *N*-methylmorphinan, levorphanol, levallorphan, and *N*-phenethyllevorphanol respectively. Levallorphan is an antagonist and not an agonist. *N*-Methylmorphinan is the weakest agonist since it lacks the important phenol group. *N*-Phenethyllevorphanol is the strongest agonist. The extra aromatic ring interacts with an extra hydrophobic binding region in the binding site by means of van der Waals interactions. This leads to an increase in agonist activity.		
Correct	**a.#**	Structure A.
incorrect	**b.#**	Structure B.
incorrect	**c.#**	Structure C.
incorrect	**d.#**	Structure D.
END OF QUESTION		

Type: matching	
Title: Chapter 24 – Question 36	
36)	To which group of compounds do the following structures belong?



A B C



D

Feedback: Benzomorphans lack two of the rings originally present in morphine. Morphinans are structures lacking one of the rings originally present in morphine. 4-Phenylpiperidines are simpler structures lacking three of the rings present in morphine. Enkephalins are endogenous pentapeptides that are produced in the brain.
The structures (A-D) are metazocine, levorphanol, pethidine, and Met-enkephalin respectively.

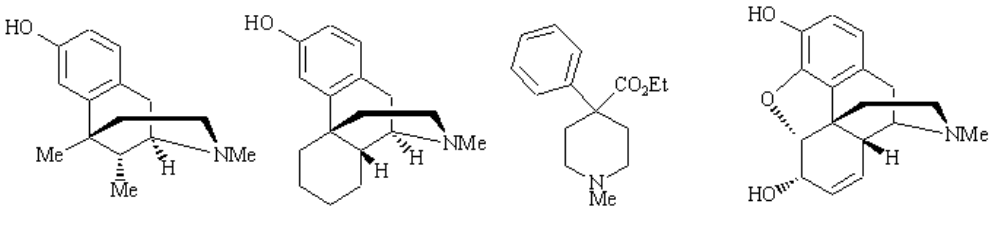
a.#	Benzomorphans	=	Structure A.
b.#	4-Phenylpiperidines	=	Structure C.
c.#	Enkephalins	=	Structure D.
d.#	Morphinans	=	Structure B.

END OF QUESTION

Type: multiple choice

Title: Chapter 24 - Question 37

37) Which of the following structures is thought to have a different binding mode in analgesic binding sites when compared with the other three?

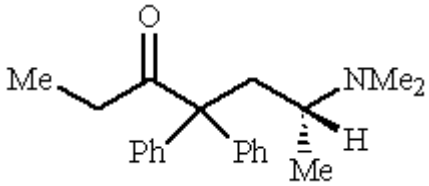


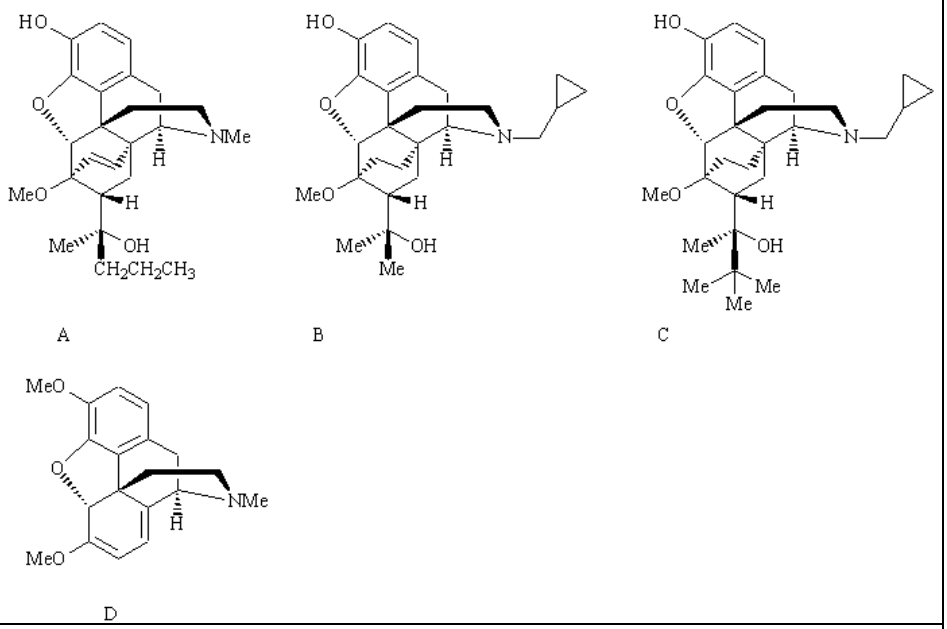
A B C D

Feedback: The structures (A-D) are metazocine, levorphanol, pethidine, and morphine respectively.
The *N*-substituents that lead to antagonism or increased activity in more complex opioids such as structures A, B, and D do not have these effects in the 4-phenylpiperidine series. This indicates that 4-phenylpiperidines such as structure C are binding to the binding site in a different manner.

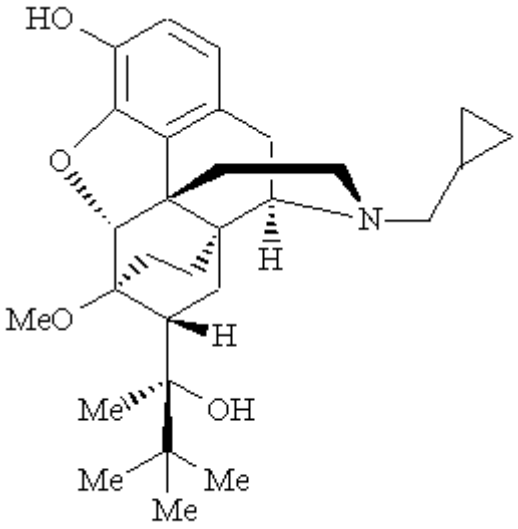
incorrect	**a.#**	Structure A.
incorrect	**b.#**	Structure B.

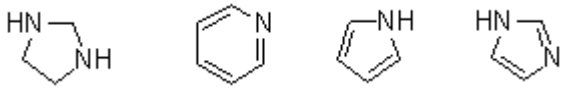
Correct	**c.#**	Structure C.
incorrect	**d.#**	Structure D.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 24 – Question 38		
38)	Methadone was synthesised in Germany during the second world war as an analgesic. Which of the following statements is false?  	
Feedback: Methadone is comparable in activity to morphine.		
Correct	**a.#**	It is 5 times more active than morphine.
incorrect	**b.#**	It is used as a substitute for heroin to wean addicts off the latter drug.
incorrect	**c.#**	It contains an asymmetric centre with one enantiomer being more active than the other.
incorrect	**d.#**	It is orally active.
END OF QUESTION		

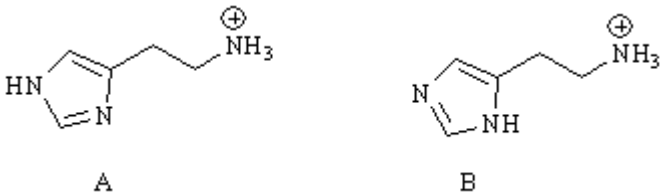
Type: matching			
Title: Chapter 24 – Question 39			
39)	What are the following molecules used for?  		
Feedback: The structures (A-D) are etorphine, diprenorphine, buprenorphine, and thebaine respectively. Buprenorphine, etorphine, and diprenorphine are all examples of a group of compounds called the oripavines. Buprenorphine is used as an analgesic in medicine. Etorphine is used as a sedative in veterinary medicine and diprenorphine is used as an antidote to etorphine. They are synthesised from the natural product thebaine.			
a.#	An analgesic in human	=	Structure C.

	medicine		
b.#	The starting material for the synthesis of oripavines	=	Structure D.
c.#	A powerful sedative in veterinary medicine	=	Structure A.
d.#	An antidote to etorphine	=	Structure B.
END OF QUESTION			

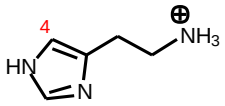
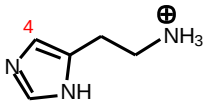
Type: multiple choice		
Title: Chapter 24 – Question 40		
40)	Buprenorphine is used clinically as an analgesic.  Which of the following statements is false?	
Feedback: Buprenorphine is **slow** to bind to analgesic receptors, and slow to depart. Less buprenorphine is required to interact with a certain number of analgesic receptors than morphine, and so lower doses can attain the same level of analgesia. However, the agent cannot reach the same maximum levels of analgesia as morphine and so it is less useful for severe pain. Buprenorphine has some side effects although they are less serious than those of morphine. It cannot be taken orally and has to be injected.		
incorrect	**a.#**	It is 100 times more active than morphine as an agonist.
incorrect	**b.#**	It is 4 times more active than nalorphine as an antagonist.
incorrect	**c.#**	There is a low risk of respiratory depression.
Correct	**d.#**	It is quick to bind to analgesic receptors, but slow to depart.
END OF QUESTION		

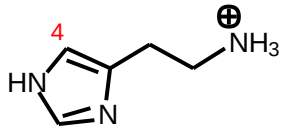
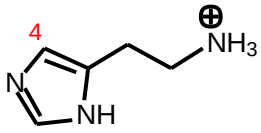
Type: multiple choice		
Title: Chapter 25 - Question 01		
01)	What sort of heterocyclic ring is present in histamine?	
Feedback: The ring systems given are as follows 		
incorrect	**a.#**	Imidazolidine.
incorrect	**b.#**	Pyridine.
incorrect	**c.#**	Pyrrole.
Correct	**d.#**	Imidazole.
END OF QUESTION		

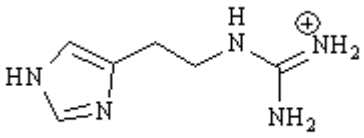
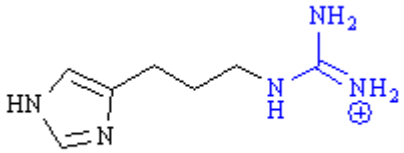
Type: multiple choice		
Title: Chapter 25 - Question 02		
02)	Which ailment is not treated by conventional histamine antagonists acting on the H_1 receptor?	
Feedback: Anti-ulcer agents acting as histamine antagonists target H_2-receptors on parietal cells.		
incorrect	**a.#**	Hay fever.
incorrect	**b.#**	Insect bites.
Correct	**c.#**	Ulcers.
incorrect	**d.#**	Asthma.
END OF QUESTION		

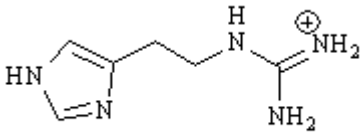
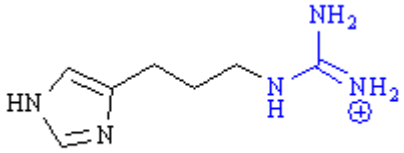
Type: multiple choice		
Title: Chapter 25 - Question 03		
03)	There are two possible tautomers for histamine.  A B Tautomer A is more stable than tautomer B. Why?	
Feedback: The side chain is electron withdrawing due to the positively charged ammonium group at the end of the chain. Since this is an inductive effect, it is felt more at the π nitrogen than the τ nitrogen since the former is nearer the side chain. Inductive effects decrease with the number of bonds involved. This means that the π nitrogen is going to be less basic and less likely to use its lone pair of electrons to form a bond to hydrogen.		
Correct	**a.#**	The side chain has an inductive electron-withdrawing effect on the ring making the π nitrogen less basic than the τ nitrogen.
incorrect	**b.#**	The side chain has an inductive electron-donating effect on the ring making the π nitrogen less basic than the τ nitrogen.

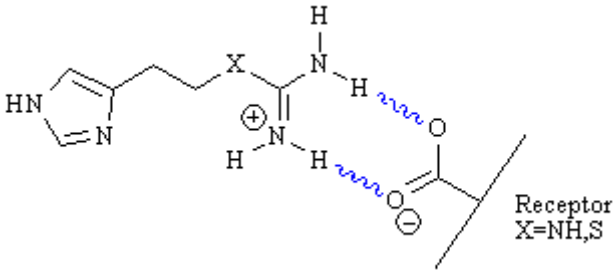
incorrect	**c.#**	The side chain has an inductive electron-withdrawing effect on the ring making the τ nitrogen less basic than the π nitrogen.
incorrect	**d.#**	The side chain has an inductive electron-donating effect on the ring making the τ nitrogen less basic than the π nitrogen.
END OF QUESTION		

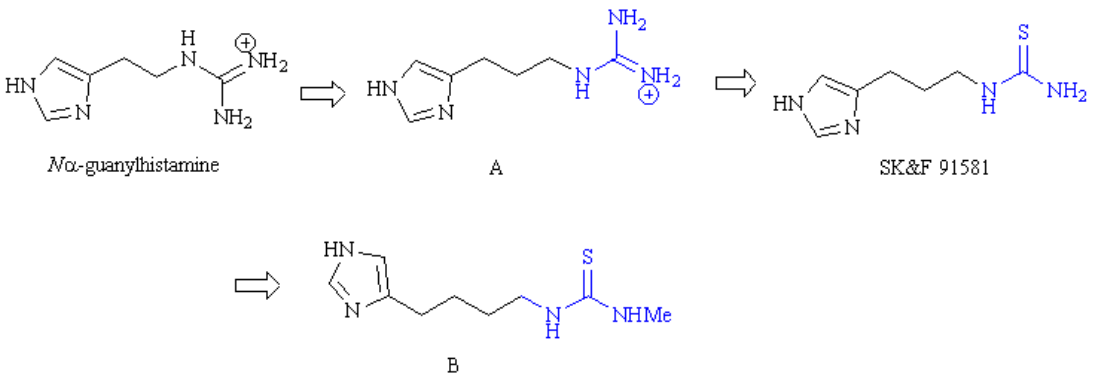
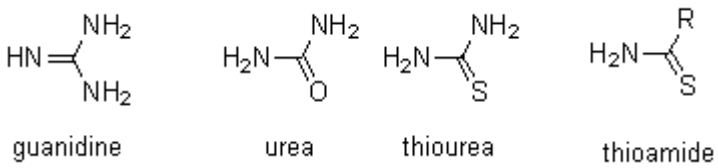
Type: multiple choice		
Title: Chapter 25 - Question 04		
04)	There are two possible tautomers for histamine.  A  B What would be the effect on the tautomer population if a methyl group were to be placed at position 4 of the ring?	
Feedback: The methyl group has an inductive effect which is electron donating. The τ nitrogen is closer to this substituent than the π nitrogen and so the τ nitrogen experiences this inductive effect to a greater extent. This makes the τ nitrogen more basic and more likely to use its lone pair of electrons to bond to hydrogen.		
incorrect	**a.#**	The relative population of B would increase.
incorrect	**b.#**	There would be no difference.
Correct	**c.#**	The relative population of A would increase.
incorrect	**d.#**	It is not possible to predict the effect.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 25 - Question 05		
05)	There are two possible tautomers for histamine. A methyl group at position 4 plays an important role in favouring tautomer A through an inductive effect.  A  B What other important effect arises from the presence of a methyl group at position 4?	
Feedback: It introduces selectivity for the H_2 receptor and is present in anti-ulcer drugs such as cimetidine.		
incorrect	**a.#**	It introduces selectivity for the H_1 receptor.
Correct	**b.#**	It introduces selectivity for the H_2 receptor.
incorrect	**c.#**	It protects the molecule from metabolism.
incorrect	**d.#**	It increases the ability of the compound to cross cell membranes.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 25 - Question 06		
06)	*N*α-Guanylhistamine was the first lead compound for the development of H_2-antagonists.  *N*α-guanylhistamine  A What strategy was used in developing the guanidine structure (A)?	
Feedback: Extension indicates the addition of a new functional group to find an extra binding region. Here, the functional group is already present and has been moved further away from the imidazole ring by chain extension. Substituent variation and isosteric replacement indicate the replacement of a group with another group which has not occurred here.		
incorrect	**a.#**	Extension.
Correct	**b.#**	Chain extension.
incorrect	**c.#**	Substituent variation.
incorrect	**d.#**	Isosteric replacement.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 25 - Question 07		
07)	*N*α-Guanylhistamine was the first lead compound for the development of H_2-antagonists.  *N*α-guanylhistamine  A What was the rationale for the strategy used in developing the guanidine structure (A)?	
Feedback: The antagonist binding region is further away from the imidazole binding region than the agonist binding region. Therefore, extending the distance between the guanyl group and the imidazole ring increases the chances that the molecule will bind as an antagonist rather than as an agonist.		
incorrect	**a.#**	To reduce the inductive effect of the guanidine unit on the imidazole ring.
incorrect	**b.#**	To increase the flexibility of the chain by allowing more single bond rotations.
incorrect	**c.#**	To move the binding group closer to the agonist binding region and away from the antagonist binding region.
Correct	**d.#**	To move the binding group closer to the antagonist binding region and away from the agonist binding region.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 25 - Question 08		
08)	During the search for H_2 antagonists, the following binding theory was proposed between a guanidine unit and a carboxylate unit in the binding site. What was it called?  	
Feedback: The binding theory was called the chelation binding theory and involves the guanidine and carboxylate ion interacting through ionic and hydrogen bonding interactions.		
incorrect	**a.#**	Ionic binding theory.
incorrect	**b.#**	Hydrogen bonding theory.
Correct	**c.#**	Chelation binding theory.
incorrect	**d.#**	Complexation binding theory.
END OF QUESTION		

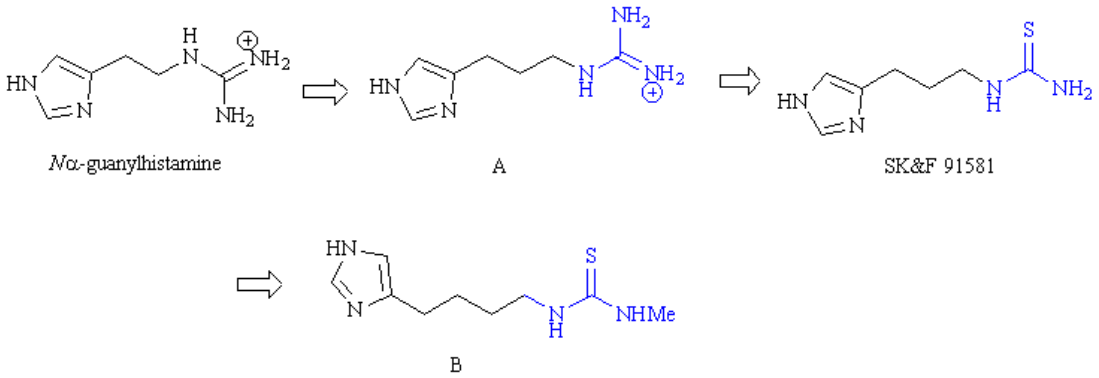
Type: multiple choice		
Title: Chapter 25 - Question 09		
09)	The following structures show some of the important molecules leading to the discovery of structure B.  What sort of functional group has been introduced in SK&F 91581?	
Feedback: The functional groups described in the options are as shown below:  		
incorrect	**a.#**	Guanidine.
incorrect	**b.#**	Urea.
Correct	**c.#**	Thiourea.
incorrect	**d.#**	Thioamide.

END OF QUESTION

Type: multiple choice

Title: Chapter 25 - Question 10

10) The following structures show some of the important molecules leading to the discovery of structure B.



A functional group has been introduced in SK&F 91581. What was the rationale for introducing this functional group?

Feedback: Structure B is burimamide. The rationale was to replace the basic guanyl group with a neutral group such that it would interact by hydrogen bonding rather than ionic bonding. A neutral group was chosen that was as similar as possible to the guanyl group in terms of size and shape.

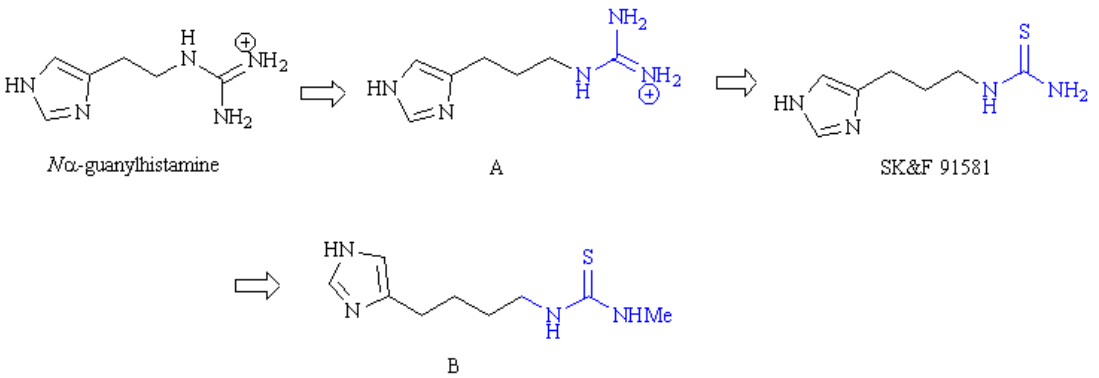
incorrect	**a.#**	To reduce the number of possible hydrogen bond donors present.
Correct	**b.#**	To replace an ionic group with a neutral group.
incorrect	**c.#**	To change the shape of the terminal functional group.
incorrect	**d.#**	To change the size of the terminal functional group.

END OF QUESTION

Type: multiple choice

Title: Chapter 25 - Question 11

11) The following structures show some of the important molecules leading to the discovery of structure B.

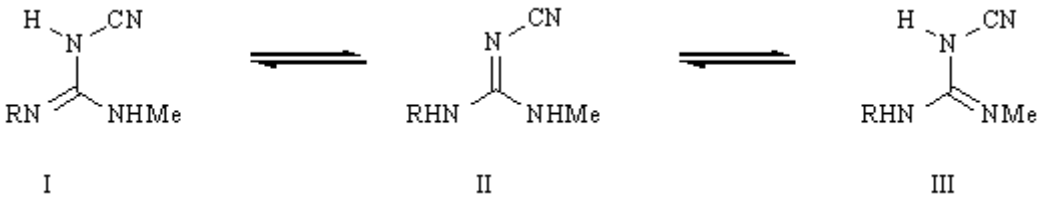


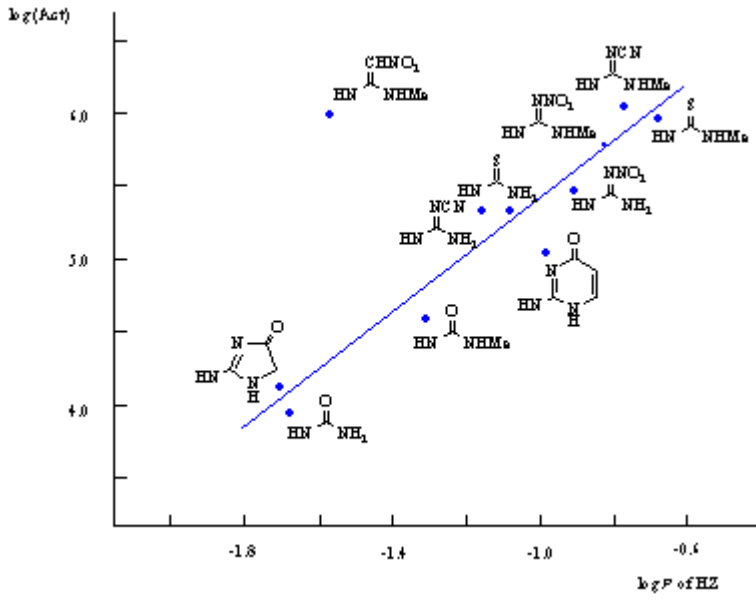
What was the rationale for the strategy used in developing structure B from SK&F 91581?

Feedback: Structure B is burimamide. The antagonist binding region is further away from the imidazole binding region than the agonist binding region. Therefore, extending the distance between the thiourea group and the imidazole ring increases the chances that the molecule will bind as an antagonist rather than an agonist.

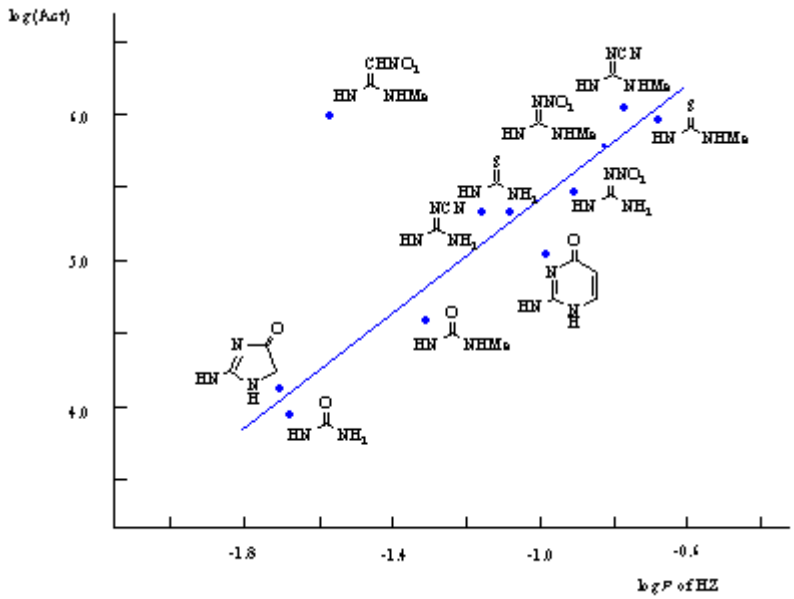
incorrect	**a.#**	To reduce the inductive effect of the terminal functional group on the imidazole ring.
incorrect	**b.#**	To increase the flexibility of the chain by allowing more single bond rotations.

incorrect	**c.#**	To move the terminal functional group closer to the agonist binding region and away from the antagonist binding region.
Correct	**d.#**	To move the terminal functional group closer to the antagonist binding region and away from the agonist binding region.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 25 - Question 12		
12)	The following diagram shows the various tautomers of the terminal functional group in structure cimetidine Which is considered to be the most stable tautomer?  	
Feedback: Since the electron-withdrawing effect of the cyanide group is inductive, it is experienced most by the nitrogen directly attached to cyanide. As a result, this nitrogen will be the least basic of the three nitrogens present in the guanidine moiety and the least likely to bond to hydrogen.		
incorrect	**a.#**	Tautomer I.
Correct	**b.#**	Tautomer II.
incorrect	**c.#**	Tautomer III.
incorrect	**d.#**	They are all equal in stability.
END OF QUESTION		

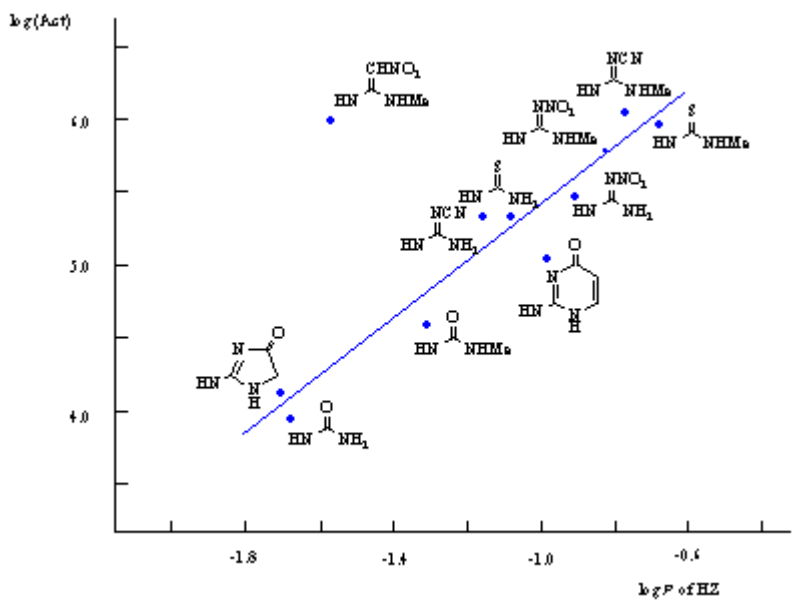
Type: multiple choice	
Title: Chapter 25 - Question 13	
13)	The following graph measures log(activity) versus log *P*.  What does the graph show about activity?

Feedback: As log *P* increases, activity increases. Log *P* is a measure of hydrophobicity and so increased activity is associated with increasing hydrophobicity.		
Correct	**a.** #	Activity increases with hydrophobic character.
incorrect	**b.** #	Activity decreases with hydrophobic character.
incorrect	**c.** #	Activity increases with basicity.
incorrect	**d.** #	Activity increases with the size of the group.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 25 - Question 14	
14)	The following graph measures log(activity) versus log *P*.  The nitroketeneaminal group does not fit the straight line. Which of the following statements is true?

Feedback: The log *P* value for this group is relatively low and the compound should have a lower activity than it actually has.		
Correct	**a.#**	The activity is too great for the hydrophobicity of the group.
incorrect	**b.#**	The activity is too great for the size of the group.
incorrect	**c.#**	The activity is too great for the basicity of the group.
incorrect	**d.#**	The activity is too small for the hydrophobicity of the group.
END OF QUESTION		

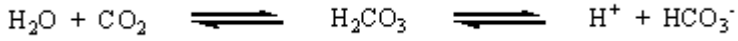
Type: multiple choice	
Title: Chapter 25 - Question 15	
15)	The following graph measures log(activity) versus log *P*.

	 What physical feature other than log *P* is important to the activity of cimetidine analogues?	
Feedback: To be precise, the orientation of the dipole moment is important to the activity rather than the size of the dipole moment.		
incorrect	**a.#**	Hydrophobicity.
incorrect	**b.#**	Basicity.
incorrect	**c.#**	Molecular weight.
Correct	**d.#**	Dipole moment.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 25 - Question 16		
16)	A QSAR study of cimetidine analogues with different planar groups produced the following equation; log (Activity) = 9.12 cosθ + 0.6log*P* -2.71. What is the relevance of the cosθ entry?	
Feedback: The cosθ entry quantifies the orientation of a dipole moment. θ is the deviation in angle of the dipole moment from the ideal orientation of 30˚.		
Correct	**a.#**	It quantifies the orientation of a dipole moment .
incorrect	**b.#**	It quantifies the magnitude of a dipole moment.
incorrect	**c.#**	It quantifies the orientation of hydrogen bonds.
incorrect	**d.#**	It quantifies the strength of hydrogen bonding.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 25 - Question 17	
17)	A QSAR study of cimetidine analogues with different planar groups produced the following equation; log (Activity) = 9.12 cosθ + 0.6log*P* -2.71.

	What is the ideal value of θ?	
Feedback: At 0˚, cosθ is equal to a maximum value of 1. At any other value, the cosθ value will be a fraction. This means that the calculated activity will be smaller.		
Correct	**a.#**	0˚
incorrect	**b.#**	45˚
incorrect	**c.#**	60˚
incorrect	**d.#**	90˚
END OF QUESTION		

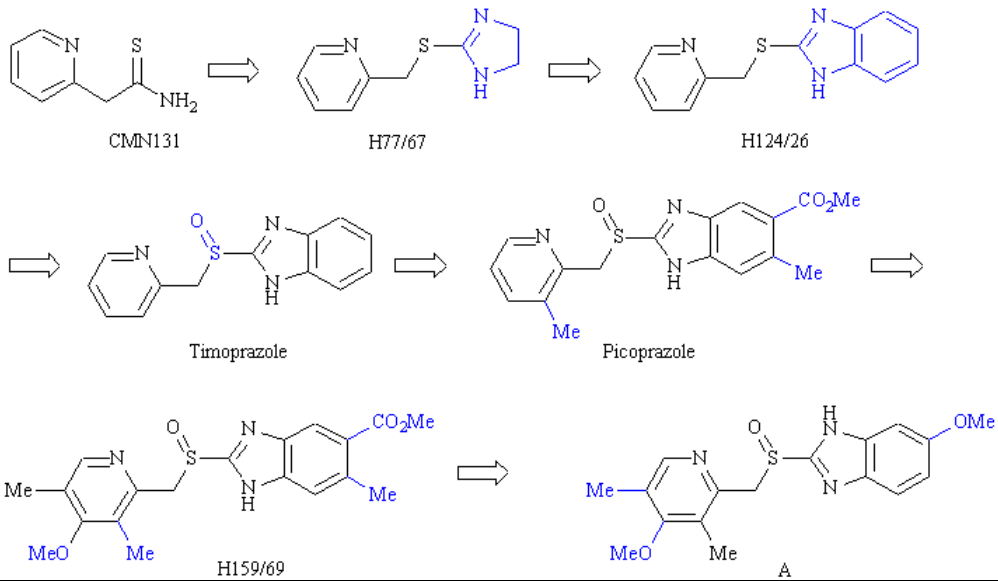
Type: multiple choice		
Title: Chapter 25 - Question 18		
18)	What enzyme catalyses the following reaction? 	
Feedback: Carbonic anhydrase catalyses the formation of carbonic acid from water and carbon dioxide. The acid dissociates to give a proton and a hydrogen carbonate ion.		
incorrect	**a.#**	Carbonic synthase.
incorrect	**b.#**	Carbonic ligase.
Correct	**c.#**	Carbonic anhydrase.
incorrect	**d.#**	Carbonic hydrolase.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 25 - Question 19		
19)	What reaction is catalysed to power the proton pump?	
Feedback: Hydrolysis of ATP to ADP involves the splitting of the terminal phosphate group from the triphosphate chain of ATP, generating energy which can be used to power the proton pump.		
Correct	**a.#**	ATP to ADP.
incorrect	**b.#**	ATP to AMP.
incorrect	**c.#**	ATP to cyclic AMP.
incorrect	**d.#**	ATP to adenosine.
END OF QUESTION		

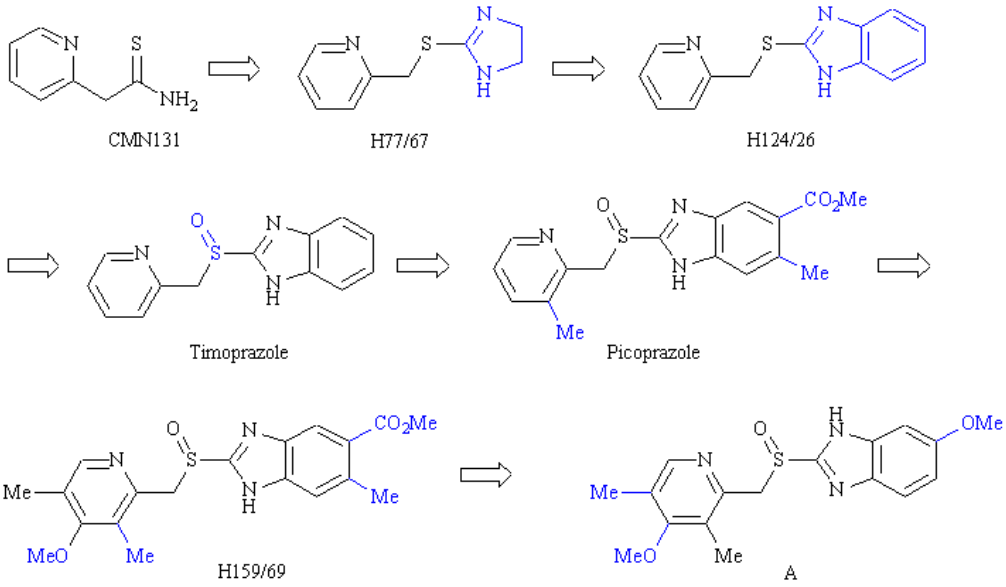
Type: multiple choice		
Title: Chapter 25 - Question 20		
20)	The proton pump inhibitors are prodrugs. Where do they get activated?	
Feedback: The proton pump inhibitors are activated by strong acidic conditions found only in the canaliculae of parietal cells.		
incorrect	**a.#**	In the liver.
incorrect	**b.#**	In the blood supply.
incorrect	**c.#**	In parietal cells.
Correct	**d.#**	In the canaliculae of parietal cells.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 25 - Question 21		
21)	What is the typical pK_a for proton pump inhibitors?	
Feedback: The proton pump inhibitors are weak bases and require strong acidic conditions before they protonate and become activated.		
Correct	**a.#**	4
incorrect	**b.#**	6
incorrect	**c.#**	7.4
incorrect	**d.#**	9
END OF QUESTION		

Type: multiple choice		
Title: Chapter 25 - Question 22		
22)	What conditions are required to activate proton pump inhibitors?	
Feedback: The proton pump inhibitors are weak bases and require strong acidic conditions before they protonate and become activated.		
Correct	**a.#**	Strongly acidic conditions.
incorrect	**b.#**	Weakly acidic conditions.
incorrect	**c.#**	Neutral conditions.
incorrect	**d.#**	Slightly basic conditions.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 25 - Question 23	
23)	The following diagram shows the development of the first commercially available proton pump inhibitor omeprazole (A). During this study, timoprazole was found to be a promising looking candidate. Addition of methyl and methoxy substituents to the pyridine ring increases the activity of timoprazole. What effect do these groups have? 
Feedback: The activation of the drug under acidic conditions requires the pyridine nitrogen to act as a base. The addition of electron-donating substituents increases the basicity of the pyridine nitrogen.	

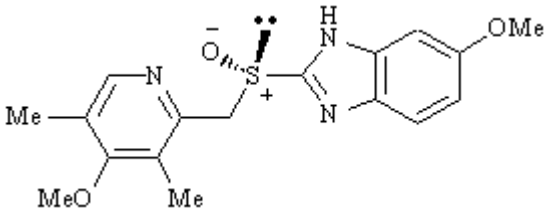
incorrect	**a.#**	They fit into hydrophobic binding pockets in the active site and increase binding strength.
Correct	**b.#**	They are electron-donating groups and increase the basicity of the pyridine ring.
incorrect	**c.#**	They act as steric shields to prevent metabolism at exposed regions of the pyridine ring.
incorrect	**d.#**	They block the drug binding to alternative targets.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 25 - Question 24	
24)	The following diagram shows the development of the first commercially available proton pump inhibitor omeprazole (A). During this study, it was found that the methyl and methoxy substituents on the pyridine ring had an important role in increasing activity. What effect would you predict if the coloured substituents on the pyridine ring of omeprazole (A) were swapped such that the methoxy group was at the *meta* position relative to the nitrogen of the ring ? 

Feedback: The methoxy substituent can interact with the pyridine ring by resonance. When the substituent is located at the *para* position, a resonance structure can be drawn where a negative charge is placed on the pyridine nitrogen, thus increasing its basic character and the activity of the compound. If the substituent is at the *meta* position, this resonance structure is no longer possible. This means that the pyridine ring will be less basic and the structure will be less active.

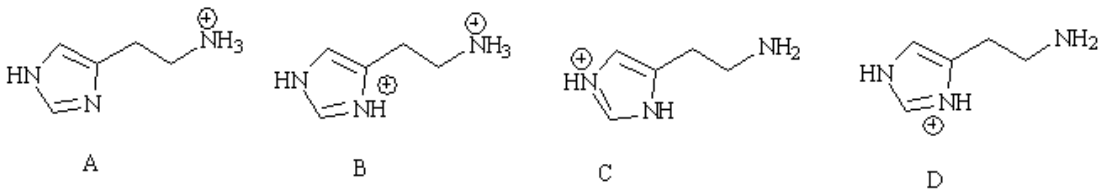
incorrect	**a.#**	Activity would be unaffected.
incorrect	**b.#**	Activity would increase.
Correct	**c.#**	Activity would decrease.
incorrect	**d.#**	It is impossible to predict what would happen.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 25 - Question 25	
25)	The following structure (esomeprazole) has been marketed as the more active enantiomer of omeprazole.

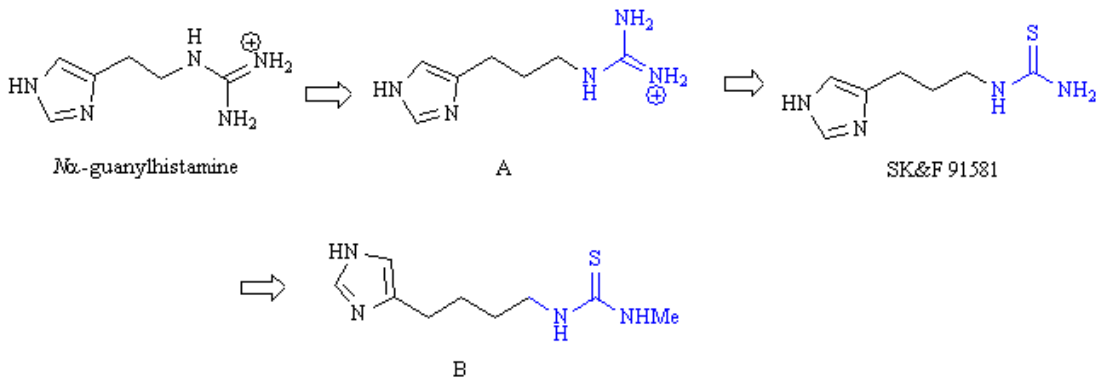
	 What is the name of the strategy where a clinically important racemic drug is replaced with the active enantiomer of the drug?

Feedback: Chiral switching has been used on several drugs which were previously marketed as racemates.		
incorrect	**a.#**	Patent switching.
incorrect	**b.#**	Enantiomeric switching.
incorrect	**c.#**	Asymmetric switching.
Correct	**d.#**	Chiral switching.
END OF QUESTION		

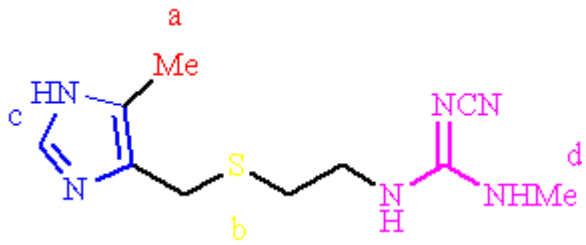
Type: multiple choice		
Title: Chapter 25 - Question 26		
26)	What type of triple therapy is most commonly used to treat ulcers caused by *Helicobacter pylori*?	
Feedback: The proton pump inhibitor serves to reduce the quantity of hydrochloric acid released into the stomach by parietal cells. The antibacterial agents are used to combat the bacterium *Helicobacter pylori* which is known to cause inflammation and cell damage in the stomach.		
incorrect	**a.#**	An H_2 antagonist, proton pump inhibitor and an antibacterial agent.
incorrect	**b.#**	An H_2 antagonist, and two antibacterial agents.
Correct	**c.#**	A proton pump inhibitor and two antibacterial agents.
incorrect	**d.#**	Two proton pump inhibitors and an antibacterial agent.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 25 - Question 27		
27)	Which of the following structures represents histamine at physiological pH?  A B C D	
Feedback: The side chain nitrogen is fully ionised and the heterocyclic nitrogens are not ionised.		
Correct	**a.#**	Structure A.
incorrect	**b.#**	Structure B.
incorrect	**c.#**	Structure C.
incorrect	**d.#**	Structure D.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 25 – Question 28		
28)	Several binding regions were proposed to be present in the binding site of the H_2 receptor. Which of the following statements is incorrect?	
Feedback: The imidazole binding region is the same for agonists and antagonists.		
Correct	**a.#**	There are two binding regions for the imidazole ring of histamine analogues, one which is occupied by agonists and one which is occupied by antagonists.
incorrect	**b.#**	There is a binding region which interacts ionically with the α-nitrogen of histamine and results in agonist activity.
incorrect	**c.#**	There is a binding region further away from the imidazole ring that produces an antagonist effect if occupied.
incorrect	**d.#**	The α-nitrogen of histamine can only bind to the agonist binding region while the guanyl group of *N*α-guanylhistamine can bind either to the antagonist binding region or the agonist binding region.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 25 - Question 29		
29)	The following structures show some of the important molecules leading to the discovery of structure B.  Nα-guanylhistamine A SK&F 91581 B Which of the following statements concerning structure B is untrue?	
Feedback: It established that binding to the antagonist binding region involved hydrogen bonding and not ionic bonding.		
incorrect	**a.#**	It established the existence of H_2-receptors.
incorrect	**b.#**	It was a good antagonist at H_2 receptors with no agonist activity.
incorrect	**c.#**	It inhibited gastric acid release from parietal cells.
Correct	**d.#**	It indicated that binding to the antagonist binding region involved ionic bonding and not hydrogen bonding.
END OF QUESTION		

Type: multiple choice	
Title: Chapter 25 - Question 30	
30)	The following structure is cimetidine. Which electron-donating group is present to favour the active tautomeric form of the imidazole ring shown?

Feedback: The methyl group has an inductive electron-donating effect on the imidazole ring. This inductive effect is more pronounced at the τ-nitrogen than the π-nitrogen since the former is closer to the substituent. The inductive effect makes the τ-nitrogen more basic and favours the tautomer where this nitrogen is bonded to a hydrogen.

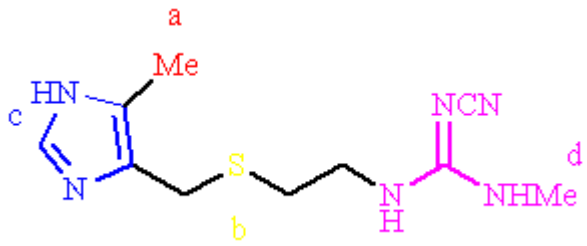
Correct	**a.#**	Group a.
incorrect	**b.#**	Group b.
incorrect	**c.#**	Group c.
incorrect	**d.#**	Group d.

END OF QUESTION

Type: multiple choice

Title: Chapter 25 - Question 31

31) The following structure is cimetidine. Which electron-withdrawing group is present to favour the active tautomeric form of the imidazole ring shown?



[

Feedback: The sulfur has an inductive electron-withdrawing effect on the imidazole ring. This inductive effect is more pronounced at the π-nitrogen than the τ-nitrogen since the former is closer to the substituent. The inductive effect makes the τ-nitrogen more basic and favours the tautomer where this nitrogen is bonded to a hydrogen.

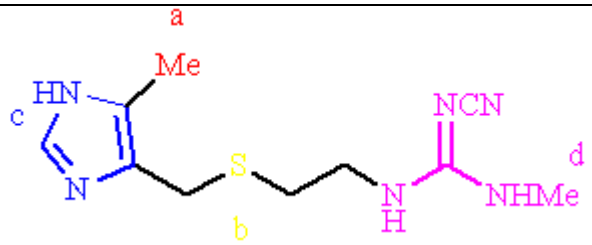
incorrect	**a.#**	Group a.
Correct	**b.#**	Group b.
incorrect	**c.#**	Group c.
incorrect	**d.#**	Group d.

END OF QUESTION

Type: multiple choice

Title: Chapter 25 - Question 32

32) The following structure is cimetidine. Which group is present to interact with a hydrogen bonding region not used by histamine?

Feedback: Group d is a naturally occurring moiety which is neutral and interacts with a binding region through hydrogen bonding. It was used to replace a thiourea group in previous structures since the latter was responsible for toxic side effects.

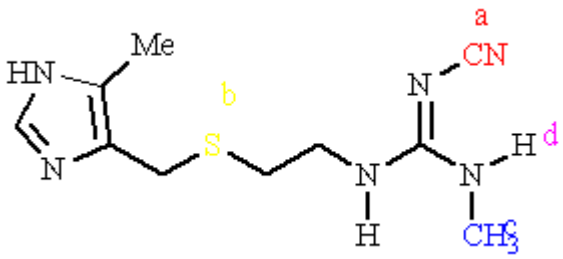
incorrect	**a.#**	Group a.
incorrect	**b.#**	Group b.
incorrect	**c.#**	Group c.
Correct	**d.#**	Group d.

END OF QUESTION

Type: multiple choice

Title: Chapter 25 - Question 33

33) The following structure is cimetidine. The guanidine moiety at the end of the side chain is neutral which is unusual since guanidine moieties are usually basic. Which one of the coloured groups shown is responsible for this?



Feedback: The cyanide group has a strong inductive effect which is electron withdrawing. As a result the basicity of the guanidine group is reduced and it is less likely to ionise.

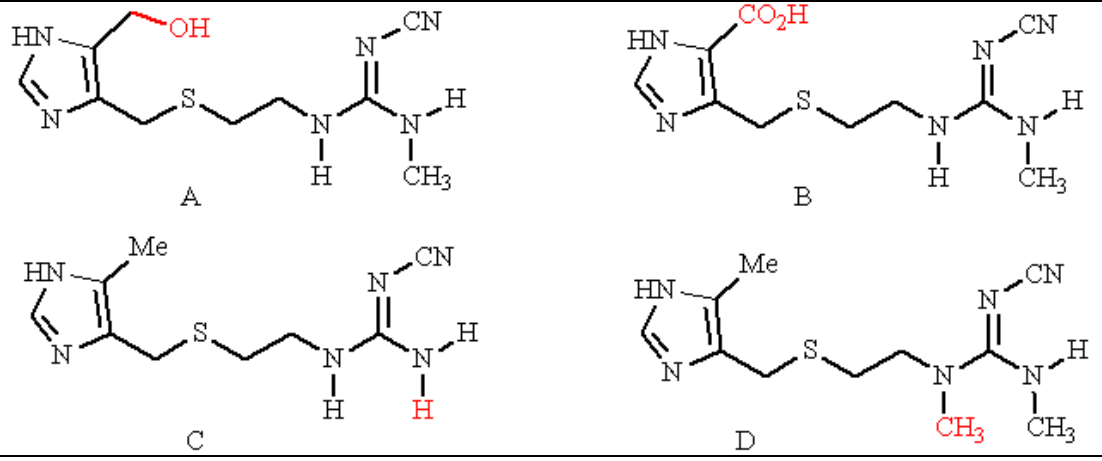
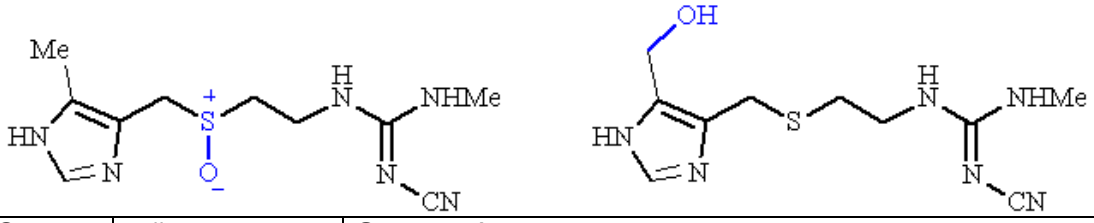
Correct	**a.#**	Group a.
incorrect	**b.#**	Group b.
incorrect	**c.#**	Group c.
incorrect	**d.#**	Group d.

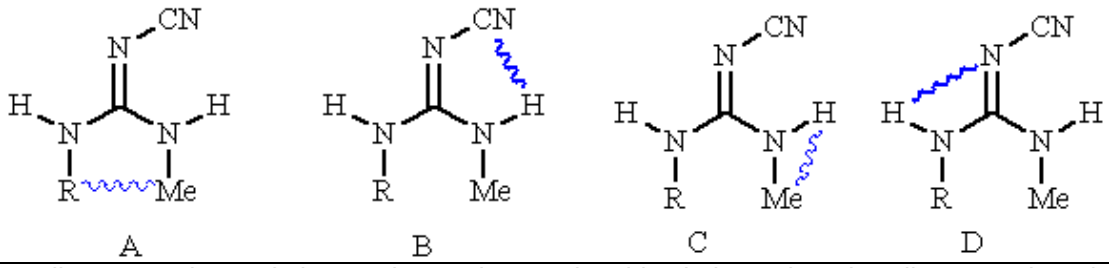
END OF QUESTION

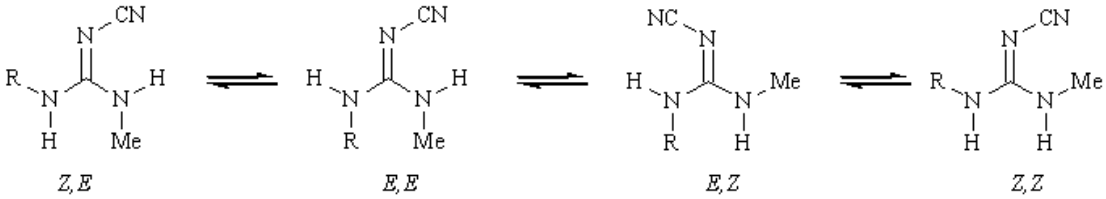
Type: multiple choice

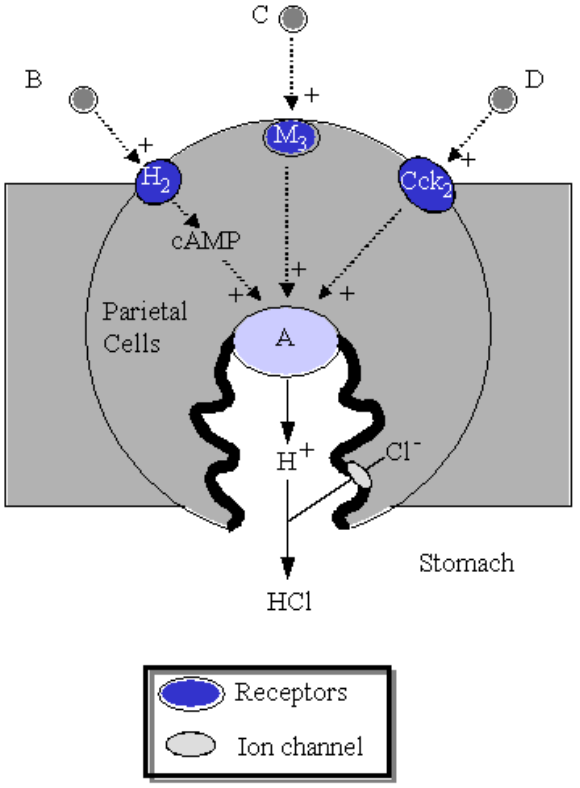
Title: Chapter 25 - Question 34

34) Which of the following structures is a metabolite of cimetidine?

		
Feedback: The metabolites of cimetidine are shown below. 		
Correct	**a.#**	Structure A.
incorrect	**b.#**	Structure B.
incorrect	**c.#**	Structure C.
incorrect	**d.#**	Structure D.
END OF QUESTION		

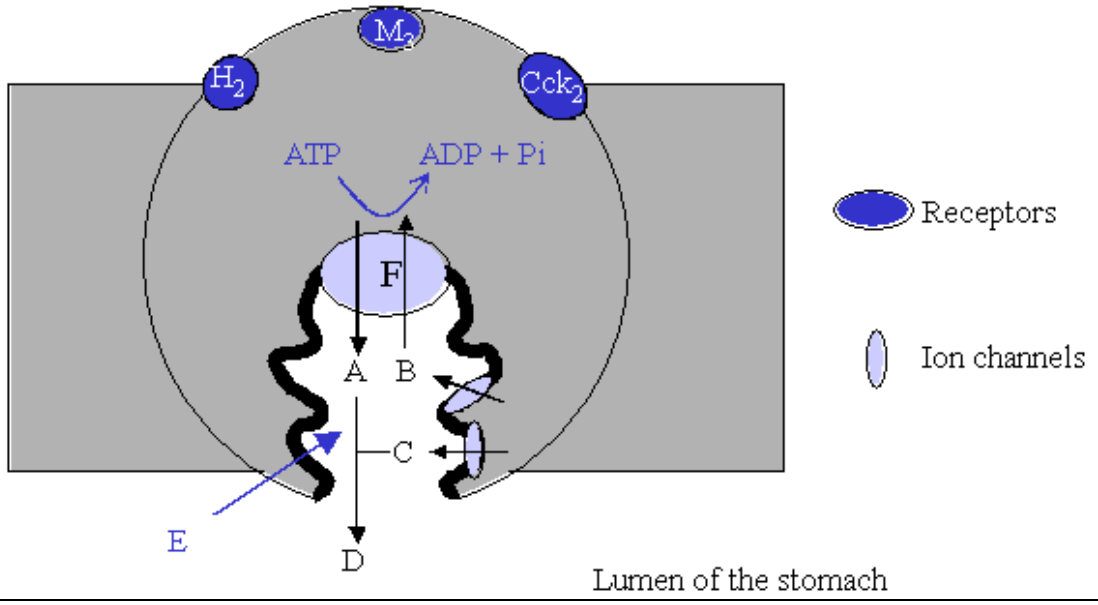
Type: multiple choice		
Title: Chapter 25 - Question 35		
35)	The following conformation for the cyanoguanidine group of cimetidine is not favoured due to a bad steric interaction. Which of the following diagrams illustrates this? 	
Feedback: In diagram A, the methyl group is too close to the side chain. In the other diagrams, there is very little steric interaction since one of the groups involved is a small hydrogen atom.		
Correct	**a.#**	Diagram A.
incorrect	**b.#**	Diagram B.
incorrect	**c.#**	Diagram C.
incorrect	**d.#**	Diagram D.
END OF QUESTION		

Type: multiple choice		
Title: Chapter 25 - Question 36		
36)	The following diagram shows various conformations for the cyanoguanidine group of cimetidine. A study was carried out to find out the most stable conformations.  *Z,E* *E,E* *E,Z* *Z,Z* Which of the following statements is true?	
Feedback: The study showed that the E,Z and Z,E tautomers were favoured and that the Z,Z and E,E tautomers were not. This demonstrated that two hydrogen bonding interactions were taking place to two different hydrogen bonding groups in the binding site, thus disproving the chelation bonding theory.		
incorrect	**a.#**	The study showed that only the *Z,E* tautomer is favoured.
incorrect	**b.#**	The study established that there is only one hydrogen bonding interaction between the guanidine unit and the binding region.
incorrect	**c.#**	The study showed that only the *E,Z* tautomer is favoured.
Correct	**d.#**	The study disproved the chelation theory of hydrogen bonding.
END OF QUESTION		

Type: matching	
Title: Chapter 25 – Question 37	
37)	The following diagram shows a parietal cell releasing HCl into the stomach. Identify A-D. 
Feedback: The proton pump is responsible for pumping protons out of the parietal cell and potassium ions into the parietal cell. Acetylcholine is a neurotransmitter that interacts with cholinergic receptors on parietal cells to increase the release of hydrochloric acid into the stomach. The hormone gastrin and the local hormone histamine have the same effect when they interact with their corresponding receptors.	

a.#	Gastrin	=	D.
b.#	Histamine	=	B.
c.#	Proton pump	=	A.
d.#	Acetylcholine	=	C.
END OF QUESTION			

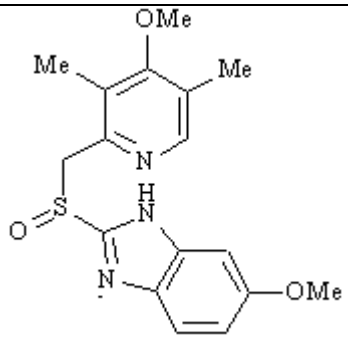
Type: matching	
Title: Chapter 25 - Question 38	
38)	The following diagram illustrates a parietal cell where feature F is the proton pump. What are the species labelled A-D?



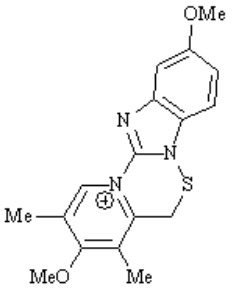
Feedback: Feature F is the proton pump responsible for pumping protons into the lumen of the stomach. Potassium ions are brought into the parietal cell in exchange and the process is powered by the hydrolysis of ATP. Chloride and potassium ions depart the parietal cells through ion channels which are separate from the proton pump.

a.#	Hydrochloric acid	=	D.
b.#	Chloride ions	=	C.
c.#	Potassium ions	=	B.
d.#	Protons	=	A.
END OF QUESTION			

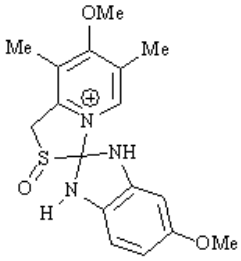
Type: multiple choice	
Title: Chapter 25 - Question 39	
39)	The following structure is the proton pump inhibitor omeprazole



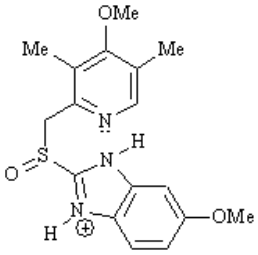
Omeprazole undergoes a series of arrangements in the acid conditions of the canaliculus before reacting with cysteine residues of the proton pump. Which of the following structures is the structure responsible for that alkyation?



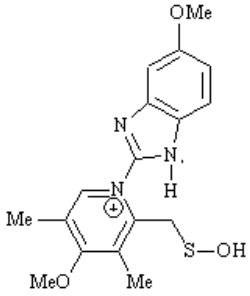
A



B



C



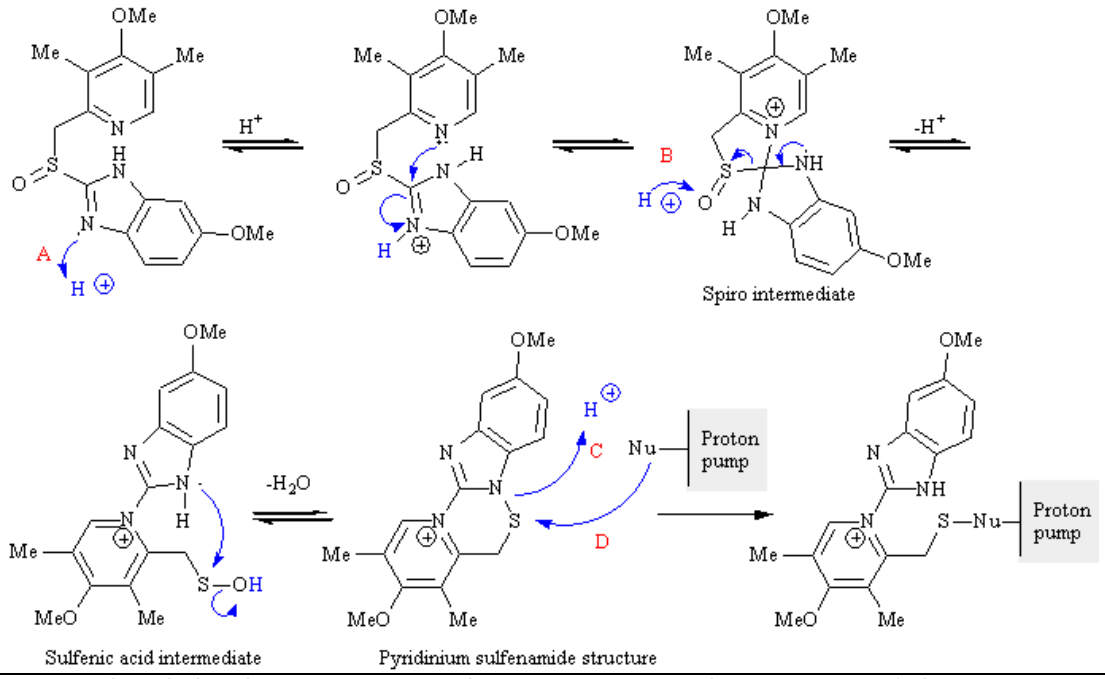
D

Feedback: The thiol group of cysteine acts as a nucleophile and reacts with structure A to produce a disulfide covalent bond between the amino acid and the drug. The other structures shown are formed in the mechanism by which omeprazole is converted to structure A.

Correct	**a.#**	Structure A.
incorrect	**b.#**	Structure B.
incorrect	**c.#**	Structure C.
incorrect	**d.#**	Structure D.

END OF QUESTION

Type: multiple choice	
Title: Chapter 25 - Question 40	
40)	The following mechanism shows how proton pump inhibitors are activated. Which arrow is incorrect?



Feedback: Arrow B is pointing the wrong way. Curly arrows represent the movement of electrons. A proton has no electrons and so it never makes sense to draw an arrow from a proton.

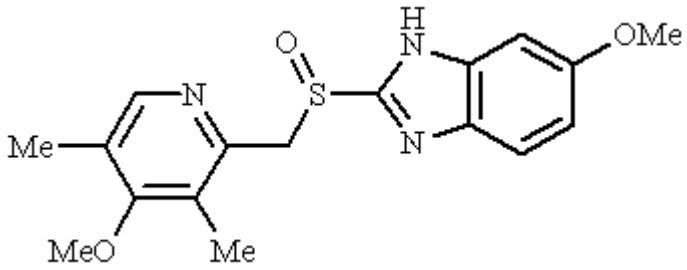
incorrect	**a.#**	A.
Correct	**b.#**	B.
incorrect	**c.#**	C.
incorrect	**d.#**	D.

END OF QUESTION

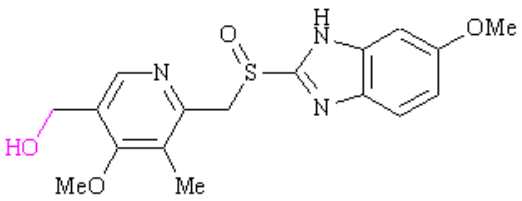
Type: multiple choice

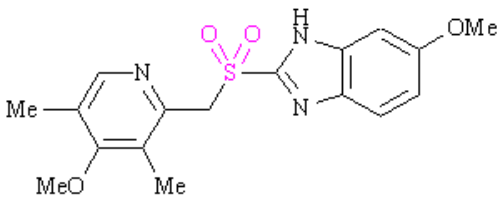
Title: Chapter 25 - Question 41

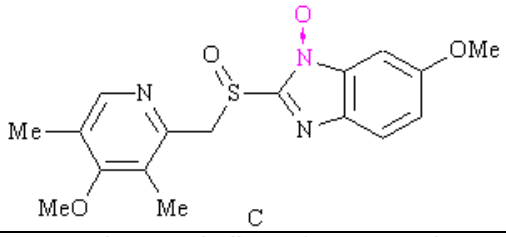
41) The following molecule is omeprazole, an important proton pump inhibitor.

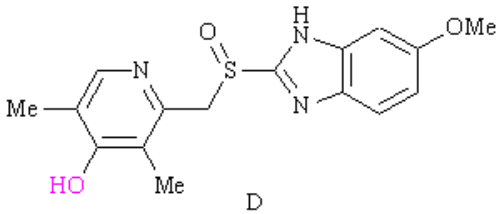


Which of the following structures is a metabolite?

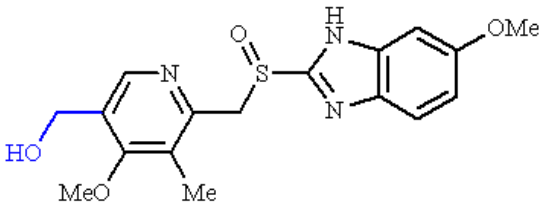


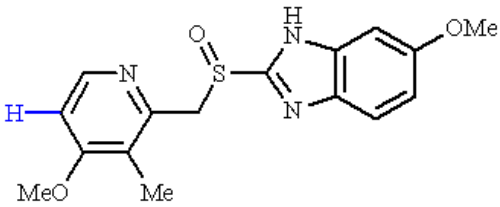






Feedback: The two major metabolites of omeprazole are shown below.





Correct	**a.#**	Structure A.
incorrect	**b.#**	Structure B.
incorrect	**c.#**	Structure C.
incorrect	**d.#**	Structure D.

END OF QUESTION